in the soil and air. They are aerobic or facultatively anaerobic, spore-bearing organisms. *Bacillus anthracis* is responsible for diseases such as anthrax and *Bacillus botulinium* causes food spoilage. *Bacillus* thuringiens synthesises a toxin that is active against insects. *Bacillus subtilis* is used for genetic engineering and many cloning vectors are available.

BACK (DORSUM): In humans, it is the posterior portion between the neck and the pelvis. The back is supported by the spinal column, which is bound together by ligaments and supported by muscles that also control posture and movement.

BACK BUTTON: A basic control used by a variety of technologies. In a Web browser, a back button allows an end user to navigate to a previously viewed Web page. A back button is also a feature of smartphone software and other consumer oriented technologies.

BACK DONATION: A form of chemical bonding in which a ligand forms a sigma bond to an atom or ion by donating a pair of electrons and the central atom donates electrons back by overlap of its *d*-orbitals with empty *p*- or *d*-orbitals on the ligand. It is very common in transition metal carbonyls (CO), nitrosy (NO), cyanides (CN) and phospines (PR_3). The ligands are usually referred to as pi acceptors.

BACK ELECTROMOTIVE FORCE (BACK EMF; COUNTER-ELECTROMOTIVE FORCE): An electromotive force set up in an electrical circuit by a changing electromagnetic field due to a pulsating current or an alternating current to oppose the flow of current within the circuit. Back emf is also the voltage that occurs in an electric motor where there is a relative motion between the armature of the motor and the external magnetic field. This voltage opposes the original applied voltage and hence the back emf. It is the effect of Lenz's Law of electromagnetism. Also, in an electric cell, polarisation causes a back emf. In polarisation, a layer of neutral hydrogen is formed near the copper rod thus partially insulating it. This results in weakening the action of the cell by increasing its internal resistance, resulting in a back emf, in opposition to the conventional current flow (from positive to negative poles), due to copper and zinc. The effect of the back emf is to slow down the growth of the current to its final value.

BACK ELECTRON TRANSFER: A term often used to indicate thermal reversal of excited state electron transfer restoring the donor and acceptor in their original oxidation level. In using this term one should also specify the resulting electronic state of the donor and acceptor.

BACK SCATTER COEFFICIENT: The number of back scattered electrons (BSEs) generated per primary electron for a given specimen and experimental condition. It depends on (mean) atomic number of the excited area of the sample, angle between electron beam and sample surface, primary electron energy and thickness of the sample.

BACK SUBSTITUTION: A method that is used in solving a system of equations that is in echelon form. The last equation in the echelon equation is used to solve for the unknown variable appearing in it. The other unknown variables are set equal to the parameters taking arbitrary values. This solution can be substituted into the previous equation, which can then likewise be solved for the first unknown appearing in it. The process continues in the same manner until all the unknown variables are computed.

BACK TITRATION: A titration, which is done in reverse. Instead of titrating the original sample, a known excess of standard reagent is added to the solution and the excess is titrated. A back titration is useful if the endpoint of the reverse titration is easier to identify than the endpoint of the normal titration, as with precipitation reactions. Back titrations are also useful if the reaction between the analyte and the titrant is very slow or when the analyte is in a non-soluble solid.

BACK-END SYSTEM: Any software performing either the final stage in a process or responses to what the front end has initiated. The task is performed in the background without the user being aware of it. Back-end systems deal with databases and data processing components, so as to launch the operating system's programs in response to front-end system requests and operations.

BACK-HACK: The process of identifying attacks on a system and, if possible, identifying the origin of the attacks. Back hacking can be thought of as a kind of reverse engineering of hacking efforts, where security consultants and other professionals try to anticipate attacks and work on adequate responses.

BACK-SCATTERED ELECTRON IMAGING: In scanning electron microscopy, the use of a special detector to produce an image from the back-scattered electrons given off from below the surface of the specimen rather than the secondary electrons used in conventional imaging. The image obtained consequently includes subsurface information.

BACK-SCATTERED ELECTRONS (BSEs): All primary high-energy electrons originating in the electron beam, which are scattered out of the original direction and retransmitted through the surface of the solid. In practice, they are electrons emitted from the surface of a solid under electron bombardment which have a kinetic energy in the range between 50 eV and excitation energy (E_0). BSEs are used to detect contrast between areas with different chemical compositions.

BACKBITING: A rearrangement that can occur in some polymerisation reactions involving free radicals. A radical that has an unpaired electron at the end of the chain changes into a radical with the unpaired electron elsewhere along the chain, the new radical being more stable than the one from which it originates. For example, the radical $RCH_2CH_2CH_2\text{-}CH_2CH_2CH_2^{\cdot}$ may change into $RCH_2CH^{\cdot}CH_2CH_2CH_2CH_3$. The rearrangement is equivalent to a hydrogen atom being transferred within the molecule. The new unpaired electron initiates further polymerisation, with the production of polymers with butyl ($CH_3CH_2CH_2CH_2\text{–}$) side chains.

BACKBONE: 1. Also known as **vertebral column or spinal column**, it is a lineup of bones separated by intervertebral disks and held together by muscles and tendons that extend from the cranium to the coccyx or the end of the tail enclosing and protecting the spinal cord. 2. In telecommunications, **backbone network** is the part of the computer network infrastructure that interconnects different networks and provides a path for exchange of data between these networks. In other words, it is the central network to which smaller networks, normally of lower speed, connect or a wide-area network (WAN) such as a public packet-switched datagram network. They are designed to transfer network traffic at high speeds and to maximize the reliability and performance of large-scale, long-distance data communications. A backbone may interconnect different local area networks in offices, campuses or buildings. It may consist of a set of nodes and their interconnecting links that form a central, high-speed network interconnecting other typically lower-speed, networks or client nodes. In a local area network multiple-bridge ring configuration, the rings are connected by means of bridges or routers.

BACKBULB: An old pseudobulb, which is usually leafless. Pseudobulbs often have viable buds and may produce a new shoot, when severed from the main part of the plant.

BACKCROSS: A cross between an individual and one of its parents. It is a breeding method used to determine the genetic makeup (genotype) of an individual organism. It can also be defined as a genetic cross in which a heterozygous organism is crossed with one of its homozygous parents or a mating between individuals of the parental generation (P) and the first generation of offspring (F1) in order to identify hidden recessive alleles. A backcross performed to ascertain the genotype of an individual is called a **testcross**.

BACKDOOR: A technique in which a system security mechanism is bypassed undetectably to access a computer or its data. The backdoor access method is sometimes written by the programmer who develops the program.

BACKGROUND: The term employed to designate the value indicated by a radiation measuring device in the absence of the source whose radiation is to be measured, when the device is placed under its normal conditions of operation.

BACKGROUND PROCESS: 1. A process that does not require operator intervention but can be run by the computer while the workstation is used to do other work. 2. In the AIX operating system, a mode of program execution in which the shell does not wait for program completion before prompting the user for another command.

BACKGROUND RADIATION: Low intensity ionising radiation present on the Earth's surface and in the atmosphere. It is radiation arising from a radioactive material other than the one directly under consideration. Alternatively, it is a radiation, which originates from the source and reaches the detector when no analyte is present. Background radiation may be caused by radiation from space, the decay of natural radioisotope and fallout from the testing of nuclear weapons in the 1960s. It is believed that this background radiation may cause thousands of cancer deaths a year and also plays a part in genetic changes.

BACKING STORE: A supplementary computer memory, usually in the form of magnetic disks, in which data and programs are held permanently for reference. It is less costly and can hold more information than semiconductor RAM but with lesser speed of access to information than in RAM.

BACKLINK: An incoming link from an external website to specific webpage. For example, if 100 other websites link to a published webpage, then that webpage has 100 backlinks. Links to the page from within the main website are not included in the backlink total.

BACKOUT PLAN: An information technology governance integration approach that specifies the processes required to restore a system to its original or earlier state, in the event of failed or aborted implementation.

BACKPORT: To retroactively supply a fix, or a new feature, to a previous version of a software product at the same time (after) supplying it to the current version.

BACKRONYM: An existing word which has been turned into an acronym by inventing a phrase whose initial letters match the word. An example is SPECTRE: Special Executive for Counter-intelligence, Terrorism, Revenge and Extortion. In this case, "SPECTRE" is an existing word, and "Special Executive for Counter-intelligence, Terrorism, Revenge and Extortion" is the invented phrase.

BACKSCATTER (BACKSCATTERING): In physics, it is the deflection of radiation or particles by electromagnetic or nuclear forces through angles greater than 90° to the initial direction of travel. In the assay of radioactivity, it applies to the scattering of radiation into the radiation detector from any material except the sample and the detector. In light scattering, it is said to occur when the scattering angle is 180°. Conversely, forward scattering occurs when the angle is 0°.

BACKSPACE: An operation carried out on the keyboard of a computer, which allows the cursor to move one space to the left by removing characters in a word processing document.

BACKUP (CACHE): A resource that can be used as a substitute in the event of a fault in a component or system. A backup file is a copy of a file taken and kept separately, in case the original is lost.

BACKWARD COMPATABILITY: A software program or hardware device that is compatible with previous versions of operating systems. A new standard product or model is considered backward compatible when it is able to read, write or view older formats.

BACKWARD DIFFERENCE: A finite difference operator, with the symbol ⍰, with an equation of the form $\nabla f(x) = f(x) - f(x-h)$, where h is a constant denoting the difference between successive points of interpolation or calculation.

BACKWARD ERROR ANALYSIS: In numerical linear algebra, it is a form of error analysis, which is intended to replace all errors made in the course of solving a problem by an equivalent perturbation of the original problem. Bounds for the perturbations are determined and these can be inserted into standard results, thus producing a bound for the error in the numerical solution.

BACKWARD INDUCTION: A form of induction in which the inductive step is an argument to the effect that what fails at step $k + 1$ must fail at or before step k.

BACONER: A pig bred and reared specially for bacon. **Bacon** is the cured and sometimes smoked meat from the back, sides and belly of a pig.

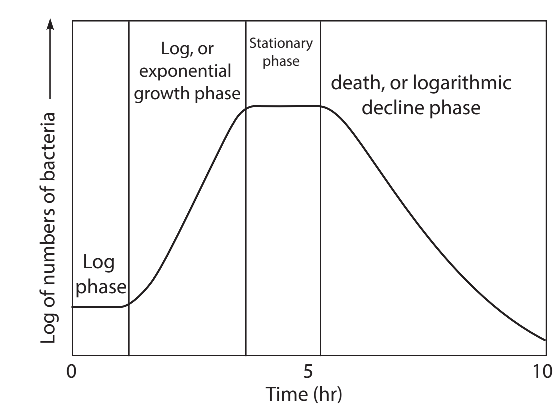
BACTEREMIA (SEPTICAEMIA): The temporary presence of bacteria in the bloodstream. It may be followed by the development of embolic infections such as arthritis, meningitis, endocarditis including liver and lung abscesses. Bacteremia occurs for a few hours after many surgical procedures and may also occur with infections such as tonsillitis.

BACTERIA: A diverse group of microscopic unicellular organisms with prokaryotic cells. A bacterial cell may be spherical *(coccus)*, rod-like *(bacillus)*, spiral *(spirillum)*, comma-shaped *(vibrio)*, corkscrew *(spiro chaetae)* or filamentous. They lack a distinct nuclear membrane or nucleus and have a cell wall of a unique composition. They usually reproduce by binary fission. Some are parasitic, however most bacteria are saprophytic and perform essential roles in the ecosystem like digesting cellulose in the intestine of herbivores, decomposing waste or dead matter and improving soil fertility by processing nitrogen compounds or fixing nitrogen. They are also used in industry such as in cheese making, manufacture of dairy products, preparation of antibiotics and other pharmaceutical items, retting of fibres and in the treatment of sewage.

BACTERIAL ARTIFICIAL CHROMOSOME (BAC): An engineered DNA molecule used to clone DNA sequences in bacterial cells such as E. coli. BACs are often used in connection with DNA sequencing. Large segments of another species DNA, ranging from 100,000 to 300,000 base pairs can be inserted into BACs. The BACs, with their inserted DNA, are then taken up by bacterial cells. As the bacterial cells grow and divide, they amplify the BAC DNA, which can then be isolated and used in sequencing DNA.

BACTERIAL DISEASE: A disease caused by bacteria or their products. It is the most common infectious diseases and includes minor skin infections to bubonic plague and tuberculosis. Many bacteria have no effect and some are beneficial, while only a small number leads to disease. This may be as a result of bacterial growth, the inflammation in response to it or of toxins such as tetanus, botulism and cholera. Bacteria may be contracted from the environment, other animals or humans or from other parts of a single individual.

BACTERIAL GROWTH CURVE: A graph representing the changes in size of a bacterial population in culture over time. The bacteria are cultured in sterile nutrient medium incubated at the optimum temperature for growth.



Bacterial Growth Curve

BACTERIAL PHOTOSYNTHESIS: Photosynthesis performed by certain bacteria, for example, cyanobacteria (blue-green algae) just like green plants and algae. It is light-dependent, the bacteria use

B

B

chlorophyll-type molecules, called bacteriochlorophyll, to collect sunlight as an energy source to convert carbon dioxide to glucose, which is used for energy production and biosynthesis. Some other bacteria also use bacteriochlorophyll to collect sunlight to make photosynthesis, but unlike cyanobacteria, do not generate oxygen from the oxidation of water. These bacteria are called **anoxygenic photosynthetic bacteria** which include the purple phototrophic bacteria, green sulfur bacteria, green filamentous bacteria, the heliobacteria and the cyanobacteria.

BACTERIAL SPORES: A bacteria with a thick outer wall, enabling it to survive hostile environments otherwise not conducive to bacterial growth and reproduction. Endospores form within special vegetative cells called sporangia in response to adverse changes in the environment. The original cell replicates its genetic material and one copy of this becomes surrounded by a tough coating. The outer cell then disintegrates, releasing the spore that is now well protected against a variety of trauma, including extremes of heat and cold, radiation and an absence of nutrients, water or air.

BACTERIAL TRANSFORMATION: The incorporation of genetic material into a bacterium from DNA in the surrounding medium.

BACTERICIDAL: Capable of killing bacteria using a substance called bactericide. Examples of bactericide are antibodies, antiseptics, antibiotics and disinfectants.

BACTERIOCHLOROPHYLL: A form of chlorophyll, with the chemical formula $C_{52}H_{70}O_6N_4Mg$ that is found in photosynthetic bacteria, notably the green and purple bacteria.

BACTERIOCIN: Protein or peptidic toxins synthesised by bacteria which are lethal to other strains of bacteria.

BACTERIOLOGY: A specialised branch of biology that deals with the study of different forms of bacteria including identification, classification and using DNA composition, movement, growth, respiration, habitat, reproduction, excretion, irritability, nutrition and function in nature.

BACTERIOPHAGE (PHAGE): A virus that attacks bacteria. It infects quickly, multiply within and destroys the host cells. Bacteriophages are often named according to the bacteria they infect, for example, **actinophages**, which infect bacteria of the order Actinomycetales and **coliphages**, which infect various strains of Escherichia coli. Some temperate phages such as the lambda virus have been shown to transfer genes from one bacterium to another in a process termed **transduction**. A particular phage can usually infect only one or a few related species of bacteria. Such viruses are now of use in genetic engineering. They are used experimentally to identify and to control manufacturing processes such as cheese production.

BACTERIORHODOPSIN: A red-pigmented membrane-bound protein synthesised by light-loving (halopilic) archaebacterium that traps light energy to release adenosine triphosphate (ATP) without chlorophyll.

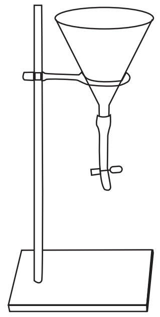
BACTERIOSTATIC: Capable of inhibiting or slowing down the growth and reproduction of bacteria without killing them. Examples are some antibiotics such as amphotericin B and polymyxin B.

BACTEROID: Any of various structurally modified bacteria such as those occurring on the root nodules of leguminous plants. It undergoes an increase in size and a change in shape to a characteristic X- or Y-shaped cell. Bacteroids lose their capacity for autonomous growth and are dependent on the host. It is in this metamorphosed state that the bacteria carry out nitrogen fixation.

BADLAND: Barren landscape cut by erosion into a maze of ravines, pinnacles, gullies and sharp edged ridges. They can be created by overgrazing and are so called because of their total lack of value for agriculture and their inaccessibility. This type is most common in arid or semiarid regions.

BAEOCYTES: Small and spherical reproductive cells formed by cyanobacteria through multiple fission.

BAERMANN FUNNEL: A device used to extract organisms such as nematodes, algae and protozoans from a soil sample or plant material. The organisms move away from the strong light and high temperature of one side and collect near the top of the funnel from where they can be released into the collecting jar.



Baermann funnel

BAEYER STRAIN THEORY: A theory in chemistry in which the four valences of carbon are normally directed symmetrically in space making angles of 109° 28′ with one another and deflection of these directions produces strain in the molecule such as in the formation of rings.

BAEYER TEST: A qualitative test for unsaturated compounds, named after the German organic chemist Adolf von Baeyer, in which alkaline potassium permanganate is used. For example, alkenes are oxidised to glycols and the permanganate loses its colour.

BAEYER-VILLIGER OXIDATION: A type of rearrangement reaction, usually an esterification reaction that is accompanied by an insertion of an oxygen atom between carbon-carbon bonds in which one of the carbons bears carbonyl oxygen by the addition of peroxy acid.

BAEYER, JOHANN FRIEDRICH WILHELM ADOLF VON (1835 - 1917): A German chemist who was the 1905 recipient of the Nobel Prize in Chemistry in recognition of his services in the advancement of organic chemistry and the chemical industry. He was born in Berlin and studied chemistry with Robert Bunsen at Heidelberg. Baeyer's chief achievements include the synthesis and description of the plant dye indigo, the discovery of the phthalein dyes and was the first to propose the correct formula for indole in 1869. His contributions to theoretical chemistry include the strain theory of triple bonds and strain theory in small carbon rings.

BAEYER, KARL ERNST VON (1792-1876): An Estonian-born German biologist who studied medicine and comparative anatomy before becoming professor of Zoology at Konigsberg University in 1817. Considered a founder of modern embryology, he discovered the notochord as well as the mammalian ovum.

BAFFLE CHAMBER: A part of the exhaust system of a furnace, fitted with metal or ceramic plates at an angle with the movement of emissions, where coarse particulate matter settles, so it is not discharged into the environment and can be gathered and disposed of safely in other ways.

BAG FILTER: A large bag constructed of a suitable fabric that is often tubular in shape, into which a particle-containing air stream flows.

BAGASSE: Fibres left after crushing sugar cane. Bagasse is used as a biofuel, to manufacture pulp for paper, used as a source of cellulose for manufacturing animal feeds and for manufacturing packaging products such as containers, plates and bowls.

BAGHOUSE: An installation which contains many bag filters in parallel so that the resistance to air flow in a large installation is not seriously increased by the addition of these controls.

BAGNOLD NUMBER: The ratio of grain collision stresses to viscous fluid stresses in a granular flow with interstitial Newtonian fluid. It was first identified by Ralph Alger Bagnold.

BAGOT: A rare breed of goat with long hair, usually with black head, neck and shoulders and the rest of the body white. Both sexes have horns.

BAIL: 1. A milking shed that can be moved from one place to another. 2. A bar separating horses in a shed.

BAINBRIDGE MASS SPECTROGRAPH: An instrument used to accurately determine atomic masses. It consists primarily of a velocity selector and a magnetic chamber.

BAINITE TRANSITION: A term that is sometimes used in metallurgy for the martensitic transition that describes the transition between a face-centred-cubic lattice and a body-centred-tetragonal lattice. An example is the transition between the face-centred-cubic lattice of austenite and the body-centred-tetragonal lattice of amartensite.

BAIRE CATEGORY: A measure of the size of sets in a topological space, named after the French analyst, René Baire (1874 - 1932). A set A, in a metric space is called nowhere dense if the closure $\bar{A}$ has an empty interior $(\bar{A})^{o} = \phi$. A set A that can be written as a countable union of nowhere dense sets is said to be first category or meagre and any other set is second category. The complement of a first category set is called **residual**. The rationals form a first category subset of the reals, as does the Cantor ternary set.

BAIRE CATEGORY THEOREM: The theorem, proved by René Baire (1874 - 1932), French analyst, in his 1899 doctoral thesis that every complete metric space is a Baire space.

BAIRE SET: A member of the smallest σ-algebra related to the continuous functions on the space containing all closed, compact subsets of a topological space.

BAIRE SPACE: A topological space with the properties that a countable family of open dense subsets of the space is itself dense in the space; named after the French analyst, René Baire (1874 - 1932). For example, any regular locally compact space is a Baire space.

BAIT PLANT: A plant, which is grown near or among crops in order to attract pests and direct them away from the crops.

BAKELITE: A trade name for certain phenol-formaldehyde resins, first introduced in 1909 by a Belgian US chemist, Leo Hendrick Baekland, hence Bakelite. The monomers are phenol and methanal hydrate. It was used as a substitute for hard rubber, amber or celluloid; for insulating electrical apparatus and for the manufacture of certain machinery gears, buttons, billiard balls, pipestems and umbrella handles, etc.

BAKER'S TRANSFORMATION: The transformation of the unit square with Lebesgue measure that is given analytically by $T(x, y) = (2x, y/2)$ for $0 \le x < 1/2$ and $T(x, y) = (2x - 1, 1/2[y + 1])$ for $1/2 \le x < 1$. This corresponds to first deforming the unit square into a rectangle twice as wide and half as tall and then cutting this rectangle along the line $x = 1$ and placing the right-hand part above the left to reform a square.

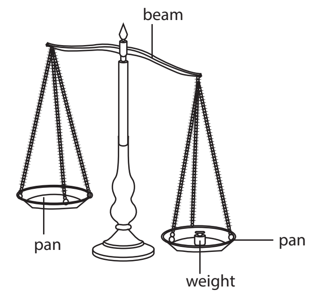
BAKER'S YEAST: Strain of the yeast, *Saccharomyces cerevisiae*, which is used in bread making. Fermentation of the sugar in the wheat dough produces carbon dioxide to cause the dough to rise.

BAKING: 1. The cooking of food such as bread by dry heat in an oven in which the action of the dry convection heat is modified by steam. 2. The process in which the carbonaceous binder, usually coal tar pitch or petroleum pitch, as part of a shaped carbon mix is converted to carbon, yielding a rigid carbon body by the slow application of heat. The process can take as little as 14 days in coarse-grained, electrothermic grades (low binder level) and as long as 36 days in ultra-fine-grained speciality grades (high binder level). The final baking temperature can be in the range of 1100 – 1500 K, depending on the grade.

BAKING POWDER: A mixture of sodium bicarbonate and solid tartaric acid used in baking as a raising agent. When added to flour, the presence of water and heat causes carbon dioxide to be released, which makes the dough rise and become light during baking.

BAKING SODA (SODIUM HYDROGEN CARBONATE; BICARBONATE OF SODA): A sodium salt of carbonic acid, with the chemical formula $NaHCO_3$. It occurs most commonly as a crystalline heptahydrate, which readily effloresces to form a white powder, the monohydrate. It can be extracted from the ashes of many plants. It is synthetically produced in large quantities from salt and limestone in a process known as the Solvay process. It is used to neutralise the acidic effects of chlorine and raise pH, as a water softener during laundry, a descaler, etc. It is also used in the manufacture of glass, used as an electrolyte, as a food additive, etc.

BALANCE: 1. A device that measures an object's mass or weight by setting up equilibrium between forces. Examples are the beam, leno balance, electric/digital balance and the spring balances. The simple beam balance, for example, has a central pivot, a beam and two scales or pans. 2. State of quality and calm between items. 3. Attainment of equilibrium in the orientation of the body that is detected and maintained by the semi-circular canals of the ear in coordination with the cerebellum of the brain.



Beam balance

BALANCE OF NATURE: A concept in ecology that describes the inherent equilibrium in most ecosystems, with plants and animals interacting so as to produce a stable, continuing system of life on earth. Organisms in the ecosystem are adapted to each other. For example, plants produce the oxygen needed by animals, while plants use the waste products of animal respiration, such as carbon dioxide, as a raw material in photosynthesis.

BALANCE RATION: In agriculture, it is the amount of feed given to an animal with the necessary nutrients in the right proportions needed by the animal for normal growth and optimum performance.

BALANCE WHEEL: A wheel in a machine, especially in an analogue clock, that regulates the rate of movement of the main mechanism.

BALANCED: Of a set, being a subset B of a vector space with the property that tx belongs to the set B whenever x belongs to B and $|t| \le 1$ (in modulus). An example of a balanced set is the unit disk in the Cartesian plane.

BALANCED BLOCK DESIGN: A type of block design in which all the blocks are of equal size and all treatments occur equally often.

BALANCED CHEMICAL EQUATION: A chemical equation in which the total number of atoms and charges of each element on the left hand side is equal to the total number of the same atoms and charges on the right hand side. For example, the reaction between tetraoxosulfate(VI) acid (H_2SO_4) and sodium hydroxide (NaOH) to form sodium tetraoxosulfate(VI) (Na_2SO_4) and water can be written as: $NaOH + H_2SO_4 \rightarrow H_2O + Na_2SO_4$. This is an incomplete equation, since the same number of atoms does not appear on both sides of the reaction (for example, one sodium atom appears in the reactants, but two sodium atoms appear in the products). Such a reaction could not actually occur. Correcting this fault is called balancing the equation. Placing a 2 in front of the NaOH balances the number of sodium atoms, as well as the hydrogen and oxygen atoms. The balanced equation therefore is: $2NaOH_{(aq)} + H_2SO_4 \rightarrow 2H_2O_{(l)} + Na_2SO_{4(aq)}$. This equation implies that two molecules of NaOH are necessary to react with each molecule of H_2SO_4. Photosynthesis can be represented by the following chemical equation: $6CO_2 + 6H_2O$ ------> $C_6H_{12}O_6 + 6O_2$.

BALANCED DIET: A meal containing the right amounts and types of nutrients for healthy living and growth in animals or the correct nutritional components required for health, generally used in reference to humans and domesticated animals. In humans, this diet includes carbohydrates, protein, fat, vitamins, water, minerals and roughage.

BALANCED FORCES: Two forces acting in opposite directions on an object, which are equal in magnitude. Anytime there is a pair balanced forces acting alone on an object, it is in equilibrium and the object stays still or continues to move at the same speed and in the same direction. For example, a person standing on the floor has two forces acting upon him, which are the force of gravity which exerts a downward force and the floor that exerts an upward force. Since these two forces are of equal magnitude and in opposite directions, they balance each other and the person is at equilibrium and stays at rest unless acted upon by an unbalanced force.

BALANCED GROWTH: In microbiology, it is a microbial growth where all cellular constituents are synthesised at constant rates, in relation to each other.

BALANCED POLYMORPHISM: The co-existence of two or more distinctly different forms of plant or animal species in a stable ratio or proportion in a population. Example is the existence of equal proportions of sicklers, carriers and normal individuals for the sickle cell trait.

BALANCED RATION: The feed or feed mixture which contains all the essential nutrients in the right quality and quantity as needed by the animals for maintenance, growth, work and production. In order to meet the nutritive requirements, the amount of feed offered to an animal in twenty four hours is called ration. It should possess the following characteristics: contain all the essential nutrients like protein, fats, carbohydrates, minerals and vitamins in the right proportion as needed by the body, be well balanced and economical, contain enough amount of crude fibre in order to stimulate the wall of gastrointestinal tract for maximum secretion and excretion of digestive juices, be nontoxic, be easily available locally and be easily digestible and palatable to animals.

BALANCED RISK: An attempt to balance the risk of exposure to ionizing radiation with the benefits of its use in diagnosis and treatment of illness.

BALATA: A natural rubber-like material very similar to gutta percha. It is obtained from the latex of a tropical American tree, *Manilkara bidentata* and used in the production of machinery belts and golf ball.

BALBIANI RINGS (CHROMOSOME PUFF): Large swollen region of polytene chromosomal puffs found primarily in the salivary glands of diptera. They are sites of RNA transcription produced by repeated rounds of DNA synthesis without mitosis and are bundles of chromonemata filaments that are not separated.

BALDNESS (ANDROGENETIC ALOPECIA; ANDROGENIC ALOPECIA): The loss of hair from the upper scalp, common in older men, but which can occur from early twenties and progresses at different rates in different men. In men it is also called **male pattern hair loss**. It also affects women. Its onset and extent are influenced by genetic make-up and the level of male sex hormones. It is sex-linked; the gene is located on the sex chromosome.

BALDWIN EFFECT: Originally called "organic evolution" by US psychologist Mark Baldwin, who in 1896, proposed that acquired behavioural traits, can be transmitted to subsequent generations without contradicting Darwin's theory of natural selection in favour of Lamarckism. It has been invoked to explain the rapid evolution of complex behaviours, cognitive skills and language in primates, especially humans.

BALDWIN'S RULES: A set of empirical rules for certain formations of 3- to 7-membered rings. The predicted pathways are those in which the length and nature of the linking chain enable the terminal atoms to achieve the proper geometries for reaction. The disfavoured cases are subject to severe distortions of bond angles and bond distances.

BALEEN (WHALEBONE): Horny plates found at the lateral sides of each upper jaw of toothless whales forming a filter, which is used for feeding by filtering tiny crustaceans called krill by forcing out large quantities of seawater swallowed into the mouth.

BALEENS (MYSTICETES): Large whales of the suborder Mysticeti, characterised by baleen plates used for feeding, two blowholes, symmetrical skull, and lower jawbones that are solid and do not join in the middle. They include blue whales, finback whales, gray whales, humpback whales, bowhead whales, etc. The females are larger than the males. They lack teeth and use the baleen plates to filter great quantities of krill (small fish, crustaceans or plankton) in one gulp.

BALL: 1. A spherical object or entity. In mathematics, it is the space inside a sphere. 2. A set in a metric space consisting of all the points whose distance from a given point is less than a given constant in an open ball; in a closed ball, the distance is less than or equal to a given constant. An open ball is an open set in the metric space and the open ball centred on a with radius ⍰ is often denoted N(a, ⍰) or N⍰ (a, ⍰). A closed ball is a closed set and is denoted B(a, ⍰) or $B(a)$ or otherwise. A ball is sometimes called a disk, especially in the complex plane or a sphere. 3. A solid spherical or pointed projectile, such as one shot from a cannon. 4. A round mass of material that has been pressed together, rolled or wound into shape. 5. A rounded part or protuberance, especially of the body.

BALL BEARING: A type of bearing for rotating machinery. It consists of two concentric rings of steel, which together form bearing races. A number of steel balls are positioned between the bearing races to maintain a separation. The inner ring is fixed to a shaft and the outer one to a support. As one of the bearing races rotates it causes the balls to rotate. Because the balls are rolling they have a much lower coefficient of friction than if two flat surfaces were sliding against each other. Its purpose is to reduce rotational friction and support radial and axial loads.

BALL LIGHTNING: A white or reddish luminous sphere, about 50 cm in diameter that sometimes appears at ground level in a thunderstorm. It moves slowly and usually disappears without detonating.

BALL VALVE: A valve that regulates the flow of a fluid by the action of fluid presure raising a ball to open a hole for the fluid to flow and blocking the hole to stop the flow.

BALL-AND-SOCKET JOINT: 1. Also called enarthrosis, cotyloid joint, enarthrodial joint or spheroid joint, it is a joint in which the ball-shaped surface of a bone fits into a cavity or socket of another, allowing considerable free movement in any direction. An example is the joint between the pelvis and the femur. 2. In engineering, also called ball joint, it is a mechanical device that permits rotary movement in all directions through the movement of a ball in a socket.

BALLED AND BURLAPPED: Plants that have been dug out of the ground with a ball of soil around the roots, wrapped in burlap and tied for transport. In landscape architecture, this is a method of preparing a plant or tree for transplantation.

BALLING GUN: A device used to place a pill in an animal's throat. It consists of a tube from which the air is partially exhausted. The ball is held on the end of the tube by atmospheric pressure and is released by a piston when fairly within the oesophagus.

BALLISTIC GALVANOMETER: A moving coil galvanometer designed for measuring charge by detecting a surge in current. It has a coil which rotates between magnets but its rotating coil has a large moment of inertia. It is used to measure quantity of charge rather than currents, for the large moment of inertia permits the passage of a quantity of charge before the coil moves significantly. The passage of the charge produces an impulse, a momentary torque, which causes the coil to swing slowly to some maximum position. It is often used to standardise capacitors.

BALLISTIC MISSILE: A missile that has a ballistic trajectory over most of its flight path, regardless of whether or not it carries a weapon. A ballistic missile is initially powered and guided but falls under gravity onto its target.

BALLISTIC TRAJECTORY: Is the path followed by an object that is being acted upon only by gravitational forces and the resistance of the medium through which it passes.

BALLISTIC PENDULUM: 1. A device used to measure the velocity of a projectile such as a bullet from which it is possible to calculate the velocity and kinetic energy. They have been largely rendered obsolete by modern chronographs, which allow direct measurement of the projectile velocity. It is still useful because of its simplicity and value in demonstrating properties of momentum and energy. 2. A mass of relatively soft material which is suspended from a horizontal bar. The angle through which this mass is displaced when struck by a projectile in flight enables the momentum and hence the velocity of the projectile to be calculated.

BALLISTICS: The study of the motion and impact of projectiles such as bullets, bombs and missiles, especially those that have a parabolic flight path from one point on the Earth's surface to another. **Interior ballistics** deals with the propulsion and the motion of a projectile within a gun or firing device. **Exterior ballistics** is concerned with the motion of a projectile while in flight and includes the study not only of the flight path of bullets but also of bombs, rockets and missiles.

BALLISTITE: An explosive that contains cellulose nitrate and nitroglycerin that burns with relatively little smoke and used as a propellant.

BALLISTOSPORE: A spore that is violently projected, for example, the basidiospores of the Hymenomycetes.

BALLOON: A large inflatable and flexible bag filled with gas, such as helium, hydrogen, nitrous oxide or air to make it rise in the air, often carrying a basket for passengers. Some balloons are purely decorative, while others are used for specific purposes such as meteorological observation, medical treatment, military defence or transportation. A balloon's properties, including its low density and relatively low cost, have led to a wide range of applications.

BALMER SERIES (BALMER LINES): Visible atomic spectrum of hydrogen, consisting of a unique series of energy emission levels (transition involving the second energy levels) which appears as lines at four wavelengths: 410 nm, 434 nm, 486 nm and 656 nm. The Balmer series is calculated using the Balmer formula, an empirical equation discovered by Johann Jakob Balmer (1825-1898), Swiss mathematician and physicist in 1885. The wavelength may be represented by $\frac{1}{\lambda}=R\left(\frac{1}{2^2}-\frac{1}{n^2}\right)$ where n = 3, 4, 5, etc. and R is the Rydberg constant, whose value is 1.097 x 10^7 m^{-1}.

BALSAM: Any of several plants and shrubs, especially of the genus *Commiphora* from which oily, gummy aromatic resins containing benzoic acid and cinnamic acid are extracted and used as a flavour in medicines. Some of the plants from which the resins are extracted include Balsam fir, Canada balsam, Balsam poplars, Peru balsam, etc.

BAMBARA BEAN (BAMBARA GROUNDNUT; STONE GROUNDNUT; *Vigna subterranea*): An annual creeping, leguminous plant with trifoliate leaves and deep taproot system that originated in West Africa and is cultivated in the dry tropical regions in Southern Africa, Zimbabwe, America, Asia and Australia. It thrives in hot, dry, marginal soils, better than most crops such as sorghum, maize and peanuts. The lateral roots contain nitrogen-fixing bacteria in the nodules. Flowers are borne in papilionaceous racemes; when fertilised, the flower is drawn below the soil through a tunnel to produce fruits which are dry indehiscent pods containing 2 or 3 pods, with seeds that are round, smooth and very hard. Seed colours are cream, black and red. Like other lugumes, it improves soil fertility and a good crop in a crop rotational plan. The bambara-bean is the third most commonly eaten legume after cowpea and groundnut. The seeds are a potentially valuable protein source for livestocks and humans.

BAMBOO: A tall plant of the grass family with hard hollow jointed stems called culms. It is classified in the division *Magnoliophyta*, class Lilopsida order *Cyperales*, family Gramineae. It grows mainly in tropical and subtropical parts of Asia, Africa and America. The most common bamboo is *Bambusa arundinacea*. Bamboo is one of the fastest-growing plants on Earth with reported growth rates of 100 cm in 24 hours. Some of the largest timber bamboo can grow over 30 metres tall and 15-20 cm in diameter. The vast majority of herbaceous bamboos flower annually, most of the woody bamboos flower very infrequently, about once every 20 to 120 years and may die in part or completely due to some possible causes. There are three categories of flowering in bamboo which largely depend on species and circumstances. They are: continuous flowering, sporadic flowering and gregarious flowering. Bamboo is used as wood for construction work, furniture, utensils, fibre, paper, fuel, etc; and also used for decorative purposes, both in gardens and in art. Bamboo sprouts are eaten as a vegetable.

BANACH ALGEBRA: An algebra over the real or complex field that is also a complete normed space and that satisfies the inequality $\|xy\| \le \|x\|.\|y\|$ for all elements of the space.

BANACH LIMIT: Any translation-invariant positive linear functional on the vector space of all bounded sequences that sends a constant sequence to that constant value. Such limits must assign each convergent sequence its correct limit and can be shown to exist in various non-constructive fashions.

BANACH SPACE: A complete normed space. The vector space of continuous functions on a compact set in norm is a Banach space.

BANACH-ALAOGLU THEOREM (ALAOGLU THEOREM): The theorem that in a dual Banach space the unit ball is weak star compact and, more generally, that the polar of a neighbourhood of the origin in a topological vector space is also weak star compact.

BANACH-TARSKI THEOREM: The seemingly paradoxical result that if A and B are bounded subsets of Euclidean space of three or more dimensions and both sets have interior points, then A can be finitely decomposed and reassembled using rigid motions to form a set congruent to B. In particular, a solid sphere may be so transformed into two spheres each the size of the original.

BANACH-STEINHAUS THEOREM (UNIFORM BOUNDED PRINCIPLE): The theorem that a pointwise-bounded family of continuous linear operators between a Banach space and a normed space is equicontinuous (uniformly bounded), if for each x in the unit ball, $\sup_i \|T_i(x)\|$ is finite, then actually $\sup_i \|T_i\|$ is finite.

BANACH, STEFAN (1892 – 1945): A Polish mathematician who is well known for his discovery of functional analysis. He became a professor at the University of Lvov in 1927; and made major contributions to the theory of topological vector spaces; and also contributed to measure theory, integration and orthogonal series.

BANANA (*Musa paradisiaca*): A tropical herbaceous flowering plant belonging to the family Musaceae, that grows to about 6-7.5 metres tall. The plant is not really a tree, but a large herb. It has an underground stem with adventitious roots that has abundant food for the plant. The main stem called pseudostem is very fleshy and consists mostly of water, formed by the tightly packed overlapping leaf sheaths. Around the base, it has other stems called suckers that grow into other plants. When the main plant produces its fruit and dies, another sucker replaces it. The plant has large spirally arranged leaves that are closely rolled up one over the other, in which a bud develops to produce other leaves. They unfurl, as the plant grows. The sizes of the leaves vary from variety to variety and can reach about 3 metres in some varieties. It bears male and female flowers in a spike at the tip of the stem, covered by purple bracts that turn downwards and opens, showing the flowers appearing in groups called hands. The female flowers develop into bananas. Some hands of sterile flowers appear above the female hands, which are shed. The male flowers are borne in the upper rows. The fruit is covered with a rind which may be green, red or yellow. The fruit can be eaten fresh or cooked or processed into starch, chips, puree, beer and vinegar or dehydrated to produce dried fruit. The flowers of the plant may be used as a vegetable. Fresh leaves have high protein content and can be fed to cattle. The uses for the leaves include lining pots or wrapping food, etc.

BANANA BOND (BENT BOND): An informal description for the type of electron deficient three-centre two-electron bond holding the B-H-B bridges in boranes and similar compounds.

BAND: 1. A specific range of wavelengths or frequencies of electromagnetic radiation. 2. In telecommunications, it is also known as **frequency band**, which is a specific range of frequencies in the radio frequency (RF) spectrum, which is divided among ranges from very low frequency (VLF) in the range of 3 kHz to 30 kHz (wavelengths from 10 to 100 kilometres) to extremely high frequency (EHF) in the range of 30 to 300 gigahertz. 3. A group of very closely spaced electron energy levels, the distribution and nature of which determine the electrical properties of a material.

BAND GAP (GAP ENERGY): The energy difference measured in electron volts (eV) between two allowed bands of electron energy in a solid that is located between the valence band and conduction band. In a semiconductor, such as silicon, it represents the energy difference between the highest valence band and the lowest conduction band. It is the amount of energy required to free an outer shell electron from its orbit about the nucleus to a free state and thus promote it from the valence to the conduction level. A material is considered to be a semiconductor if the value of the energy band is below 2eV; above this value, it is an insulator. Therefore, semiconductors have a very small band gap, whilst insulators have a wider band gap.

BAND OF STABILITY: A band with nonradioactive nuclides in a graph of atomic number (number of protons) against number of neutrons. It is a way of viewing which isotopes of the elements are stable. In such a graph the stable isotopes make a thick curved line on the graph which is called the band of stability. The lighter elements tend to have the number of neutrons equal to number of protons. The heavier elements tend to have more neutrons than protons. The band of stability helps to predict the behaviour of unstable radioactive isotopes as well, because you can predict which decay process will move the element closer to the band of stability so it will have the ideal ratio of neutrons to protons.

BAND SPECTRUM: A molecular spectrum characteristc of molecular gases or chemical compounds consisting of groups or bands of closely spaced lines. In other words, they are absorption lines occurring across a limited range of frequencies with each line representing an increment of energy due to a change in the rotational state of a specific molecule.

BAND THEORY: The theory, which states that in crystals, electrons fall into allowed energy bands between which lies forbidden bands. Forbidden bands are ranges of energy between the valence and conduction bands. Band theory modifies the free electron treatment by including a regular periodic potential resulting from the positive ions in the lattice. It accounts for many of the electrical and thermal properties of solids and forms the basis of the technology of devices such as semiconductors, metallic conductors and capacitors. As electrons in an atom move from one energy level to another, so can electrons in a solid move from an energy level in one band to another in the same band or in another band. Any solid material that is composed of bands (which are valence band, conduction band and energy gap or band gap) are involved in conduction of electrons. The **valence band** is an electron band located beneath the energy gap or band gap in a solid material and contains the valence electrons of atoms. It is the outermost electron shell of atoms, where the electrons are too tightly bound to the atom to carry electric current. However, in a semiconductor, with a rise in temperature some of the electrons in the valence band gain thermal energy and cross the energy gap and move into the adjacent conduction band. This creates vacancies in the valence band and the material conducts electricity. The **conduction band** is located above the valence band and separated by band gap or energy gap in a solid material. It is the range of permissible energy values required to free an electron bound to an atom to move to conduct electricity. In metallic conductors there is zero band gap and the valence and conduction bands overlap. The **energy gap** or **band gap** is the energy difference measured in electron volts (eV) between the bottom of the conduction band and the top of the valence band of the electrons in a insulators and semiconductors. A material is considered to be a semiconductor if the value of the energy band is below 2eV; above this value, it is an insulator. Therefore, semiconductors have a very small band gap, whilst insulators have a wider band gap.

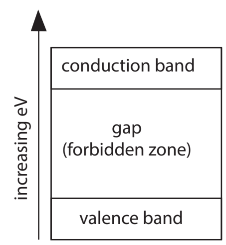


Fig I:
The energy bands of glass (an insulator)

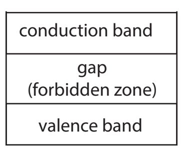


Fig II:
The energy bands of a semiconductor

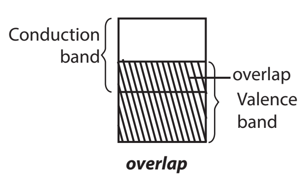


Fig III:
The energy bands of a metal conductor

BAND THEORY OF METALS: A theory that relates to the bonding and properties of metallic solids, by postulating the existence of energy bands. In a metal, the outer orbitals of a very large number of atoms overlap to form a very large number of molecular orbitals that are delocalised over the metal. As a result, a large number of energy levels are crowded together into bands, leading to a decrease or disappearance of the energy gap, making electrons move freely from one energy level to the next.

BANDED IRON FORMATION (BIF): A type of sedimentary rock consisting of alternating layers of iron-rich and iron-poor rocks. Most deposits of this type were laid down about 2.5 billion years ago at a time when Earth's atmosphere still consisted mainly of nitrogen and carbon dioxide. Oxygen given off as a waste product by early photosynthetic bacteria immediately bonded with the abundant dissolved iron to yield minerals like magnetite and hematite. Banded iron formations (BIFs) occur worldwide and give rich iron ore deposits in parts of Minnesota and Michigan.

BANDED MATRIX: A matrix in which the elements are zero except in a band of diagonal.

BANDING: 1. The smearing of grease bands round the trunks of fruit trees to trap insects. 2. In agriculture, applying fertiliser below and to the side of seed at planting, or applying herbicides to a defined width that includes the planted row. 3. Attaching a metal or plastic band to the legs of birds and other animals for identification. 4. A chromatically, structurally or functionally differentiated strip or stripe in or on an organism. 5. A cordlike tissue that connects or holds structures together. 6. A specific range of wavelengths or frequencies of electromagnetic radiation. 7. A range of very closely spaced electron energy levels in solids, the distribution and nature of which determine the electrical properties of a material.

BANDPASS: The range of frequencies that can pass through a filter such as one in an electrical circuit.

BANDPASS FILTER: An optical device which permits the transmission of radiation within a specified wavelength range and does not permit transmission of radiation at higher or lower wavelengths. It can be an interference filter. In other words, it is a filter that transmits a continuous range (band) of wavelengths but reflects or absorbs wavelengths above and below this range. Bandpass filters are specified by the **full width at half maximum** (FWHM) of their transmission peak: wideband (greater than 60 nm FWHM), medium band (20 to 60 nm FWHM) and narrowband (less than 20 nm FWHM).

BANDWIDTH: 1. A range of radio frequencies used in radio or telecommunications transmission and reception. It is the numerical difference between the upper and lower frequencies of a band of electromagnetic radiation, especially an assigned range of radio frequencies. 2. For analogue signals such as human voice it is the difference between the highest and lowest frequency components, measured in hertz (cycles per second). For example, human voice, which produces analogue sound waves, has a typical bandwidth of three kilohertz between the highest and lowest frequency sounds it can generate. 3. In data communication and computer networks, a bandwith of a communication link is the amount of bits that can be transmitted across the links in one second. It is measured in bits per second (bit/s). Also, it is the amount of information available or consumed data communication resources expressed in bits per second (bit/s), kbit/s, Mbit/s, Gbit/s, etc.). For example, a modem with a bandwidth of 56 kilobits per second (Kbps) can transmit a maximum of about 56,000 bits of digital data in one second. 4. In linear algebra, it is the width of the non-zero terms around the diagonal of a matrix.

BANG-BANG PRINCIPLE: The principle applicable to linear time problems in control theory that asserts that the optimal solution is 'bang-bang' in that the control mechanism is either fully on or fully off and has only finitely many switchings. These bang-bang controls arise as extreme points of the feasible controls.

BANNER: The usually largest upper petal of a papilionaceous flower, as occurs in sweet peas and peas.

BANNER AD: A long, rectangular graphic image that can be placed just about anywhere on a Web page. Most banner ads are 468 pixels wide by 60 pixels high. They may contain text, images or sometimes animations that make it hard to focus on the page's content. When a user clicks the advertisement, he or she is redirected to the advertiser's website.

BANTAM: Dwarf variety of poultry, especially chickens. Etymologically, the name is derived from the city of Bantam, currently known as Banten Province in Indonesia.

BAOBAB: A tree belonging to the family *Bombacaceae* and classified as *Adansonia digitata*, common in the savanna regions of Africa, Asia and Australia. The trunk has a thickness of up to 7 metres in diameter and bears gourd-like fruits. Products derived from it include edible pulp of the fruit, the edible leaves, bast fibre, medicine, soap and tannin. The stem, when cut yields clear drinking water.

BAR (BARYE): 1. A centimetre–gram–second (cgs) unit of pressure, which is equal to 100,000 (10^5) pascals, 100,000 (10^5) newtons per square metre or 1,000,000 (10^6) dynes per square centimetre. The International System of Units (SI) is the currently used system of units, and pascal (Pa) has replaced the bar. 2. The small superscript symbol, ‒, used to distinguish entities otherwise indicated by the same letter, such as vectors and scalars or to indicate the complex conjugate of a complex number, the closure of a topological set or a statistical mean. An example is $\bar{A}$. . 3. A subset, S, of a spread in a finite-width tree, such that every continuation of the infinite sequence assigned to a given node has an element in S; a bar for a tree is a bar for the root of the tree. Intuitively, S constitutes a bar to progress up the tree from the given node in the sense that there is no branch that avoids S. 4. A bar structural element whose length is much greater than its greatest cross-sectional diameter. Under compression, a bar is called a **column** or **strut**. Under bending, a bar is called a **beam**. Under torsion, a bar is called a **shaft**; and under tension, it is called a **tie**.

BAR CHART (BAR GRAPH): A block diagram showing quantities in the form of bars. It is a way of displaying data in statistics, using horizontal or vertical bars. The heights or lengths of the bars are proportional to the quantities they represent.Bar charts are best used when dealing with discrete variables or discrete data. The space between each bar is always half of the bar's thickness.

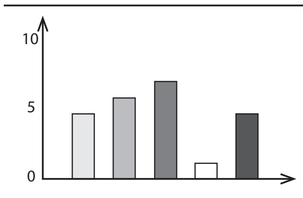


Fig. I: Vertical bar chart/graph

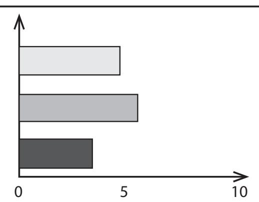


Fig. II: Horizontal bar chart/graph

BAR INDUCTION: An intuitionistically valid form of induction based on the principle for finitary spreads that if Q is a subset of the spread containing a bar P and if, whenever all immediate descendants of a sequence a belong to Q then so does a, then the empty sequence also belongs to Q.

BAR MAGNET: A permanent magnet in the form of a bar with opposite magnetic poles at either end.

BAR OF SANIO (CRASSULA): One of a pair of bar-like thickenings made up of primary wall and intercellular material found adjacent to the bordered pits of most gymnosperm tracheids.

BARB: 1. Any of the stiff processes forming a row on each side of the main shaft of a feather. They form the expanded part or vane of the feather. 2. In botany, it is the short, sharply hooked bristle or hairlike projection. 3. A sharp point projecting in reverse direction to the main point of a weapon or tool, as on an arrow or fishhook.

BARBADOS NUT (PHYSIC NUT; PURGING NUT; *Jatropha curcas*): A semi-evergreen poisonous, flowering plant in the spurge family, Euphorbiaceae. It is native to tropical America, but cultivated in tropical and warmer subtropical regions around the world. It is a shrub that grows to about 6 metres in height and very tolerant to arid conditions. The bark is thin, greenish and exudes copious amounts of watery sap when cut. The leaves are pale green with alternate to subopposite leaf arrangement. The small, yellow to green flowers are borne in axils of the leaves and are mostly hidden by foliage. The round fruit is small and capsule-like, about 2.5-4 centimetres in diameter, green and fleshy when immature, becoming dark brown when ripe, splitting to release 2 or 3 black seeds each. All parts of the plant are considered toxic, particularly the seeds, which contain toxalbumin or curcin, which is very poisonous. The seeds contain oil that can be processed to produce high quality fuel (biodiesel) which can be used as diesel oil. The plant is an ornamental one. Its parts such as the roots, stems, leaves, seeds have been used traditionally to treat various ailments.

BARBATE: Tufted or bearded with long and stiff hairs.

BARBED: Having rigid, short, reflexed points, like the barb of a fishhook.

BARBED WIRE: Any wire with sharp points, used for fencing in order to discourage or prevent livestock from touching it.

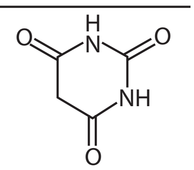
BARBELLATE: In botany, it refers to a structure with short barbs or stiff hairs. **Barbellulate** describes very tiny short stiff hairs or barbs.

BARBELS: Tactile organs protruding about the mouths of certain fishes, notably the barbels (Barbus) of the family *Cyprinidae*. In some fishes, they contain the taste buds, and assist them to locate food in waters where visibility is poor.

BARBICELS: Tiny, curved, hairlike structures growing from barbules, that connect adjacent barbules by means of hooks (barbicels) and grooves, to form the firm, mesh-like structure of the feather vane of birds. Down feathers do not have barbicels.

BARBIER-WIELAND DEGRADATION: The stepwise degradation of a carboxylic acid to the next lower homologue, RCH_2COOCH to RCOOH. Starting from an ester, it can be converted to tertiary alcohol using a Grignard reagent, PhMgX and acid, HX: $RCH_2COOCH_3 \rightarrow RCH_2C(OH)Ph_2$. The secondary alcohol formed is then dehydrated using ethanoic anhydride ($CH_3COOCOCH_3$), $RCH_2C(OH)Ph_2 \rightarrow RCH{=}CPh_2$. The alkene is then oxidised with chromic acid: $RCH{=}CPh_2 \rightarrow RCOOH + Ph_2CO$.

BARBITURATE: A highly addictive drug with sedative and hypnotic properties that consist of any salt or ester of barbituric acid ($C_4H_40_3N_2$). Examples include butobarbital for treating insomnia and thiopental for inducing general anaesthesia. Due to their side effects and increased risk of dependence and abuse, they are listed as controlled substances and are largely replaced by tranquillisers and are prescribed for only certain conditions.



Chemical Structure of Barbiturate

BARBS: The series of branches of feathers that are attached to the main shaft of a feather to form its vanes. Projecting from the barbs are the barbules (minute hairs in some feathers interlocked by hooks and grooves, forming a flat vane).

BARBULE: Any of the minute hairs projecting from each side of the barb of a feather interlocked by hooks and grooves to form a flat vane making the feather solid and stiff.

BARCAN FORMULA: In logic, the expression of modal logic $(\forall x) \Box Fx \rightarrow \Diamond (\forall x) Fx$ that asserts that if everything has some necessary property, then it is necessary that everything has that property, hence, since $\Diamond(\forall x) Fx = -\Diamond (\exists x) - Fx$, it is not even possible that something (else) might exist that would lack that property. (Here $\Box$ is the 'necessity" and $\Diamond$ the 'possibility' operator).

BARCODE: This refers to a printed machine-readable lines of different widths and thicknesses representing decimal digits. When read, it helps the user to identify a product or object. **Barcode reader** (BCR) or **scanner**, also known as a **point of sale scanner** (POS) is a device that is used to read a barcode on an object. It can also print out the details of the product. This is commonly used in supermarkets to read and log the price of a product into the computer system of the supermarket.

BARDEEN, JOHN (1908 – 1991): US Physicist who worked at Harvard, Minnesota University and Bell Telephone Laboratories before becoming a professor at the University of Illinois in 1951. At Bell, with Walter Brattain (1902 – 1987) and William Shockley (1910 – 1989), they developed the point-contact transistor; for which they shared the 1956 Nobel Prize for Physics. In 1956, with Leon Cooper (1930 -) and John Schreiffer (1931 -), he formulated the BCS theory of superconductivity, for which they shared the 1972 Nobel Prize for Physics.

BARE ROOTS: Roots of plants that are exposed with the soil shaken off or washed off by erosion.

BARE SOIL: A soil without any vegetation including shrubs or trees.

BAREBONES PC: A computer that has minimal components. A barebones system includes a case, motherboard, CPU, hard drive, RAM and power supply. Most barebones systems are sold as kits, with the components assembled by the user.

BARFOED'S REAGENT: A mixture of ethanoic (acetic) acid and Copper(II) ethanoate (acetate), used to show the presence of strongly reducing sugars in solution. It consists of a 0.33 molar solution of neutral copper acetate in 1% acetic acid solution or it may be prepared by dissolving 70 g copper acetate monohydrate and 9 mL glacial acetic acid in water to a final volume of one litre.

BARFOED'S TEST: A biochemical test to detect the presence of monosaccharide sugar in solution, devised by the Swedish physician, Christen T. Barfoed (1815–99). It is based on the reduction of copper(II) acetate to copper(I) oxide (Cu_2O), which forms a brick-red precipitate: $RCHO + 2Cu^{2+} + 2H_2O \rightarrow RCOOH + Cu_2O + 4H^+$. A mixture of Barfoed's reagent is added to the test solution and boiled. If any monosaccharide sugar is present in the solution a red precipitate of copper(I) oxide is formed. If any disaccharide sugars are present the reaction will be negative as they are weaker reducing agents.

BARITE (BARYTES): A common barium sulfate mineral, with the chemical formula $BaSO_4$. It is used as a source of barium compounds.

BARIUM: A soft, silvery-white, reactive metallic element, belonging to Group II elements of the Periodic Table, with the chemical symbol Ba, atomic number 56 and relative atomic mass 137.34. It occurs in nature as sulfates and carbonates. It is used in alloys, pigments and safety matches.

BARIUM CARBONATE: An insoluble white compound, with the chemical formula $BaCO_3$ and melting point of 174 °C (447 K). It decomposes on heating to give barium oxide and carbon dioxide. It occurs naturally as the mineral witherite and can be prepared by adding an alkaline solution of a carbonate to a solution of a barium salt. It is used as a raw material for making other barium salts, as a flux for ceramics, in television picture tubes and as a raw material in the manufacture of certain types of optical glass.

BARIUM CHLORIDE: A white crystalline solid, with the chemical formula $BaCl_2$. It is used in aqueous solution to test for the presence of the sulfate ion. The anhydrous compound has two crystalline forms: an alpha form, which transforms at 962 °C (1235 K) to a beta form.

BARIUM NITRATE: A compound that, like barium chloride, can be used to test for the presence of the sulfate ion as it readily dissolves in water. Its chemical formula is $Ba(NO_3)_2$. It occurs naturally as the rare mineral nitrobarite.

BARIUM OXIDE: A solid that is whitish or yellowish with the chemical formula BaO. It is obtained by heating barium in oxygen or by the thermal decomposition of barium carbonate or nitrate. Barium oxide is used in the manufacture of lubricating-oil additives.

BARIUM PEROXIDE (BARIUM DIOXIDE): A dense off-white solid, with the chemical formula BaO_2, prepared by heating barium oxide in oxygen and insoluble in water. It is used as a bleaching agent and with acids hydrogen peroxide is formed. It is also used in fireworks as an oxidiser.

BARIUM SULFATE: A white crystalline insoluble powder, with the chemical formula $BaSO_4$, density 4.5 g/cm^3, melting point 1345 °C (1618 K) and boiling point 1600 °C (1873 K). It occurs naturally as the mineral barites or heavy spar and can be prepared as a precipitate by adding sulfuric acid to barium chloride solution. It is used as a pigment and in medicine to provide a contrast medium for X-rays of the stomach and intestine. Although barium compounds are extremely poisonous, the sulfate is safe to use because it is very insoluble.

BARK: The protective outer layer on the stems and roots of trees, woody vines and shrubs. It consists of two layers: the outer bark, which is dead and peels off and the inner bark, which contains living tissues, the cambium and phloem. Successive layers of bark are formed inside one another as the plant grows and includes parts such as cork cells or phellem (a layer of dead protective tissue between the bark and cadmium in woody plants); cork cambium or phellogen (a layer of plant tissue outside of the true cambium, giving rise to cork tissue); phelloderm (a layer of thin-walled cells produced by the inner surface of the cork cambium); and the secondary phloem (the food-conducting tissue of vascular plants, consisting of sieve tubes, fibres, parenchyma and sclereids). As the thickness of the bark increases the outer layers may either become fissured as in elm or are shed as scales as in plane or rings as in birch.

BARKHAUSEN EFFECT: The magnetisation of a ferromagnetic substance by an increasing magnetic field, which takes place in discontinuous steps, discovered by German physicist Heinrich Barkhausen in 1919. If the magnetisation is within a coil connected to an amplifier and loudspeaker, the sudden changes create a series of clicks or, a sound when there is a rapid change.

BARLEY: Cereal plant of the genus *Hordeum*, in the family Poaceae or Gramineae and its edible grain, grown in temperate regions. The three cultivated species are *Hordeum vulgare*, *Hordeum distichum* and *Hordeum irregulare*. The grain is used for making malt, beer, whisky and also used as animal feed.

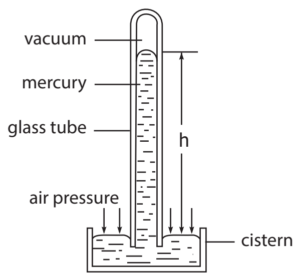
BARLEY YELLOW DWARF: A plant disease caused by the barley yellow dwarf virus, spread by aphids, causing poor root development in barley plants. It is the most widely distributed viral disease of cereals. It affects barley, oats, wheat, maize, triticale and rice.

BARN: 1. A unit of area, with the symbol b or bn sometimes used to measure cross section in nuclear interactions involving incident particles. One barn is equal to 10^{-28} m^2 (10^{-24} cm^2). For example, the elastic scattering cross section for thermal neutrons on hydrogen is 20.47 barn. 2. A building for storing hay or housing horses and cattle.

BARNACLE: A marine crustacean in the subclass *Cirripedia* that in the adult stage form a hard shell and remain attached to submerged surfaces, such as rocks, ships' bottoms and certain whales. The barnacle that fixes itself to ships is the goose barnacle, which has a long muscular stalk and a shell composed of five plates. The acorn barnacles, which cover rocks, breakwaters, etc, just below high-water mark, are similarly constructed, but they have no stalk.

BAROGRAPH: A meteorological instrument that records on paper variations in atmospheric pressure over a period.

BAROMETER: An instrument used to measure atmospheric pressure, due to the weight of the column of air above it. It was invented in 1643 by the Italian scientist Evangelista Torricelli, using a column of water in a tube 10.4 m long. The two types now in use are mercurial barometers, which give very accurate readings but are costly and they need special care in handling; and the aneroid barometer, which gives less precise measurements, but easier to use. The mercury barometer consists of a hollow glass tube, about 0.9 m long, closed at one end and filled with mercury. The open end is placed in an open vessel of mercury called a cistern. The mercury column in the tube will drop until the weight (pressure) of the air pushing down on the mercury in the cistern is equal to the weight of the mercury remaining in the tube. The higher the atmospheric pressure, the higher the mercury will stand in the tube. Similarly, the lower the atmospheric pressure, the lower the mercury will stand. The aneroid barometer consists of a closed sealed capsule with flexible sides. Any change in pressure alters the thickness of the capsule.



Mercury Barometer

BAROPHILES (PIEZOPHILES): Organisms that thrive under conditions of high hydrostatic pressure, for example, deep in the ocean or far underground. Examples of such organisms are deep sea bacteria or archaea.

BARORECEPTOR: A receptor located in the blood vessels of several mammals that responds to changes in blood pressure.

BAROTAXIS (BAROTAXY): 1. The reaction of living tissue to changes in pressure. 2. In biology, the stimulation of living matter by change of the pressure relations under which it exists; or the responsive movement of a free-moving organism or cell toward or away from an external stimulus, such as light. 3. A directed reaction of a motile organism to a mechanical pressure stimulus.

BAROTOLERANT: An organism that can tolerate high hydrostatic pressure, although it will grow better under normal pressure.

BAROTROPIC: A state of the atmosphere where isotherms are parallel to isobars. Therefore, density is a function of pressure, with pressure also depending on density. A **barotropic fluid** is a fluid for which density is a function of only the pressure. A **barotropic region** is one of uniform temperature distribution, where no fronts occur, generally found in the tropics. **Barotropy** is the state of a fluid in which surfaces of constant density or temperature are coincident with surfaces of constant pressure; it is the state of zero baroclinity.

BARR BODY: A condensed X chromosome named after a Canadian anatomist M.L Barr (1908-1995), who identified it in 1949, which is present in female mammals with non-dividing nuclei used to confirm the sex of athletes in sex determination tests.

BARREL: 1. A unit of a liquid capacity, the value of which depends on the liquid being measured. For example, a barrel of petroleum contains 159 litres while a barrel of alcohol contains 109 litres. 2. A closed, convex, absorbing, balanced subset of a normed space or a topological vector space.

BARRELLED (BARRELLED SPACE): A topological vector space in which every barrel contains a neighbourhood of the origin. Every Banach space and every Frechet space is barrelled.

BARRIER FUNCTION (INTERIOR PENALTY FUNCTION): For a set, S, with non-empty interior, a non-negative function, B that is continuous over the interior of S and approaches infinity as the boundary is approached from the set.

BARRIER ISLAND: A long island of sand lying offshore and parallel to the coast that serves to protect the shore from erosion.

BARRIER PENETRATION (TUNNELING): In quantum mechanics, the passing of a particle through a seemingly impenetrable barrier without a cause that is explainable by classical physics.

BARRIER REEF: A coral reef that lies offshore separated from mainland by a shallow lagoon. The most famous is the Great Barrier Reef in northwest Australia.

BARRIERS TO GENE FLOW: Factors such as geographic, mechanical and behavioural isolating mechanisms that restrict gene flow between populations, leading to populations with differing allele frequencies. An example is the Great Wall of China, which has hindered the gene flow of native plant populations.

BARROW: A castrated male of the swine family.

BARROW, ISAAC (1630 – 1677): An English mathematician who published a method for finding tangents in 1670. His method is now used in differential calculus. It is believed that he may have been the first to appreciate that the problems of finding tangents and areas under curves are inversely related. Newton succeeded him in Cambridge after resigning his chair.

BARYCENTRE: 1. Also known as **centroid**, it is the centre of mass of two or more bodies that are orbiting each other or the point around which they both orbit. For example, Earth-Moon system moves in the Sun's gravitational field as though both masses were located at a centre of mass some 4700 km from the Earth's geometric centre. 2. The centroid of a set. When the set is a k-dimensional simplex, the barycentre has all barrycentric coordinates equal to $1/(k + 1)$.

BARRYCENTRIC COORDINATES: The unique set of non-negative coefficients, λ_i with $\sum_i \lambda_i = 1$ that identifies a given point, x, in a simplex of $n + 1$ points, p_i that do not all lie in the same hyperplane, as a convex combination $x = \sum_i \lambda_i p_i$.

BARYOGENESIS: The creation of matter in excess of antimatter in the early universe. Only the relatively few unmatched matter particles survived to make up all subsequent structures.

BARYON: A hadron or subatomic particle made up of three quarks. Protons, neutrons and hyperons are all baryons. Baryons are involved in strong interactions with other particles and have a half-integral spin. All baryons other than protons and neutrons are short-lived even though there are a lot of possible baryons that could be formed.

BARYON NUMBER: A quantum number, with the symbol B, which is equal to the difference between the number of baryons and the number of antibaryons in a system of subatomic particles. It remains the same throughout any reaction. Baryons have a baryon number of +1, while antibaryons have a baryon number of -1. Quarks and antiquarks have baryon numbers of $+^1/_3$ and $-^1/_3$, respectively (baryons consists of three quarks). Mesons, bosons and leptons all have baryon numbers of 0.

NAME	Q	B	S	QUARK CONTENT
Proton	+1	1	0	uud
Neutron	0	1	0	uud
Lambda	0	1	-1	uds
Sigma plus	+1	1	-1	uus
Sigma minus	-1	1	-1	dds
Sigma zero	0	1	-1	uds
Xi minus	-1	1	-2	dss
Xi zero	0	1	-2	uss

Baryon Number Table

BARYTES: An orthorhombic mineral form of barium sulfate, with the chemical formula $BaSO_4$. It is the chief ore of barium.

BASAL BODY (BASAL GRANULE; KINETOSOME): A structure at the base of a cilium or flagellum, consisting of nine triplet microtubules arranged in a circle with no central microtubule. It serves as a nucleation site for the growth of the axoneme microtubules.

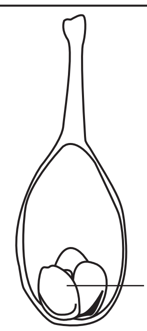
BASAL CELL CARCINOMA (RODENT ULCER): A malignant tumour that spreads slowly usually occurring on areas of the skin that are often exposed to the Sun, such as the face and neck, at the edge of the eyelids, lips or nostrils. It is caused mainly by long-term exposure to ultraviolet (UV) radiation from the sunlight damaging the DNA in the skin cells.

BASAL GANGLIA: Small masses of nervous tissue within the brain that connect the cerebrum with other parts of the nervous system. They are involved with the subconscious regulation of voluntary movements.

BASAL MEDIUM: A growth medium that allows the culture of many types of micro-organisms which do not require special nutrient supplements.

BASAL METABOLIC RATE (BMR): The rate at which energy is used by an organism at complete rest. This energy is used to maintain the vital involuntary activities, such as breathing, maintenance of heat, heartbeat and blood circulation and the activities of the nervous system and internal organs. In humans it is measured by the heat given off per unit time and expressed as the calories released per kilogram of body weight or per square metre of body surface per hour. The BMR is an important diagnostic tool, especially in determining disorders of the thyroid gland.

BASAL PLACENTATION: A form of placentation seen in bistorts (Polygonum) in which ovules are attached to the base of the ovary of a simple or compound ovary, as shown in the diagram. Examples are found in composite flowers such as sunflower, ray flower and dandelion.



Basal placentation

BASALT: A dark glass-like extrusive igneous rock. It is the main type of rock under the sediments on ocean floors and it is also found on the Moon.

BASE: 1. In chemistry, it is a substance which reacts with an acid to form salt and water only or according to the Brønsted-Lowry theory it is a proton (hydrogen ion) acceptor, while the Lewis theory defines it as an electron pair donor. Bases are usually metal oxides or hydroxides. A soluble base is referred to as an alkali if it contains and releases hydroxide ions ($OH-$) quantitatively. Examples of common bases are sodium hydroxide and ammonia. Metal oxides, hydroxides and especially alkoxides are basic and counter anions of weak acids are weak bases. A base accepts protons from hydronium ions or donates hydroxide ions to a solution, thus lowering the concentration of hydronium ions and raising the pH. However, an acid donates protons to a solution or accepts $OH-$, thus increasing the concentration of hydronium and lowering the pH. The pH level of a basic solution is higher than 7. A strong base is a base which hydrolyses completely, raising the pH of the solution toward 14. Common examples of strong bases are the hydroxides of alkali metals and alkaline earth metals like NaOH and $Ca(OH)_2$. Bases turn red litmus paper blue, phenolphthalein pink, keep bromothymol blue in its natural colour of blue and turns methyl orange yellow. Bases feel slimy or soapy on fingers, due to saponification of the lipids in human skin; however, concentrated bases are caustic on organic matter and react violently with acidic substances. Aqueous solutions or molten bases dissociate in ions and conduct electricity. 2. In mathematics, also called **radix**, it is the number of different single digit symbols used in a particular number system. In the usual decimal counting system of numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 the base is 10. 3. A number that when raised to a potential power or when multiplied by itself a number of times has a logarithm equal to the power. For example,
$10^3 = 10 \times 10 \times 10 = 1{,}000$. 4. The side or face of a geometric figure, usually the one at the bottom, to which an altitude is or is thought to be drawn. 5. The number of distinct residues in a system of modular arithmetic. 6. A substructure of a given mathematical structure from which the whole structure can be generated. 7. A base for a topology, in particular, a collection of open sets such that every member of the topology is a union of members of the collection. 8. A **base at a point**, **local base**, a **base for the neighbourhood system**, more particularly, a subcollection of neighbourhoods of the given point with the property that every neighbourhood of the point contains a member of the subcollection. 9. In botany, it is the bottom or lower portion of the leaf blade nearest to the point of attachment.

BASE ADDRESS: A unique location in primary storage or main memory that serves as a reference point for other memory locations called absolute addresses. In other words,it is an address that serves as a reference point for other addresses. For example, a base address could indicate the beginning of a program. The address of every instruction in the program could then be specified by adding an offset to the base address.

BASE ANALOGUE (BASE ANALOG): A chemical, usually a purine or pyrimidine that can substitute for a normal nucleobase in nucleic acids. A common example is 5-bromouracil (5BU).

BASE CLAUSE: The initial instance from which a generalisation is proved by mathematical induction; the statement that defines the first element of the infinite sequence generated by the induction.

BASE COMPOSITION: In biochemistry, it is the proportion of total bases consisting of 'guanine plus cytosine' or 'thymine plus adenine' base pairs.

BASE COURSE: A layer of specified or selected material of planned thickness constructed on the subgrade or subbase for the purpose of serving one or more functions such as distributing load, providing drainage, or minimising frost action, and other such purposes.

BASE FIELD: The underlying field of a vector space or other structure, for example, the polynomials over a given base field.

BASE IONISATION CONSTANT (BASE DISSOCIATION CONSTANT): The equilibrium constant K_b for the base ionisation; or a reaction in which a weak base is in equilibrium with its conjugate acid in aqueous solution. For example, ammonia ionizes in water as follows:

$$NH_{3(aq)} + H_2O_{(l)} \rightleftharpoons NH_4{}^+{}_{(aq)} + OH^-{}_{(aq)}$$

$$K_b = \frac{[NH_4{}^+{}_{(aq)}][OH^-{}_{(aq)}]}{[NH_{3(aq)}]}.$$

The base-ionisation equilibrium constant is the mathematical product of the equilibrium concentrations of the products of this reaction divided by the equilibrium concentration of the original base. In other words, Kb is the product of the concentrations of the products over the concentration of the reactants (not including water). As the Kb value of a base increases, so does the strength of the base. For a strong base: $K_b > 1$ and for a weak base: $K_b < 1$. The value is sometimes expressed as the logarithm of its reciprocal, called pK_b. Therefore pK_b = -logK_b. The smaller the value of pK_b, the stronger the base. K_b is a better measure of the strength of a base than pH because adding more water to the base solution will not change the value of the equilibrium constant K_b, but it will change the H^+ ion concentration. Water is not included in the base-ionisation equilibrium expression because the $[H_2O]$ has no effect on the equilibrium.

BASE METAL: A common and fairly cheap metal that oxidises or corrodes relatively easily during use. It reacts variably with diluted hydrochloric acid (HCl) to form hydrogen. Examples include iron, nickel, lead and zinc. Copper is regarded as a base metal because it oxidises relatively easily, although it does not react with HCl. The term is commonly used in opposition to noble metal.

BASE NUMBER (BASIC NUMBER): The number of chromosomes in a genome, usually between six and thirteen and is represented by the symbol x. In diploid organisms the number of chromosomes in the genome is equivalent to the haploid number n and so $2n = 2x$. However in tetraploid organisms $2n = 4x$, in hexaploid organisms $2n = 6x$, etc.

BASE PAIRING: A hydrogen bonding between any two complementary nitrogenous bases of genetic materials based on a rule that is applied as follows: A with T: the purine adenine (A) always pairs with the pyrimidine thymine (T); C with G: the pyrimidine cytosine (C) always pairs with the purine guanine (G). The bonding forms the basis of the genetic code of heredity.

BASE PEAK: The peak in a mass spectrum corresponding to the separated ion beam which has the greatest intensity. This term may be applied to the spectra of pure substances or mixtures.

BASE PERIOD: In statistics, the period used as a standard for comparison for some variable, such as, for example, consumer prices; 100 is usually taken as the index number for the variable in the base period, so that an index of 150 for a given period indicates that prices then were one and a halftimes those of the base period.

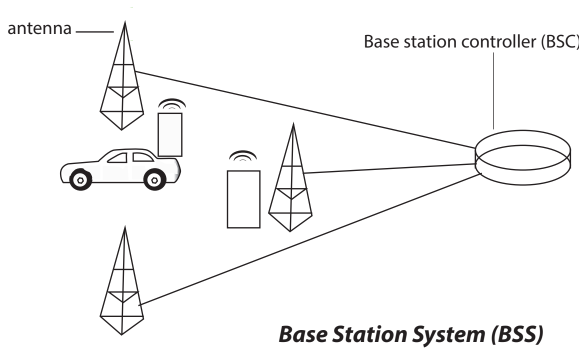
BASE QUANTITY (FUNDAMENTAL QUANTITY): One of the quantities that in a system of quantities are conventionally accepted as functionally independent of one another. It is a quantity whose magnitude is defined without reference to other quantities. In the SI, these are length, mass, time, electric current, temperature, amount of substance and luminous intensity. All other physical quantities (and units) are regarded as being derived from these base quantities and base units.

BASE RATIO: The ratio of the molar quantities of purine and pyrimidine bases in DNA and RNA. The observation that the amount of adenine (A) always equals the amount of thymine (T) and similarly the amount of guanine (G) equals that of cytosine (C) provided one of the first indications that adenine always pairs with thymine and guanine with cytosine in the complementary polynucleotide chains. A comparison of base pair ratios, A + T:G + C or percentage G + C of the total, shows a constant figure within a species but differences between species.

BASE SATURATION PERCENTAGE: The extent to which the adsorption complex of a soil is saturated with exchangeable cations other than hydrogen and aluminium. It is expressed as a percentage of the total cation exchange capacity. A low percentage base saturation means acidity, whereas a percentage base saturation approaching 100 will result in neutrality or alkalinity.

BASE STACKING: The close packing of the planes of base pairs, commonly found in DNA and RNA structures. Base stacking interactions in DNA and RNA are due to dispersion attraction, short-range exchange repulsion and electrostatic interactions which also contribute to stability.

BASE STATION (ACCESS POINT): A device, such as a wireless router, that allows wireless devices to connect to a network based on the Wi-Fi standard. Some common Wi-Fi configurations include 802.11b and 802.11g. It connects users to other users within the network and also can serve as the point of interconnection between the WLAN and a fixed wire network.



Base Station System (BSS)

BASE STATION CONTROLLER (BSC): An automatic controller that permits one or more base transceiver stations (BTS) in a wireless network to communicate with a mobile switching centre. In a GSM network, its role is to allocate Base Transceiver Stations (BTS) radio channels to mobile subscribers, perform paging (finding the cell in which a mobile user is resident) and perform handoff of mobile users.

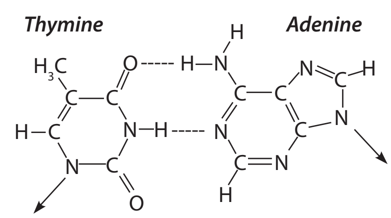
BASE TEN BLOCKS: Mathematical manipulatives representing ones (cubes), tens (sticks), and hundreds (flats). It is used for teaching children to develop understanding for basic mathematical concepts like place value, addition, subtraction, number sense, and counting.

BASE UNIT (BASIC UNIT; FUNDAMENTAL UNIT): Any of a set of seven International System of Units (SI) without combination from which all units of physical quantities can be derived.The seven S.I. base units are as follows: metre with the symbol m for length, kilogram with the symbol kg for mass, second with the symbol s for time, ampere with the symbol A for electric current, kelvin with the symbol K for temperature, candela with the symbol cd for luminous intensity, and mole with the symbol mol for amount of substance. These seven units are called base units because none of them can be expressed as combinations of the other six. Some combination base units are given special names, for example, the kilogram metre per second squared ($kg{\cdot}m{\cdot}s^{-2}$) is called the newton (N) and the kilogram metre squared per second squared ($kg{\cdot}m^2{\cdot}s^{-2}$) is called the joule (J).

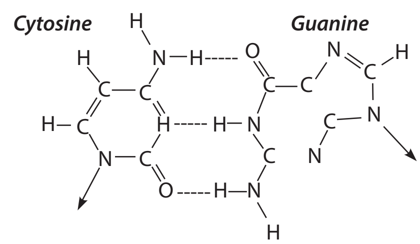
BASE-PAIR: The pair of nitrogenous bases, consisting of a purine linked by hydrogen bonds to a pyrimidine, that connects the complementary strands of DNA or of hybrid molecules joining DNA and RNA. The base pairs are adenine-thymine and guanine-cytosine in DNA and adenine-uracil and guanine-cytosine in RNA.

BASE-PAIR SUBSTITUTION: A point mutation involving the replacement or substitution of a single base nucleotide with another nucleotide of the genetic material, DNA or RNA. When a purine-pyrimidine pair is substituted by the other purine-pyrimidine pair it is called **transition**; when a purine-pyrimidine pair is replaced by one of the two pyrimidine pairs, it is called **transversion**. Normally, when a stop codon occurs at the end of a gene, it stops protein synthesis, but, when it occurs in an abnormal position, it can result in a truncated and nonfunctional protein. Gene mutations are most commonly caused by environmental factors such as chemicals, radiation and ultraviolet light from the sun can cause mutations. They are also caused by by errors made during mitosis and meiosis. Point mutations can cause several genetic disorders such as sickle cell anaemia, cystic fibrosis, Tay-Sachs disease, Huntington disease, haemophilia and some cancers .

BASE-PAIRING RULE (BASE-PAIRING PRINCIPLE): The rules of base pairing (or nucleotide pairing) in the formation of nucleic acids (DNA and RNA) with the requirement that adenine must always pair with thymine (or uracil RNA) and guanine with cytosine which constitute the basis of genetic code of heredity.



Base pairing between thymine and adenine



Base pairing between cytosine and guanine

B

BASE-10 BLOCKS (MULTIBASE ARITHMETIC BLOCKS, MAB): A set of four different types of blocks used to represent ones, tens, hundreds, and thousands in the base-ten place-value system. These blocks include the following: Unit (1 block made up of 1 unit); Rod or Long (10 blocks made up of ten units); Flats (100 blocks made up of 100 units); and Blocks representing 1000 blocks made up of 1000 units. When used together, it can help to teach and understand mathematical concepts like place value, addition, subtraction, multiplication, division, volume, perimeter, and area.

BASEBAND PROCESSOR: A chip in wireless transmission devices such as mobile phones that performs signal processing and implements the device's real time radio transmission operations.

BASEBAND SIGNAL: The band of frequencies in an analogue or digital waveform occupied by all transmitted signals and used to transmit along a pathway or processed by an electronic circuit. Baseband signals can be composed of a single frequency or group of frequencies or in the digital domain composed of a data stream sent over an unmultiplexed channel. Examples of an analogue baseband signal may be audio or composite video. Examples of a digital baseband signal may be Ethernet signals operating over a Local Area Network (LAN). In terms of bandwidth, baseband is the highest frequency used by the bandwidth or the upper bound of the bandwidth.

BASEFLOW (DRY-WEATHER FLOW; GROUND WATER FLOW): A stream flow which results from precipitation that infiltrates into the soil and eventually moves through the soil to the stream channel.

BASELINE: A set of data used for comparison with subsequent data. Baseline data can be used to judge whether an experimental intervention is successful.

BASEMENT MEMBRANE (BASAL LAMINA; BASEMENT LAMINA): A thin sheet of connective tissue that underlies and supports the cells of an epithelium, separating this from underlying tissue.

BASIC: 1. In chemistry, a solution containing an excess of hydroxide ions. 2. In computer science, it is an acronym for "Beginners All Purpose Symbolic Instructional Codes", which is a type of general purpose high-level programming language whose design philosophy emphasises ease of use. The programming language is written in familiar English words that can be combined to make up programs.

BASIC ANHYDRIDE: The oxide of a metal that reacts with water to form a base. For example, sodium oxide is basic. When hydrated, it forms sodium hydroxide.

BASIC FACTS (BASIC MATHEMATICS FACTS): In mathematics, it refers to a number, a fact or idea that can be instantly recalled without having to use any special operations to recall them. For example, in basic classes some additions
(such as 1 + 1 = 2, 2 + 2 = 4, 1 + 0 = 1, etc.); subtractions (such as 1 - 1 = 0, 2 - 1 = 1, etc.); and multiplications (1 - 10 multiplication tables), etc., are memorised forever so that there is no need to use any special operations to recall them.

BASIC FEASIBLE SOLUTION: A feasible solution in linear programming that corresponds to an extreme point of the feasible set. The term is used because the solution corresponds to a basis in the simplex tableaux.

BASIC INPUT/OUTPUT SYSTEM (BIOS): A chip located on computer motherboards that contains instructions and set-up for how the system should boot and how it operates. It is a program a personal computer's microprocessor uses to get the computer system started after it is turned on. The CPU accesses the BIOS even before the operating system is loaded. The BIOS then checks all the hardware connections and locates all devices. If there is no problem with the system, the BIOS loads the operating system into the computer's memory and finishes the boot-up process. It also manages data flow between the computer's operating system and attached devices such as the hard disk, video adapter, keyboard, mouse and printer.

BASIC LOCAL ALIGNMENT SEARCH TOOL (BLAST): A computational method or algorithm that is widely used in bioinformatics for comparing sequences of bases or of amino acids by the process of local alignment.

BASIC OXIDE: A metallic compound that is a base or that forms a hydroxide when combined with water. For example, calcium oxide (CaO) is a base that is combined with water to form calcium hydroxide ($Ca(OH)_2$.

BASIC OXYGEN PROCESS (LINZ-DONAWITZ PROCESS): A widely used process for making steel. The basic-lined converter has no tuyeres. A stream of pure oxygen at high pressure is blown through a lance onto the surface of molten pig iron (low in phosphorus) mixed with scrap to help remove impurities. It has replaced the Bessemer process and open-hearth process. It was named after Linz and Donawitz, towns in Austrian.

BASIC PLUMAGE: Feathers that occur in birds that moult only once a year, which they do not shed throughout the year.

BASIC ROCK: An igneous rock rich in calcium or iron and magnesium, but containing a relatively low amount of silica.

BASIC SALT: Type of salts containing hydroxide (OH^-) ions formed by the neutralisation of a strong base and a weak acid. Their conjugate base from the weak acid hydrolyses to form a basic solution. Examples are sodium carbonate (Na_2CO_3) and basic lead carbonate ($2PbCO_3.Pb(OH)_2$).

BASIC SLAG: Slag formed from a basic flux such as calcium oxide, containing calcium silicate, phosphate and sulfide. It is used to remove acid impurities and also used as fertiliser if the phosphorus content is high enough.

BASIC STAINS: Coloured dyes or pigments used in histology and microscopy to improve contrast between various cells and tissue components. Examples include eosin, iodine and safranin.

BASIC-OXYGEN PROCESS: A method of making high-grade steel from carbon-rich molten pig iron. It involves blasting of oxygen at high supersonic speed into molten pig iron to lower the carbon content.

BASICITY: The number of replaceable hydrogen atoms in an acid. For example, nitric acid is monobasic, sulfuric acid is dibasic and phosphoric acid is tribasic.

BASIDIOCARP: The reproductive organ of all basidiomycete fungi except the Ustilaginales and Uredinales, which bears the basidia. It may be mushroom shaped, bracket shaped, club shaped or a hollow sphere or cylinder. Much of the basidiocarp is composed of sterile pseudoparenchymatous tissue. The basidia are borne in a fertile layer, called the **hymenium**.

BASIDIOMYCOTA (BASIDIOMYCETES): A large phylum of filamentous fungi, formerly classified as a class or a subdivision containing mushrooms, puffballs, stinkhorns, bracket fungi, etc. They are characterised by hyphae and possess gills on their fruiting bodies. They reproduce sexually through the formation of specialised club-shaped end cells (basidia) that normally bear four external meiospores.

BASIDIOSPORE: The characteristic spore produced by specialised fungal cells called basidia during sexual reproduction and borne externally on a basidium by fungi belonging to the phylum Basidiomycota.

BASIDIUM (BASIDIA): A specialised cell in fungi of the phylum Basidiomycota, in which nuclear fusion and meiosis occur. This results in the formation of four basidiospores, which are borne externally by the basidium at the tips of sterigmata.

BASIFIXED: Describing an anther that is joined to the filament at its base.

BASILAR ARTERY: An artery in the base of the brain, formed by the union of the two vertebral arteries. The basilar artery extends from the lower to the upper border of the pons Varolii and then divides to form the posterior cerebral arteries.

BASIN: 1. An oval or circular depression, or dip, in the Earth's surface, with sides higher than the bottom. Basins are formed by forces above the ground (like erosion) or below the ground (like earthquakes). Basins are either filled with water or empty. Basins can be formed on the floor of an ocean, containing mountains, valleys, and plains. 2. The drainage area of a stream, river, or lake surrounded by an area of highland. Basins are usually low-lying areas in which thick layers of sediment have accumulated.

BASIN INVERSION (INVERSION): The relative uplift of a sedimentary basin or similar structure as a result of crustal shortening.

BASIN IRRIGATION: One of the methods of irrigating land in which basins are created and filled with water and used to irrigate the field. Although this kind of irrigation is not suitable for crops such as potatoes, cassava, carrots and maize, it is especially suitable for water-loving crops such as paddy rice as its roots are submerged in water.

BASIPETAL: Describing movement, differentiation, etc. occurring from apex to base. For example, the direction of proto- and metaxylem formation is from the apex or nodes downwards, while the transport of auxin in the stem is generally towards the base.

BASIS: 1. Any set of vectors that determines a space as the set of sums of their multiples. It is also called **Hamel basis**, especially when the basis vectors are orthogonal. 2. In a Euclidean space, a maximal set of mutually orthogonal vectors in terms of which all the elements of the space are uniquely expressible, at the number of which is the dimension of the space; thus the vectors x, y and z, in the positive directions of the coordinate axes, form a basis of the three-dimensional space, whose members can be written as linear combinations $ax + by + cz$. 3. Any linearly independent subset of a vector space that generates the space. The cardinality of such a subset is the dimension of the space. For example, the dimension of the vector space of all polynomials over a field is $\aleph_0$ and the set with elements $1, x, x^2, \ldots, x^n, \ldots$, forms a basis; the vectors (1,0,0), (0,1,0), (0,0,1) are a basis for three-dimensional Euclidean space. 4. In a free module, any linearly independent set that spans the module. 5. Also called **Schauder basis**, in a separable normed space, a sequence of vectors $\{x_i\}$ in terms of which each element can be expressed uniquely as an infinite combination: $x = \sum_{i=1}^{\infty} v_i x_i$.

BASIS THEOREM: The theorem that any linearly independent set of d vectors is a basis of a vector space of finite dimension d.

BASIS FUNCTION: A one-electron function used in the expansion of the molecular orbital function. Basis functions are commonly represented by atomic orbitals centred on each atom of the molecule.

BASISCOPIC: A structure facing towards the base such as the sori of most ferns.

BASKET CELL: A type of cell found in the cerebral cortex. They form a layer next to the purkyne cells, from which they receive stimuli.

BASOPHIL (BASOPHIL GRANULOCYTES): A type of white blood cell that has a lobed nucleus surrounded by granular cytoplasm. It represents about 0.01% to 0.3% of circulating white blood cells. They are produced continually by stem cells in the red bone marrow and move about in an amoeboid fashion.

BASOV, NIKOLAI GENNEDIYEVITCH (1922 – 2001): Russian physicist best known for the development of the maser, the precursor of the laser. In 1955, while working as a research student with Aleksandr Prokhorov (1916-2000) at the Soviet Academy of Sciences, he devised a microwave amplifier based on ammonia molecules. The two scientists shared the 1964 Nobel Prize with America physicist, Charles Townes (1915 -), who independently developed a maser.

BAST (SECONDARY PHLOEM): A type of phloem derived from the secondary meristems of a vascular plant, which is a distinctive feature of most dicot plants. The formation of the secondary phloem implicates secondary growth in vascular plants.

BAT: A nocturnal flying mammal of the order *Chiroptera*, having membranous wings that extend from the forelimbs to the hindlimbs or tail and anatomical adaptations for echolocation, by which they navigate and hunt prey. Bat is the only mammal that can fly. It feeds on fruits and insects and lives on trees and in caves.

BATCH: 1. In computing, it refers to a set of jobs that are processed together by a computer with no input from individual terminals. 2. A set of jobs to be executed together by a machine. Alternatively, it is a quantity of material which is known or assumed to be produced under uniform conditions.

BATCH CULTURE (BATCH-CULTURE): In biotechnology, a technique used to grow biological material which consists of a large-scale closed system with a fixed volume of nutrient culture medium under specific environmental conditions such as nutrient type, temperature, pressure and aeration. During the incubation period, the culture declines after a certain time interval, when the culture is harvested.

BATCH FILE: A text file that contains a sequence of commands for a computer operating system.

BATCH OPERATION: The operation of an analytical instrument in such a way that one or more analytical procedures must be completed for a sequence of samples before the next sequence can be started. The term batch usually implies a sequence of a variably sized group of samples, the size of which is not related to a particular type of instrument.

BATCH PROCESSING: 1. An industrial process in which enzymes are dispersed throughout a substrate. At the end of the batch, the enzyme has to be separated from the product and the fermentation vessel must be emptied and cleaned before the next batch. 2. A system of processing a series of jobs all at one time on a computer without manual intervention. Once the job begins, it continues until it is done or until an error occurs.

BATCH REACTOR: A chemical reactor in which the reactants and catalyst are introduced in the desired quantities and the vessel is then closed to the delivery of additional material.

BATESIAN MIMICRY: A type of similarity of one organism to another in which a harmless species looks like a different species that is poisonous or otherwise harmful to predators, because a predator that has learned to avoid an organism with a given warning system will avoid all similar organisms, thus making the resemblance a protective mechanism. It was named after the British naturalist, Henry Walter Bates (1825-1892), who first described it.

BATESON, WILLIAM (1861-1926): A British geneticist who worked at the University of Cambridge and buttressed the works of Gregor Mendel in genetics by studying the inheritance of comb shapes in chickens and also established that some traits are controlled by more than one gene. He refuted T.H Morgan's explanation of linkage and coined the discipline genetics in 1905.

BATH: A container of liquid in which something is washed or dipped in chemical or industrial processes.

BATHOCHROMIC SHIFT (RED SHIFT): The displacement of a particular spectral band to a longer wavelength (low energy) as a result of chromophoric substitution such as introduction of hydroxy or methoxy group into the chromophore or solvent and medium changes.

BATHOLITH: A large intrusive igneous rock, having a surface area in excess of 100 sq km and normally surrounded by metamorphic rocks. They are formed either as one large mass or many smaller masses at great depths in the Earth's crust and are exposed at the surface only after considerable erosion of the overlying mountain mass.

BATHYAL ZONE (BATHYPELAGIC ZONE): An open-ocean zone that extends from a continental slope down about 2000 metres. The zone is dark with less than 1% of sunlight, as such no photosynthesis occurs. Because of the lack of light, some species which live there do not have eyes, except the viperfish and the frill shark. Some marine lives such as squid, large whales, octopuses, sponges, brachiopods, sea stars, and echinoids are found there. It is the feeding grounds of some of the world's largest whales.

BATRACHOTOXIN: A very powerful poison secreted from the skin of South American arrow-poison frogs of the genus *Dendrobates*. It is a neurotoxin that can block the transmission of nerve signals to the muscles permanently. It also has a severe effect on the heart, where it permanently interferes with conduction, causing arrhythmias, ventricular fibrillation and eventual cardiac failure. It is used by Indians in western Colombia for poisoning blowgun darts for hunting.

BATTERY: A group of electric cells which converts chemical energy into electrical energy. It is made up of two or more cells each containing two conducting electrodes, one positive and one negative immersed in an electrolyte in a container. Examples are the lead-acid battery, nickel-cadmium battery (NiCd or NiCad) and nickel–iron battery (NiFe battery).

BATTERY CAGE (BATTERY FARM): An industrial agricultural confinement system or an intensive method of keeping poultry in separate cages, used primarily for egg-laying hens.

BAUD: A unit for measuring signal speed in a computer communications system. The baud rate of a data communications system is the number of symbols per second transferred. A symbol may have more than two states, so it may represent more than one binary bit. A binary bit always represents exactly two states.

BAUHINIA: A genus of tropical or subtropical leguminous flowering plants which has more than 200 species in the subfamily Caesalpinioideae of family Fabaceae. The plants can be small trees, shrubs or woody vines. The plants do well in acidic and salty soils either in full sun or under partial sun. The trees can grow to between 6-12 metres high with branches spreading between 3-6 metres outwards with lobed leaves. The various species produce flowers, whose colours are in the shades of red, purple orange and are scented. The plants are propagated by seeds or cutting. The plants are widely cultivated as ornamental ones. Medicinally, its various parts have been used to treat various ailments such as diabetes, infections, pain and inflammation. *Bauhinia blakeana*'s flower is the emblem of Hong Kong that appears on the regional flag.

BAUXITE: The main ore from which aluminium is extracted. It contains aluminium oxide (Al_2O_3). The major impurities in bauxite are clay minerals, iron oxides and silica.

BAY: 1. A wide inlet of the sea, enclosed by a wide inward-curving stretch of coastline. Bays may be formed by continental drift, glacial and river erosion, by the folding of the earth's crust and also by coastal erosion due to waves and currents. A large bay may be called a gulf, a sea, a sound or a bight. 2. A part of a building marked off by vertical elements, such as columns or pilasters: 3. A reddish-brown animal, especially a horse having a black mane and tail.

BAYES' THEOREM (BAYES' RULE; BAYES' LAW): The equation used in probability which allows the user to calculate the probability of an event occurring provided that another event whose probability is already known has occurred. Mathematically, Bayes' theorem gives the relationship between the probabilities of A and B, $P(A)$ and $P(B)$ and the conditional probabilities of A given B and B given A, $P(A|B)$ and $P(B|A)$, respectively. In its most common form, it is expressed as $P(A|B)=\frac{P(B|A)P(A)}{P(B)}$, the meaning of this statement depends on the interpretation of probability ascribed to the terms. Bayes' theorem was named after the Reverend Thomas Bayes (1702–61), who studied how to compute a distribution for the probability parameter of a binomial distribution. It has applications in fields such as science, engineering, medicine and law.

BAYESIAN: In statistics, of a theory, presupposing known prior probabilities that may be subjectively assessed; these can be revised in the light of experience in accordance with Bayes's theorem. A hypothesis is thus confirmed by an experimental observation that is likely given the truth of the hypothesis and unlikely given its falsehood. It was named after the English probability theorist and theologian, Thomas Bayes (1702-61).

BAYESIAN FILTER (BAYESIAN SPAM FILTERING): A very powerful technique for dealing with spam that uses Bayesian logic, also called Bayesian analysis, to evaluate the header and content of an incoming e-mail message and determines the probability that it constitutes spam.

BAYESIAN INFERENCE: A method of statistical inference that estimates the probability that a hypothesis or an event may occur based on the frequency at which the event had occurred in the past. It is the application of Bayes' theorem to update beliefs.

BAYESIAN LOGIC (BAYESIAN ANALYSIS): A branch of logic applied to decision-making and inferential statistics that deals with probability inference, using the knowledge of prior events to predict future events.

BAYESIAN STATISTICS AND PROBABILITY: A method of interpreting probability in terms of a number indicating how much confidence is placed in a proposition or how true it is. Bayesian statistics is that subset of the entire field of statistics in which the evidence about the true state of the world is expressed in terms of degrees of belief or more, specifically, Bayesian probabilities; and the starting point in modern Bayesian probability theory is that probability is interpreted as a degree of belief. Such an interpretation is only one of a number of interpretations of probability and there are many other statistical techniques that are not based on "degrees of belief". The general set of statistical techniques can be divided into a number of activities, including Statistical inference, Statistical modelling, Design of experiments and Statistical graphics. Many of these have special "Bayesian" versions and using Bayesian probability allows a researcher to judge the amount of confidence that one has in a particular result.

BEACH DEPOSITS: Concentrations of mineral formed by the grinding action of waves, winds, etc.

BEACH MORNING GLORY (GOAT'S FOOT; *Ipomoea pes-caprae*): A salt-tolerating tropical creeping vine belonging to the family Convolvulaceae, native to tropical coastlines around the world, growing on the upper parts of sandy beaches and is resistant to salty air. The stems grow up to 5 metres long and have deep, fleshy tap-roots. It is a creeping plant which binds soil particles together due to its woody rootstocks. It possesses two large lobed leaves which reduce the impact of rain drops on the sandy beach. The pink to rose purple flowers are trumpet-shaped. The fruit is a woody capsule. The hardy nature of the plant enables it to serve as a pioneer species in a biological succession, growing on dunes.

BEAD VINE (CORAL BEAD PLANT; CORAL BEAN; CRABS EYE; LICORICE VINE; *Abrus precatorius*): A legume, belonging to the Fabacea family, with a long slender climbing stem that twines around hedges, shrubs and trees. It is native to India and found in the tropics and subtropical regions of the world. It has slender branches and a cylindrical wrinkled stem with a smooth-textured brown bark. The leaves are glabrous with long internodes and are alternate compound paripinnate with stipules; and pinnately compound, oblong and obtuse leaflets. The flowers are small and pale, violet to pink with a short stalk, arranged in clusters. The fruit is a pod, flat, oblong and truncate-shaped with a sharp, deflexed beak and silky-textured. The seedpod curls back when it opens to reveal the seeds, which are black and red although other colours are available. Its leaves are used as a vegetable in central and east Africa. Its seeds are toxic due to the presence of Abrin; and are used as beads and in making necklaces and percussion instruments.

BEAD-ROD MODEL: A model, which simulates the hydrodynamic properties of a chain macromolecule. It consists of a sequence of beads, each of which offers hydrodynamic resistance to the flow of the surrounding medium and is connected to the next bead by a rigid rod which does not. The mutual orientation of the rods is random.

BEAD-SPRING MODEL: A model, which simulates the hydrodynamic properties of a chain macromolecule. It consists of a sequence of beads, each of which offers hydrodynamic resistance to the flow of the surrounding medium and is connected to the next bead by a spring which does not contribute to the frictional interaction but which is responsible for the elastic and deformational properties of the chain. The mutual orientation of the springs is random.

BEADLE, GEORGE WELLS (1903 - 1989): United States of America (U.S.A) geneticist, who worked on the fruit fly, *Drosophila melanogaster* at Thomas Hunt Morgan's laboratory at the California Institute of Technology and realised that genes must influence heredity chemically. He established the one gene-one enzyme theory using mould culture working together with Edward Tatum at Stanford University. He shared the 1958 Nobel Prize for Physiology with Edward Tatum and Joshua Lederberg.

BEAK (BILL; ROSTRUM): The hard horny projecting structure forming the upper and lower mandibles of birds. The upper and lower mandibles are covered with a thin keratinized layer of epidermis known as the **rhamphotheca**. In most species, two holes known as nares lead to the respiratory system. Beaks can vary significantly in size, shape and colour and are used for eating, grooming, manipulating objects, killing prey, fighting, probing for food, courtship and feeding the young.

BEAKER: 1. A flat-bottomed glass container, usually cylindrical, used in laboratories to hold liquid samples and measure rough volumes of liquid. 2. An open glass or plastic container with lip for pouring.

BEAM: 1. In physics, it is a collection of two or more rays moving together in the same direction or towards the same point away from the same point. Categories of beam include: (i) parallel rays in which the rays do not meet; (ii) convergent beam in which the rays meet at a particular point; (iii) divergent beam in which the rays move from a particular point. 2. A narrow stream of moving particles or other radiation especially from a light source, the Sun or the Moon. An example is alpha radiation. 3. A series of radio signals used to guide ships or aircraft. 4. A long piece of wood, metal, concrete, etc, usually horizontal and supported at both ends, that carries the weight or part of a building or some other structure. An example is a cantilever that projects beyond a fulcrum and is supported by a balancing member or a downward force behind the fulcrum or a beam that is supported at one end and carries a load at the other end or along its length. 5. Any of the horizontal cross bars of a ship, joining the sides and supporting the deck or the breadth of a ship at its width point. 6. A horizontal bar or weighing slab from which the pans hang.

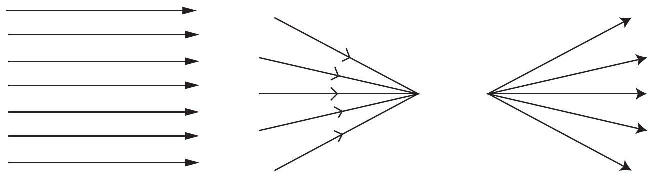
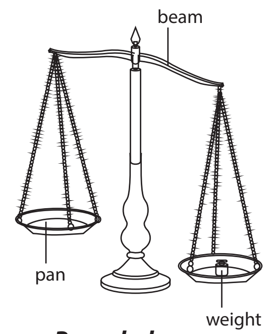


Fig. 1: Parallel rays Fig. 2: Convergent rays Fig. 3: Divergent rays

Categories of rays

BEAM BALANCE: An instrument for measuring the mass of an object. In its traditional form, it consists of a pivoted horizontal lever of equal length arms, called the beam, with a weighing pan, suspended from each arm. The unknown mass is placed in one pan and standard masses are added to the other pan until the beam is as close to equilibrium as possible. In precision balances, a slider mass is moved along a graduated scale. The slider position gives a fine correction to the mass value. Very precise measurements are achieved by ensuring that the balance's fulcrum is essentially friction-free, by attaching a pointer to the beam which amplifies any deviation from a balance position; and finally by using the lever principle, which allows fractional masses to be applied by movement of a small mass along the measuring arm of the beam. Balances are different from scales, in that a balance measures mass, whereas a scale measures weight. Although a balance technically compares weights, not masses, the weight of an object is proportional to its mass and the standard weights used with balances are usually labelled in mass units. Balances are used for precision mass measurement, because unlike spring scales their accuracy is not affected by differences in the local gravity.



Beam balance

BEAM CURRENT: The number of primary electrons reaching the surface of the specimen per unit time expressed as electrical current. Its recommended symbol and unit are i_B and A, respetively.

BEAM HOLE: A hole through the shielding of a nuclear reactor to enable a beam of neutrons or other particles to escape for experimental purposes.

BEAMFORMING: A type of radio frequency management (RF) in which an access point uses multiple antennas to send out the same signal. By sending out multiple signals and analysing the feedback from clients, the wireless LAN infrastructure can adjust the signals it sends out and determine the best path the signal should take in order to reach a client device. In a sense, beamforming shapes the RF beam as it traverses the physical space of the enterprise.

BEAMING: In infrared transmission, it refers to the communication of data between wireless devices using a beam of infrared light. This beam, invisible to humans, is used in many devices, such as television remote controls and garage door openers. Infrared transmission is often used to transfer information between computing devices, for example, beaming data between a handheld device, such as a personal digital assistant (PDA) and a laptop computer. Some portable computing devices are equipped with infrared ports and entire networks can be set up using only infrared transmission for communication. Infrared transmission requires a clear line of sight and therefore not applicable where there are physical objects such as walls in between the devices to be connected.

BEAMSPLITTER: An optical element that allows part of an electromagnetic wave to pass through while reflecting the other part.

BEAMWIDTH: The angular extent of the beam of a radio telescope measured in degrees of arc. Generally an antenna with low gain has a large beam width, receiving signals well from a number of different directions.

BEAN: The small, seeds of leguminous trees and shrubs and various leguminous plants of the family Leguminosae (pulse family) belonging to many and variable genera and species. They are generally warm-season annuals, although the roots of tropical species tend to be perennial that grow erect or as vines. The seeds are typically kidney-shaped, growing in long pods and most species are edible.

BEAN SPROUTS: Young seedlings of bean, often eaten raw, especially in Chinese dishes.

BEAR: 1. Any of various usually omnivorous mammals of the family Ursidae. It is a large strong wild animal with a massive thick-furred body and short tail that lives especially in colder parts of Europe, Asia and North America. 2. To give birth to young; or to produce fruits or flowers.

BEARDED: 1. Having many awns as, for example, the ears of the bearded fescue (*Festuca ambigua*). 2. Having any other kind of stiff hairs as, for example, the collection of hairs on the lower petals of the flag iris (*Iris pseudacorus*).

B

BEARING: 1. A device designed to reduce the friction between the fixed and rotating parts in a machine. For example, a motor has a ball-bearing at each end of the rotor around which it rotates. 2. In mathematics, the angle measured in a clockwise direction from the north line or North Pole to a point. A bearing is used to represent the direction of one point relative to another point. Three digits (000^o) are used to represent a bearing and to differentiate it from an ordinary angle.

BEARING CAPACITY: The average load per unit area that is required to rupture a supporting soil mass.

BEAT FREQUENCY: The absolute value of the difference in frequency of two waves. It is the fluctuation produced in musical acoustic when two notes of nearly equal pitch or frequency are heard together. Beats result from the interference between the sound waves of the notes. The beat frequency equals the difference in frequency of the notes. Given oscillations at two frequencies, f_1 and f_2, their beat frequency is $f_{beat} = | f_1 - f_2 |$, where $f_1 > f_2$.

BEATERS: Steel bars on the drum of a combine harvester.

BEATS: 1. An inter-phase effect between two waves of slightly different frequency, for example, between two sound waves at a point in a medium or between two alternating currents at a point in a circuit. Beats occur within two sound waves of similar frequencies superimpose. As the waves move in and out of phase with each other, they interface first constructively and then destructively. The ear hears a single note whose loudness varies regularly at the beat frequency, which equals the difference between the frequencies of the two waves. 2. A periodic increase and decrease in loudness heard when two notes of slightly different frequencies are sounded at the same time.

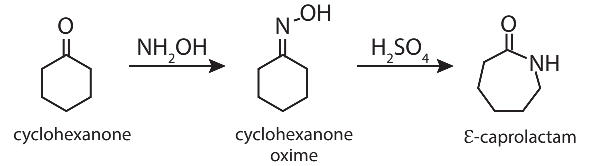
BEATTIE-BRIDGMAN EQUATION: An equation of state that relates the pressure, volume and temperature of a gas and the gas constant. It uses empirical constants to assess the reduction in the effective number of molecules due to various types of molecular aggregation. It is expressed as: $P = \frac{RT}{V^2}(1-\varepsilon)(V+B) - \frac{A}{V^2}$, where *P* is the pressure, *V* the volume, *R* the gas constant, *T* the thermodynamic temperature and *A*, *B* and ε are constants related to five empirical constants, A_o, B_o, *a*, *b* and *c* by $A = A_o\left(1 - \frac{a}{V}\right)$, $B = B_o\left(1 - \frac{b}{V}\right)$ and $\varepsilon = \frac{C}{VT^3}$.

BEAUFORT SCALE (BEAUFORT WIND FORCE SCALE): An international system of recording wind velocity, usually made at 10 metres above ground level; and indicated by numbers ranging from 0 for calm to 12 for hurricane. The Beaufort number 0 means calm and wind speed of less than 1 km/h; and 12 is a hurricane with wind speed of more than 118 km/h and waves over 13 m and greatly reduced visibility. It was invented in 1806 by Admiral Sir Francis Beaufort (1774–1857), British hydrographer and a naval officer.

BEAUMONT PERIOD: A period of forty-eight consecutive hours during which the minimum temperature is 10°C and the relative humidity is 75% or more. This period has been widely used as a criterion for issuing warnings of the onset of potato blight (*Phytophthora infestans*).

BECKLIN-NEUGEBAUER OBJECT: A point in the star-forming part of the Orion Nebulla that produces infra red radiation, discovered in 1967 by Eric Becklin and Gerry Neugebauer at the California Institute of Technology in 1966. It is the brightest object in the sky apart from the Sun at wavelengths of around 10 microns.

BECKMANN REARRANGEMENT: A catalysed rearrangement of an oxime to an amide, named after the German chemist Ernst Otto Beckmann (1853–1923). Oximes formed from ketones by reacting a ketone with hydroxylamine undergo rearrangement to form the corresponding amide catalysed by phosphorus pentachloride, sulfuric acid, etc. It is used in the manufacture of nylon and other polyamides.



Beckmann Rearrangement

BECKMANN THERMOMETER: A sensitive thermometer with an adjustable range for measuring small changes in temperature. It is used particularly for measuring depression of freezing point or elevation of boiling point of liquid when solute is added in order to find relative molecular masses. It consists of mercury in glass thermometer with the scale covering only 5 or 6 calibrated in hundredth of a degree.

BECQUEREL: The SI unit for the measurement of radioactivity. Its symbol is Bq and it is named after Henri Becquerel, who shared a Nobel Prize with Pierre and Marie Curie for their work in discovering radioactivity. One Bq is defined as the activity of a radioisotope decaying at the rate of 1 nucleus disintegration per second. The Bq unit is equivalent to s^{-1}.

BECQUEREL, ANTOINE HENRI (1852–1908): A French physicist, whose early researches were in optics. In 1896 he accidentally discovered radioactivity in fluorescent salts of uranium. Later he showed that it consists of charged particles that are deflected by a magnetic field. For this work he was awarded the 1903 Nobel Prize for Physics, which he shared with Pierre and Marie Curie.

BED: 1. A single sedimentary rock unit with a distinct set of physical characteristics or contained fossils, readily distinguishable from those above and below. 2. The bottom of the sea, a river or lake. 3. A garden plot. 4. A unit layer 1 cm or more thick that is visually or physically more or less distinctly separable from other layers above and below in a stratified sequence.

BEDBUG: A wingless blood-sucking insect, of the family Cimicidae. The common bed bug is classified as *Cimex lectularius* and is about 6 mm long with a flat reddish body. It infests dwellings and beddings. It is not known to transmit any disease in humans.

BEDDING: Materials such as straw used for animals to sleep on.

BEDDING PLANT: A plant suitable for planting in a garden bed, irrespective of their growing habits. Examples are perennials such as hyacinths and tulips, chrysanthemums, castor oil plant, petunias, begonias, pelargoniums and calceolarias.

BEDROCK: The layer of rock beneath soil and other surface materials such as gravel or alluvium. It can extend hundreds of metres below the surface of the earth, toward the base of Earth's crust. It may be igneous, sedimentary, or metamorphic origin and forms the upper surface of the rocky foundation that composes the earth's crust. Where there is high erosion, bedrock may become exposed to the surface. An exposed portion of bedrock is called an outcrop.

BEE: Any of 3000 species of bees divided into 19 families in the order Hymenoptera. It is a large insect of about 2 mm – 4 cm long; solitary or social in habit and living on nectar and pollen to produce wax and honey. Some species of bees have stingers. Bees are dependent on pollen as a protein source and on flower nectar or oils as an energy source. Adult females collect pollen primarily to feed their larvae. The pollen they inevitably lose in going from flower to flower is important to plants because some pollen lands on the pistils of other flowers of the same species, resulting in cross-pollination. They are the most important pollinating insects and their interdependence with plants makes them good examples of the type of symbiosis called mutualism. Most bees have specialised branched or feathery body hairs that help in the collection of pollen. Female bees, like many other hymenopterans, have a defensive sting. Some bees produce honey from flower nectar. Honey bees and stingless bees commonly store large quantities of honey, which is exploited by beekeepers, who harvest the honey for human consumption.

BEEF: The flesh or meat from slaughtered mature cattle such as ox, a bull or cow. Veal is the flesh of calves or immature cattle. The best beef is obtained from steers or castrated males and heifers or female cows that have not calved.

BEEHIVE: 1. A naturally enclosed structure in which some honey bee species of the subgenus *Apis* live and raise their young. It also refers to man-made structures in which bees are kept for such purposes as to produce honey, beeswax, pollen, royal jelly, pollinating fruits and vegetables, raising queens and bees for sale or for scientific. The internal structure consists of densely-packed matrix of hexagonal cells made of beeswax or honeycomb. The bees use the cells to store honey and pollen as food and to house the eggs, larvae and pupae. 2. A **beehive shelf** is an equipment usually made of pottery, used in the laboratory to support a receiving jar or tube while a gas is being collected over water with pneumatic trough.

BEEKEEPER (APIARIST): A person who keeps bees for such purposes as to produce honey, beeswax, pollen, royal jelly, pollinating fruits and vegetables, raising queens and bees for sale or for scientific research.

BEEP CODE: The audio signal given out by a computer to announce the result of a short diagnostic testing sequence the computer performs when first powering up (called the Power-On-Self-Test or POST). The Power-On-Self-Test is a small program contained in the computer's Basic Input/Output Operating System (BIOS) that checks to make sure necessary hardware is present and required memory is accessible. If everything tests out correctly, the computer will typically emit a single beep and continue the starting-up process. If something is wrong, the computer will display an error message on the monitor screen and announce the errors audibly with a series of beeps that vary in pitch, number and duration. This is especially useful when the error exists with the monitor or graphic components.

BEER: An alcoholic drink made from the yeast fermentation of malted cereal grains such as barley, to which hops and water have been added.

BEER'S LAW (BEER-LAMBERT LAW; BEER–LAMBERT–BOUGUER LAW): It states that the fraction of incident light absorbed by a solution at a given wavelength is related to the thickness of the absorbing layer and the concentration of the absorbing substance. It is commonly written in the form $A = \varepsilon cl$, where A is the absorbance, c is the concentration in moles per litre, l is the path length in centimetres and ε is a constant of proportionality known as the molar absorptivity. The law is accurate only for dilute solutions; deviations from the law occur in concentrated solutions because of interactions between molecules of the solute, the substance dissolved in the solvent.

BEESWAX: A yellowish wax made by bees for building honey combs. It is also used for making wood polish.

BEET: The beet (*Beta vulgaris*) is a plant in the Chenopodiaceae family which is now included in Amaranthaceae family. It is best known in its numerous cultivated varieties, the best known of which is the purple root vegetable known as the beetroot or garden beet. However, other cultivated varieties include the leaf vegetables, chard and spinach beet, as well as the root vegetables sugar beet, which is important in the production of table sugar and mangelwurzel, which is a fodder crop. Three subspecies are typically recognised. All cultivated varieties fall into the subspecies *Beta vulgaris* subspecies *vulgaris*.

BEET SUGAR: An alternative name for sucrose, with the chemical formula $C_{12}H_{22}O_{11}$. It is made from *Beta vulgaris*, the common beet. It is formed through a process of photosynthesis in the leaves and it is then stored in the root. The syrup is extracted from the roots and the sugar is made by crystallisation.

BEETLE: Any of the several types of insects, belonging to the order Coleoptera, the largest group of animals on Earth. They are found in all major habitats, except marine and the Polar Regions. They come in a variety of shapes and colours and can range from 0.4 to about 80 millimetres in length. They are generally herbivores, scavengers or predators, although some adult beetles do not feed at all. They are characterised by modified outer pair of wings that forms a hard covering or exoskeleton, particularly on their forewings known as elytra (singular elytron). Beetles have three major body segments: the head, with a single pair of antennae and usually a pair of compound eyes; the thorax, which typically bears two pairs of wings and three pairs of legs; and the abdomen, where the reproductive and digestive organs are housed. Beetles have chewing jaws called mandibles. Two pairs of finger-like appendages, the maxillary and labial palpi, are found around the mouth in most beetles, serving to move food into the mouth.

BEGGIATOALES: An order of filamentous or unicellular bacteria that move by gliding over the substrate. Many of its members, the filamentous sulfur bacteria, grow by oxidising sulfides. The genus Beggiatoa has certain features in common with the blue-green alga, Oscillatoria.

BEGONIA: Any of flowering, mostly succulent plants in the family Begoniaceae with over 1000 species, planted as ornamentals. They do well in tropical and subtropical areas; but are tender plants that cannot withstand drought and require rich, well-drained damp soil and shade for protection from the strong sun. The leaves of begonia are large, alternate, usually asymmetrical and varied in shape and colour. The plants are monoecious with male flowers and female flowers occurring separately on the same plant. The male flowers have lots of stamens and the female flowers have ovaries below their petals. The flowers have colourful sepals ranging from pink, white, scarlet or yellow without petals. The fruits are winged capsules containing lots of minute seeds. The plants can be propagated by seeds or cuttings from the stem or leaf and may be planted in flowerbeds and in outdoor or indoor containers. The three main types of begonias include the following: rhizomatous, with fleshy stems and roots creeping along the soil, grown mostly for foliage and also for flowers; fibrous-rooted begonia having fibrous roots and include the semperflorens or wax begonias, cane-like begonias and other varieties; and the tuberous-rooted type of begonia with fleshy, round tuberous root, grown from tubers as a bedding plant and also as a greenhouse plant. It goes dormant in the winter. Some others are described commonly as cane-like, hardy, shrublike, thick-stemmed and trailing.

BEHAVIOUR: Observable actions of an organism in response to stimuli from its external or internal environment. For example, a hen may let out a warning cry on seeing a bird of prey (external stimulus); or begin to search for food on feeling hungry (internal stimulus.

BEHAVIOUR THERAPY (BEHAVIOURAL THERAPY): In psychology, it is the application of behavioural principles that is used in the treatment of anxiety and mood disorders such as phobias, obsessions, bipolar disorders and schizophrenia; as well as autism, substance abuse and eating disorders. It also helps to cope with difficult situations and in the management of pain for people with some chronic diseases.

BEHAVIOURAL ECOLOGY: The study of the ecological function of animal behaviour in its evolution, survival and adaptation, based on Darwinian fitness (reproductive success).

BEHAVIOURAL GENETICS: The study of the importance of genetic make up of an organism in relation to environmental factors that influence animal behaviour.

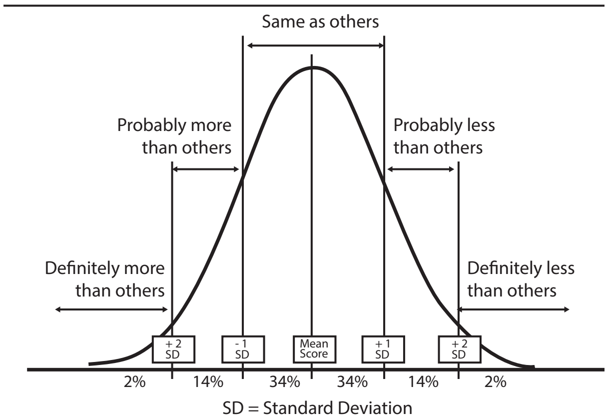
BEHAVIOURISM (BEHAVIOURAL PSYCHOLOGY; BEHAVIOURISM SCHOOL OF PSYCHOLOGY): A school of psychology that originated in the early twentieth century from the USA and established by John B. Watson (1878-1958), an

American psychologist that is concerned with observable behaviour and quantifiable aspects of behaviour as opposed to subjective phenomena, such as emotions or motives and asserts that all human activities can ultimately be explained in terms of conditioned reactions or reflexes and habits formed in consequence that can be objectively and scientifically measured.

BEILSTIN'S TEST: A chemical test for the presence of a halogen such as chlorine, iodine and bromine in a compound. A cleaned copper wire is heated in a flame of a burner to form a coating of copper(II) oxide. It is then dipped in the test sample and once again heated in the flame. An alternative wet test for halide is the sodium fusion test.

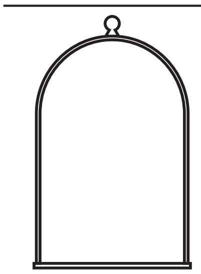
BEL: A unit of the power of a radiation such as sound or a signal such as audio signal compared to some standard. The commonly used unit for sound or audio signal is the decibel (1 bel = 10 decibels).

BELL CURVE (GAUSSIAN DISTRIBUTION; NORMAL DISTRIBUTION;): A probability distribution, $f(x)$, of a random variable, x that is assumed by many statistical methods. This is given by $f(x)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right]$, where μ is the mean and σ^2 is the variance. All normal distributions are symmetric and have bell-shaped density curves with a single peak.



A graph of the Bell curve, the mean is zero

BELL JAR: A bell-shaped glass cover used to enclose equipment in experiments and prevent gases from escaping or entering.



Bell Jar

BELL METAL: A hard alloy of copper (80%) and 20% tin used in making bells, cymbals and related instruments due to its low damping qualities.

BELL NUMBER: The number of ways that n distinguishable objects such as differently coloured balls can be grouped into sets such as buckets if no set can be empty. For example, if there are three balls, coloured red (R), green (G) and blue (B), they can be grouped in five different ways: (RGB), (RG)(B), (RB)(G), (BG)(R) and (R)(G)(B), so that the third Bell number is 5. Starting with $B_0 = B_1 = 1$, the first few Bell numbers are 1, 2, 5, 15, 52, 203, 877, 4140, 21147, 115975, ...

BELL WETHER: A sheep with a bell hung round its neck, which leads a flock.

BELL-SHAPED DISTRIBUTION: A probability distribution whose entire shape is a vertical cross-section of a bell. Examples include the normal distribution and the Student's t-distribution. In probability theory, the normal or Gaussian distribution is a continuous probability distribution that has a bell-shaped probability density function, known as the Gaussian function or the bell curve.

BELL'S THEOREM (BELL'S INEQUALITY): A theorem that states that no physical theory of local hidden variables can ever reproduce all of the predictions of quantum mechanics. It has important implications for physics and the philosophy of science as it proves that every quantum theory must violate either locality or counterfactual definiteness.

BELLADONNA (DEADLY NIGHTSHADE): Common names for ***Atropa belladonna***, a poisonous perennial herbaceous plant in the nightshade family, Solanaceae that grows up to 1.5 metres. It has purplish-red bell-shaped flowers which bear black berries. All parts of the plant contain tropane alkaloids. Atropine, a poisonous crystalline alkaloid, with the chemical formula $C_{17}H_{23}NO_3$ is isolated from its leaves and root. Medicinally, the uses of atropine include to check secretions and spasms, to relieve pain or dizziness and as a cardiac and respiratory stimulant.

BELLY (ABDOMEN): In mammals, the part of the body between the thorax (chest) and pelvis containing the stomach, alimentary canal and digestive organs.

BELLMAN'S PRINCIPLE OF OPTIMALITY: The fundamental principle of dynamic programming stating that the optimal solution to an n-step dynamic process must come from an optimal solution of the (n-1)-step process that commences with the optimal outcome of the first step. This principle can be extended to permit the recursive solution of many dynamic programming problems. Bellman's principle of optimality is not the same as Lewis Carroll's 'Bellman's principle' that what is said thrice is true.

BELOW: Less than; the limit of a function from below is the left-hand limit written variously as $\lim_{x\uparrow a} f(x)=\lim_{x\to a} f(x)=f(a-)$, the one-sided limit where x is restricted to values less than a.

BELT: A loop of flexible material such as strong cloth used to transfer force or goods for example, in a conveyor belt in a factory or an endless moving strap used to connect wheels and so drive machinery or carry things along.

BELTSVILLE SMALL WHITE: A breed of turkey that weighs only 3.6 to 5.5 kilograms at maturity.

BENARD CELL (BÉNARD CELL): Ordered hexagonal convection cells or atomic-molecular structures that spontaneously form in viscous mediums, such as whale or silicon oil, when placed on a hot plate and heated past a bifurcation point into a turbulent flow regime. It was named after the French physicist Henri Bénard, who conducted the first experiments on them in 1900. Further analysis of Bénard cells in the 1970s showed that as the system moves farther away from equilibrium, it reaches a critical point of instability, at which the ordered hexagonal pattern emerges and during convection, heat is transferred by the coherent motion of large numbers of molecules. It was used as an example of dissipative structures, since it is a type of ordered structure that forms according to an entropy generation version of the second law of thermodynamics. This concept later developed into a thermodynamic model of the evolution of life. It must be noted that a Rayleigh–Bénard convection is a type of natural convection, occurring in a plane horizontal layer of fluid heated from below, in which the fluid develops a regular pattern of convection cells known as Bénard cells while a Bénard–Marangoni convection refers specifically to the effects of surface tension.

BENCHMARK: 1. A standard against which something can be measured or assessed. 2. In computing, it is a standard test to measure the performance of a piece of equipment, hardware or software, usually consisting of a standard program or suite of programs.

BENCHMARK DOSE, BMD (BENCHMARK CONCENTRATION, BMC): A dose or concentration that produces a specific change in response rate of an adverse effect compared to the response in unexposed subjects. It can be used in health risk assessment of chemical substances instead of the no-observed-adverse-effect-level

(NOAEL). The adverse effect is called the **benchmark response**, abbreviated **BMR**.

BENCHMARKING (ENVIRONMENTAL BENCHMARKING): A tool for analysing environmentally related practices and indicators that lead to superior environmental performance, while also enhancing economic performance. It includes all areas of activities and not restricted solely to those activities that have an obvious environmental impact. Therefore, it may include an assessment of environmental management systems (EMS), management performance, environmental accounting, resource and waste management, product environmental quality, environmental education and training, customer relations and emergency response.

BENDING MOMENT: The algebraic sum of the moments of all the vertical forces to either side of the point or section of a horizontal beam under load. The bending moment at any point is calculated by multiplying a force by the distance between that point and the force. It is a calculation used to identify where the greatest amount of bending takes place. The bending moment that bends the beam convex downward is positive and one that bends it convex upward is negative.

BENDS: A severe pain and difficulty in breathing, experienced by deep-sea divers who come to the surface too quickly.

BENEDICK EFFECT: The converse of the Thomson effect, in which a difference in voltage between two points produces a difference in temperature between these points.

BENEDICT'S SOLUTION (BENEDICT'S REAGENT): A chemical reagent named after an American biochemist, Stanley Rossiter Benedict (1884 — 1936). It is a deep-blue alkaline solution used to test for the presence of the aldehyde functional group, –CHO or to test for reducing sugars such as monosaccharides, the disaccharides, lactose and maltose. It consists of a mixture of copper(II) sulfate, sodium carbonate and sodium citrate dissolved in water. A brown, red or yellow colour is formed with the solution. It reacts chemically like Fehling's solution.

BENEDICT'S TEST: A qualitative test for reducing sugars such as monosaccharides, the disaccharides, lactose and maltose in solution. A high concentration of reducing sugars induces the formation of a red precipitate, while a lower concentration produces a yellow precipitate. The substance to be tested is heated with Benedict's solution. Formation of a brick-red precipitate indicates presence of the aldehyde group. Since simple sugars such as glucose give a positive test, the solution is used to test for the presence of glucose in urine, a symptom of diabetes mellitus.

BENEFICATION: A variety of processes such as crushing, grinding and magnetic separation used to separate ore into the valuable components and the waste material.

BENIOFF ZONE (WADATI-BENIOFF ZONE): A long narrow region, usually adjacent to a continent, along which earthquake foci lie on a plane which dips downwards at about 45° and along which the oceanic lithosphere is thought to be descending into the earth's interior. The Benioff zone extends from the surface to a depth of about 700 km. Earthquakes can be produced in a Benioff zone by a slip along the subduction thrust fault or by a slip on faults within the downgoing plate as a result of bending and extension as the plate is pulled into the mantle.

BENITOITE: A rare glassy, blue to violet mineral, barium titanium silicate with the chemical formula $BaTi(SiO_3)_3$, Mohs scale hardness 6 - 6.5 and relative density 3.6. Benitoite typically occurs with an unusual set of minerals, along with minerals that make up its host rock. Such minerals include natrolite, neptunite, joaquinite, serpentine and albite. It forms hexagonal system tabular, triangular crystals and is a valuable gem when transparent and without flaws.

BENSON-CALVIN-BASSHAM CYCLE (CALVIN CYCLE): The metabolic pathway found in the stroma of the chloroplastin photosynthetic organisms, in which carbon enters in the form of CO_2 and leaves in the form of sugar. It is named after the three discoverers but is commonly abbreviated to **Calvin cycle**.

BENT: 1. Old dry stalks of dead grass. 2. Stiff-stemmed grasses of the genus *Agrostis*, found in hill pastures and tolerant of poor conditions.

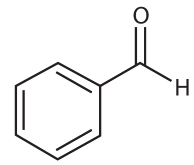
BENT'S RULE: It states that in a molecule, smaller bond angles are formed between electronegative ligands since the central atom, to which the ligands are attached, tends to direct bonding hybrid orbitals of greater *p* character towards its more electronegative substituents.

BENTHIC ZONE: One of the two basic subdivisions of a body of water such as an ocean or a lake. It is the lowest level and includes the sediment surface and some sub-surface layers. Organisms living in this zone are called benthos, some of which are permanently attached to the bottom.

BENTHOS: Flora and fauna occurring on the bottom of a lake or sea. They include the starfish, oysters, clams, sea cucumbers, brittlestars and anemone. Most of them feed on food as it floats by or scavenge for food on the ocean floor. Life in the benthos region is organised by size. **Macrobenthos** which include oysters, starfish, lobsters, sea urchins, shrimp, crabs and coral, are organisms that are larger than one millimetre, **Meiobenthos** which include diatoms and sea worms are between one tenth and one millimetre in size and **Microbenthos** which are smaller than one tenth of a millimetre include the diatoms, ciliates and bacteria.

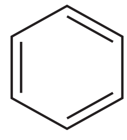
BENTONITE: A fine-grained clay formed from volcanic ash by the hydration (formation of hydrates or absorption of water) and loss of bases and perhaps silica from tiny particles of volcanic glass present in the ash. There are different types of bentonites, which are named after the main element in the material. Examples are potassium (K), sodium (Na), calcium (Ca) and aluminium (Al). Sodium bentonite, which expands considerably when saturated with water, is used diversely, as in the making of paper and sealing of dams. Calcium bentonites are used in the making of fuller's earth.

BENZALDEHYDE (BENZENECARBALDEHYDE; PHENYLMETHANAL): A yellowish volatile oily liquid with the chemical formula C_6H_5CHO, density 1.0415 g/ml (liquid), melting point -26 °C (247 K) and boiling point 178.1 °C (451.25 K) with an almond-like smell, which occurs in almond kernels. It is used in flavourings, perfumery and the dyestuffs industry.



Chemical structure of benzaldehyde

BENZENE (BENZOL; CYCLOHEXA-1,3,5-TRIENE): An aromatic hydrocarbon, colourless and highly flammable liquid with a sweet smell and a relatively high melting point. Its chemical formula is C_6H_6, density 0.8765 g/cm^3, melting point 5.5 °C (278.7 K) and boiling point 80.1 °C (353.3 K). It is obtained from crude oil. It is a known carcinogen, which can cause cancers. Its use as an additive in gasoline is now limited, but it is an important industrial solvent and precursor in the production of drugs, plastics, synthetic rubber and dyes.



Chemical structure of benzene

BENZENE 1,4-DIOL (HYDROQUINONE): A white crystalline aromatic compound, with the chemical formula $C_6H_4(OH)_2$, density 1.3 gcm^{-3} (solid), melting point 172 °C (445 K) and boiling point 287°C (560 K). It is normally used for making dyes and also used as a bleaching agent, photographic developer, in paints, in motor vehicle oils and in medicines.

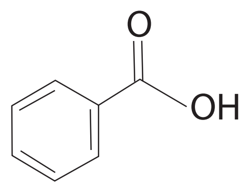
BENZENE HEXACHLORIDE, BHC: Any of several stereoisomers of 1,2,3,4,5,6-hexachlorocyclohexane formed by the light-induced addition of chlorine to benzene with the molecular formula $C_6H_6Cl_6$.

B

An example is lindane, also known as gamma-hexachlorocyclohexane (γ-HCH) or gammaxene, used as an insecticide. Lindane is carcinogenic in animals and also probably in humans and its use is very much restricted because of its toxicity and persistence in the environment.

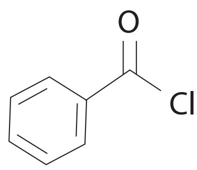
BENZENE RING: A hexagonal ring molecular structure common to benzene and its derivatives, in which six carbon atoms are bonded in a hexagon by alternating single and double bonds between them and with each carbon atom bonded to a hydrogen atom or to other atoms or groups of atoms.

BENZENECARBOCYLIC ACID (BENZOIC ACID; CARBOXYBENZENE; DRACYLIC ACID): A white crystalline compound with the chemical formula $C_7H_6O_2$, density 1.27 g/cm^3, melting point 121.2 °C (394.35 K) and sublimes at 100 °C (373.15 K). It is only slightly soluble in water but is soluble in most organic solvents and reacts with bases to form the corresponding benzoate salts. It occurs naturally in some plants and used as a preservative for food.



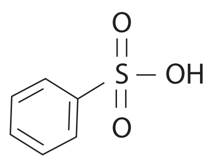
Chemical Structure of Benzenecarbocylic Acid (Benzoic Acid)

BENZENECARBONYL CHLORIDE (BENZOYL CHLORIDE): A colourless fuming liquid with an irritating odour, with the chemical formula C_6H_5COCl. It is used to introduce benzenecarbonyl groups into molecules and production of dyes, perfumes, pharmaceuticals and resins.



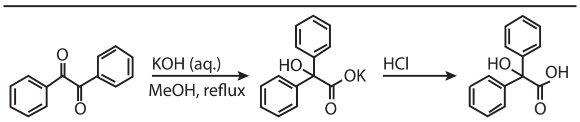
Chemical Structure of Benzene Carbonyl Chloride

BENZENESULFONIC ACID: A colourless deliquescent solid, with the chemical formula $C_6H_5SO_3H$, molar mass 158.18 gmol^{-1}, density 1.32 g/cm^3, melting point 44°C for the hydrate and 51°C for the anhydrous, boiling point 190°C. It is a colourless crystalline solid; made by treating benzene with concentrated sulfuric acid. Its alkyl derivatives are used as detergents. Certain pharmaceutical drugs are prepared as salts of benzenesulfonic acid and are known as **besylates** or **besilates**. An example is amlodipine besylate, used in treating hypertension and other heart diseases.



General structure of benzenesulfonic acid

BENZILIC ACID REARRANGEMENT: An organic rearrangement reaction in which benzil (1,2-diphenylethan-1,2-dione) is treated with hydroxide and then acid to give benzilic acid (2-hydroxy-2,2-diphenylethanoic acid), first performed by Justus Liebig in 1838. In the reaction a phenyl group ($C_6H_5^-$) migrates from one carbon atom to another. This type of reaction is displayed by 1,2-diketones in general and the reaction product is an ⍰-hydroxy-carboxylic acid.

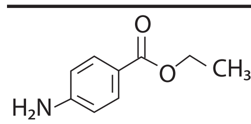


Benzilic Acid Rearrangement

BENZINE: A colourless highly flammable liquid mixture of hydrocarbons, chiefly alkanes such as pentane and hexane, obtained from petroleum and used in dry-cleaning. It is a good solvent for organic substances such as fats, oils and resins and also used in the preparation of certain dyes and paints.

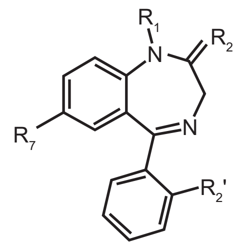
BENZOATE: A salt or ester of benzoic acid, containing the group C_6H_5COO- or the ion $C_6H_5COO^-$.

BENZOCAINE (ETHYL 4-AMINOBENZOATE): An anaesthetising drug, used in some throat lozenges and skin creams. Its chemical formula is $C_9H_{11}NO_2$ and a molar mass of 165.189 g/mol.



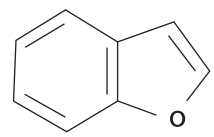
Chemical Structure of Benzocaine

BENZODIAZEPINES: A group of structurally related drugs which affect the central nervous system of animals. They induce sleep and are used in treating cases like insomnia, depression and anxiety, convulsions, epilepsy and alcohol withdrawal. They are addictive and can be dangerous when abused; hence they are sold only as prescription drugs. Examples include valium (diazepam) and rohypnol (flunitrazepam).



Chemical Structure of Benzodiazepines

BENZOFURAN (COUMARONE): A heterocyclic compound having a benzene ring fused to a five-membered furan ring. Its molecular formula is C_8H_6O, molar mass 118.13 g mol^{-1}, melting point −18 °C (255 K) and boiling point 173 °C (446 K). Benzofuran is a component of coal tar; and can also be obtained by dehydrogenation of 2-ethylphenol. Its derviatives have shown antifungal, antimicrobial and as antagonists for the H3 receptor and angiotensin II.



Chemical Structure of Benzofuran

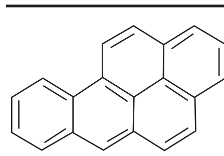
BENZOL: Unrefined benzene or crude benzene, containing toluene, xylene and other hydrocarbons, obtained from coal tar or coal gas; it can also be obtained by dehydrogenation of 2-ethylphenol. It is used as a fuel.

BENZOYL: In organic chemistry, it is the functional group with the formula C_6H_5CO-. Almost invariably, benzoyl is a portion of a molecule. Benzoyl should not be confused with benzyl, which has the formula $C_6H_5CH_2$. Benzoyl chloride is a favoured source of benzoyl groups.

BENZOYLATION: A chemical reaction in which a benzyl group (C_6H_5CO) is introduced into a molecule.

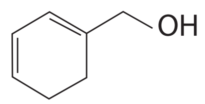
BENZOYLECGONIN: A primary metabolite of cocaine consisting of ecgonine benzoate. It is the compound tested in cocaine urinalysis and can be detected in urine some days after the cocaine has cleared out. It is tested by immunoassay or by gas chromatography/mass spectrometry (GC-MS).

BENZPYRENE [BENZOPYRENE; BENZOAPYRENE; BENZOEPYRENE; BENZO[a]PYRENE (BaP)]: A class of highly carcinogenic polycyclic aromatic compound with the molecular formula $C_{20}H_{12}$, molar mass 252.3 gmol^{-1}, density 1.24 g/cm^3, melting point 179 °C and boiling point 495 °C. It consists of a benzene ring fused to a pyrene molecule and occurs in coal tar, cigarette smoke, grilled or smoked fish and meat, the incomplete combustion of organic materials such as petroleum products, wood and as a by-product of many industrial processes. In coal tar, it occurs together with other related pentacyclic aromatic species such as picene, benzofluoranthenes and perylene. They intercalate into DNA, interfering with transcription and are treated as pollutants and carcinogens.


General structure of benzopyrene

BENZYL ALCOHOL (PHENYLCARBINOL): A colourless liquid slightly aromatic liquid with the chemical formula $C_6H_5CH_2OH$, density 1.044 g/cm^3, melting point -15 °C and boiling point 205 °C. It is partially soluble in water (4 g/100 mL) and completely miscible in alcohols and diethyl ether. It occurs naturally in many plants and is commonly found in fruits and teas. It is also found in a variety of essential oils including jasmine, hyacinth and its esters are found in flowers and used in perfumes. Benzyl alcohol is prepared by the hydrolysis of benzyl chloride using sodium hydroxide:

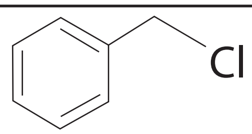
$C_6H_5CH_2Cl + NaOH \rightarrow C_6H_5CH_2OH + NaCl$.



General structure of benzyl alcohol

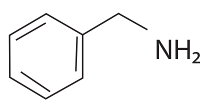
It can also be prepared via a Grignard reaction by reacting phenylmagnesium bromide (C_6H_5MgBr) with formaldehyde, followed by acidification. It is also used in colour-film developing and in dyeing. It is used as a solvent for inks, paints, lacquers and epoxy resin coatings. It is also a precursor to a variety of esters, used in the soap, perfume and flavour industries.

BENZYL CHLORIDE: An organic compound, with the chemical formula $C_6H_5CH_2CI$, molar mass 126.58 g/mol, density 1.100 g/cm^3, melting point -39 °C and boiling point 179 °C. It is a colourless liquid with a sharp odour; obtained industrially by passing chlorine through toluene that has been heated to 90 - 100 ° C in the reaction $C_6H_5CH_3 + Cl_2 \rightarrow C_6H_5CH_2Cl + HCl$. It is not water-soluble and miscible in alcohol, chloroform and other organic solvents. When heated with water, benzyl chloride gradually hydrolyses to form benzyl alcohol. Benzyl chloride is used to obtain benzyl alcohol and, more importantly, benzyl cellulose, which is widely used in the production of plastics, films, electrical insulating materials and lacquers.



Chemical Structure of Benzyl Chloride

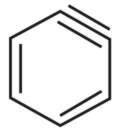
BENZYLAMINE (1-AMINOTOLUENE; PHENYLMETHYLAMINE; 1-PHENYLMETHANAMINE): A colourless liquid, with the chemical formula $C_6H_5CH_2NH_2$, density 0.981 g/ml, melting point 10 °C (283 K) and boiling point 185°C (458 K) consisting of a benzyl group ($C_6H_5CH_2$), attached to an amine functional group. It behaves in the same way as primary aliphatic amines. It is created from a benzyl group and an amine group in many ways such as by hydrogenation of benzonitrile; by the reaction of benzyl chloride and ammonia; by the reaction between benzyl bromide and acetamide; etc. It is most commonly used as a base substance for medicinal and industrial purposes such as the production of nylon fibres, cationic paints, pharmaceuticals, vitamin H (biotin), photographic chemicals, plastics processing industry, etc.



Chemical Structure of Benzylamine

BENZYLIC INTERMEDIATES: Carbanions, carbenium ions or radicals, derived formally by detachment of one hydron, hydride or hydrogen, respectively, from the CH_3 group of toluene or such substitution derivatives, for example, benzyl radical ($C_6H_5CH_2\cdot$).

BENZYNE (DIDEHYDROBENZENE): A transient and highly reactive species due to the nature of its triple bond, formed by removing two atoms of hydrogen from benzene, with the chemical formula C_6H_4. It is a member of a class of compounds known as arynes. It is a chemical species whose structure consists of an aromatic ring in which four carbon atoms are bonded to hydrogen atoms and two adjacent carbon atoms lack substitutents. Benzyne is, like benzene, resonance stabilised.



Chemical Structure of Benzyne

BERGIUS PROCESS: A process of making hydrocarbon mixtures for fuels from coal by heating powdered coal mixed with iron and hydrogen at a pressure of about 200 atmospheres. It was first developed by Friedrich Bergius in 1913 and in 1931 awarded the Nobel Prize in Chemistry for his development of high pressure chemistry.

BERGIUS, FRIEDRICH KARL RUDOLF (1884 -1949): German chemist known for the Bergius process for producing synthetic fuel from coal. He won the Nobel Prize in Chemistry in 1931, together with Carl Bosch in recognition of contributions to the invention and development of chemical high-pressure methods.

BERIBERI: A nutritional disorder that results in inflammation of the nerve endings and mostly occurring in the polished rice-eating communities in the tropics; caused by a deficiency of thiamine (vitamin B_1). It mainly affects the nervous system and usually results in death from heart failure. When rice is polished the thiamine-rich coat is removed.

BERKELIUM: A radioactive metallic synthetic transuranic element belonging to the actinoids with the symbol Bk and atomic number 97. Berkelium was the fifth transuranic element to be synthesised by bombarding americium-241 with alpha particles (helium ions) at the University of California at Berkeley.

BERLESE FUNNEL (BERLESE TRAP; TULLGREN FUNNEL): In biology, it is an ordinary funnel into which a sample of soil or leaf litter containing living organisms such as insects, mites and other invertebrates are trapped. It works by placing a funnel above a collecting vessel and placing a light on top of the funnel to illuminate the sample that is placed in the collecting vessel. As living organisms in the sample move away from the source of light and heat, they end up falling into the collecting vessel where they are trapped and used for studies.

BERNOULLI FAMILY: An extraordinary Swiss family from Basel that produced eight outstanding mathematicians within three generations. Together with Isaac Newton, Gottfried Leibniz, Leonhard Euler and Joseph Lagrange, the Bernoulli family dominated mathematics and physics in the 17th and 18th centuries, making important contributions to differential calculus, geometry, mechanics, ballistics, thermodynamics, hydrodynamics, optics, elasticity, magnetism, astronomy and probability theory.

BERNOULLI TRIAL: An experiment in which there are two possible independent outcomes, named after Swiss mathematician and physicist, Jacob Bernoulli (1654-1705). It usually denotes success and failure, with the properties that the probability of occurrence of each outcome is the same in each trial and the occurrence of one excludes the occurrence of the other in any given trial. For example, in tossing a coin, the outcome has only two possibilities: either the coin lands head or tail. Let p be the probability of success in a Bernoulli trial and q, the probability of a failure, then the probability of failure, q is given by: $q = 1 - p$. Random variables describing Bernoulli trials are often encoded using the convention that 1 = "success", 0 = "failure". Therefore, the probability distribution of such a random variable (termed a Bernoulli distribution) is given by $P(x = r) = {}^nC_r P^r q^{n-r}$.

BERNOULLI EQUATION: A differential equation of the form $\frac{dy}{dx} + \phi y = \psi y^n$, where ϕ and ψ are functions of x alone. It may be written in linear form by the change of variable $z = y^{1-n}$ and is closely related to the Jacobi equation.

BERNOULLI NUMBER: The sequence $\{B_n\}$ of coefficients of the power series, defined by $\frac{z}{e^z - 1} + \frac{z}{2} = \sum_{m=0}^{\infty} B_{2m} \frac{z^{2m}}{(2m)!}$ These numbers allow evaluation of even values of the zeta function.

BERNOULLI, DANIEL (1700-82): Swiss mathematician and physician, the son of the mathematician Johann Bernoulli. As a teenager he obtained degrees in philosophy and logic, medicine and mathematics in Basel. He was called to the St. Petersburg Academy of Science in 1725, where he spent eight years and wrote important texts on the theory of mechanics, including a first version of his famous treatise on hydrodynamics. In 1733, he became a professor of botany, physiology and physics in Basel. He developed the theory of watermills, windmills, water pumps and water propellers. He was the first to differentiate between hydrostatic and hydrodynamic pressure. His "Bernoulli Principle on Stationary Flow" has remained the general principle of hydrodynamics and aerodynamics up to date and forms the basis of modern aviation.

BERNOULLI, JAKOB (1654-1705): Swiss analyst and physicist, a member of the Bernoulli family and the most outstanding, after whom a large variety of results in analysis and statistics are named.

BERNOULLI'S PRINCIPLE: The principle, named after Daniel Bernoulli (1700-82), a Swiss mathematician and physicist that states that the pressure in a moving fluid decreases as the speed increases. The principle can be applied to various types of fluid flow; for example, it explains the pressure difference on each surface of an aerofoil, which gives a lift to the wing of an aircraft.

BERNOULLI'S THEOREM (BERNOULLI'S LAW): A law which states that at any point in a pipe through which a fluid is flowing the sum of the pressure energy, the kinetic energy and the potential energy of a given mass of the fluid is constant. In a horizontal flow the sum of kinetic energy and pressure energy is constant or $\frac{p}{\rho}+\frac{1}{2}V^2 = K$, where P is the static pressure, ρ is the density, V is the air speed and K is a constant. From this law, it follows that where there is a velocity increase in a fluid flow, there must be a corresponding pressure decrease.

BERNSTEIN MODE: A class of modes of plasma oscillation that occur at harmonics of the cyclotron frequency.

BERNSTEIN POLYNOMIALS: The sequence of polynomials defined for a given continuous function f on the interval [0,1] by the formula $B_n(f)(x) = B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} x^k (1-x)^{n-k}$. The polynomials $B_n(f)$ converge in uniform norm to f; yielding a proof of the Weierstrass approximation theorem.

BERRY: A many-seeded fleshy indehiscent fruit such as a strawberry, grape, tomato or banana. The epicarp usually forms a tough outer skin, especially in the pepo and hesperidium and the mesocarp becomes massive and fleshy. The mesocarp and endocarpore are fused. The epicarp and mesocarp are free and may be highly coloured to attract the animals that act as agents of dispersal.

BERRY MECHANISM (BERRY PSEUDOROTATION MECHANISM): A mechanism by which ligands in trigonal bipyramidal complexes can interchange between axial and equatorial positions. It involves intermediate formation of a square pyramidal conformation.

BERRY PHASE: A crucial concept in many quantum mechanical effects, including quantum computing. For example, it modifies the motion of vortices in superconductors and the motion of electrons in nanoscale electronic devices. As a result of wave-particle duality, all particles have wavelike properties, but the Berry phase is a special type of phase that a particle acquires if it is forced to slowly rotate.

BERRY'S THEOREM: The semantic paradox discovered by the English librarian, G. G. Berry that results from classifying the positive integers in terms of the smallest number of syllables of ordinary English needed to describe them.

BERTHELOT EQUATION: An equation of state that relates the pressure, volume and temperature of a gas and the gas constant. It is given by: $PV = RT\left\{1 + \frac{9}{128}\cdot\frac{PT_c}{P_cT}\left(1 - \frac{6T_c^2}{T^2}\right)\right\}$, where p is the pressure, V the volume, R the gas constant and T the thermodynamic temperature and pressure of the gas.

BERTHELOT, PIERRE EUGÈNE MARCELLIN (1827 - 1907): French chemist, historian, philosopher and public servant, noted for the Thomsen-Berthelot principle of thermochemistry; and one of the first scientists to use the word "synthesis" to describe the production of organic compounds from their elements. He synthesized many organic compounds from inorganic substances and showed that organic compounds such as phenol could be synthesized from acetic acid which, in turn, could be prepared from other chemicals including carbon tetrachloride. The fundamental conception of all Berthelot's chemical work was that all chemical phenomena depend on the action of physical forces that can be determined and measured. He stated that chemical phenomena are not governed by any peculiar laws special to themselves, but are explicable in terms of the general laws of mechanics that are in operation throughout the universe. He performed experiments to determine gas pressures during hydrogen explosions using a special chamber fitted with a piston and was able to distinguish burning of mixtures of hydrogen and oxygen from true explosions. He is considered as one of the greatest chemists of all time.

BERTHOLLIDE COMPOUND (NON-STOICHIOMETRIC COMPOUND): A chemical compound typically having a formula $TiO_{1.8}$ in which the elements do not combine in simple well-defined ratios. For example, rutile (titanium(IV) oxide) is often deficient in oxygen. Often, they are solids that contain crystallographic point defects, such as interstitial atoms and vacancies, which result in excess or deficiency of an element, respectively.

BERTRAND'S METHOD: A method devised in 1852 to find simple algebraic integrals of motion for Hamiltonian systems with two degrees of freedom.

BERYL: A transparent precious stone composed of beryllium, aluminium and silicate. Its chemical formula $Be_3Al_2(SiO_3)_6$. It occurs in hexagonal prisms and constitutes the chief source of beryllium. It is usually green, pink, yellow or blue due to impurities and valued as gems.

BERYLLIUM: The chemical element with the symbol Be and atomic number 4. It is found in Group 2 of the periodic table. It is a divalent element that occurs naturally only in combination with other elements in minerals. The more common beryllium containing minerals include: bertrandite $Be_4Si_2O_7(OH)_2$, beryl ($Al_2Be_3Si_6O_{18}$), chrysoberyl (Al_2BeO_4) and phenakite. Beryllium is most commonly extracted from the mineral beryl, which is either sintered using an extraction agent or melted into a soluble mixture. As a free element it is a steel-grey, strong, lightweight and brittle alkaline earth metal. Unlike most metals, beryllium is virtually transparent to x-ray and hence it is used in radiation windows for x-ray tubes. Beryllium alloys are used in the aerospace industry as lightweight materials for high performance aircraft, satellite and spacecraft. It is also used as an alloy with copper to make spark-proof tools.

BERYLLIUM BRONZE: A hard, lightweight, silver white or grey metallic element, with the chemical formula $Be_3Al_2(SiO_3)_6$ and chemical symbol Be. It is used to make sturdy, light alloys and to control the speed of neutrons in nuclear reactors and also used in making ceramic materials. It is a member of the Group II elements in the Periodic Table (the alkaline earth metals).

BERYLLIUM HYDROXIDE: A white crystalline compound with the chemical formula $Be(OH)_2$ which is precipitated from solutions of beryllium salts by adding alkali. It is amphoteric and dissolves in excess alkali to give beryllates.

BERYLLIUM OXIDE: An insoluble solid compound, with the chemical formula BeO, density 3.01 g cm^{-3}, melting point 2507 °C (2780 K) and boiling point 3900 °C (4173 K). It occurs naturally as bromellite and can be made by burning beryllium in oxygen. It is an important amphoteric oxide, reacting with acids to form salts and with alkalis to form compounds known as beryllates. Beryllium oxide is used in the production of beryllium and beryllium–copper refractories, transistors and integrated circuits.

BERZELIUS, JÖNS JACOB (1779-1848): Swedish chemist and professor in Stockholm (1807-32), regarded as one of the founders of modern chemistry. He introduced the system of symbols in use today, grouped chemistry into organic and inorganic and introduced much of the basic vocabulary of chemistry in use today such as allotrope, catalysis, halogen, isomer, polymer and protein; he determined atomic weights, created the modern system of chemical symbols, the theory of electrochemistry; discovered the elements cerium, selenium and thorium and isolated silicon, zirconium and titanium. He also contributed to the classical techniques of analysis and investigated isomerism and catalysis, both of which he named. He published more than 250 original research papers.

BESSEL FUNCTION: A special function that results from solving the wave equation in cylindrical coordinates. It is used in branches of mathematical physics. The Bessel function is given by

$$J_n(x) = \frac{x^n}{2^n n!}\left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2.4(2n+2)(2n+4)} - \dots\right].$$

BESSEL INEQUALITY: The Fourier series inequality stating that the sum of the squares of the moduli of the Fourier coefficients of a function f on the interval [0, 2π) is dominated by the integral of the squared function and so satisfies $\sum_{n=0}^{\infty}|c_n|^2 \le \int_0^{2\pi} f(x)^2\,dx.$

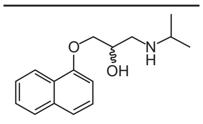
BESSEMER PROCESS: A process for converting pig iron from a blast furnace into steel. It involves the removal of impurities from the iron by oxidation with air being blown through the molten iron. The oxidation also raises the temperature of the iron mass and keeps it molten.

BEST APPROXIMATION: In a metric space, a point of a specified set that is closest to a given point that usually lies outside the set. For example, in the simplest Chebyshev approximation one seeks the polynomial closest in Chebyshev norm to a prescribed continuous function.

BET ISOTHERM: An isotherm that takes into account the possibility that monolayer in the Langmuir adsorption isotherm can act as a substrate for further adsorption.

BETA: 1. The second letter of the Greek alphabet, denoted by "Β" for upper case and "β" for lower case. 2. In astronomy, it is a star that is usually the second brightest of a constellation. 3. In chemistry, it is used to denote one of the possible positions of an atom or group in a compound; or to denote one of two or more isomeric compounds. 4. A software still under development that is made available to a select few called **closed beta** or to the general public called **open beta** for testing. Beta testing allows developers to find problems they may have missed and fix them before the program is released. 5. In social animals, it is the second highest ranked dominant individual in a group. 6. In statistics, also known as **probability of a Type II error**, it is the probability in a hypothesis test concerning the value of a parameter accepting the null hypothesis when the alternate hypothesis is, in fact true. The value of beta depends upon the true parameter value.

BETA BLOCKER (BETA-ADRENOCEPTOR ANTAGONIST): Any of a group of drugs used to treat hypertension, cardiac arrhythmias, angina pectoris, migraine headaches and other disorders related to the sympathetic nervous system. They bind preferentially to beta adrenoceptors and hence block their stimulation by the body's own neurotransmitters, adrenaline and noradrenaline.



Chemical Structure of Beta Blocker

BETA CELLS (BETA-CELLS, B-CELLS): A type of cell in the pancreas located in the islets of Langerhans making up 65-80% of the cells in the islets. Beta cells store and release insulin, a hormone that controls the level of glucose in the blood; release C-peptide, a byproduct of insulin production into the bloodstream; and also produce amylin, which functions as part of the endocrine pancreas and contributes to glycemic control.

BETA DECAY: The spontaneous emission of beta particle from a nucleus. Beta decay is accompanied by neutrino emission and a release of energy. It is any of three processes of radioactive decay: electron emission, positron emission and electron capture in which a nucleus emits an electron or beta particle. With this sort of decay the electrons or beta particles are known as beta radiation. Beta decay is a slower process than gamma or alpha decay, but capable of penetrating hundreds of times farther than alpha particles. There are two types of beta decay where the beta particle is either an electron, called beta-minus (β^-) decay or a positron called beta-plus (β^+) decay. In beta-plus decay, a proton in an atomic nucleus decays into a neutron, a positron and a neutrino. The positron and neutrino are emitted from the nucleus, while the neutron remains by the reaction: ${}^{Z}_{A}X \rightarrow {}^{Z}_{A-1}Y + e^{+} + \text{neutrino}$ where X is the parent atom, Y is the daughter atom, Z is the atomic mass of X, A is the atomic number of X. The electron and antineutrino are emitted from the nucleus, while the proton remains. The atomic number of the atom is thereby increased by 1. An example of beta-minus decay is the decay of Carbon-14 into Nitrogen-14, used in carbon dating.

BETA DISTRIBUTION: A distribution often used as a prior distribution for a proportion. The probability density function, for a random variable X having a beta distribution is $f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}, 0 < x < 1,$ where α and β are positive parameters and B is the beta function. Beta distribution has a variety of shapes on the values of α and β.

BETA DIVERSITY (β-DIVERSITY; BETA RICHNESS): The measure of biodiversity, which is the rate, magnitude and direction of change in species diversity between different habitats or communities.

BETA FUNCTION: The function $B(p, q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx$ that has the relation to the gamma function that $B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$.

BETA HAEMOLYSIS (β-HAEMOLYSIS): A clear zone of haemolysis seen around a bacterial colony growing on blood agar.

BETA IRON: A non-magnetic allotrope of iron that exists between 768 °C (1 041.15 K) and 900 °C (1 173.15 K).

BETA PARTICLE (β-PARTICLE): An electron emitted from a nucleus by radioactive decay at a high speed of up to 98% of the speed of light, emerging with a variable energy. It has a greater penetrating power than alpha particles.

BETA RADIATION (BETA RAY; β-RADIATION): A stream of beta particles moving at high speed of up to 98% of the speed of light from certain sources. It was discovered by the British physicist Ernest Rutherford in 1899. They carry negative charges and energy but they can be stopped by elements with a low atomic number, such as aluminium sheet of 5 mm thick. Beta radiation can penetrate human skin to the germinal layer. Examples of some pure beta emitters are strontium-90, carbon-14.

BETA SHEET (β-PLATED SHEET): A structure that occurs in many globular proteins in which extended polypeptide chains lie parallel to each other and are linked by hydrogen bonds between the N-H and C=O groups.

BETA-CLEAVAGE (β-CLEAVAGE): A characteristic fragmentation of the molecular ion derived from some organic compounds, most notably alcohols, ethers and amines, in which the bond connecting alpha- and beta-carbons break.

BETACYANIN (BETALAINS): A class of nitrogen-containing pigments containing the red betacyanins such as betanidin and the beetroot pigment) and the yellow betaxanthins, for example, indicaxanthin from the fruits of *Opuntia ficusindica*. Alternatively, a group of red and yellow indole-derived pigments found mainly in plants of the order chenopodiales, which include the goosefoot, cactus and Portulaca families. They are most often noticeable in the petals of flowers, but may colour the fruits, leaves, stems and roots of plants that contain them.

BETAINE: 1. A sweet-tasting alkaloid, with the chemical formula $C_5H_{11}NO_2$ that occurs in the sugar beet and other plants and in animals. 2. **Betains** are a group of chemical compounds that resemble betaine and are slightly basic zwitterions.

BETATRON: A particle accelerator for producing high energy electron mainly for research purposes.

BETAXANTHIN: Water soluble, nitrogen-containing pigments varying from yellow to red.

BETWEEN: Of one element of an ordering with respect to two others, being a member of a chain the first and last elements of which are respectively the two given elements. For example, the integer a lies between b and c if and only if either $b < a < c$ or $c < a < b$; a point A lies between two others, B and C, if and only if they are ordered B A C by some suitable relation, such as to the right of.

BETWEEN-SUBJECTS DESIGN: In statistics, an experimental design that is concerned with measuring the value of a dependent variable for distinct and unrelated groups subjected to each of the experimental conditions.

BEZOUT'S LEMMA (BEZOUT'S IDENTITY): The generalisation, to polynomials over a field, of a result known to Euclid for integers, that if d is the greatest common divisor of f and g, it can be written as $d = af + bg$ for two other polynomials a and b.

BEZOUT'S THEOREM: The result that two algebraic plane curves with degree m and n respectively and with no common component have exactly mn points of intersection, counting multiplicity and points at infinity.

BHABHA SCATTERING: The exchange of a virtual photon by an electron and a positron. In classical electrodynamics, the resulting effect is known as Coulomb attraction.

Bi- (Bis): A prefix denoting two or twice.

BI-ORTHOGONAL: Of two sequences (a_n) and (b_n) in Hilbert space such that (a_n, b_m) is unity when $m = n$ and zero otherwise.

BIAL'S REAGENT: A mixture of orcinol in concentrated hydrochloric acid with 10% iron(III) chloride solution. It is used to detect the presence of pentose sugars in a test solution. A green colour after boiling indicates a positive reaction.

BIAS: 1. In statistics, it is a measure of how closely the mean value in a series of replicate measurements approaches the true value. It is the difference between the expected value of a statistic and the true value of the corresponding parameter. If the mean of a large number of unbiased estimation is calculated, the correct value is found. The bias indicates the distance of the estimator from the true value of the parameter. 2. The expected value of (T - θ), for T an estimator of the parameter θ.

BIASED: Of a sample, having a distribution that is not determined only by the population from which it is drawn, but also by some property that influences the distribution of the sample. For example, an opinion poll might be biased by geographical location.

BIASED LINEAR PULSE AMPLIFIER: A pulse amplifier which, within the limits of its normal operating characteristics, has a constant gain for that portion of an input pulse that exceeds the threshold value and that produces no output for pulses whose amplitude is below the threshold.

BIASED SAMPLE: A sample that does not fairly represent the total population from which it was selected. A sample is biased if every member of the population does not have the same chance of being selected for the sample.

BIAXIAL CRYSTAL: A birefringent crystal of low symmetry in which the index ellipsoid has three unequal axis along which the polarised vectors of a monochromatic light beam will travel with the same speed or along which no double refraction occurs. It is formed by substances such as sulfur, mica and turquoise, which may be monoclinic orthorhombic or triclinic.

BICARBONATE (HYDROGEN CARBONATE): A salt of carbonic acid in which one hydrogen atom has been replaced, usually by a metal. A **Bicarbonate ion** is the chemical compound containing the bicarbonate radical, HCO^{-}_3 or with the empirical formula HCO^{-}_3 and a molar mass of 61.0168g/mol. It consists of one central carbon atom surrounded by three oxygen atoms in a trigonal planar arrangement, with a hydrogen atom attached to one of the oxygens. It carries a negative one formal charge. It is the conjugate base of carbonic acid H_2CO_3 and the conjugate acid of CO_3^{2-}. A bicarbonate salt forms when a positively charged ion attaches to the negatively charged oxygen atoms of the ion, forming an ionic compound. In inorganic chemistry, it is an intermediate form in the deprotonation of carbonic acid. Many bicarbonates are soluble in water at standard temperature and pressure. Bicarbonate is alkaline and it is a vital component of the pH buffering system of the human body. The most common salt of the bicarbonate ion is sodium bicarbonate, $NaHCO_3$, which is used as baking powder. The flow of bicarbonate ions from rocks weathered by the carbonic acid in rainwater is an important part of the carbon cycle.

BICARBONATE INDICATOR (HYDROGENCARBONATE INDICATOR): A hydrogen index (pH) indicator used to detect levels of carbon dioxide. The indicator is used in photosynthesis and respiration experiments to determine whether carbon dioxide is being liberated or not. The indicator is sensitive to colour change as the concentration of the gas increases; and the initial red colour changes to yellow as the pH becomes more acidic.

BICARBONATE IONS: Chemical compound such as sodium bicarbonate (baking soda) containing the bicarbonate radical, HCO^{-}_3.

BICARBONATE OF SODA (SODIUM BICARBONATE; SODIUM HYDROGEN CARBONATE): A sodium salt of carbonic acid, with the chemical formula $NaHCO_3$. It occurs most commonly as a crystalline heptahydrate, which readily effloresces to form a white powder, the monohydrate. It can be extracted from the ashes of many plants. It is synthetically produced in large quantities from salt and limestone in a process known as the Solvay process. It is used to neutralise the acidic effects of chlorine and raise pH, as a water softener during laundry, a descaler, in the manufacture of glass, as an electrolyte, as a food additive, in medicine to treat acid indigestion, in making effervescent salts and beverages, in artificial mineral water, in pharmaceuticals and in fire extinguishers.

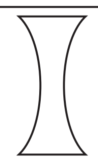
BICARPELLATE: An organ with two carpels as found in some flowers.

BICEPS: Any muscle having two heads or fixed ends of attachment, notably the biceps brachii at the front of the upper arm and the biceps femoris in the thigh. The biceps brachii bends the elbow when it contracts and works in coordination with the triceps brachii, an extensor. It originates in the shoulder area and the heads of the biceps merge partway down the arm to form a rounded mass of tissue linked by a tendon to the radius, the smaller of the two forearm bones. The biceps femoris has two parts, one of which forms part of the hamstrings muscle group.

BICHROMATE (DICHROMATE): A compound, a salt of dichromic acid containing the divalent negative ion, $Cr_2O_7^{2-}$, with a characteristic orange-red colour.

BICOLLATERAL BUNDLE: A vascular bundle in which the phloem occurs both externally and internally with respect to the xylem, with all tissues lying on the same radius and with the internal phloem lying next to the pith. It occurs in the stems of some dicotyledons, for example, members of the *Solanaceae*. The presence of bicollateral bundles distinguishes solanaceous plants from members of the *Scrophulariaceae*.

BICONCAVE: A lens with two concave faces or two curved sides rounded inward on both sides.



Biconcave lens

BICONDITIONAL STATEMENT (EQUIVALENCE): A compound statement that says one sentence is true if and only if the other sentence is true. Its symbol is ↔ or ⇔. Symbolically, this is written as p ↔ q or p ⇔ q. For example, "A triangle has two equal sides if and only if it has two equal angles" is a biconditional statement.

BICONTINUOUS: Of a function, continuous and possessing a continuous inverse. A continuous bijection whose domain is compact is necessarily bicontinuous, when the range is a Hausdorff space.

BICRENATE: Doubly crenate, as when the teeth of a crenate leaf are also crenate.

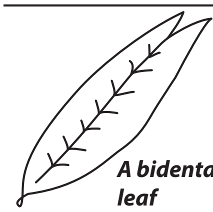
BICUSPID: With two cusps or points such as bicuspid tooth or bicuspid valve. **Bicuspids** or **premolars** are the two sets of four teeth, in each jaw in humans. They are flat teeth with pronounced cusps that grind and mash food. A **bicuspid valve** also known as **mintral valve** or **left atrioventricular** is a valve of the heart, composed of two triangular flaps, that is located between the left atrium and left ventricle that regulates blood flow between these chambers. **Bicuspid aortic valve disease** (**BAVD**) is a congenital condition of the aortic valve, where two of the three normal flaps or leaflets of the aortic valve fuse during development to become bicuspid instead of the normal tricuspid or three flap valve. With this deformity, the valve doesn't function perfectly, but it may function adequately for years without causing symptoms or obvious signs of a problem.

BICYCLE: A pedal driven two-wheeled vehicle used for transportation as well as for exercises for fitness and sports. It consists of a metal frame mounted on two large wire spooked wheels with handle bars in front for steering and usually two pedals by which it is driven. They are energy efficient, non-polluting form of transport.

BICYCLE REARRANGEMENT: A photochemical rearrangement of unsaturated substrates by group migration over a π perimeter following the movement of a bicycle pedal.

BIDENT: A two-year-old sheep.

BIDENTATE: 1. In biology, having two teeth or toothlike parts or processes, as shown in the diagram. 2. In chemistry, of a ligand, having two atoms from which electrons can be donated to the central coordinated atom.



A bidentate leaf

BIDIRECTIONAL REPLICATION: Replication of deoxyribonucleic acid in both directions away from the origin, as opposed to replication in one direction only, called **unidirectional replication**.

BIEBERBACH'S CONJECTURE: The conjecture, proved by Louis de Branges in 1985 that if S is the class of normalised injective holomorphic function so that S consists of one-to-one holomorphic functions from the unit disk with power series of the form $z + a_2z^2 + ... + a_nz^n + ...$ that for $|z| \leq 1$, and for every function in S, the coefficients satisfy $|a_n| \leq n$ for all n.

BIENNIAL PLANT: A plant, usually herbaceous that lives for two years only or a type of plant that requires two growing seasons to complete its lifecycle. It grows vegetatively in the first year and the photosynthates are stored in perennating organs, which are used during the second year in the production of flowers and seeds. Biennials are. Examples are cabbage and carrot.

BIENNIAL ROTATION: In agriculture, it is growing two different crops such as maize and soybeans, or rice and soybeans in alternating years.

BIFACIAL: With the opposite surfaces different in colour and texture as seen in Dicotyledonous leaves.

BIFARIOUS: In botany, it refers to plant parts having two vertical rows arranged on either side of a central axis.

BIFEROUS: Appearing twice annually.

BIFID: Deeply two-lobed or two-cleft, usually from the tip as found in the styles of carpels of some flowers.

BIFLOROUS: Plants flowering in the spring and in the autumn.

BIFOLIATE: In botany, it refers to two leaflets or two leaves.

BIFUNCTIONAL CATALYSIS: Catalysis, usually for hydron transfer by a bifunctional chemical species involving a mechanism in which both functional groups are implicated in the rate-controlling step, so that the corresponding catalytic coefficient is larger than that expected for catalysis by chemical species containing only one of these functional groups.

BIFURCATE: Generally, to separate or fork into two parts or branches. In botany, it refers to structure or organ divided into two branches or two-forked. **Bifurcation** is also a phenomenon that can occur in a dynamical system in which some solutions of a certain type of system, such as laminar flow in a fluid, suddenly changes to another type, such as turbulent flow, when one of the parameters that characterises the system reaches a critical value. The term bifurcation has other applications in law, hydrology, fluid dynamics, mathematics, economics, chemistry, anatomy and physiology. In each application, bifurcation refers to the splitting into two of a certain element or system, such as the splitting of a single hydrogen atom into two hydrogen bonds.

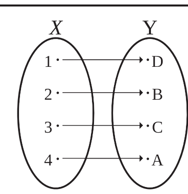
BIG BANG (BIG BANG THEORY): A hypothetical explosive event that marked the origin of the universe as we know it. It refers to the idea that the universe is believed to have emerged from a single point approximately 13.7 billion years ago and was originally extremely hot and dense and has since cooled by expanding to the present diluted state and continues to expand today. The following evidence supports the Big Bang Theory: observation of the expansion of the universe, the existence and near uniformity of the cosmic microwave background radiation and the relative proportion of the element in the early universe as observed in young galaxies. In order to recreate the conditions immediately after the big bang, a 5 billion dollar instrument called the Large Hadron Collider (LHC) was designed and installed near the Swiss-French border. In 2008, two beams of protons were fired into a 27 kilometre long tunnel which houses the LHC, with the first beam moving clockwise and the second moving anticlockwise together with cataclysmic force and they are expected to collide. The beams successfully completed their lapse contrary to the fear that it might create a black-hole which will end the world. According to the best available measurements as of 2010, the initial conditions occurred around 13.3 to 13.9 billion years ago.

BIG O-NOTATION (O-NOTATION): In computer science, the term is used to describe the performance or complexity of an algorithm.

BIGEMINATE (BIJUGATE): 1. Twice divided into equal parts. 2. In botany, having a forked petiole and a pair of leaflets at the end of each division, as found in a decompound leaf.

BIGENERIC: Describing a hybrid derived from an intergeneric cross, for example, Triticale derived from a cross between *Triticum* and *Secale*.

BIJECTION (ONE-ONE BIJECTION): A mathematical mapping between two spaces in which every element in each space corresponds to only one element of the other space for mapping in either direction. In a bijective mapping, two sets X and Y are called bijective if there is a bijective map from X to Y. A function f from set X to Y is bijective, also called **one-to-one**, **into, onto**, if, for every y in B, there is exactly one x in A such that $f(x) = y$. A function f is **injective**, if and only if anytime $f(x) = f(y)$, $x = y$. Alternatively, f is bijective if there is a **one-to-one correspondence** between those sets and therefore, both injective and surjective. A function f from set X to Y is surjective or onto, if and only if for every y in B, there is at least one x in A such that $f(x) = y$, therefore, f is surjective if and only if $f(A) = B$.



A diagram indicating bijection funtion. The elements in X are the domain elements and the elements in Y are the co-domain.

BIJECTIVE: Of a function, relation, etc., constituting a bijection, being both injective and surjective.

BILABIATE: Having two lips such as the corolla of some irregular flowers, for example, the corollas of the snapdragon.

BILATERAL DIAGRAM: A device invented by Lewis Carroll to represent the different logical states that two objects with two properties can take. Each cell in a four-square array represents one of the four possible object/property states and is covered by a red counter if the state is present and by a grey counter if it absent.

BILATERAL SHIFT: The linear operator defined on a space of (square summable) doubly infinite sequences $\left\{x_n\right\}_{n=-\infty}^{\infty}$ by $(Sx)_n = x_{n-1}$.

BILATERAL SYMMETRY: A type of arrangement of the parts and organs of an organism or a body part, along a central axis, so that the body is divided into two halves that are mirror images of each other along one plane only. Thus most leaves can only be divided into similar halves by cutting along the midrib. Bilateral symmetry in flowers is usually termed **zygomorphy**. The flowers of relatively advanced angiosperm families, such as Scrophulariaceae, are often zygomorphic.

BILE: A greenish-yellow alkaline bitter fluid produced by the liver, composed of water, bile acids and their salts, bile pigments, cholesterol, fatty acids and inorganic salts, which is stored in the gall bladder and released into the intestine. It helps in the digestion of fats and of the fat-soluble vitamins (A, D, E and K) through the intestinal wall by emulsifying it.

BILE ACIDS: A group of steroid acids present in the biles of mammals. In humans, the commonest is cholic acid ($C_{24}H_{40}O_5$), which is conjugated by its carboxyl group to the amino groups of the amino acids glycine and taurine. The bile acids are emulsifiers for fats and fat-soluble vitamins, thus promoting their absorption by the small intestine. Bile salts are bile acids compounded with a cation, usually sodium.

BILE DUCT: The tube through which bile passes from the liver or the gall bladder to the duodenum.

BILE SALTS: Derivatives of cholesterol with detergent properties, which are salts of cholic and deoxycholic acid and their glycine and taurine conjugates. It is secreted in the bile that aid in the solubilisation of lipid molecules in the digestive tract.

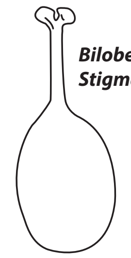
BILHARZIA (SCHISTOSOMIASIS): A chronic illness that results from infection of the blood by parasitic flatworms of the genus *Schistosoma* (blood flukes). It causes debilitation and can cause liver and intestinal damage. It is most common in Asia, Africa and South America, especially in areas where the water harbours freshwater snails, the alternate host.

BILIARY SYSTEM: The bile-producing system consisting of the liver, gallbladder and associated ducts that empty into the larger intrahepatic bile ducts.

BILINEAR FUNCTION: A function that is made up of two functions. It is linear with respect to each variable independently. For example $f(u, v) = 4uv + 8u$.

BILIPROTEIN: 1. Also known as **phycobilin**, it refers to any of the blue or red water-soluble pigments found in the blue-green and red algae that absorb light for photosynthesis. Structurally they are very similar to the porphyrin part of the chlorophyll molecule, except that they contain no magnesium. The three classes of phycobilins are the phycoerythrins, phycocyanins and allophycocyanins. 2. Green chromoproteins found in many insects, such as grasshoppers and also in the eggshells of many birds. The biliproteins are derived from the bile pigment biliverdin, which in turn is formed from porphyrin. Biliverdin contains four pyrrole rings and three of the four methine groups of porphyrin.

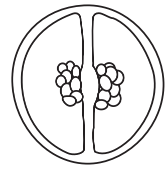
BILLION: The cardinal number represented by a 1 followed by nine zeros (1,000,000,000 or 10^9), it is equivalent to a thousand million.



BILOBED: Divided into two lobes as found in the stigma of carpels in some flowers.

BILOCELLATE: Divided into two secondary locules, as when a main locule of an ovary is partitioned into two cavities.

BILOCULAR (BILOCULATE): In botany, having two locules as found in some ovaries.


A bilocular ovary

BIMAGIC SQUARE: A magic square, in which replacing each integer or number by its square in a magic square produces another magic square. A **magic square** is a square array of numbers consisting of the distinct positive integers 1, 2, ..., arranged such that the sum of the numbers in any horizontal, vertical or main diagonal line is always the same number. If, in addition to being bimagic, the integers in the square can be cubed and the resulting square is still magic, the square is said to be **trimagic**. To date the smallest bimagic square seems to be of order 8, while the smallest trimagic square is of order 32.

3	10	15	6	3	10	15	6	3	10	15	6
16	5	4	9	16	5	4	9	16	5	4	9
2	11	14	7	2	11	14	7	2	11	14	7
13	8	1	12	13	8	1	12	13	8	1	12
3	10	15	6	3	10	15	6	3	10	15	6
16	5	4	9	16	5	4	9	16	5	5	9
2	11	14	7	2	11	14	7	2	11	14	7
13	8	1	12	13	8	1	12	13	8	1	12
3	10	15	6	3	10	15	6	3	10	15	6
16	5	4	9	16	5	5	9	16	5	4	9
2	11	14	7	2	11	14	7	2	11	14	7
13	8	1	12	13	8	1	12	13	8	1	12

An example of a bimagic square

BIMESTRIAL: Occurring every two months or lasting two months.

BIMETALLIC STRIP: A strip consisting of two dissimilar metals of different coefficients of thermal expansion riveted or welded together so that the strip bends on heating. They expand by different amounts when the temperature rises. If one end is fixed the other end can be made to open or close an electric circuit and it is used for temperature measurement and control. The diagrams show strips of brass and iron welded together. The metal brass has a greater linear expansivity than the metal iron. Therefore, when the strip is heated, brass will expand more than iron and the strip will curl with brass on the outside (fig. a). The reverse is true when the strip is cooled, as brass then shrinks more than iron (fig. b). Bimetallic strips are used in devices such as thermometers and thermostats.

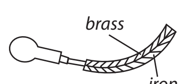


Fig I: Bimetallic strip

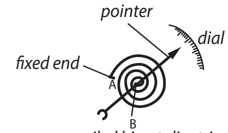


Fig II: Bimetallic strip thermometer

BIMODAL: Having two modes. For example, in statistics, a bimodal distribution is a continuous probability distribution with two different modes.

BIMODAL DISTRIBUTIONS: A data distribution in statistics that has two peaks.

BIMODAL NETWORK: Polymer network comprising polymer chains having two significantly different molar-mass distributions between adjacent junction points.

BIMOLECULAR REACTION: An elementary step in a chemical reaction involving two molecules.

BIMORPH CELL: A device made up of two plates of piezoelectric plates such as Rochelle salt joined together so that an applied voltage causes one to expand and the other to contract, so that the cell bends in proportion to the applied voltage. It is used in phonograph pickups and microphones.

BINARY: 1. Describing a compound or alloy formed from two elements. 2. The principle behind digital computers, in which all input to the computer is converted into binary numbers made up of the two digits 0 and 1 (bits). For example, when you press the "A" key on your keyboard, the keyboard circuit generates and transfers the number 01000001 to the computer's memory as a series of pulses with different voltages. The bits are stored as charged and uncharged memory cells or as microscopic magnets on disk and tape. Display screens and printers convert the binary numbers into visual characters.

BINARY ACID: A binary compound in which acidic hydrogen atom is bonded to an atom other than oxygen. Examples are hydrogen chloride (HCl) and hydrogen sulfide (H_2S).

BINARY CODE: The method of writing all numbers using only two digits, 0 and 1. It is used in digital computers.

BINARY CODED DECIMAL: A number in binary or base two code, but representing decimal place-value notation. Such a number is written in groups of four bits, each group being the binary number equal to the corresponding digit of the given decimal number. Thus, 0110 1001 0011 represents 693, since $0110_2 = 6_{10}$, $1001_2 = 9_{10}$ and $0011_2 = 3_{10}$.

BINARY COMPOUND: A compound formed from just two elements. Examples are sodium chloride (NaCl), sodium fluoride (NaF) and magnesium oxide (MgO). A binary compound can be either molecular or ionic.

BINARY DATA: A data set whose variable can take only two values, 0 and 1, even if the variable is not quantitative. Examples are gender, the presence or absence of a characteristic and the success or failure of an experiment. These variables are called **dichotomous variables**.

BINARY DIGIT: They are values made up of 0 and 1. They are normally used in binary arithmetic and also used in computer and electronics for instructions. Binary digits are measured in bits and bytes.

BINARY FILE: Any file format that is not plain text and can be read only by a computer. In this format, each character position can hold any one of 256 different binary codes. Examples of binary files are executable programs (.EXE or .COM), multimedia, compressed versions of other files and graphics files. Binary files cannot be sent directly over the Internet, for example, as e-mail, as the Internet transmits only ASCII files that use only seven bits and has a size limit of 64Kb per message, whereas binary files can contain ASCII and many more codes because they use all eight bits of the byte. Therefore, binary files must be coded into ASCII for transmission, splitting them into smaller packages as necessary. Programs used for such purposes include UUencode, base64 and BinHex.

BINARY FISSION (PROKARYOTIC FISSION): A form of asexual reproduction used by all prokaryotes, some protozoa and some organelles within eukaryotic organisms in which a single-celled organism divides into two daughter cells, with each having the potential to grow to the size of the original cell.

BINARY LARGE OBJECT (BLOB): A database field that holds any binary data, including text, images, audio or video. This is different than most other data types used in databases, such as integers, floating point numbers, characters and strings, which store letters and numbers. Since blobs can store binary data, they can be used to store images or other multimedia files. For example, a photo album could be stored in a database using a blob data type for the images and a string data type for the captions.

BINARY LINE SEARCH (BISECTION METHOD; DICHOTOMOUS LINE SEARCH): The iterative method of finding the maximum of a unimodal function that proceeds at each iteration by excluding one half of the remaining interval by testing two new function values, both near the midpoint of the present interval, one above and one below.

BINARY NOTATION (BINARY NUMBER SYSTEM): The place-value notation to base 2, in which numbers are expressed by sequences of the digits 0 and 1, for example, 101101. This differs from the decimal numbers, which has 10 numerals (0,1,2,3,4,5,6,7,8 and 9) to represent any value. In other words, it involves counting in twos, where every two is 10_2. For example, the number 8 is represented by 1000 in the binary system of numeration or 1000_{two} or 1000_2. Since 0 and 1 can correspond to off and on conditions in an electric circuit, the binary notation is widely used in computers and is the basis of all digital computing.

BINARY OPERATION: The combination of two terms to produce a third; the arithmetic operations +, -, ×, ÷ are binary since each can combine any two terms to produce a third. For example, $2 + 5 = 7$, $5 - 2 = 3$, $3 \times 4 = 12$ and $6 \div 3 = 2$. Combinations of these arithmetic operations can also be used to define new binary operations. For instance, if on any set of numbers, the operation "⋆" is defined by the expression $x \star y = 2x + y$, the value of $(2 \star 3) \star 5$ will be as follows: $[2\times2 + 3] \star 5 = 7 \star 5 = [2\times7 + 5] = 19$.

BINARY PREFIX: A set of prefixes for binary powers to indicate multiplication by a power of 2 used in data processing and data transmission contexts. For example, in citations of main memory or RAM capacity, "gigabyte" customarily means 1073741824 bytes. This is a power of 2, specifically 2^{30}, so this usage is referred to as a binary unit or binary prefix. The prefixes for binary multiples are as follows: **kibi** has a factor of 2^{10} with the symbol K derived from kilobinary $(2^{10})^1$; **mebi** has a factor of 2^{20} with the symbol Mi derived from megabinary $(2^{10})^2$; **gibi** has a factor of 2^{30} with the symbol Gi derived from gigabinary $(2^{10})^3$ giga $(10^3)^3$; **tebi** has a factor of 2^{40} with the symbol Ti; **pebi** has a factor of 2^{50} with the symbol Pi; and **exbi** has a factor of 2^{60} with the symbol Ei; It was suggested by the International Electrotechnical Commission (IEC), to eliminate the ambiguity with the use of the kilo, mega, giga, etc, in computing. For example, one kibibit (1 Kibit) = 2^{10} bit = 1024 bit, in SI unit = one kilobit (1 kbit) = 10^3 bit = 1000 bit; one mebibyte (1 MiB) = 220 B = 1 048 576 B, in SI unit = one megabyte (1 MB) = 10^6 B = 1 000 000 B; and one gibibyte (1 GiB) = 2^{30} B = 1 073 741 824 B, in SI unit = one gigabyte (1 GB) = 10^9 B = 1 000 000 000 B, etc.

BINARY REGRESSION: Regression in which the response variable can take only two values, 0 and 1.

BINARY RELATION: A relation explicitly involving ordered pairs.

BINARY SEARCH: In computer science, it is a search algorithm that rapidly locates any particular element in a large ordered set in which each element in the set is given a key, whose number is always a power of 2. For example, if there are 32 elements in the list, they might be numbered 0 through 31 (binary 00000 through 11111). By repeatedly dividing the set in half according to how the required numbered key or element compares with the middle one, it discards the half of the set in which the required number or element does not appear, thereby reducing the number of elements left to search by half. This process is repeated, each time selecting half of the list and discarding the unwanted half until the required element is found, which is the only one remaining.

BINARY STARS: A pair of stars revolving about a common centre of gravity, in which their relative sizes and brightness and the distance between them vary widely. An example is Alpha Centauri in the Southern Hemisphere.

BINARY SYSTEM: A numeration system that uses binary notation, it uses only two different digits, 0 and 1.

BINARY TREE: A tree in which every node has at most two successors.

BINATE: In botany, it refers to organs, for example leaves that occur or are borne in two parts or in pairs.

BIND: In logic, to bring (a variable) within the scope of an appropriate quantifier.

BINDING (ACTIVE): Of a constraint given by a weak inequality, satisfied as equality by a given point. For example, the constraint $x^2 + y^2 \leq 2$ is binding at (1, 1), since $1^2 + 1^2 = 2$, but is not binding at (0, 1).

BINDER: 1. A coal tar pitch or petroleum pitch (but may include thermosetting resins or mesophase pitch powders) which, when mixed with a binder coke or a filler, constitutes a carbon mix. This is used in preparation of the formation of shaped green bodies and subsequently carbon artifacts. 2. An additive used to hold the solid stationary phase to the inactive plate or sheet in thin-layer chromatography.

BINDER COKE: A constituent of a carbon or ceramic artifact resulting from carbonisation of the binder during baking.

BINDING ENERGY: 1. Energy required to separate a particle from a system of particles or to disperse all the particles of a system. 2. **Nuclear binding energy** is the energy required to separate an atomic nucleus into its constituent protons and neutrons. 3. **Electron binding energy or ionisation potential**, is the energy required to remove an electron from an atom, molecule or ion. The binding energy of a single proton or neutron in a nucleus is about a million times greater than that of a single electron in an atom.

BINDING ENERGY PER NUCLEON (BE/A): The ratio of the binding energy to the number of nucleons of the atom. The greater the binding energy per nucleon, the more stable the atom is.

BINDING FRACTION: The ratio of the binding energy of a nucleus to the atomic mass number.

BINDING SITE: 1. A specific region (or atom) in a molecular entity that is capable of entering into a stabilising interaction with another molecular entity. An example of such an interaction is that of an active site in an enzyme with its substrate. Typical forms of interaction are by hydrogen bonding, coordination and ion pair formation. Two binding sites in different molecular entities are said to be complementary if their interaction is stabilising. 2. **Binding sites** are areas on the ribosome within which tRNA-amino acid complexes fit during protein synthesis.

BINGHAM NUMBER: A dimensionless number, which is used to study the flow of Bingham plastics. Bingham plastic is a viscoplastic material that behaves as a rigid body at low stresses but flows as a viscous fluid at high stress. An example is toothpaste, which does not extrude until a certain pressure is applied to the tube, which can then be pushed out as a solid plug.

BINGHAM PLASTIC: A viscoplastic material that behaves as a rigid body at low stresses but flows as a viscous fluid at high stress. It is used as a mathematical model of mud flow in drilling engineering and in the handling of slurries. Toothpaste is an example, which does not extrude until a certain pressure is applied to the tube and then, it can be push out as a solid plug.

BINOCULAR: Any optical instrument consisting of two similar telescopes, designed to serve both the observer's eyes at the same time or a magnifying optical instrument for viewing an object with both eyes. Binoculars consist of two telescopes containing convex lenses and prisms, which together produce stereoscopic, seemingly three-dimensional effects as well as magnifying the image. Examples are field glasses and opera glasses.

BINOCULAR VISION: The ability to form clear and distinct images of objects in the retina of both eyes. Owing to parallax, the images in the two eyes are slightly different, which enables the observer to perceive what is seen in three dimensions and so to judge distance, size and shape. Only humans and some other higher animals possess binocular vision. Infants lack binocular vision. Adults without binocular vision experience distortions in depth perception and visual measurement of distance.

BINOMIAL: 1. A mathematical expression that is the sum of two monomials or made up of two terms linked by a plus or minus sign. An example is $3a + 2b$. 2. A system of naming or identifying organisms by using a two part Latin or scientific name.

BINOMIAL CLASSIFICATION (BINOMIAL NOMENCLATURE; LINNEAN SYSTEM): System of naming or identifying organisms using a two-part Latin known as a binomen. The first name identifies the genus with the first letter being capitalised. The second name identifies the species, with the first letter not being capitalised. For example, domestic dog, a species of the genus Canis of the dog family Canidae is identified as *Canis familiaris*; wolf, another species is known as *Canis lupus*; and the jackal, another subspecies is called *Canis aureus*. Human being is known as *Homo sapiens*, the only surviving species of the genus Homo. The system was developed by Carolus Linnaeus (1707-1778).

BINOMIAL COEFFICIENTS: Coefficients or positive integers of the terms that arise in any binomial expansion that equal the number of combinations of r items that can be selected from a set of n items. It is usually written $\begin{pmatrix} n \\ r \end{pmatrix}$, ${}^nC_r$, or C(n, r) and read as "n combination r." For the binomial expression, $(1 + x)^2 = 1 + 2x + x^2$, 2 is the coefficient of x and 1 is the coefficient of x^2. The general coefficient of the binomial expansion $(a+b)^n = \frac{n!}{(n-r)!r!} a^{n-r} b^r$. Binomial coefficients also represent an entry in Pascal's triangle. Arranging binomial coefficients into rows for successive values of n and in which r ranges from 0 to n, gives Pascal's triangle.

BINOMIAL DISTRIBUTION: Statistical distribution generated by a binomial expansion and the probability of an event occurring. For the probability, p of an event happening in a single attempt, the probability q that it will not happen is 1 - p. If there are n attempts the probability of an event happening b times is given as ${}^nC_b.p^b.q^{n-b}$ with the distribution as $p^n + {}^nC_1.p^{n-1}q + {}^nC_2.p^{n-2}q^2 + \ldots$.

BINOMIAL EXPANSION: The result of multiplying out a binomial expression supplied to a given power or multiplying out n binomial expressions. For example, the binomial expression $(1 + x)^2$ has the expansion $(1+x)(1+x) = (1+x)^2 = 1 + 2x + x^2$.

BINOMIAL EXPERIMENT: An experiment consisting of a fixed number of independent Bernoulli trials. The outcome of the experiment, Y is the random variable of the number of successes, the binomial distribution.

BINOMIAL TABLE: It gives the values for the distribution function of a random variable that follows a binomial distribution.

BINOMIAL TEST: A parametric hypothesis for making inferences about the probability of a success p, in a series of independent repeated Bernoulli trials. The test statistic is the observed number of successes, B. In test statistic for small sample size, the values of B can be obtained from tables and for large sample size is given by $Z = \frac{B - np_0}{[np_0(1-p_0)]^{\frac{1}{2}}}$, where p_0 is the hypothesised value of p. When the hypothesis is true z has an asymptotic standard normal distribution.

BINOMIAL THEOREM: A rule for the expansion of a binomial expression $(x + y)$, consisting of two variables raised to any positive integer power. Examples of the powers of binomial include the following: $(x + y)^0 = 1$, $(x + y)^1 = x + y$, $(x+y)^2 = x^2 + 2xy + y^2$, $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$, $(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$,
Another form of binomial theorem is given as:

$$(a+b)^n = \sum_{r=0}^{n} \overset{n_{c_r} a^n - rb^r}{\begin{pmatrix} n \\ r \end{pmatrix}} .$$

Binomial theorem is also given by: $(a+b)^n = a^n + {}^nc_1 a^{n-1}b + {}^nc_2 a^{n-2}b^2 + \ldots + b^n$. This formula holds for only positive integer ($n \in z^+$) and

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

This formula also holds for all real numbers $(n \in \mathfrak{R})$. The series is valid for -1 < × < 1. Binomial theorem can also be given by the following: $(a+b)^n = a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + \ldots b^n$. This formula holds for only positive integer $n(\in z +)$ and

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)n(n-2)}{3!}x^3 + \ldots$$

The last formula holds for positive and negative integers of n, i.e., ($n \in z$). Before the formula is used, the expression must be stated in the form $(n + x)^n$.

BINORMAL: The vector perpendicular to both the tangent and the norm to a curve at a point in three-space, given by the vector product B = T × N, where T is the tangent vector and N is the principal normal vector.

BIOACCUMULATION (BIOLOGICAL ACCUMULATION; BIOMAGNIFICATION): An increase in the concentration of chemicals, such as pesticides, in organisms that live in environments contaminated by a wide variety of organic compounds.

BIOACTIVATION: A metabolic process in which a chemically reactive substance is produced from a relatively inactive precursor.

BIOAUGMENTATION: A type of bioremediation in which cultured microorganisms, such as bacteria are added to a contaminated area to metabolise and get rid of the contaminants. The process can be used to treat sewage or contaminated water. It can also be used to treat soil.

BIOASSAY (BIOLOGICAL ASSAY; BIOLOGICAL STANDARDISATION): A controlled experiment for the quantitative estimation of a substance by measuring its effects in a living organism. For example, certain fungi are used to find the concentration of a particular vitamin, such as thiamin, in a substance as the growth of the fungus is directly proportional to the concentration of the vitamin. It is used in the development of new drugs and in monitoring environmental pollutants. It has been used extensively to measure the effects of different auxins and gibberellins on plant growth.

BIOASTRONAUTICS: The study of the biological, behavioural and medical effects of space flight upon humans and other living organisms such as animal or plant life.

BIOASTRONOMY (ASTROBIOLOGY): The study of the origin, evolution and distribution of life in the universe. Bioastronomy became the name of the International Astronomical Union's Commission 51, was established in 1982 as "Bioastronomy: Search for Extraterrestrial Life" and was renamed simply "Bioastronomy" in 2006. An IAU international symposium on bioastronomy is held every three years.

BIOAVAILABILITY: 1. The extent of absorption of a substance by a living organism compared to a standard system; or the degree to which a drug or other substance becomes available to the target tissue after administration. 2. The ratio of the systemic exposure from extravascular exposure to that following intravenous exposure.

BIOCATALYST: An enzyme or enzyme complex consisting of or derived from, an organism or cell culture that catalyses metabolic reactions in living organisms and/or substrate conversions in various chemical reactions.

BIOCHEMICAL CYCLE (BIOGEOCHEMICAL CYCLE): The flow of a chemical element or molecule through the biological and physical environment of an ecosystem; in effect, recycling the element. For example, water is always recycled through the water cycle by evapotranspiration, condensation and precipitation and falling back to Earth. Examples of biochemical cycles are the carbon, hydrogen, oxygen, nitrogen, sulfur and phosphorus cycles.

BIOCHEMICAL EVOLUTION: The changes that occur at the molecular level in organisms over a period of time. These include deletions, additions or substitutions of single nucleotides through the rearrangement of part of genes, to the duplication of entire genes or even whole genomes.

BIOCHEMICAL FUEL CELL: A system that exploits biological reaction for the conversion of biomass to electricity: the fuel source is bioorganic matter such as industrial waste and sewage; air is the oxidant at the cathode and microorganisms catalyse the oxidation of the bioorganic matter at the anode.

BIOCHEMICAL GENETICS: The branch of genetics concerned with inheritance at the molecular level. It includes the study of the structure of DNA, its replication, the genetic code and transcription and translation in protein synthesis and the regulation of gene expression.

BIOCHEMICAL MUTANT: An organism possessing a mutation that causes a particular enzyme either not to be synthesised or to be defective in some respect. The reaction that it catalyses therefore does not take place and, if this reaction is an intermediate step in a metabolic pathway, the subsequent steps in the pathway are also prevented. Such mutants can be induced by various methods, such as irradiation with x-rays and have proved particularly useful in elucidating the stages of a number of pathways.

BIOCHEMICAL OXYGEN DEMAND (BIOLOGICAL OXYGEN DEMAND; BOD): The amount of oxygen taken up by aerobic microorganisms to decompose organic waste matter in water such as that polluted by sewage. It is used as a measure of the degree of water pollution.

BIOCHEMICAL PATHWAY (METABOLIC PATHWAYS): A series of enzyme-catalysed reactions occurring within a cell that results in the conversion of an original product into a final one.

BIOCHEMICAL TAXONOMY (CHEMOTAXONOMY): A method of biological classification of plants and microorganisms based on similarities in the structures of certain compounds among the organisms being classified.

BIOCHEMISTRY: The study of metabolism in prokaryotic and eukaryotic cells. Alternatively, it is chemistry as it applies to living matter or the study of the chemical substances and processes that occur in plants, animals and microorganisms. Examples include the chemical reactions by which proteins and all their precursors are synthesised, how food is converted to energy, how energy is stored and released and how hereditary characteristics are transmitted.

BIOCHIP: 1. An integrated circuit whose electrical and logical functions are performed by protein molecules appropriately manipulated. Living cells rely on tiny electrical signals to work. Scientists believe that by redesigning cell structures they can use living material to process data in the way that a computer chip does. 2. The tiny conventional microchip that can be linked to organic material to control its growth and operation. Another name for this science is molecular electronics.

BIOCIDE: A chemical, which is used to kill or control living organisms usually in a selective way. Biocides used on plants are called herbicides, those used on fungi are called fungicides and those used on pests are known as pesticides.

BIOCLIMATE CHART: A chart used to determine the corrective measures needed to restore the feeling of comfort at any point outside the comfort zone.

BIOCOENOSIS (BIOCOENOSE; BIOCENOSE; BIOTIC COMMUNITY; BIOLOGICAL COMMUNITY; ECOLOGICAL COMMUNITY): All the organisms living in a particular place at a particular time. It is a diverse group of species or organisms with its own distinct habitat, interacting to form an ecological community.

BIOCOMPATIBILITY: The ability of biologic materials to exist or perform in harmonious combination.

BIOCONCENTRATION: The uptake of a substance from water alone. By contrast, bioaccumulation refers to uptake from all sources combined such as water, food and air.

BIOCONJUGATE: A molecular species produced by living systems of biological origin when it is composed of two parts of different origins, for example, a conjugate of a xenobiotic with some groups, for example, glutathione, sulfate or glucuronic acid, to make it soluble in water or compartmentalised within the cell.

BIODEGRADABLE: Any organic material that can be decomposed by natural living organisms such as bacteria and fungi. Biodegradable materials include sewage, trees, grass, waste vegetables, meat and other foodstuffs, while examples of non-biodegradable substances include glass, heavy metal and most types of plastics.

BIODEGRADATION: Breakdown of a substance catalysed by enzymes *in vitro* or *in vivo*. This may be characterised for purposes of hazard assessment as: (i) **Primary biodegradation**, which is the alteration of the chemical structure of a substance resulting in loss of a specific property of that substance. (ii) **Environmentally acceptable biodegradation** to such an extent as to remove undesirable properties of the compound. This often corresponds to primary biodegradation but depends on the circumstances under which the products are discharged into the environment. (iii) **Ultimate biodegradation** is the complete breakdown of a compound to either fully oxidised or reduced simple molecules such as carbon dioxide/ methane, nitrate/ammonium and water. Products of biodegradation can be more harmful than the substance degraded.

BIODIESEL: A liquid biofuel, which is mainly derived from vegetable oils such as oilseed rape, sunflower, and groundnut, certain residues, animal fats and oils, and from algae.

BIODIGESTER (ANAEROBIC DIGESTER): A device that uses bacteria to break down organic matter such as agricultural waste, waste food, compost, manure or sewage waste in the absence of oxygen to produce biogas (mostly methane (CH_4) and carbon dioxide (CO_2) and other trace gases) in a process called anaerobic fermentation. It includes an oxygen-free closed tank with a decomposition space for the organic materials and a valve system that can be opened to release the methane gas and control the pressure within the tank. The organic waste decomposes in the bottom of the device as bacteria break down the waste material. When fully decomposed, the waste is used as compost and fertiliser. Methane gas, carbon dioxide and other gases rise to the surface and can be collected from the valve to be used as a fuel source in many applications. For example, domestically, it can be used for heating and lighting. The gas can also be used in devices such as boilers to produce steam. It can also be refined into bio-methane for use in motor vehicles, for generation of electricity and to produce heat. Often this requires that it is purified to remove carbon dioxide and the water content. This is a renewable energy that is commonly used.

BIODIVERSITY (BIOLOGICAL DIVERSITY): The variety and complexity of species (plants, animals, fungi and micro-organisms) present and interacting in an ecosystem and the relative abundance of each. Much of this diversity is found in the world's tropical areas, particularly in the forest regions. It can be measured by the numbers and types of different species or the genetic variations within and between species.

BIODIVERSITY GRADIENT: The gradual reduction in biomass species numbers that occurs with increasing latitude.

BIODIVERSITY HOTSPOT: A relatively small area with an exceptional concentration of species. For an area to be labelled a biodiversity hotspot it must contain a minimum of 1,500 species of vascular plants and must have lost at least 70 percent of its original habitat, as a result most of the living species disappeared.

BIODYNAMICS: The study of the forces acting upon bodies in motion or in the process of changing motion, as they affect living things.

BIOELECTRONICS: The application of biomolecular principles to microelectronics such as in biosensors and biochips.

BIOELEMENTS: Any chemical element that is found in molecules and compounds, that makes up a living organism. Examples are oxygen, hydrogen, carbon and nitrogen, which are the most common bioelements found in the human body.

BIOENERGY: Energy that is derived from biological matter such as plants and animals but which has not undergone a geological process. Carriers of bioenergy may be solid such as wood, straw, liquid (e.g., biodiesel, bioethanol) or gaseous (e.g., methane).

BIOENERGETICS: 1. The study of the processes such as photosynthesis, by which living cells use, store and release energy that occur in living organisms. The amount of energy that an organism takes in from food or sunlight is measured and divided into the amount used for growth of new tissues, that lost through death, wastes and in plants transpiration; and that lost to the environment as heat through respiration, etc. 2. A combination of therapies, such as breathing and body exercises and the free expression of feelings and impulses, aimed at relieving tension and release physical and emotional energy.

BIOENGINEERING: The application of engineering knowledge to medicine and biology. It includes the fabrication and use of artificial tissues organs and organ components to replace parts of the body that are damaged, lost or malfunctioning. An example is the fitting of artificial limbs or heart valves on a person or animal.

BIOETHANOL: A liquid biofuel, mainly produced by the fermentation of plant starches derived from plants or products such as cereals, milk (lactose), potatoes, sugar beet, sugarcane and wine.

BIOFEEDBACK: The use of monitoring devices that display information about the operation of a bodily function that is not normally consciously controlled. This helps an individual to control muscle tension, pain, body temperature, brain waves and other bodily functions and processes through relaxation, visualisation and other cognitive control techniques.

BIOFILM: A colony of bacteria and other microorganisms that adhere to a substrate, enclosed and protected by secreted slime. They readily form on any surface, whether living or nonliving, where there is moisture and supply of nutrients and important components of aquatic and terrestrial ecosystem, providing nutrients for small organisms at the base of the food chain.

BIOFUEL: A wide range of fuels which are in some way derived from biomass and an alternative to fossil fuels. The term covers solid biomass, liquid fuels and various biogases. Examples are firewood, biogas, biodiesel and bioethanol.

BIOGAS TECHNOLOGY: The production of a mixture of different gases, mainly methane (CH_4) and carbon dioxide (CO_2) and other trace gases by the breakdown of organic matter in the absence of oxygen. Such organic matters include agricultural waste, manure, plant material, sewage, green waste or food waste. The gas produced is known as biogas that may be used as a fuel source in many applications, for example, domestically for cooking or lighting. It can also be used directly in heat and power gas engines or it can be refined into bio-methane for use in motor vehicles, for generation of electricity and to produce heat.

BIOGENESIS: The principle that a living organism can only arise from other living organisms similar to itself and can never originate from non-living material.

BIOGENETIC LAW (RECAPITULATION THEORY): The principle that the stages that an organism goes through during embryonic development reflect the stages of that organism's evolutionary development.

BIOGENETICS: The branch of biology concerned with manipulation of genetic material in order to alter the hereditary traits of a cell organism or population.

BIOGENIC AMINE: Any amine produced by living organisms that act primarily as neurotransmitters and are capable of affecting mental functioning and of regulating blood pressure, body temperature and other bodily processes. Examples are adrenaline (epinephrine), noradrenaline (norepinephrine), dopamine, serotonin and histamine.

BIOGEOCHEMICAL CYCLE (BIOCHEMICAL CYCLE): The flow of a chemical element or molecule through the biological and physical environment of an ecosystem; in effect, recycling the element. For example, water is always recycled through the water cycle by evapotranspiration, condensation and precipitation and falling back to Earth. Examples of biochemical cycles are the carbon, hydrogen, oxygen, nitrogen, sulfur and phosphorus.

BIOGEOCHEMISTRY: A branch of geochemistry that studies the circulation of chemical elements such as carbon and nitrogen between the cells of living organisms and geological materials of a given region. For example, the analysis of carbon in bone gives biogeochemists information on how the animal lived, its diet and its environment.

BIOGEOGRAPHY: The study of the geographic distribution of living and fossil species of plants and animals on Earth as consequences of ecological and evolutionary processes.

BIOHAZARDS: Biological substances such as medical waste or samples of microorganisms, virus or toxin that can be harmful to people, animals or the environment.

BIOINFORMATICS: The use of information technology in the collection, storage and analysis of DNA and protein sequence data.

BIOINORGANIC CHEMISTRY: Biochemistry involving compounds that contain metal atoms or ions. It involves the study of naturally occurring inorganic elements in biology and the use of metals in biological system, for example, as probes and as cofactors in some proteins. Two common examples of bioinorganic compounds are haemoglobin, which contains iron and chlorophyll, which contains magnesium. Many organisms require a number of other elements such as metal ions that are important in a number of biochemical processes, including respiration, metabolism, cell division, muscle contraction, nerve impulse transmission and gene regulation.

BIOINSECTICIDE (BIOLOGICAL INSECTICIDES; BIOPESTICIDES): A naturally based substance that is used to kill or inhibit the activity of insect pests. It includes the following: (i) biochemical pesticides, made from naturally occurring substances such as baking soda, canola oil, neem oil, cayenne pepper, etc.; (ii) microbial pesticides which use microorganisms such as viruses, protozoa, or oomycetes as the active ingredient in a preparation; and (iii) by genetic modification of plants to naturally resist insect pests.

BIOISOSTERE: A compound resulting from the exchange of an atom or group of atoms with another, broadly similar, atom or group of atoms.

BIOLISTICS: A technique for introducing genetic material into living cells, especially plant cells in which DNA-coated microscopic particles are fired at high speed into the cells using a special gun.

BIOLOGICAL ASSIMILATION (ASSIMILATION; BIO-ASSIMILATION): A combination of two processes to supply animal cells with nutrients or tissues. Assimilation occurs in every cell of the body to help develop new cells. The first process is absorption of vitamins, minerals and other chemicals from food within the gastrointestinal tract. In humans this is achieved by chemical breakdown by enzymes and acids and physical breakdown by oral mastication and stomach churning. The second process is the chemical alteration of substances in the bloodstream by the liver or cellular secretions resulting in the absorbed food reaching the cells via the liver. Examples of assimiliation in organisms are: (i) **Photosynthesis** in which carbon dioxide and water are transformed into a number of organic molecules in plant cells. (ii) **Nitrogen fixation** from the soil into molecules by symbiotic bacteria which live in the roots of certain plants, such as Leguminosae. (iii) The absorption of nutrients into the body after digestion in the intestine and its transformation in biological tissues and fluids.

BIOLOGICAL CLOCK (BODY CLOCK): The biological rhythm or internal mechanism that automatically regulates periodic changes in behaviour or physiology in animals and plants with daily, monthly or seasonal cycles or with stages of development and aging.

BIOLOGICAL CONTROL: A method of controlling pests using natural agents such as predators, breeding resistant crop strains, inducing sterility in pests, fungi and other micro-organisms rather than the use of chemicals. Natural enemies of insect pests, also known as **biological control agents** include predators, parasitoids and pathogens. Biological control agents of plant diseases are most often referred to as **antagonists**. Biological control agents of weeds include herbivores and plant pathogens. Predators, such as birds, lady beetles and lacewings, are mainly free-living species that consume a large number of prey during their whole lifetime. For example, the moth *Cactoblastis cactorum* has been used in Australia to help control the prickly pear cactus *(Opuntia vulgaris*). The caterpillars bore into the pads of the cactus, severely affecting its growth. **Parasitoids** are species whose immature form develops on or within a single insect host, ultimately killing or fatally infecting the host. Many species of wasps and some flies are parasitoids. **Pathogens** are disease-causing organisms including bacteria, fungi and viruses. They kill or debilitate their own host and are relatively specific. The three main branches of biological control are **Classical biological control,** which is the control of pests introduced from another region through importing specialised natural enemies of the pest from its native range, **Conservation biological control** which manipulates the environment to favour natural enemies of the pest and **Augmentation biological control** which occurs when the number of biological control agents is supplemented. **Inoculation** is the introduction of a small number of individuals of the biological control agent, while **inundation** is the introduction of vast numbers of individuals. The benefits of biological control are that it can provide fairly permanent regulation of devastating agricultural and environmental pests that may be difficult or impossible to manage with more traditional chemical means, however, biological control agents may negatively affect native species directly or indirectly.

BIOLOGICAL DETERGENTS: Laundry detergents containing enzymes obtained from bacteria adapted to live in hot springs, which break down stains caused by proteins in milk, blood, egg, etc. The stains are converted by the enzymes into soluble substances which can be washed away.

BIOLOGICAL EFFECT MONITORING: Continuous or repeated measurement and assessment of early biological effects of exposure to a substance. It normally involves measuring biochemical responses, for example, plasma and erythrocyte cholinesterase activity in workers exposed to organophosphorus pesticides; or measuring increases in urinary protein following exposure to cadmium. **Biological monitoring or biomonitoring** refers to the continuous or repeated measurement and assessment of chemicals or their metabolites in tissues, secreta, excreta, expired air or any combination of these in exposed workers.

BIOLOGICAL EXPOSURE INDEX (BEI): Reference values used for guidance to assess biological monitoring results. BEIs represent the levels of determinants that are most likely to be observed in specimens collected from a healthy worker who has been exposed to chemicals to the same extent as a worker with inhalation exposure to the threshold limit value (TLV).

BIOLOGICAL FITNESS (DARWINIAN FITNESS): The ability of an organism to survive and reproduce. It is a reproductive success as the more offspring an organism produces during its lifetime, the greater its biological fitness.

BIOLOGICAL HALF-LIFE: The time taken for half a substance such as a metabolite, drug, signalling molecule, radioactive nuclide or other substance to be removed from the body by biological processes.

BIOLOGICAL INDICATOR (BIOINDICATOR): Any biological species or group of species that is very sensitive to factors in its environment and used to monitor the health of an environment or ecosystem. For example, lichens can be biological indicators because they are sensitive to sulfur dioxide in the air.

BIOLOGICAL MAGNIFICATION (BIOAMPLIFICATION; BIOMAGNIFICATION): A term used to refer to the process in which certain substances such as pesticides or heavy metals move up the food chain, work their way into rivers or lakes and are eaten by aquatic organisms such as fish, which in turn are eaten by large birds, animals or humans. The substances become concentrated in tissues or internal organs as they move up the chain.

BIOLOGICAL MEDIUM: One of the major components of an organism such as blood, fatty tissue, lymph nodes or breath, in which chemicals can be stored or transformed.

BIOLOGICAL OXIDATION: A biochemical reaction involving the transfer of an electron from one organic compound to another organic compound or to oxygen. When a compound loses an electron or is oxidised, another compound gains the electron or is reduced. Oxidation-reduction (redox) reactions represent the main source of biological energy. Oxidation and reduction reactions occur simultaneously and one does not occur without the other. The principal sources of reductants for animals are the numerous breakdown products of the major foodstuffs: carbohydrates, fats and proteins. Energy release from these substances occurs in a stepwise series of hydrogen and electron transfers to molecular oxygen.

BIOLOGICAL OXYGEN DEMAND (BIOCHEMICAL OXYGEN DEMAND, BOD): The amount of oxygen taken up by aerobic microorganisms that decompose organic waste matter in water such as that polluted by sewage. It is used as a measure of the amount of organic water pollution.

BIOLOGICAL SHIELD: A protective cover surrounding a reactor that is impervious to radiation, usually consisting of a thick wall of steel and concrete that is intended to protect personnel from the effects of radiation.

BIOLOGICAL SPECIES CONCEPT (BSC): The concept of a species composed of related organisms that share common characteristics and whose members can interbreed successfully but cannot reproduce with other groups. This concept became popular during the late 19th and early 20th centuries, largely replacing the typological species concept. Sexual reproduction forms an important pert of this concept.

BIOLOGICAL WARFARE (BIOLOGICAL WEAPONS; BIOWEAPONRY; GERM WARFARE): The military use of microorganisms such as bacteria, viruses, fungi, the agents for anthrax and botulism to induce disease or death among humans, livestock and crop plants.

BIOLOGICAL WEATHERING (ORGANIC WEATHERING): A form of weathering caused by the activities of living organisms or by chemicals formed from plants and animals. Examples include the growth of roots or the burrowing of animals into a rock.

BIOLOGY: The study of all living things or the basic science of plants and animals, including the study of their structure, function origin and ecology, classification, interrelationships and distribution. Some of the various fields include cytology (the study of cells), zoology (the study of animals), ecology (the study of the environment) and botany (the study of plants). A **biologist** is a scientist who studies living organisms such as plants and animals. Most biologists specialise in the study of a certain type of organism or in a specific activity including the following: Aquatic biologists study micro-organisms, plants and animals living in water; biochemists study the chemical composition of living things; botanists study plants and their environments; microbiologists investigate the growth and characteristics of microscopic organisms such as bacteria, algae or fungi; physiologists study life functions of plants and animals, both in the whole organism and at the cellular or molecular level, under normal and abnormal conditions; biophysicists study how physics, such as electrical and mechanical energy and related phenomena, relating to living cells and organisms; zoologists and wildlife biologists study the origin, behaviour, diseases and life processes of animals and wildlife; ecologists investigate the relationships among organisms and between organisms and their environments, examining the effects of population size, pollutants, rainfall, temperature, altitude, etc.

BIOLUMINESCENCE: The production and emission of light without heat by living organisms. Examples of bioluminescent organisms are glow-worms, fireflies, bacteria, fungi and some deep sea fish. In biomedical research, the enzymes and other proteins associated with bioluminescence have been developed and exploited as markers or reporters of other biochemical processes.

BIOMARKER (BIOLOGICAL MARKER): A normal metabolite that, when present in abnormal concentration in certain body fluids such as blood or urine, can indicate the presence of a particular disease or toxicological condition. For example, a high concentration of sugar in the urine or blood can indicate a defective pancreas, causing diabetes mellitus.

BIOMASS: 1. Also known as **standing crop**, it is the total weight or volume of either all the living organisms or of one species present at any one time in a community. Measurements of biomass can be used to study interactions between organisms, the consequences of those interactions and variations in population numbers, etc. 2. Any plant material, such as trees, grass, waste vegetables; manure or sewage which might be used as a resource. Biomass can be used as a direct source of energy such as by burning wood or straw, decomposed to produce fuels or as a feed stock for the chemical industry. Generally, the term is intended to refer to materials that do not directly go into foods or consumer products but may have alternative industrial uses.

BIOMASS PYRAMID (ECOLOGICAL PYRAMID OF BIOMASS): A graphical representation that shows the relative masses of consumers and food that are available at the same time. There are two types of biomass pyramids: the upright and the inverted. The upright type occurs in most ecosystems and results when the combined weight of producers is larger than the combined weight of consumers. An inverted type results when the combined weight of producers is smaller than the combined weight of consumers.

BIOMASS-ENERGY (BIOENERGY): Plant matter expressed on a dry weight basis (after removal of all water) that can be converted to an energy or fuel source by either direct or indirect methods. The most popular indirect method is conversion to ethanol.

BIOMATHEMATICS (MATHEMATICAL BIOLOGY): The application of mathematical processes, techniques and tools in analyzing or modelling natural and biological processes or (simply) the application of mathematics to the field of biology. Being an interdisciplinary field biomathematics employs theoretical analysis, mathematical models and abstractions of living organisms to investigate the principles that govern the structure as well as the development and behavior of systems, while experimental biology deals with conducting experiments to prove and validate scientific theories. When systems are modelled or described in a quantitative rather than experimental manner, their behavior can be better simulated, hence some properties that might not be evident to the biological experimenter can be evident to or predicted by the bio-mathematician.

BIOME (BIOTIC REGION): A very large community of vegetation which is defined by climate and covering a large area of the Earth, characterised by specific types of plants and animals. Examples are tropical rain forests, tundra, deserts and savanna grasslands. A biome may include lakes and oceans, which are influenced more by terrestrial rather than climatic factors.

BIOMECHANICS: The application of the principles of mechanics to living organisms, particularly those that have coordinated movements. It also deals with the properties of biological materials such as blood and bone. Biomechanics includes research and analysis of the mechanics of living organisms and the application of engineering principles to and from biological systems.

BIOMEDICAL TECHNOLOGY: The application of health care theories to develop methods, products and tools to diagnose and treat various diseases in order to maintain or improve health.

BIOMEDICINE: The combined discipline of biology and medicine for the analysis of human tolerances of and protection against, environmental variances.

BIOMETRIC AUTHENTICATION: The automatic identification of living individuals by using their physiological and behavioural characteristics. Biometric authentication uses a biometric sampling device to access certain physical characteristics through fingerprints, face recognition, voice recognition, retina patterns, pictures, weight, or other means that are stored in a database. These user characteristics are unique to that individual alone, and cannot be duplicated as the sampling device is tamper-proof. These characteristics can be retrieved by a **biometric device** which identifies a personality with the various characteristics stored in the database.

BIOMETRICS: A field of science that uses computer technology to identify people based on physical or behavioural characteristics, such as fingerprints or voice scans.

BIOMETRY: The application of mathematical and statistical concepts to the analysis of biological phenomena.

BIOMIMETIC: 1. A term used to refer to a laboratory procedure designed to imitate a natural chemical process. 2. A compound that mimics a biological material in its structure or function.

BIOMINERALS: Minerals formed from the remains or as a result of the metabolic activity, of long-dead organisms. **Biomineralisation** is the process by which living organisms produce minerals, often to harden or stiffen existing tissues. Such tissues are called mineralised tissues. Examples include silicates in algae and diatoms, carbonates in invertebrates and calcium phosphates and carbonates in vertebrates.

BIOMOLECULES: Any organic molecule that is produced by a living organism and involved in the maintenance and metabolic process of living organisms. Examples include carbohydrate, lipid and protein.

BIOMONITORING: 1. The use of living organisms to test the suitability of effluents for discharge into receiving waters and to test the quality of such waters downstream from the discharge. 2. Analysis of blood, urine, tissues, etc. to measure chemical exposure in humans.

BIONICS: The application of biological methods and systems found in nature to the study and design of engineering systems and modern technology. Examples include radar, inspired by the echolocation system of bats or the development of associative memories in computers as in the human brain and the functioning of the eye used to design charge coupled device (CCD) cameras.

BIOPAK: A container for housing a biological organism in a habitable environment.

BIOPHYSICS: The study of the physical aspects of biology, including the application of physical laws and the techniques of physics to study biological phenomena.

BIOPOIESIS (ABIOGENESIS; AUTOGENESIS): The hypothesis that living matter developed from complex organic molecules that are themselves nonliving but self-replicating. It is the process by which life is assumed to have begun on Earth.

BIOPOLYMER: A natural polymer in a living organism that is formed by linking together several smaller molecules. Examples are cellulose, protein, chitin or DNA. A biopolymer can also be synthesised in a laboratory.

BIOPSY: The examination of cells, fluids or tissues removed medically from a living body to diagnose a disease.

BIOREACTOR (INDUSTRIAL FERMENTOR): A large fermentation stainless steel tank used to grow microorganisms such as bacteria or yeast in the biotechnological production of substances such as pharmaceuticals, antibodies or vaccines or for the bioconversion of organic waste.

BIOREGENERATION: The renewal or restoration of life-supporting resources using biological processes. It usually describes processes in a closed system.

BIOREMEDIATION: The use of agents, such as bacteria, fungi, or green plants to decontaminate soil and groundwater, as well as clean up oceans after oil spills and other environmental disasters. There are various types of bioremediation; examples include the following: (i) **microbial bioremediation** which uses microorganisms such as bacteria to break down contaminants or, for example, to remove agricultural pests or counteract diseases of trees, plants in soil; (ii) **mycoremediation** that uses fungi to break down contaminants such as pesticides, hydrocarbons, and heavy metals; (iii) **phytoremediation** uses plants to bind, extract, and clean up pollutants such as pesticides, petroleum hydrocarbons, metals, and chlorinated solvents; (iv) **bioventing** in which contaminants or organic constituents adsorbed into the unsaturated zone of soil are biodegraded by aerating the subsurface to enhance biological activity of naturally occurring micro-organisms in the soil; and (v) **bioaugmentation** in which cultured microorganisms, such as bacteria are added to a contaminated area to metabolise and get rid of the contaminants.

B

BIORHYTHM (BIOLOGICAL RHYTHM): A roughly periodic change in the behaviour or physiology of an organism that is generated and maintained by a biological clock and often synchronised with daily, monthly or yearly environmental changes.

BIOS: Abbreviation for **Basic Input/Output System**, it is a chip located on computer motherboards that contains instructions and setup for how the system should boot and how it operates. It is a program a personal computer's microprocessor uses to get the computer system started after it is turned on. The CPU accesses the BIOS even before the operating system is loaded. The BIOS then checks all the hardware connections and locates all devices. If the system is trouble free, the BIOS loads the operating system into the computer's memory and finishes the boot-up process. It also manages data flow between the computer's operating system and attached devices such as the hard disk, video adapter, keyboard, mouse and printer.

BIOSENSOR: A device that uses immobilised agents, usually enzymes, antibiotics or whole cells to detect or measure a chemical compound. It is used in tests to diagnose medical conditions such as blood pressure. It consists of a biological recognition element and a transducer capable of detecting the biological reaction and converting it into a signal which can be processed.

BIOSPHERE (ECOSPHERE): The whole of the region of the Earth's surface, including air, land, surface rocks and water, within which life occurs and which biotic processes in turn alter or transform.

BIOSTATISTICS: The scientific field in which statistical methods are applied in the fields of demography, biomedical sciences and clinical trials for the proper interpretation of scientific data.

BIOSTRATIGRAPHY: The characterisation of rock strata based on the fossils they contain. The fossils are useful because sediments of the same age can look completely different because of local variations in the sedimentary environment. For example, one section might have been made up of clays and marls while another has more chalky limestones, but if the fossil species recorded are similar, the two sediments are likely to have been laid down at the same time.

BIOSTROME: A layer of rock composed of the remains of various fossilised animals, such as crinoids and coral.

BIOSTIMULATION: A process which helps catalyse the activity of microorganisms involved in biodegradation. This can be done by addition of various forms of rate limiting nutrients and electron acceptors, such as phosphorus, nitrogen, oxygen, or carbon.

BIOSYNTHESIS: The production of more complex molecules or organic chemicals from simple inorganic ones by living cells. An example is the conversion of carbon dioxide and water to glucose by plants during photosynthesis.

BIOSYNTHETIC PATHWAY: A series of enzymatic reactions in which more complex molecules are built from simpler ones. Biosynthetic pathways are energy requiring, this energy usually being supplied either as phosphate bond energy by adenosine triphosphate (ATP) or as reducing power by NADPH. There are three stages of biosynthesis: (i) The first stage involves the manufacture of simple organic molecules from inorganic molecules like carbon dioxide and water. (ii) The simple molecules from the first stage are then converted in the second stage to building blocks for the many biological macromolecules. (iii) Finally macromolecules are synthesised from their constituent subunits. The major biosynthetic pathways in plants are shown in the diagram.

BIOSYSTEMATICS: The study of variation and relationship in populations rather than individuals. The genetic and evolutionary nature of groups is assessed by the examination of cytology, comparative morphology and ecology. Biosystematics is often described as the taxonomic application of genecology or, occasionally, as experimental taxonomy.

BIOT-SAVART LAW: A principle in electromagnetism that states that the magnetic flux density, B of a magnetic field produced by a current, I in a long straight conductor is directly proportional to I and inversely proportional to the perpendicular distance, r from the conductor. Its vector equation can be represented as $d\vec{B} = \frac{\mu_o I d\vec{L} \times 1_{\vec{r}}}{4\pi r^2}$.

BIOTA: All the organisms living in a particular geographic region, including plants, animals and microorganism. The biota of the Earth lives in the biosphere.

BIOTAXIS: 1. The selection and arrangement process of living cells; or the ability of cells to develop into certain forms and arrangements. 2. Also called **biotaxy**, it is the classification of living beings according to their anatomical characteristics.

BIOTECHNOLOGY: The development of technologies for the application of biological processes to the production of materials of use in medicine and industry or the industrial use of living organisms to manufacture food, drug or other products. For example, the brewing and baking industries have long relied on the yeast microorganism for fermentation purposes, whereas the dairy industry employs a range of bacteria in cheese and yoghurt production.

BIOTELEMETRY: The electronic measuring, transmitting and recording of qualities, properties and actions of a living organism's basic physiological functions, such as heart rate and muscle activity, usually by means of radio transmissions from a remote site or with a device attached to or embedded in the animal. Biotelemetry may include behavioural, physiological, physical or environmental data.

BIOTIC: Pertaining to or referring to living organisms in the environment that adversely affect other living organisms. Examples are insects, disease pathogens, nematodes, and weeds.

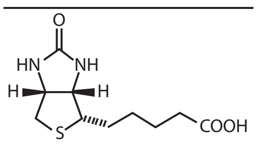
BIOTIC DISEASES: Diseases caused in plants by living organisms such as fungi, bacteria, viruses, nematodes, insects, mites and animals. **Abiotic diseases** in plants are caused by non-living factors, such as drought stress, sunscald, freeze injury, wind injury, chemical drift, nutrient deficiency or improper cultural practices, such as overwatering or planting too deep.

BIOTIC ENVIRONMENT: The part of an organism's environment that consists of other living organisms which interact with each other and their abiotic environment. This may affect it in many ways, for example, as competitors, predators, parasites, prey or symbionts.

BIOTIC FACTOR: Living variables affecting an ecosystem. An example is the changing population of elephants and its effects on the African Savanna.

BIOTIC POTENTIAL (INTRINSIC RATE OF INCREASE): The capacity of a population of organisms to increase at maximum rate under optimum environmental conditions.

BIOTIN (VITAMIN B7; VITAMIN H; COENZYME R; BIOPEIDERM): An organic compound in the vitamin B complex, with the chemical formula $C_{10}H_{16}O_3N_2S$ and molar mass 244.31 g mol^{-1}, which is essential for growth and well-being in animals and some microorganisms. It is the coenzyme for various enzymes that catalyse the incorporation of carbon dioxide into various compounds, synthesise fatty acids and convert amino acids to glucose in the body. It is widely distributed in nature and is especially abundant in egg yolk, beef, liver and yeast.



Chemical Structure of Biotin

BIOTITE: An important rock-forming silicate mineral, a member of the mica group of minerals, in common with which it has a sheet like crystal structure. It is usually black, dark brown or green in colour.

BIOTOPE: A region that has a characteristic uniform environmental condition and consequently a particular type of fauna and flora.

BIOTRANSFORMATION (BIOCONVERSION): Any chemical conversion of substances that is mediated by living organisms including biodegredation.

BIOTROPHIC: Close associations seen between two different organisms, that work mutually to benefit each other.

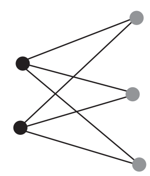
BIOTYPE: A group of individuals or organisms within a species having the same genetic makeup.

BIOVENTING: A type of bioremediation procedure in which contaminants or organic constituents adsorbed into the unsaturated zone of soil are biodegraded by aerating the subsurface to enhance biological activity of naturally occurring micro-organisms in the soil.

BIPALMATE: In biology, it refers to a leaf structure in which the leaves and branches are palmately branched. A palmately shaped leaf has several lobes whose midribs spread out from one point.

BIPARTITE: 1. In mathematics, containing two separate sets of variable values; made up of two continuous, but non-intersecting, curves. 2. In botany, divided into two parts nearly to the base, as seen in the leaves of many passion flowers. 3. A graph is bipartite if its vertices can be partitioned into two disjoint subsets U and V such that each edge connects a vertex from U to one from V. A bipartite graph is a complete bipartite graph if every vertex in U is connected to every vertex in V. If U has n elements and V has m, then we denote the resulting complete bipartite graph by Kn,m.

(Fig I)
Bipartite Leaf

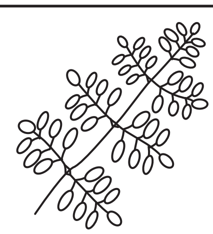


(Fig II)
Complete Bipartite Graph

BIPEDALISM: A type of locomotion, in which an organism stands on two legs and move about using only the two legs. Birds and humans are examples.

BIPETALOUS: Flowers with two petals.

BIPHOTONIC EXCITATION (TWO-PHOTON EXCITATION): Simultaneous (coherent) absorption of two photons of same or different wavelengths, in which the energy of excitation is the sum of the energies of the two photons.



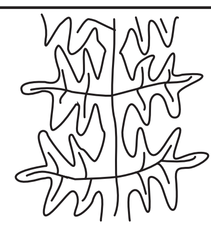


Fig. I: Bipinnate lealets ***Fig. II: Bipinnatifid leaf***

BIPINNATE: In botany, a pinnate leaf having the leaflets further subdivided into smaller leaflets, as seen in the sensitive plant (*Mimosa pudica*). **Bipinnatifid** leaf is twice pinnately cleft, as shown in the diagram; doubly pinnatifid.

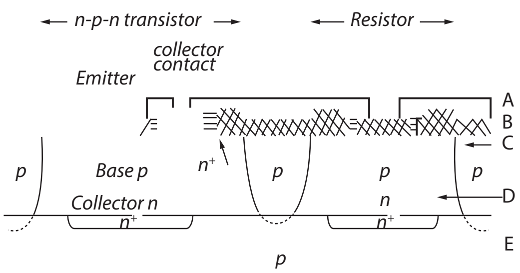
BIPOLAR DISORDER (MANIC DEPRESSION): A mental disorder characterised by recurring periods of high (mania) or low (depression) moods. In high moods, the symptoms are similar to that of mania and include behaviours such as irritability, excessive happiness, over-reaction to the point of physical exhaustion, difficulty in concentrating, reduced sleep, overly optimistic, involvement in dangerous activities, increased sexual desire, etc. In low moods the symptoms include depressed behaviours such as sadness, loss of energy, loss of interest or pleasure in most activities, lacking self-confidence and feeling hopeless, trouble concentrating, crying, irritability, insomnia, thoughts of death and attempting suicide, etc. The true cause of the disease is not known, but factors such as genetic and defective neurotransmitter can be contributors.

BIPOLAR NEURON: A neuron that has two processes, an axon and a dendron, extending in different directions from its cell body. Many sensory neurons are bipolar specialised and used for the transmission of special senses such as smell, sight, taste, hearing and vestibular functions.

BIPOLAR OUTFLOW: Two continuous flows of gas from the poles of a star. It is associated with protostars or the formation of young stars or with evolved post-AGB stars, often in the form of bipolar nebulae.

BIPOLAR SET: The set of vectors, denoted S^{00} or S^{∞} that are polar to the polar set of a given set, S, of vectors in a Hilbert space; this coincides in the real setting with the convex hull of S and zero.

BIPOLAR TRANSISTOR: A type of transistor in which both electrons and holes play an essential part. There are two categories of bipolar transistors, which are PNP and NPN transistors. An example is the junction transistor. The diagram below shows a bipolar integrated circuit.



A: Metal Interconnections, B: Insulating layer
C: Insulating diffusions, D: n-type epitaxial layer
E: p-type subsrate

Bipolar integrated circuit

BIPOLARON: A bound pair of polarons with the mutual attraction resulting from a lattice distortion.

BIPRISM: A glass prism with apex angle only a little less than 180°, which produces a double image of a point source, giving rise to interference fringes on a nearby screen.

BIQUADRATE: Quartic, raised to the fourth power.

BIQUADRATIC: 1. Of or relating to the fourth power. 2. A **biquadratic equation** or a **quartic equation** is a polynomial equation of the 4th degree; or a mathematical statement that two expressions are equal. Its highest order term is of the 4th power and only even powers occur with non-zero coefficients, such as $x^4 + 3x^2 - 5 = 0$. Such equations may be solved by the quadratic formula.

BIQUADRATIC EQUATION (QUARTIC EQUATION): A polynomial equation of the 4th degree; or a mathematical statement that two expressions are equal.

BIRADICAL (DIRADICAL): A radical that has two unpaired electrons at different points in the molecule, so that the radical centres are independent of each other.

BIRAMOUS APPENDAGE: A type of appendage that is characteristic of arthropods of the phylum Crustacea. It forks from the basal protopodite to form two branches, the inner endopodite and the outer expodite.

BIRD: A member of the vertebrate class Aves. Birds form the biggest group of land vertebrates and are characterised by warm blood, feathers, fore limb modified to form wings, breathing through lungs and egg-laying by the females.

BIRD POLLINATION (ORNITHOPHILY): Pollination by pollen carried by birds. It is an important form of pollination for many tropical species, the most common pollinators being hummingbirds in South America and sunbirds in Africa. Bird-pollinated flowers are often red in colour. Pollination by insects (entomophily) is believed to be derived from ornithophily.

BIREFRINGENCE (DOUBLE REFRACTION): An optical property in which a ray of light passing through anisotropic materials, such as calcite, quartz, ice, sugar and topaz is split into two rays travelling at different velocities and in different directions. It was first described by the Danish scientist Rasmus Bartholin in 1669, who saw it in calcite. One ray is refracted at an angle as it travels through the medium, while the other passes through unchanged. The effect is now known to also occur in certain plastics, magnetic materials, various noncrystalline materials and liquid crystals. The splitting occurs because the speed of the ray through the medium is determined by the orientation of the light compared with the crystal lattice of the medium. Plastics composed of long-chain molecules become anisotropic when stretched or compressed. Solutions of long-chain molecules become birefringent when they flow.

BIRGE-SPONER EXTRAPOLATION: A method used to calculate the heat dissociation of a molecule by extrapolation from observed band spectra.

BIRKELAND-EYDE PROCESS: A process for the fixation of nitrogen by passing air through an electric arc to produce nitrogen oxides. It was introduced in 1903 by the Norwegian chemists Kristian Birkeland (1867-1913) and Samuel Eyde (1866-1940). The process is economic only if cheap hydroelectricity is available.

BIRKHOFF ERGODIC THEOREM (STRONG/POINTWISE ERGODIC THEOREM): The theorem that, for any measure-preserving transformation, T, on a measure space and an integrable f, the Cesaro means of $f(T^n x)$ converge almost everywhere to an invariant function f^* for which $f^*(Tx) = f^*(x)$; when the underlying space has finite measure, f and f^* have the same integrals.

BIRKHOFF, GEORGE DAVID (1884-1944): American analyst and topologist who became president of the American Mathematical Society and the Association for the Advancement of Science and influenced an entire generation of American mathematicians. Although his own major work was in the application of analysis to dynamics, he also contributed to the study of difference equations, constructed a relativistic theory of gravity independently of Einstein and devised a mathematical theory of 'aesthetic measure'.

BIRKHOFF'S THEOREM: The theorem that every doubly-stochastic matrix can be expressed as a convex combination of permutation matrices.

BIRTH: The act of producing live young from within the body of female animals. After going through pregnancy term, the process of labour and birth is divided into three stages: (i) The first stage begins by having contractions that cause progressive changes in the cervix and ends when the cervix is fully dilated. This stage is divided into two phases, **Early labour**, when the cervix gradually effaces or thins out and dilates or opens; and **Active labour**, in which the cervix begins to dilate more rapidly and contractions are longer, stronger and closer together. (ii) The second stage of labour begins once the cervix is fully dilated and ends with the birth of the baby. (iii) The third stage begins right after the birth of the baby and ends with the delivery of the placenta.

B

BIRTH CANAL: The passage, extending from the uterus through the cervix, vagina and vulva through which the baby passes during childbirth.

BIRTH CONTROL (FAMILY PLANNING): A method that allows a human couple to decide when to have and how many children to have. It includes the use of contraceptives, avoiding sexual intercourse when the female is ovulating and sterilisation.

BIRTH DEFECT: A structural, functional or metabolic abnormality present at birth that results in physical or mental disability, some of which may be fatal. There are more than 4,000 known birth defects, which may be caused by genetic or environmental factors.

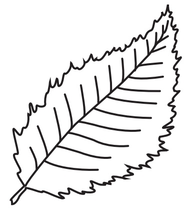
BIRTH RATE (NATALITY): The births in a population per unit time, usually expressed as the number of births per year per thousand of the population.

BISECT: 1. To divide a line or angle into two equal parts. 2. A profile of plants and soil showing the vertical and lateral distribution of roots and tops in their natural position.

BISECTION METHOD (BINARY LINE SEARCH; DICHOTOMOUS LINE SEARCH): The iterative method of finding the maximum of a unimodal function that proceeds at each iteration by excluding one half of the remaining interval by testing two new function values, both near the midpoint of the present interval, one above and one below.

BISECTOR (BISECTRIX): A straight line that divides another line or an angle into two equal parts. For example, the perpendicular bisectors of the sides of any triangle are concurrent.

BISERRATE: 1. Doubly serrated or having saw-like teeth on two sides as the margin of a serrate leaf, shown in the diagram. 2. In zoology, serrated on both sides, as the antennae of some insects. 3. In entomology, having two small triangular teeth placed close together, like the teeth of a saw.



A biserrate leaf

BISEXUAL: 1. In biology, it is any organism capable of producing both male and female gametes. In plants and microorganisms, this is often referred to as **monoecism**. Monoecious plants have both the male and female reproductive systems (stamens and pistils) present on the same plant. In multicellular animals, bisexuality is usually called **hermaphroditism**. Examples of animals which are mostly hermaphroditic are invertebrates such as worms, bryozoans, trematodes (flukes), snails, slugs, and barnacles. 2. In humans, also called **pansexual** or **omnisexual**, it is being romantically or sexually attracted to both males and females.

BISHOP-PHELPS THEOREM: A theorem that in a Banach space the support points of (closed) support or boundary hyperplane are dense in the boundary of the set and are themselves norm dense in the dual space. This may fail in an incomplete normed space if the set is merely closed and bounded.

BISMUTH: A hard, brittle, pinkish or greyish-white metallic element, with the chemical symbol Bi and atomic number 83. It is the last of the stable elements and it is a poor conductor of heat and electricity. Of all the metals, it is the most naturally diamagnetic and only mercury has a lower thermal conductivity. It is used in making alloys of low melting point and in medical compounds to soothe gastric ulcers. Bismuth compounds are used in cosmetics and in medical procedures. Bismuth's commercial importance includes replacing lead alloy, because of the toxicity of lead.

BISMUTH(III) CHLORIDE OXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for bismuth oxychloride, with the chemical formula BiOCl. It is a white powder, which is insoluble in water and soluble in acid. It is toxic if ingested. It is used in pigments and cosmetics.

BISMUTHANES: The saturated hydrides of tervalent bismuth, having the general formula Bi_nH_{n+2}. Hydrocarbyl derivatives of BiH_3 belong to the class bismuthines.

BISTABILITY: The ability of a system to exist in two steady states. Bistability is a necessary condition for oscillations to occur in chemical reactions. The two steady state of a bistable system are not states of thermodynamic equilibrium, as they are associated with conditions far from equilibrium.

BISTABLE CIRCUIT: A transistor switching circuit which can stay in either of its two states, astable and monostable. In one state, it can be considered to represent 0 (false/off) and in the other state it represents 1 (true/on).

BISULFATE (HYDROGEN SULFATE): A salt formed when one hydrogen atom is removed from sulfuric acid by reaction with a metal, metal salt or organic group, chemical formula HSO_4^-.

BISULFITE (HYDROGEN SULFITE): A salt containing the monovalent group $-HSO_3$ or the ion HSO_3^-. Bisulfite salts such as sodium bisulfate is used in almost all commercial wines to prevent oxidation and preserve flavour. It is also used in fruit canning to prevent browning caused by oxidation and to kill microbes; and also used as a mild reducing agent in organic synthesis in particular in purification procedures.

BIT: 1. An abbreviation for binary digit, normally applied to a single digit of binary notation. 2. The smallest unit of data in a computer, capable of storing a single unit of information, consisting of an element of its physical structure capable of being in either of two states, such as a switch with on and off positions or a microscopic magnet capable of alignment in two directions. 3. A single binary digit either, 0 or 1. 4. The amount of information that is required to express choice between two possibilities. 5. A metal mouthpiece used to control a horse.

BIT DEPTH: In digital audio, the term refers to the potential accuracy of a particular piece of hardware or software that processes audio data. In general, the more bits that are available, the more accurate the resulting output from the data being processed.

BIT ERROR RATE (BER): In telecommunication transmission, it refers to the percentage of bits that have errors relative to the total number of bits received in a transmission, usually expressed as ten to a negative power. For example, a transmission might have a BER of 10 to the minus 6, meaning that, out of 1,000,000 bits transmitted, one bit was in error. The BER is an indication of how often a packet or other data unit has to be retransmitted because of an error. Too high a BER may indicate that a slower data rate would actually improve overall transmission time for a given amount of transmitted data since the BER might be reduced, lowering the number of packets that had to be re-sent.

BIT RATE: In data communications, it is a measure of data speed for computer modems and transmission carriers. Bit rate is measured in bits per second (bps), kilobits per second (Kbps) or megabits per second (Mbps). For example, a DSL connection may be able to download data at 768 kbps, while a Firewire 800 connection can transfer data up to 800 Mbps.

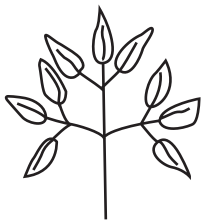
BIT STREAM (BITSTREAM): A contiguous sequence of bits (binary digits) representing a stream of data that is transmitted continuously over a communications path.

BIT STUFFING: The process of inserting non-information bits into data to break up bit patterns to affect the synchronous transmission of information. It is widely used in network and communication protocols, in which bit stuffing is a required part of the transmission process. Bit stuffing is commonly used to bring bit streams up to a common transmission rate or to fill frames. Bit stuffing is also used for run-length limited coding.

BITANGENT: A line tangent to a curve or surface at two different points.

BITCOIN: A digital currency that is not backed by any country's central bank or government but can be used worldwide for payment system. Bitcoins can be traded for goods or services with vendors who accept Bitcoins as payment.

BITERNATE: Doubly ternate or with the ternate divisions again ternately divided, as shown in the diagram. An example is when a petiole has three ternate leaflets.



Biternate leaves

BITMAP, BMP (BIT-MAPPED GRAPHIC): An image created on a visual display unit where each pixel corresponds to one or more bits in memory, the number of bits per pixel determining the number of available colours. The image is created by placing dots or bits in a row or column forming an image with several thousand small dots or bits to make the complete image that looks like a picture, when one sits a reasonable distance away from the screen. Common bitmap file types include BMP (the raw bitmap format), JPEG, GIF, PICT, PCX and TIFF.

BITNET: A computer network originally linking IBM mainframe systems located in North America and with backing from IBM.

BITTER PATTERN: A microscopic pattern that forms on the surface of a ferromagnetic material that has been coated with a colloidal suspension of small iron particles.

BITTERN: 1. The solution of salts remaining when sodium chloride or common salt is crystallised from seawater. It is a source of bromides and magnesium. 2. Common name for migratory marsh birds of the family *Ardeidae* (heron family). They have mottled brownish feathers and a booming call and are found mostly across the Americas.

BITTORRENT: A file sharing program designed to reduce the bandwidth required to transfer files by sharing files between all clients currently getting the file and continues to share the file on their computer. For example, if a user begins downloading a movie file, the BitTorrent system will locate multiple computers with the same file and begin downloading the file from several computers at once. Since most Internet Service Providers offer much faster download speeds than upload speeds, downloading from multiple computers can significantly increase the file transfer rate. It was introduced by Bram Cohen on July 2, 2001.

BITUMEN: An impure flammable, dense, highly viscous, petroleum-based hydrocarbon that is found in deposits such as oil sands and pitch lakes (natural bitumen) or is obtained as a residue of the distillation of crude oil or coal. Its density and viscosity is due to mainly large hydrocarbon molecules known as asphaltenes and resins, which are present in lighter oils but are highly concentrated in bitumen. In addition, bitumen frequently has a high content of metals, such as nickel and vanadium and nonmetallic inorganic elements, such as nitrogen, oxygen and sulfur. It is a component of asphalt and tar and used for surfacing roads; its products also include bitumen waterproofing agents that are used in water tanks and swimming pool or around the home as a sealant. It is also used in the manufacture of printing ink and rubber.

BITUMINOUS COAL (BLACK COAL): A dark brown to black coal containing a tar-like substance called bitumen. It is the most abundant form of coal, with a relatively high heat value. Its primary constituents are macerals, vitrinite and liptinite, with a carbon content of between 60 and 80%; the 20% is composed of water, air, hydrogen and sulfur. It has been used for steam generation in electric power plants and industrial boiler plants. Certain varieties are also used to make coke, a hard substance of almost pure carbon that is important for smelting iron ore. One major problem is that burning large quantities of bituminous coal that has a medium to high sulfur content contributes to air pollution and produces acid rain.

BIURET: A colourless, crystalline organic compound, with the chemical formula $NH_2CONHCONH_2$, which decomposes at 186 °C -189 °C (459.15 K - 462.15 K). It is soluble in hot water and a condensation product of urea, equivalent to two molecules of urea less one of ammonia.

BIURET TEST: A biochemical test to detect proteins in solutions, named after the substance biuret, which is formed when urea is heated. A positive result is indicated by a violet ring, due to the reaction of peptide bonds in the proteins.

BIVALENT (DIVALENT): 1. Also known as **divalent** in chemistry, it is having a valence of two. Bivalent or divalent anions are atoms or radicals with two additional electrons when compared to their elemental state with 2 more electrons than protons. A bivalent or divalent cation lacks two electrons as compared with the neutral atom. For instance, ferrous ion (Fe^{2+}) is the divalent cationic form of iron. Divalent cations such as calcium (Ca^{2+}) and magnesium (Mg^{2+}) are present in abundance in hard water and are responsible for the formation of limescale in solution. 2. A pair of homologous chromosomes united by chiasmata. Bivalents are most clearly seen during late prophase of meiosis 1.

BIVALVIA (PELECYPODA; LAMELLIBRANCHIA): A class of aquatic molluscs consisting of two flat shells hinged dorsally belonging to the phylum Mollusca. They include the oysters, mussels and clams. They are laterally compressed and possess a shell composed of two valves that hinge along their back or dorsal margin and enclose the body.

BIVANE: A wind vane used in turbulence studies to obtain horizontal and vertical components of the wind vector.

BIVARIATE: In statistics, of, relating to or involving two random variables that are not necessarily independent of one another.

BIVARIATE DATA: Data consisting of pairs (x_1, y_1), (x_2, y_2), ..., taken from a bivariate distribution.

BIVARIATE DISTRIBUTION: The joint distribution of two random variables x and y. An example is the bivariate normal distribution, given by:

$$f(x,y) = \frac{1}{2\pi\delta_1\delta_2\sqrt{1-p^2}} \exp\left\{-\frac{1}{1-p^2}\left[\frac{(x-\mu_1)^2}{\delta_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\delta_1\delta_2} \frac{2\rho(x-\mu_1)(y-\mu_2)^2}{\delta_2^2}\right]\right\},$$

where $\mu_1, \mu_2, \delta_1, \delta_2, p$ are, respectively, the mean, standard deviations and correlation of the two variables.

BIVARIATE EXPONENTIAL DISTRIBUTION: Any bivariate distribution, $f(x,y)$, that has an exponential distribution for each of its marginal distributions. An example is as follows:

$f(x,y) =[(1 + \alpha x)(1 + \alpha y)- \alpha]e^{-x-y-\alpha xy}$, $x \geq 0$, $y \geq 0$, $\alpha \geq 0$.

BJC4000: A babble jet printer made by Canon that produces 720 x 360dpi in mono and 360 x 360dpi in colour.

BLACK BODY: A hypothetical body that absorbs all the radiation falling on it regardless of frequency or angle of incidence; no radiation passes through it. Black bodies are ideal sources of thermal radiation. Black bodies do not reflect light and therefore appear black if their temperatures are low enough so as not to be self-luminous. A black body in thermal equilibrium or at a constant temperature emits electromagnetic radiation called **black-body radiation**. The radiation is emitted according to Planck's law (it has a spectrum that is determined by the temperature alone, not by the body's shape or composition).

BLACK BODY RADIATION: The electromagnetic radiation emitted by a hypothetical body called a black body, which is in thermal equilibrium or at a constant temperature. The black body absorbs all radiation incident upon it and re-radiates energy which is characteristic of this radiating system only, not dependent upon the type of radiation it absorbs or the radiation is emitted according to Planck's law (it has a spectrum that is determined by the temperature alone, not by the body's shape or composition).

BLACK CARBON: A black particulate form of pure carbon, formed through the incomplete combustion of fossil fuels, biofuel, and biomass, and is emitted in both anthropogenic and naturally occurring soot. Black carbon is considered as a major global environmental problem and a contributor to global climate change, having negative implications for both human health and the climate.

B

BLACK DAMP (CHOKE DAMP; STYTHE): Air depleted of oxygen and largely replaced by nitrogen, argon, carbon dioxide and water vapour following the explosion of firedamp in a mine.

BLACK DISEASE: A liver disease normally affecting sheep and camel. It is occasionally found in pigs and horses.

BLACK DOT: A fungal disease of potatoes that covers the skin with tiny black spots.

BLACK DWARF: A hypothetical celestial object that is cold in nature and thought to be the remains of a dead star of low mass, which is formed after a white dwarf star has radiated away all of its heat energy.

BLACK SOIL: A type of Chernozem dark fertile soil, rich in organic matter, 7% - 15% humus with a high agricultural yield. It has a large depth, often more than 1 metre. Such soils are very fertile over a long period of time, and used extensively for growing cereals or for raising livestock.

BLACK HOLE: A region in outer space, thought to form during the course of stellar evolution from which no matter or radiation can escape or an object in space that has collapsed under its own gravitational forces to such an extent that its escape velocity is equal to the speed of light. A black hole sucks in more matter, including other stars, from the space around it. Matter that falls into a black hole is squeezed to infinite density at the centre of the hole. Black holes can be detected because gas falling towards them becomes so hot that it emits x-rays. The concept was developed by the German astronomer Karl Schwarzschild in 1916 based on German-born U.S. theoretical physicist, Albert Einstein's general theory of relativity.

BLACK LEAD: A grey-black substance used in lead pencils and for polishing.

BLACK SMOKER (SMOKER): An active hydrothermal vent, usually at a mid-ocean ridge on the sea floor from which superheated water at about 350 °C, with very high concentrations of dissolved minerals such as copper, zinc, gold and manganese and other sulfur-bearing minerals or sulfides erupt at high pressure. When the saturated mineral solutions exit the vent, they cool by contact with the ocean causing the metals to precipitate, mainly as sulfide or sulfate salts forming chimney-like structures.

BLACK-EYED SUSAN VINE (*Thunbergia alata*): A herbaceous perennial creeper or climbing plant species in the Acanthaceae family. It is native to Eastern Africa and found in tropical Africa and has become naturalised in Asia, Hawaii, Australia and some other parts of the world. It has many twining stems which grow to a height of about 2.4 metres, with heart or arrow-shaped leaves. There are five petals which persist throughout the growing season. The flowers are borne singly in leaf axils with a small calyx enclosed in two large, ridged bracts. It is warm orange with a dark spot in the centre although different species may be red orange, red-orange, pale yellow, white or bright yellow. In East Africa, it is used as a vegetable or stock feed. It is cultivated as an ornamental plant in gardens and in hanging baskets. It is also used medicinally to treat many diseases such as skin problems, cellulitis, back and joint pains, eye inflammation, piles and rectal cancer. It is also used to treat gall sickness and some ear problems in cattle.

BLACK-SCHOLES MODEL: A financial model of variations over time in the price stock market. The model, proposed in 1974, is based on the notion that the underlying price variations could be modelled as Brownian motion.

BLACK, JOSEPH (1728-99): British chemist and physician, born in France and known for his discoveries of latent heat, specific heat and carbon dioxide. He studied at Glasgow and Edinburgh and his thesis in 1754 contained the first accurate description of the chemistry of carbon dioxide. He was professor of Medicine at University of Glasgow.

BLACKBODY TEMPERATURE: The effective temperature at which a blackbody emits blackbody radiation.

BLACKHEAD: A common and fatal disease, which attacks the liver and intestines of young turkeys.

BLACKLIST: In communications technology, it refers to a list of items, such as usernames or IP addresses, that are denied access to a certain system or protocol. When a blacklist is used for access control, all entities are allowed access, except those listed in the blacklist. A **whitelist** denies access to all items, except those included in the list.

BLADDER: A hollow sac-like muscular organ found in animals for storage of liquid or gas. For example, the urinary bladder, with elastic walls is used by vertebrates for storing urine and the gallbladder is used for storing bile pigment. Other examples are air bladder, gall bladder, swim bladder, etc.

BLADDERWORM (CYSTICERCUS): A larval stage of some tapeworms, of the class *Cestoda* consisting of fluid-filled sac and armed with six hooks, containing an inverted scolex. It develops in the bodies of various animals as juveniles. As adults, they live in the digestive tract of vertebrates.

BLADE: 1. The green thin expanded part of a leaf consisting of a vascular skeleton embedded in photosynthetic tissues. 2. Term applied to leaf of seedling of members of the grass family. 3. The flat wide part of an oar, a propeller, a spade, etc. 4. The sharp-edged cutting part of a tool or weapon.

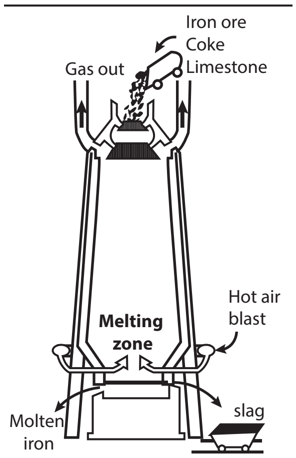
BLANCHING: In cooking, it describes a process of food preparation wherein the food substance, usually a vegetable or fruit, is plunged into boiling water at 90 °C (363.15 K) and removed after a brief, timed interval and finally plunged into iced water or placed under cold running water (shocked) to halt the cooking process. This denatures the proteins of enzymes that cause spoilage and kills many bacteria.

BLANDFORD-ZNAJEK PROCESS: An astrophysical process discovered by Roger Blandford and Roman Znajek, which provides a mechanism for the extraction of energy from a rotating black hole by an external magnetic field, making the black hole a powerful source of energy. It also explains the way quasars are powered. It requires an accretion disc with a strong polar magnetic field around a spinning black hole. The magnetic field extracts spin energy and the power can be estimated as the energy density at the speed of light cylinder times area: $P = B^2(r/r_c\}^4\ r_c c = B^2 r^4 \omega^2/c$, where B is the magnetic field strength, r_c the speed of light radius and ω the angular velocity.

BLANK VALUE: A reading or result originating from the matrix, reagents and any residual bias in the measurement device or process, which contributes to the value obtained for the quantity in the analytical procedure.

BLAST FREEZING: An industrial method of rapidly freezing substances such as food by blowing very cold air over them in order to reduce the metabolic processes.

BLAST FURNACE: A type of furnace used for continuous production of molten iron or tall oven-like chimney used for smelting iron ore to make pig iron. In a blast furnace, fuel and ore are continuously supplied through the top of the furnace, while air (sometimes with oxygen enrichment) is blown into the bottom of the chamber, so that the chemical reactions take place throughout the furnace as the material moves downward. The end products are usually molten metal and slag phases tapped from the bottom and flue gases exiting from the top of the furnace.



Blast furnace

BLAST WAVE: In fluid dynamics, it is the pressure and flow resulting from the deposition of a large amount of energy in a small very localised volume. In other words, it is an area of pressure expanding supersonically outward from an explosive core. It has a leading shock front of compressed gases. The blast wave is followed by a blast wind of negative pressure, which sucks items back in towards the centre. It is harmful especially when one is very close to the centre or at a location of constructive interference. High explosives, which detonate, generate blast waves.

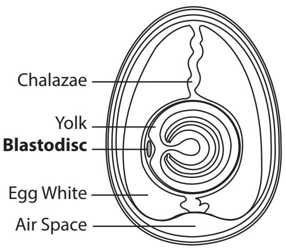
BLASTING GELATIN (GELIGNITE): A high explosive invented in 1875 by Alfred Nobel, who had earlier invented dynamite. It is made from nitroglycerine and gun cotton (cellulose nitrate) dissolved in nitroglycerine and mixed with wood pulp and sodium nitrate or potassium nitrate. It can be easily moulded because of its composition and safe to handle without protection because it burns slowly and cannot explode without a detonator.

BLASTOCOEL (BLASTOCELE; CLEAVAGE CAVITY; SEGMENTATION CAVITY): The fluid-filled cavity at the centre of a blastula, which forms during embryogenesis when a zygote or a fertilised ovum divides into many cells through mitosis.

BLASTOCYST: The developmental stage of the fertilised ovum by the time it is ready to implant. It is formed from the morula and consists of an inner cell mass, an internal cavity and an outer layer of cells (the trophoblast). The development of the human blastocyst is preceded by a zygote, the fertilised egg cell and succeeded by an embryo and arises after compaction. It comprises 70-100 cells.

BLASTODERM: 1. Also called **germinal membrane** or **germ membrane**, it is a layer of cells that surrounds the central cavity of a blastula. It forms a layer covering the yolk in yolky eggs like that of insects and birds, which later divides into the three germ layers from which the embryo develops.
2. The stage in embryogenesis, when a unicellular layer at the surface surrounds the yolk mass.

BLASTODISC: A disc-like area, of the eggs of birds and reptiles, which is free of yolk and appears as a small disc on the upper surface of the yolk mass. It undergoes cleavage, giving rise to the embryo.



Blastodisc of an egg

BLASTOMYCOSIS: A fungal infection caused by *Blastomyces dermatitidis*, which is found in moist soil enriched with decomposing organic debris. It mainly affects skin, lungs, and bones; and can cause permanent lung damage. It is transmitted by inhalation of airborne spores when contaminated soil is disturbed, and therefore affects mainly farmers, forestry workers, hunters, etc., in endemic areas, such as parts of North America, Central and South America and Africa.

BLASTOMERE: An undifferentiated cell of a cleaving embryo and of the morula and blastula stages of embryonic development.

BLASTOPORE: The opening of the archenteron in the gastrula as a central cavity that develops into the mouth and the intestinal or digestive cavity in protostomes and deuterostomes.

BLASTULA (BLASTOSPHERE): The first stage of development of an animal embryo, formed as a result of cleavage of fertilised egg, when the egg changes from a solid mass of cells to a hollow ball of cells containing a fluid filled cavity.

BLAZAR: A very active type of galaxy, which got its name from a combination of BL Lacertae object and quasar. It is a type of active galactic nucleus (AGN) that often appears as an extremely bright, starlike object characterised by rapid changes in luminosity and a flat spectrum. Blazars were first discovered in 1978 and about 200 are currently known. Originally thought to be ordinary irregular variable stars, their spectral properties now lead astronomers to consider blazars as a class of active galactic nuclei. Blazars emit radiation over a very wide range of frequencies, from radio to gamma rays, with their jets pointed at the observer. This orientation accounts for their peculiar properties, specifically the variability and intensity of their brightness and it also distinguishes blazars from another class of active galactic nucleus, quasars**.**

B

BLEACH: A substance used to remove the colour or whiten materials such as textiles and paper usually by oxidation. The two main types of bleaching agents are the oxidising bleaches, which add hydrogen or remove oxygen and the reducing bleaches, which bring about reduction. Sunlight and oxygen act as bleaches but the most commonly found bleach is sodium hypochlorite (NaClO) solution (3 – 6%). Hydrogen peroxide is used to bleach wool, silk and cotton; hypochlorites, including bleaching powder, are used for cotton; and sodium chlorite for synthetic fibres. Sulfur dioxide bleaches are impermanent. Careful control of pH is essential. **Bleaching powder** is a white powder of complex composition, made by the action of chlorine on calcium hydroxide. It contains calcium hypochlorite $Ca(OCl)_2$. The powder releases chlorine when it is treated with dilute acid.

BLEEDING OUT: Draining out blood from slaughtered animal. The animal is bled by cutting the main artery and vein to the heart; and the carcass is then hanged head-down for effective draining of blood.

BLENDE (ZINC BLENDE): The primary ore of zinc, occurring in usually yellow-brown or brownish-black crystals or cleavage masses, essentially zinc sulfide (ZnS) with some cadmium, iron and manganese.

BLENDING INHERITANCE: An assumption from an early theory that the inheritance pattern of a system involving incomplete dominance, whereby characters are inherited in heterozygous individuals that show the effects of both alleles. As a result the inherited characters in the offspring are intermediate between those of the parents.

BLIGHT: Any plant disease or disorder characterised by withering, growth cessation or death of parts without rotting, caused by fungi, insect attack or unfavourable environmental conditions.

BLIND EXPERIMENT/STUDY: An experiment in which information about the test is kept from the participants such as the subjects, researchers and the committee monitoring the experiment or any combination of them to reduce or eliminate bias, until after a trial outcome is known. Blind testing is used, for example, in clinical trials to evaluate the effectiveness of medicinal drugs; and comparative testing of commercial products to objectively assess user preferences without being influenced by branding and other properties not being tested.

BLIND SPOT (OPTIC DISK): The small, circular, optically insensitive area where the optic nerve and blood vessels pass through the retina of the eye or the position of the retina at which blood vessels and nerve fibre enter the optic nerve. No visual image can be formed because there are no rods, cones or light sensitive cells in this part of the retina.

BLINDNESS: Severe impairment or absence of vision. Blindness may be due to congenital defects such as night blindness, accident, disease or old age. The commonest cause of blindness is trachoma. It is caused by severe diabetes, glaucoma or cataract or degenerative changes associated with aging. Malnutrition, especially vitamin A deficiency may cause blindness in children. Infant blindness can result if the mother has rubella (German measles) during pregnancy.

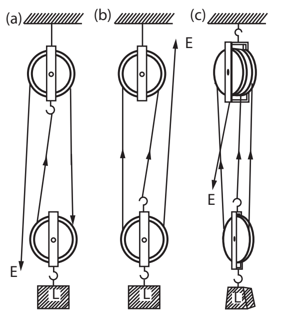
BLIZZARD: A snowstorm in which wind velocity reaches more than 14 m/s (50.4 km/h), the temperature is below freezing and visibility poor. Blizzards are common in Polar Regions.

BLOAT: The swelling or filling with liquid, air or gas, such as distension of the abdomen that results from swallowed air or from intestinal gas.

BLOCH'S THEOREM: A theorem relating to the quantum mechanics of crystals that specifies the form $\Psi(r) = \exp(ik.r)U(r)$ of the wave function and characterises electron energy levels in a periodic crystal, where K is the wave vector, which can assume any value, r is a position vector and U(r) is a periodic function that satisfies $U(r+R)=U_r$ for all vectors R of the Bravais lattice of the crystal. It is interpreted to mean that the wave function for an electron in a periodic potential is a plane wave modulated by a periodic function.

BLOCK: 1. Sets in which experimental units are grouped in such a way that the units are similar as possible within each block. The experimental error associated with the block will be smaller than that obtained if the same number of units were randomly located within the whole experimental space. The blocks are generally determined by taking into account both controllable causes related by the factors studied and causes that may be difficult or impossible to keep constant over all of the experimentally units. The variations between the blocks are then eliminated when the effect of the factors are compared. Several types of regroupings can be used to reduce the effects of one or several sources of error. A randomized block design results if there is only one source of error. 2. In data management, a block is a group of records on a storage device. Blocks are manipulated as units. For example, disk drives often read and write data in 512-byte blocks. In word processing, a block is a contiguous set of characters, often consisting of a phrase, a sentence, a paragraph or a set of paragraphs that is selected by the user for copying/pasting, cutting or moving. But a block can consist of any contiguous set of characters, whether or not it forms a logical unit of text. In network communication, a block is a group of data bits or bytes that is transferred as a standard unit. The size or length of such a block depends on the communications protocol.

BLOCK AND TACKLE: A system of two or more pulleys with a rope or cable threaded between them, usually used to lift or pull heavy loads at harbours and construction sites. The pulleys are assembled together to blocks and then the blocks are paired so that one is fixed and one moves with the load. The rope is threaded or reeved through the pulleys to provide mechanical advantage that amplifies that force applied to the rope.



E = Effort, L = Load
Block and tackle

BLOCK CIPHER: A method of encrypting text in which a cryptographic key and algorithm are applied to a block of data at once as a group rather than to one bit at a time. The main alternative method, used much less frequently, is called the stream cipher.

BLOCK COPOLYMER: A copolymer that is a block polymer (polymer composed of block macromolecules). In the constituent macromolecules of a block copolymer, adjacent blocks are constitutionally different, i.e. adjacent blocks comprise constitutional units derived from different species of monomer or from the same species of monomer but with a different composition or sequence distribution of constitutional units.

BLOCK DESIGN: 1. In statistics, a design in which groups of subjects are regarded as sufficiently homogeneous to behave in the same way, so that comparison of the application of different experimental conditions to subjects in the same group is meaningful. It is a **balanced block design** if all the blocks are of equal size and all treatments occur equally often; it is a **completely balanced block design** if in addition each treatment occurs equally often in each block, that is, if the block size is a multiple of the number of treatments. 2. A family of subsets (blocks) of a given finite set of varieties or points such that each block has the same number of members and such that each pair of points belongs to the same number of blocks. For example, {{1,2,4}, {2,3,5}, {3,4,6}, {4,5,7}, {5,6,1}, {6,7,2}}, is a block design on P = {1, ..., 7} with each pair in exactly one block. The simplest examples occur as finite geometries such as the seven-point finite projective plane. A block design on a set of v points with k blocks in which each point belongs to exactly λ blocks is called a (v, k, λ)-design; therefore, the above example is a (7, 3, 1) design.

BLOCK DIAGRAM: A simplified way of showing a circuit by giving only the main functions of elements or compounds.

BLOCK GRAPH: A graph of a set of frequencies denoted by a set of rectangles (blocks) stacked in columns whose areas are proportional to the frequencies.

BLOCK MACROMOLECULE: A macromolecule which is composed of blocks in linear sequence.

BLOCK MULTIPLICATION: Multiplication of matrices whose entries are themselves smaller matrices called blocks. Their diagonal matrices are square matrices. The block off the diagonal are zero matrices and the diagonal matrices are the square matrices. This allows one to exploit the structure of the matrix.

BLOCK-DIAGONAL MATRIX: A matrix with the only non-zero elements lying in a sequence of square matrices arranged along the main diagonal; the block-diagonal matrix, C, with two blocks, one an $m \times m$ matrix A and the other an $n \times n$ matrix B, is denoted diag [A, B] and has entries $c_{ij} = a_{ij}$, for i and $j \le n$ and $c_{ij} = b_{i-n,j-n}$ for i and $j > n$, all other entries being zero.

BLOCKING SOFTWARE: Any of various software programs that work on the World Wide Web to prevent access to categories of information considered offensive or dangerous by identifying websites with the keywords in their content or by their Internet address. It is typically used by parents or teachers to ensure that children do not see pornographic or other adult material.

BLOG (WEBLOG): A personal online journal that displays in chronological order the postings by one or more individuals and is frequently updated and intended for general public consumption. Blogs are defined by their format: a series of entries posted to a single page in reverse-chronological order. Many blogs syndicate their content to subscribers using RSS, a popular content distribution tool.

BLOOD: A fluid that circulates in the arteries, veins and capillaries of vertebrate animals. It consists of plasma, blood cells and platelets and it carries oxygen, nutrients and hormones to the tissues and removes carbon dioxide and other waste substances from the cells.

BLOOD CELL (BLOOD CORPUSCLE): Any of the three categories of cells contained in blood plasma, which are red blood cells, white blood cells and platelets. Together, they comprise a total of 45% of blood tissue, with plasma occupying 55%.

BLOOD CLOTTING (BLOOD COAGULATION): The production of a mass of semi solid material at the site of an injury that seals the wound due to the formation of insoluble fibrin protein which forms a fibrous network in which blood cells are trapped. The smallest cells in the blood are the platelets or thrombocytes, which begin the process of coagulation by clumping together and sticking to the edges of the cut. They also produce a substance that combines with calcium ions in the blood to form thromboplastin, which in turn converts the protein prothrombin into thrombin in a series of reactions. Thrombin, a proteolytic enzyme, converts fibrinogen, a protein substance, into fibrin, an insoluble protein that forms an intricate network of minute threadlike structures called fibrils and causes the blood plasma to gel.

BLOOD FLUKE (SCHISTOSOMA): A genus of trematodes of the class *Digenea*, consisting of parasitic flatworms that live in the blood vessels of man and other vertebrates. These flukes cause the disease schistosomiasis or bilharzia. They are common in tropical Africa, South America, part of the Middle East, India, Japan, China and some parts of the West Indies. The worms are wrapped together, one being male and the other a female. They affect the large intestine or the urinary bladder depending on the species. The schistosomes which affect humans in Africa are *Schistosoma mansoni* and *Schistosoma haematbium*. *Schistosoma japonicum* is found in Asia. *Schistosoma haematobium* lives in the venous plexus of the urinary bladder and *Schistosoma mansoni* lives in the mesenteric plexus of the large intestine. The eggs are passed out of the infected human through his urine of faeces into water. When the eggs reach water, they hatch out into miracidia which are free-swimming larvae which have to reach a snail host within 24 hours or die. Development of water projects for irrigation or electricity provides the habitat for the snail vector. The vector snails for *Schistosoma haematobium* belong to the genius *Bulinus* which lives in temporary water bodies such as ponds, dams and paddy fields. The vector snails for *Schistosoma mansoni* belong to the genius *Biomphalaria* which prefers permanent water in stream, irrigation schemes and lakes. They remain in the water snail for many weeks. Later the developed tailed embryos (cercariae) come out of the snail and swim freely in water until one bathes, handles or drinks the water. The embryos enter the skin of man and go in the veins and mature as schistosoma. The blood carries the flukes into the urinary bladder or bowels where they feed on more blood and lacerate the blood vessels and walls of bladder causing great loss of blood. The eggs with spiny ends are later released into the veins and passed out with urine or feaces. The individual becomes anaemic, rashes may occur at points of penetration of skin, the liver and the spleen enlarge (inflammation) including feverish conditions.

BLOOD GLUCOSE REGULATION: The biological regulation of glucose (blood sugar) concentration in the blood to limits acceptable by the body. A number of hormones, including insulin, glucagon, cortisol, growth hormone, epinephrine and norepinephrine, maintain glucose levels in the blood.

BLOOD GROUP: The various types into which an individual's blood may be classified based on the presence or absence of certain proteins (antigens) on the surface of the red blood cells. The two main antigens are A and B, giving rise to four blood groups: having A only (A), B only (B), both A and B (AB) and having neither (O). The knowledge of blood group is important for blood transfusion, to establish the legitimacy (parentage) of children and also to predict the blood group of a foetus.

BLOOD PIGMENT: Any one of a group of metal-containing coloured protein compounds whose function is to increase the oxygen-carrying capacity of blood.

BLOOD PLASMA: A colourless transparent liquid part of the blood excluding blood cells. It consists of water containing a large number of dissolved substances including proteins (mainly albumin and various globulins, including the immunoglobulins), salts, food materials, hormones, vitamins and excretory materials.

BLOOD POISONING: The infection of blood with harmful bacteria especially through a cut or wound.

BLOOD PRESSURE: The pressure exerted by the flow of blood through the main arteries in an animal. It is most commonly measured via a sphygmomanometer and recorded in either kilopascals (kPa) or in millimetres of mercury (mmHg).

BLOOD SERUM: Blood plasma from which the fibrin and clotting factors have been removed so that it cannot clot. It contains solutes of proteins, electrolytes, waste products, dissolved gases, hormones, enzymes, antibodies and water and may be used in the treatment or prevention of certain infections.

BLOOD SUGAR: The concentration of glucose in the blood, normally expressed in millimoles per litre (mmol/l). The normal range is 3.5 - 5.5 mmol/l. Blood-sugar is an important investigation in a variety of diseases, most notably in diabetes mellitus.

BLOOD TEST: The examination of blood especially for medical diagnosis or the use of agglutination in order to determine a person's blood group.

BLOOD TRANSFUSION: The transfer of blood from a healthy person, the donor into the circulation of another, who is anaemic or has lost blood through an injury or surgery to restore blood volume, increase haemoglobin levels or combat shock.

BLOOD VASCULAR SYSTEM: The tissues and organs of an animal that transport blood through the body. Examples are the heart and the blood vessels in vertebrates.

BLOOD VESSEL: A tubular structure through which the blood of an animal flows or a tube that carries blood around the body of a multi-cellular animal. Blood vessels include **veins**, which carry blood from the capillaries back toward the heart; **arteries**, which carry the blood away from the heart; and **capillaries**, which enable the actual exchange of water and chemicals between the blood and the tissues.

BLOOD-BRAIN BARRIER: The physiological mechanism that regulates the passage of substances from the blood to the cerebrospinal fluid surrounding the brain and the spinal cord. It prevents some substances, such as certain drugs, from entering the brain tissue, while other substances are allowed to enter freely.

B

BLOODSTREAM: The flow of blood circulating through the blood vessels of a person or animal.

BLOOM: 1. A rapid increase in the number of certain species of algae found in lakes, ponds and oceans. 2. A whitish powdery coating over the surface of certain fruits such as grapes, that easily rubs off when handled. It often contains yeasts that live on the sugars in the fruit. 3. The state of being in flower or the mass of flowers on a single plant.

BLOOMING: 1. The process of depositing a transparent film of a substance, such as magnesium fluoride, on a lens to reduce the reflection of light at the surface. 2. Flourishing and in exceptionally good health or condition.

BLOSSOM: The flower, especially, of a fruit tree or flowering shrub.

BLOTCHED: Marked with irregular blot or spot.

BLOTTING: A technique used for transferring DNA, RNA or protein from gels to a suitable binding matrix, such as nitrocellulose or nylon paper, while maintaining the same physical separation.

BLOW FLY: One of the group of sarcophageous flies such as bluebottles and greenbottles that lays its eggs on meat. It breeds on decomposing bodies and a study of their maggots and pupae can be used to determine the approximate time of death of a decomposed body.

BLOW-OUT: A valley or depression of sand that is common in coastal dunes that may increase to an extremely large size and several metres in depth. Blow-outs are created by the wind in areas, where the initial patch of protective vegetation is lost and there is less shelter from the strong onshore winds allowing the sand to be blown away. The loss of the initial vegetation may be due to lack of sand supply from a beach; extended droughts, by fire and sometimes by human and animal activities.

BLOWDOWN: Hydrocarbons purged during refinery shutdowns and start-ups which should be piped to storage systems for safe venting, flaring or recovery. This term also applies to the purging of water in boiler operation and serves in the control of dissolved solids in the boiler water.

BLOWHOLE: 1. A hole in a cliff or cave through which seawater spurts at high tide. Blowholes are formed by erosion. Continued hydraulic action by pounding waves on the roof of the cave forms a sort of chimney. Eventually, this breaks through the surface of the ground near the edge of the cliff. At high tide, incoming waves force water out of the top of the blowhole. 2. The nostrils (single or double) of whales, situated toward the back of the skull.

BLOWN-OUT LAND: Shallow depressions formed by the removal of all or almost all the soil or sediments by wind action. It is usually a barren, shallow depression that has no agricultural purposes.

BLOWPIPE: A narrow tapered tube through which air is blown into a flame to increase the temperature and to direct part of the flame (the oxidising or reducing zone) onto a substance undergoing chemical analysis.

BLU-RAY (BLU-RAY DISC, BD): An optical disc format that is capable of storing large amounts of video in high-definition and ultra high-definition data.

BLUBBER: The fat layer between the skin and muscle of sea mammals such as whales, dolphin, or porpoise. The fat keeps them warm in the cold water and also used to make oil that is used to manufacture soap, leather, and cosmetics.

B

BLUE LASER: A type of laser that emits electromagnetic radiation with a wavelength between 360 and 480 nanometres, developed by Blu-ray Disc Association. This was developed into a technology called blue-violet laser technology that enables extremely large-capacity recordings and high-speed data transfer rates. It is used in high-definition consumer video recording, and HD-DVD suitable for storage libraries and other applications.

BLUE SHIFT: A general displacement of spectral lines to shorter wavelengths that happens when a source of electromagnetic radiation moves toward the observer. The opposite effect is referred to as redshift. The greater the velocity of approach along the line of sight, the greater the blueshift. Individual stars within the Milky Way Galaxy frequently show blueshifts. However, with the exception of a few galaxies within the Local Group, including the Andromeda Galaxy, most extragalactic objects exhibit redshifts, due to the overall expansion of the universe.

BLUE WHALE: The largest animal in the world, the blue whale (*Balaenoptera musculus*) of the baleen whale (mysticeti) suborder that can reach up to 30 metres long and weighing over 180 metric tons (180,000 kg) as an adult. Calves are born weighing up to 2.7 metric tons and about 8 metres long. It is widely distributed and found in all the oceans. It has a bluish-grey back with white-grey spots and yellow underbelly. It has a very small dorsal fin that is located near the fluke or tail; a thin flipper 2.4 m long; and a fluke (tail) that is 7.6 m wide. It has several ventral throat grooves. It breathes air at the surface of the water through two blowholes located near the top of the head. Despite its size, it feeds almost exclusively on small crustaceans known as krill. It is toothless and uses horny plates, called baleen attached to the upper jaws to filter this tiny food by forcing out large quantities of seawater swallowed into the mouth. Blue whales live individually or in small groups called pods and frequently swim in pairs. They emit very loud, highly structured, repetitive low-frequency sounds that can travel for many miles underwater. Scientists think they use these vocalisations not only to communicate, but, along with their excellent hearing, to sonar-navigate the lightless ocean depths. Blue whales are currently classified as endangered on the World Conservation Union (IUCN) Red List.

BLUE-GREEN ALGAE (BLUE-GREEN BACTERIA, SINGULAR CYANOBACTERIA; CYANOPHYTA): A phylum of single-celled organisms that resemble bacteria in their internal cell structure and lack an enclosed nucleus and other specialised cell structures. They are prokaryotic and are the earliest known form of life on the earth. They are widely distributed in aquatic habitats, on the damp surfaces of rocks and trees and in the soil; where they are joined together in colonies or filaments; and obtain energy through photosynthesis and reproduce by simple cell division or by fragmentation of the filaments. Some fix nitrogen and thus are necessary to the nitrogen cycle, while others live a symbiotic existence; for example, living in association with fungi to form lichens. Sometimes they form algae blooms on fresh water bodies, where the water is polluted by nitrates and phosphates from fertilisers and detergents and cover the water's surface. Overtime, they give off toxins as they decay, which kill fish and other wildlife and can be harmful to domestic animals, cattle and people.

BLUE-GREY (BLUE-GRAY): A crossbred type of cattle, resulting from mating a white short horn bull with a Galloway cow.

BLUEBOTTLE: Any of several flies of the genus *Calliphora* that have a bright metallic-blue body and breed in decaying organic matter.

BLUETOOTH: A proprietary open wireless technology standard that describes how mobile phones, computers and personal digital assistants (PDAs) can be easily interconnected using a short-range wireless connection. It is used for short-range connections between desktop and laptop computers, PDAs (like the Palm Pilot or Handspring Visor), digital cameras, scanners, cellular phones and printers.

BMP: Abbreviation for **bitmap**, which is an image created on a visual display unit where each pixel corresponds to one or more bits in memory, the number of bits per pixel determining the number of available colours.

BOBBY CALF: An unwanted male calf in extreme dairy breeds such as the Channel Island breeds.

BODY: 1. The whole physical structure of a human being, an animal or the main part of something, especially a building or vehicle. 2. A subset of a vector space that has a non-empty interior. 3. A volume of continuously distributed physical matter, such as an expanse of fluid or an elastic band. It is defined formally as a smooth three-dimensional manifold that is homeomorphic to the closure of a connected open subset of Euclidean point-space.

BODY BURDEN: The total amount of substance of a chemical present in an organism at a given time.

BODY CAVITY: The internal cavity of the body of an animal which is present in most invertebrates and all vertebrates and contains the major organs.

BODY CLOCK (BIOLOGICAL CLOCK): The biological rhythm or internal mechanism that automatically regulates periodic changes in behaviour or physiology in animals and plants with daily, monthly or seasonal cycles or with stages of development and aging.

BODY FLUID: A liquid produced by the body, examples are blood, saliva, semen, vaginal secretions, milk, urine, sweat or tears.

BODY FORCE: The force that acts in every part or throughout the volume of a body. Examples are gravitational and electromagnetic forces. The effect of body forces on a sub-body is given by the integral $\int \rho b dv$ over the volume of its current configuration, where b is the body force density and ρ is the density.

BODY FORCE DENSITY: A vector field that represents the body forces per unit mass or per unit volume acting on a body, for example, the downward acceleration equal to the local gravitational constant. In other words it is the volume integral with reference to a volume of study which gives the total force acting throughout the body.

BODY MASS INDEX (BMI; QUETELET INDEX): A measure of body fat that is the ratio of the weight of the body in kilograms to the square of its height in metres (W/H^2). In adults a body mass index of 25 to 29.9 is considered an indication of overweight and 30 or more is an indication of obesity, leading to conditions such as high blood pressure (hypertension), diabetes mellitus and heart diseases.

BODY PLAN: The blueprint according to which an organism develops a predetermined number, arrangement and size of body components. The body plan is embodied and implemented by the organism's genes but can be influenced by environmental factors such as malnutrition or exposure to toxic agents. Some aspects of the body plans of an organism are symmetry, its number of body segments and number of limbs.

BODY SPIN (SPIN TENSOR; VORTICITY TENSOR): The skew, symmetric part of the velocity gradient; if Q is the body spin and L is the velocity gradient then $\Omega = 1/2(L - L^T)$. This is the local angular velocity expressed in tensor form.

BODY WAVE: A seismic wave that moves through the interior of the earth, as opposed to surface waves that travel near the earth's surface.

BODY-CENTRED CUBE: A crystal structure that is cubic, with an atom at the centre of each cube. In the crystal, each atom is surrounded by eight others lying at each corner.

BOEHMITE: A light-grey to dark reddish-brown mineral form of mixed aluminium oxide and hydroxide present in bauxite, with the chemical formula AlO(OH). It was named after the German scientist, J. Böhm (1895-1952).

BOERHAVIA DIFFUSA (PIG WEED; RED HOGWEED; RED SPIDERLING TARVINE; SPREADING HOGWEED): A perennial plant in the Nyctaginaceae, the Four O'Clock family, that grows as common weed in wet areas, especially in the rainy seasons with sunshine. It occurs throughout India, Africa, the Pacific and southern United States and has become naturalised in many places. It has a fusiform root with a stem, which is prostrate, decumbent or ascending.

The leaves are broadly ovate and obtuse at the base and arranged oppositely or sub-oppositely. The small red or white flowers are in penduncluate, glomerulate clusters arranged in slender, long stalked, axillary or terminal corymbs. The fruits are rounded above, obovoid, slightly cuneate below and glandular throughout. Medicinally, the various parts such as the roots, leaves and seeds have been used to treat various ailments such as leucorrhoea, anaemia, inflammations, heart diseases, asthma, diabetes, dyspepsia, tumours, spleen enlargement and abdominal pains. It also has antibacterial and antioxidants properties.

BOG: An area of ground saturated with water and decayed vegetation. It is a region of permanently badly drained wetland that is subject to high rainfall and has a persistently moist atmosphere. These conditions result in a climax community with no trees. The most common plants are bog mosses (*Sphagnum*), cotton grasses (*Eriophorum*), ling (*Calluna vulgaris*), cross-leaved heath (*Erica tetralix*), bog myrtle (*Myrica gale*), rushes (*Juncus*) and sedges (*Carex*).

BOG IRON: Impure iron deposits that develop in bogs or swamps by the chemical or biochemical oxidation of iron carried in solution.

BOHM DIFFUSION: The diffusion of plasma across a magnetic field.

BOHM-AHARONOV EFFECT (AHARONOV-BOHM EFFECT): An effect that occurs in small disordered metallic conductors and causes conductance fluctuations in small (non-superconducting) rings and wires. It arises from the presence of a vector potential produced by an applied magnetic field.

BOHR EFFECT: The effect of carbon dioxide on the dissociation of oxygen from haemoglobin, discovered by Christian Bohr (1855-1911), a Danish physiologist.

BOHR MODEL (BOHR ATOMIC MODEL): A model of the atom conceived by Neil Bohr (1885-1962) in 1913 based on his observations of the atomic emissions spectrum of the hydrogen atom. The model assumes the following: (i) that the electrons revolve in orbits of specific radius around the nucleus without emitting or absorbing radiation; (ii) that within each orbit, an electron has a fixed amount of energy; and (iii) that emission or absorption of energy (photons) occurs between transitions. The energy absorbed or emitted can be calculated using the relation $E_2 - E_1 = hf = \frac{h_c}{\lambda}$, where E is the energy level, h is the Planck's constant (6.63 x 10^{-34} Js, f is the frequency of the emitted photon, c is the speed of light (3.0 x 10^8 m/s) and λ is the wavelength.

BOHR RADIUS: A physical constant, approximately equal to the most probable distance between the proton and electron in a hydrogen atom in its ground state. Its symbol is a_0. Its value in metres is $a_0 \approx 5.29\times10^{-11}$ m, its value in picometres: ≈ 52.9 pm and value in angstroms: ≈ 0.53 Å.

BOHR, NEILS HENRIK DAVID (1885-1962): Danish physicist, who helped solve problems in nuclear physics and contributed to the understanding of atomic structure for which he won the 1922 Nobel Prize for Physics. He is considered one of the foremost scientists of modern physics, along with Albert Einstein, Erwin Schrödinger and Enrico Fermi. He was the son of a physiology professor and educated at the University of Copenhagen, earning his doctorate in 1911. He studied nuclear physics under British physicist, Sir Joseph John Thomson at Cambridge, England, but he soon moved to Manchester to work with another British physicist, Ernest Rutherford. In 1913 he published his explanation of how atoms, with electrons orbiting a central, achieve stability by assuming that their angular momentum is quantized. Movement of electron from one orbit to another is accompanied by the absorption or emission of energy in the form of light, thus accounting for the series of lines in the emission spectrum of hydrogen. In 1943, during World War II he worked at Los Alamos, New Mexico with Robert Oppenheimer and others to develop the atomic bomb.

BOHR'S THEORY: A theory published in 1913 by Neils Bohr (1885-1962), a Danish physicist, which explains the line spectrum of hydrogen. It assumed that a single electron of mass m travelled in a circular orbit of radius r, at a velocity v, around a positively charged nucleus. The angular momentum of the electron would then be mvr. Bohr proposed that electrons could only occupy orbits in which this angular momentum had certain fixed values, $h/2\pi$, $2h/2\pi$, $3h/2\pi$, ..., $nh/2\pi$, where h is the Planck constant. This means that the angular momentum is quantized, i.e. it can only have certain values, each of which is a multiple of n. Each permitted value of n is associated with an orbit of different radius and Bohr assumed that when the atom emitted or absorbed radiation of frequency v, the electron jumped from one orbit to another, with the energy emitted or absorbed by each jump being equal to hv. This theory gave good results in predicting the lines observed in the spectrum of hydrogen and simple ions such as He^+, Li^{2+}, etc. The idea of quantized values of angular momentum was later explained by the wave nature of the electron. Each orbit has a whole number of wavelengths around it; i.e. $n\lambda = 2\pi r$, where λ is the wavelength and n a whole number. The wavelength of a particle is given by h/mv, so $nh/mv = 2\pi r$, which leads to $mvr = nh/2\pi$. Modern atomic theory does not allow subatomic particles to be treated in the same way as large objects and Bohr's reasoning is somewhat discredited. However, the idea of quantized angular momentum has been retained.

BOHRIUM (UNNILSEPTIUM): A radioactive transactinide element, with the symbol Bh and atomic number 107. It is the heaviest member of Group VIIB first made in 1981 by Peter Armbruster and a team in Darmstadt, Germany, by bombarding bismuth–209 nuclei with chromium–54 nuclei. Its most stable known isotope, ^{270}Bh, has a half-life of 61 seconds. Chemical experiments have confirmed bohrium's predicted position as a heavier homologue to rhenium with the formation of a stable +7 oxidation state. Only a few atoms of bohrium have ever been detected.

BOIDAE: A venomous family of snakes with large, thick-body consisting of pythons and boas. Examples are the African python, *Python sebae* (the largest snake in Africa), the royal python, (*Python reguis*) and the world's largest snake, the anaconda (*Eunectes murinus*), a boa. They have recurved sharp teeth in both the lower and upper jaws. They feed mostly on mammals or birds which they kill by coiling around and constricting them to death.

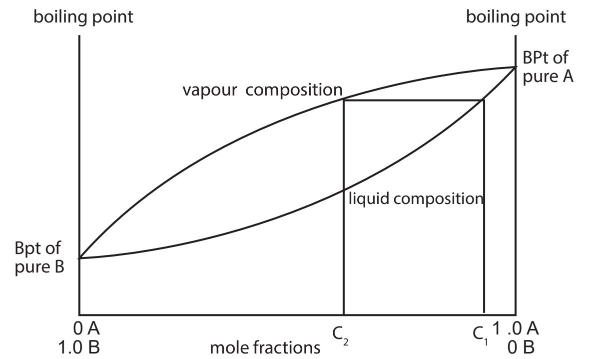
BOIGINAE: A subfamily of venomous snakes belonging to the largest snake family, Colubridae with back fangs with venom. Examples are the African beauty snake (*Psammophis sibilans*) and Boomslang (*Dispholidus typus*).

BOILER: An apparatus, in which liquid is heated to convert to vapour or steam. It consists of a furnace in which fuel is burned to transmit heat from the combustion products to the water or other liquid and a space where steam or vapour can form and collect. A conventional boiler burns a fossil fuel or waste fuel.

BOILING: A phase transition, which involves the rapid conversion of a liquid into vapour at a fixed temperature, known as boiling point. For example, the boiling point of pure water at standard atmospheric pressure and at sea level is 100 °C (373 K). Boiling takes place when the vapour pressure of the liquid equals the pressure of the atmosphere above the liquid and it takes place when the liquid reaches a certain temperature and involves the formation of vapour bubbles within the body. A liquid may also boil when the pressure of the surrounding atmosphere is sufficiently reduced, such as the use of a vacuum pump or at high altitudes.

BOILING POINT (B.P.): The temperature at which a pure liquid boils or the temperature at which the saturated vapour pressure of a liquid equals the atmospheric pressure. As a result, bubbles form in the liquid and the temperature remains constant until all the liquid has evaporated. The boiling point of pure water for example, is equal to 100 °C (373 K) at standard atmospheric pressure and at sea level. Factors such as atmospheric pressure and the level of impurities in a liquid affect its boiling point.

BOILING POINT COMPOSITION DIAGRAM: A graph showing the boiling point and vapour composition of a mixture of two liquids, which depends on the composition of the mixture. The graph has two curves: the lower one gives the boiling points at a fixed pressure for the different compositions. The upper one is plotted by taking the composition of vapour at each temperature on the boiling-point curve. For a binary mixture of A and B, a typical diagram may look like the following:



Boiling Point Composition Diagram

BOILING POINT ELEVATION: The increase in the boiling point of a solvent caused by the dissolution of a non-volatile solute such as a salt in it.

BOILING TEMPERATURE: The temperature at which the vapour pressure of liquid becomes equal to the air or applied pressure.

BOILING TUBE: A type of graduated cylindrical tube made of glass such as borosilicate that can withstand high temperatures. It is closed at one end and similar to a test tube, but is larger and wider. It is used for boiling solutions, substances, etc.

BOILING-WATER REACTOR (BWR): A type of nuclear reactor that uses water as a coolant and moderator originally designed by Allis-Chalmers and General Electric (GE). The water which passes over the reactor core to act as moderator and coolant boils within the core, by operating at a pressure of about 70 atmospheres (1 atmosphere equals 1.01325 x 10^5 newtons per square metre) that allows it to boil and form steam. The steam produced in the reactor pressure vessel is piped directly to the turbine generator, condensed and pumped back to the reactor. It is used for the generation of electrical power and is the second most common type of electricity-generating nuclear reactor after the pressurised water reactor (PWR). Its disadvantage is that any fuel leak might make the water radioactive and that radioactivity would reach the turbine and the rest of the loop.

BOILOFF: The vapour loss from a volatile liquid, for example, liquid oxygen, particularly when stored in a vehicle ready for flight.

BOK GLOBULES (THACKERAY'S GLOBULES): A near-spherical dark nebular thought to consist of gas and dust clouds collapsing under gravity.

BOLE: 1. Reddish-brown clay used as a pigment. 2. The trunk of a tree.

BOLOMETER: A sensitive instrument used to measure thermal radiation. It consists of a simple electric circuit, with a slab of material whose electrical resistance changes with temperature. During operation, radiant energy is absorbed by the slab, producing a rise in the slab's temperature and thereby a change in its resistance. The electric circuit converts the resistance change to a voltage change, which then can be amplified and observed by various, usually conventional, instruments.

BOLT: 1. To produce flowers and seeds too early, as in lettuce. 2. A flash of lightning appearing briefly as a jagged line of light. 3. A short cylindrical metal bar with a screw thread, used with a nut. 4. A small metal spike used to provide an anchor in rock faces, in climbing. 5. A sliding rod, bar or plate that ejects a used cartridge and closes the breech or a short arrow for use with a crossbow. 6. A sliding bar that fits into a socket and secures a door or gate.

BOLTER: A plant such as lettuce which grows too fast, producing a low seed yield.

BOLTING: The premature production of flowers and seeds. It is a problem in biennial crops, such as sugar beet, which shows considerably reduced yields if they 'run to seed' in the first growing season.

BOLTZMANN CONSTANT: The physical constant relating energy at the particle level with temperature observed at the bulk level, with the symbol k or k_B. It is given by the universal gas constant divided by the Avogadro's constant, $k = R/N_A$. It may be thought of, therefore, as the gas constant per molecule. It was named after Ludwig Boltzmann (1844-1906), an Austrian physicist.

BOLTZMANN EQUATION (BOLTZMANN TRANSPORT EQUATION): One of the most important equations used in the study of a group of particles in non-equilibrium statistical mechanics, especially transport properties. It describes the statistical distribution of one particle in rarefied gas and it is one of the most important equations of non-equilibrium statistical mechanics, which is the area of statistical mechanics that deals with systems far from thermodynamic equilibrium as, for example, when there is an applied temperature gradient or electric field. This equation is used to study how a gas or fluid transports physical quantities such as heat and momentum and thus to derive transport properties such as viscosity and thermal conductivity, however, the problem of existence and uniqueness of solutions to the Boltzmann equation is still not fully resolved.

BOLTZMANN FORMULA: An equation concerning the entropy S of a system derived from statistical mechanics. The formula is $S = k \ln W$, where k is the Boltzmann constant and W is the number of distinguishable ways of describing the system. It expresses in quantitative terms the concept that entropy is a measure of the disorder of a system.

BOLTZMANN, LUDWIG EDUARD (1844-1906): Austrian physicist, who laid the foundation and contributed to the fields of statistical mechanics and statistical thermodynamics and one of the most important advocates for atomic theory. Born in Vienna and educated at the universities of Vienna and Oxford, he held professorships in Graz, Vienna, Munich and Leipzig, where he worked on the kinetic theory of gases and on thermodynamics. In the 1870s Boltzmann published a series of papers that showed that the second law of thermodynamics could be explained by statistically analysing the motions of atoms. He also formulated the law of thermal radiation, named after him and the Austrian physicist Josef Stefan. He suffered from depression and committed suicide in 1906. Although his work was strongly criticised by scientists of his time, much of Boltzmann's work was substantiated by experimental data soon after his death.

BOLUS: A mass of food after it has been chewed and mixed with saliva and pushed by the tongue to the back of the mouth and into the pharynx.

BOLZA'S PROBLEM: The general problem of determining the arc, from a given class, that minimises a function of the form

$$I = \int_{x_0}^{x_1} u(y_1, ..., y_n; y_1', ..., y_n')dt + v(y_{10}, ..., y_n; y_{11}, ..., y_n),$$

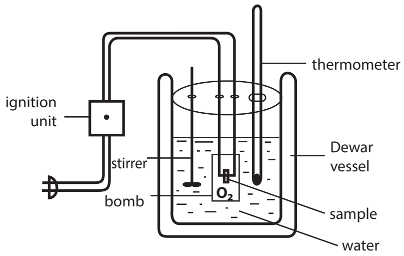
find in a class of arcs satisfying a group of differential equations which have been subjects to some constraints. It was named after the German-born American analyst, Oscar Bolza (1857 -1942).

BOLZANO-WEIERSTRASS THEOREM: A theorem used in the mathematical field of real analysis, which states that every bounded infinite subset of Euclidean space possesses a cluster point, so that every bounded infinite sequence possesses a convergent subsequence.

BOLZANO'S THEOREM (INTERMEDIATE VALUE THEOREM): A theorem, used in mathematical analysis which states that if a continuous function f having interval $[x, y]$ as its domain takes every value between $f(x)$ and $f(y)$ at each end of the interval, then it also takes values between $f(x)$ and $f(y)$ at some point within the interval. This intermediate value property, which derivatives also possess by virtue of the mean-value theorem, is also called the Darboux property. It was named after the Czech analyst Bernhard Bolzano (1781-1848).

BOMB: A device containing an explosive or incendiary material that generates and releases its energy very rapidly and made to explode when dropped, thrown or detonated by a timing device. The explosion creates a violent, very destructive shock wave, which can cause destructions to properties and injuries.

BOMB CALORIMETER: An apparatus that is used to measure the quantity of heat produced when a known amount of carbohydrate, fat, protein or some mixture of these is burned. It consists of a calorimeter with a strong-walled container constructed of a corrosion-resistant alloy, called the bomb, immersed in about 2.5 litres of water in a metal container. The sample, such as food and fuel or any organic material, is ignited by electricity and the heat generated is measured.



Bomb calorimeter

BOMB-DIGESTION: The treatment of materials which are not fully dissolved by acid-digestion at atmospheric pressure and requiring a more vigorous treatment in pressure vessels lined with polytetrafluoroethylene (PTFE) glass, silica or vitreous (glassy) carbon or in sealed silica tubes. The test sample and acids are heated in such a closed vessel, so that the digestion is carried out at higher temperature and pressure.

BOND: A strong force of attraction holding atoms together in a molecule or crystal. In other words, it is the electrostatic force of attraction between two chemical species (ions, atoms, molecules). The various types include the following: (i) **ionic** or **electrovalent bond** that is created during the formation of a compound by transfer of one or more electrons from one atom to another, the resulting oppositely charged ions being held together by attraction; (ii) **covalent bond**, in which the attractive force between atoms is created by the sharing of electrons; and (iii) **metallic bond**, which is the electrostatic force between delocalised electrons and positively charged metal ions characteristic of metals. The electrons are shared between atoms and move about in the crystal. Typically chemical bonds have energies of about 1000 kJ mol^{-1}.

BOND DISSOCIATION ENERGY (BOND DISSOCIATION ENTHALPY): The bond dissociation enthalpy of an X-Y bond is the enthalpy of the gas phase reaction in which this bond is broken to give isolated X and Y atoms: $XY_g \longrightarrow X_g + Y_g$. Bond dissociation enthalpies are always positive number because it takes energy to break bonds. It is an endothermic process.

BOND ENERGY: The amount of energy that has to be supplied to break a chemical bond between two atoms in a molecule. In other words, it is the average amount of energy absorbed to break a covalent bond in one mole of a covalently bonded molecule into its isolated gaseous species. It is a measure of the bond strength in a chemical bond.

BOND LENGTH (BOND DISTANCE): The distance between the nuclei of two bonded atoms in a molecule. Bond length is inversely related to bond order, when more electrons participate in bond formation the bond will get shorter. Bond length is also inversely related to bond strength and the bond dissociation energy, as a stronger bond is also a shorter bond. In a bond between two identical atoms half the bond distance is equal to the covalent radius. Bond lengths are measured in molecules by means of x-ray diffraction.

BOND ORDER: The difference between the number of electrons in bonding molecular orbitals and antibonding molecular orbitals, divided by two. For example, in nitrogen N≡N the bond order is 3, in acetylene H−C≡C−H the bond order between the two carbon atoms is 3 and the C−H bond order is 1. Bond order gives an indication to the stability of a bond.

B

BOND-ENERGY-BOND-ORDER METHOD: An empirical procedure for estimating activation energies, involving empirical relationships between bond length, bond dissociation energy and bond order.

BONDED PHASE: A stationary phase which is covalently bonded to the support particles or to the inside wall of the column tubing.

BONDERISING (BONDERIZING): A chemical process to help prevent the corrosion of iron, steel and other metals. The surface is sprayed with or immersed in a hot phosphate solution and a superficial insoluble phosphate layer is formed. The metal is then usually painted or lacquered.

BONDI, HERMANN (1919 - 2005): An Anglo-Austrian mathematician and cosmologist, who moved to England to attend Trinity College, Cambridge in 1937. Bondi, Thomas Gold and Fred Hoyle suggested that matter could be continuously created out of nothing to maintain the density over time and that the rate at which matter would have to be created was much too low to be observationally testable, a theory they called the steady-state theory; an alternative to the Big Bang theory. His most lasting legacy may probably be his important contributions to the theory of general relativity.

BONDING: The joining together of atoms to form molecules or crystalline salts.

BONDING MOLECULAR ORBITAL (BONDING ORBITAL): A molecular orbital formed by bonding electrons, which have lower energy than any of the atomic orbitals from which they are formed. It may be regarded as a region in which a bonding electron may be found in a molecule. It lends stability to a molecule or ion when electrons are present and results in a net attraction and chemical bonding.

BONDING NUMBER: The bonding number of a skeletal atom is the sum of the total number of bonding equivalents (valence bonds) of that skeletal atom to adjacent skeletal atoms in a parent hydride, if any and the number of attached hydrogen atoms, if any. Examples are SH_2: for S, $n = 2$; SH_6: for S, $n = 6$.

BONDING PAIR: A pair of electrons involved in a covalent bond.

BONE: The main component of the skeletal system or the hard connective tissue that makes up the skeleton of a vertebrate. It is one of the hardest animal tissues found in only vertebrates. It consists of collagen, a protein which gives it its tensile strength and calcium phosphate which gives it its hardness. Bones perform a variety of functions, including structural and mechanical; protecting vital organs; providing a site for the production of blood cells; and serving as a reserve of calcium. Each bone is a complex living organ that is made up of many cells, protein fibres and minerals. At birth, babies have about 300 bones. As they grow older, various bones fuse together and the typical adult human skeleton has 206 bones. Some individuals may have more or fewer bones because of anatomical variations. The breakdown is as follows: the axial skeleton has 80 bones composed of the skull with 28 bones and the torso with 52; the appendicular skeleton has 126 bones composed of the upper extremity with 64 (32 x 2) and the lower extremity with 62 (31 x 2). The axial skeleton runs along the body's midline axis and includes the skull, hyoid, auditory ossicles, ribs, sternum and the vertebral column. The appendicular skeleton includes the folowing regions: upper limbs, lower limbs, pelvic girdle and pectoral (shoulder) girdle. There are different types of bones, categorised by shape: flat, long, short and irregular. Flat bones are like those found in the skull, ribs, sternum and scapula. Long bones are the type found in the extremities, like the femur or thighbone in the leg or ulna in the arm. Short bones are found in the wrists and ankles and some of the irregular bones are found in the vertebrae and at the sutures in the skull.

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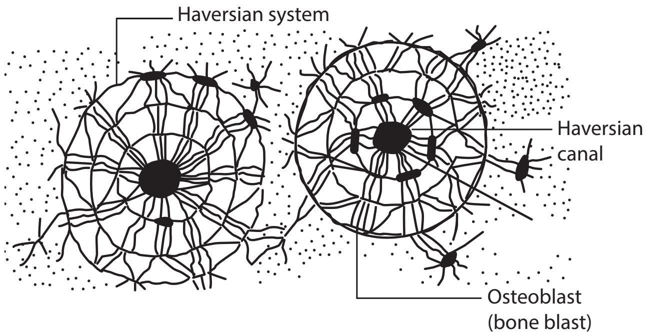
The longest bone in our bodies is the femur (thigh bone). The smallest bone is the stirrup bone inside the ear. Bones are connected to other bones at joints. The different types of joints include: fixed joints such as in the skull, which consists of many bones; hinged joints such as in the fingers and toes; and ball-and-socket joints such as the shoulders and hips. **Skeletal muscles**, also known as **striated muscles** or **striped muscles** are muscles attached to bones and are responsible for skeletal movements. The centre of the bone is filled with marrow and this is surrounded by the hardened bone tissue itself. In bone, the extracellular matrix is composed of collagen with calcium phosphate deposited in it. Vitamin C is needed for collagen synthesis. Bone tissue is composed of repeating, circular units called **Haversian systems**. Within each Haversian system, there is a central canal where blood vessels and nerves can be found. This is surrounded by concentric layers of the matrix material called **lamellae**. In lamellae are spaces called lacunae which contain the osteocytes. Osteoblast is a large cell responsible for the synthesis and mineralisation of bone during both initial bone formation and later bone remodelling. Osteoblasts form a closely packed sheet on the surface of the bone, from which cellular processes extend through the developing bone. They produce many cell products, including the enzymes alkaline phosphatase and collagenase, growth factors, hormones such as osteocalcin and collagen (the structural protein of bones, tendons, ligaments and skin). A large part of the bone formed during the growth period is destroyed by specialised cells called **osteoclasts** in the growth process.



Microscopic structure of bone

BONE DENSITY: The amount of bone tissue and minerals in a certain volume of bone.

BONE MARROW: A soft tissue in the central cavity and internal spaces of some limb bones that manufacture red and white blood cells. At birth and in young animals, the marrow of all bones is concerned with the formation of blood cells. It contains the type of stem cells, namely hemopoietic, which produces blood cells; and stromal, which produces fat, cartilage and bone. The two types of bone marrow marrows are red marrow and yellow marrow. Red marrow, sometimes also called myeloid tissue is the tissue that develops into red blood cells, white blood cells and platelets. It is present throughout the skeleton during the developmental stages of the foetus. After birth, as the child ages, the red marrow is gradually replaced by yellow marrow.

BONE MEAL: Ground animal bones used in animal feed or as a fertiliser.

BONY LABYRINTH: The complex cavity of the inner ear. It consists of the central vestibule, the semicircular canals and the cochlea.

BOOK LUNG: The main respiratory organ of many terrestrial arachnids, including scorpions and spiders. Book lungs consist of a series of soft, haemolymph-filled, overlapping flaps, covered up by a plate on the abdomen, through which oxygen is taken up and carbon dioxide given off.

BOOKMARK: A Web browser feature used to save a URL address for future reference. Bookmarks save user and browser time, which is especially useful for Web pages with long URLs.

BOOLEAN: A term used to describe a variable, function, operator, etc. that takes either of the binary digits 0 (false) and 1 (true). It is typically used in computing to record the result of a test. It was named after the British mathematician, George Boole (1815-64).

BOOLEAN ALGEBRA: 1. A form of symbolic logic devised by George Boole, which provides a mathematical procedure for manipulating logical relationships in symbolic form. For example, in Boolean algebra, $a + b$ means a or b and ab means a and b. It makes use of set theory and is extensively used by the designers of computers to enable the bits 0 and 1, as used in the binary notation, to relate to the logical functions the computer needs in carrying out its calculations. 2. The branch of logic, which uses algebraic symbols, employed in computer programs. The main operators are AND and OR which are applied to elements that are true or false. Other operators include NOT, NAND, NOR and XOR.

BOOLEAN RING: 1. A ring in which every member is idempotent; for example, if A is a matrix the A is idempotent if $A^2 = A$. 2. Less abstractly, a class of sets that is closed under finite union and relative compliment. A Boolean ring with a largest element is identifiable with a Boolean algebra.

BOOST: 1. In aeronautics, it defines the amount by which the induction pressure of a supercharged internal-combustion engine exceeds that of the ambient pressure. 2. To increase the voltage of an electric circuit. 3. To raise or lift by pushing up from behind or below. 4. An instrument usually with hot tar, used for marking sheep and often bearing the owner's initials.

BOOSTER: 1. A device for increasing voltage or power. An example is transformer. 2. A dose of injection of medicine or drug that increases the effect of an earlier one.

BOOSTER ENGINE: Extra cylinders on a steam locomotive, driving the trailing truck or a tender truck, to give more power on starting.

BOOSTER ROCKET: A rocket used to give initial speed to a missile or spacecraft.

BOOT (BOOT UP; BOOTING): The process of initialising a computer system by clearing memory and reloading the operating system; or to cause a computer system to reach a known beginning state. A **cold boot** means starting the system from a powered-down state. A **warm boot** means restarting the computer while it is powered-up. Important differences between the two procedures are; (i) in a power-up self-test, various portions of the hardware (such as memory) are tested for proper operation, during a cold boot while a warm boot does not normally perform such self-tests and (ii) a warm boot does not clear all memory. Booting is also

BOOT DISK: A removable data storage medium used to load and boot an operating system or utility program. It is a read-only medium that stores temporary files on a CD-ROM or floppy disc drive. Other boot disk mediums include USB drives, zip drives and paper tape drives. One of the most common uses of a boot disk is to start the computer when the operating on the internal hard drive does not load.

BOOT FILES: The system files needed to start Windows, whether in Normal Mode or Safe Mode. Boot files are also necessary when reinstalling Windows from a disc or flash drive, running bootable antivirus software, testing the computer's memory, partitioning the hard drive with tools like GParted, using a password recovery tool, wiping all the data from the HDD, etc. **.BOOT files** are those that end with the .BOOT suffix which are InstallShield files. These are plain text files that store installation settings for the Flexera InstallShield program, which is an application used for creating setup files for software installations.

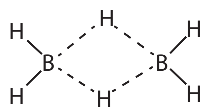
BOOT LOADER (BOOT MANAGER): A small program that places the operating system (OS) of a computer into memory. When a computer is powered-up or restarted, the basic input/output system (BIOS) performs some initial tests and then transfers control to the master boot record (MBR) where the boot loader resides.

BOOT SECTOR: A dedicated section of a hard disk, hard drive or other bootable disk drive or storage device that contains data used to boot a computer system. It includes the master boot record (MBR), which is accessed during the boot sequence and the volume boot record (VBR), which contains partition information at the beginning of each partition.

BOOTSTRAP PROTOCOL (BOOTP): A computer networking protocol that automatically configures (such as assigning an IP address to network devices) a user and has an operating system booted without user involvement.

BORAGINACEAE: A family of plants which includes a variety of shrubs, trees and herbs totalling about 2000 species in 146 genera found worldwide. These include borage, (*Borago officinalis*); bocote (*Cordia alliodora*), genera *Heliotropium* and *Symphytum*. Some of the species in the family have some medicinal properties. For example, *Heliotropium indicum* is used for the treatment of the following ailments: hypertension, glaucoma, ocular hypotension, oxidative damage, abdominal pain, convulsion, cataract, conjunctivitis, cold, and inflammatory reactions.

BORANE: Any of the boron hydrides having an unusual chemical bonding (B_nH_{n+2}) due to electron deficiency. The ability of boranes to form three-centre-two-electron bonds and normal two-centre-two-electron covalent bonds allows them to form complex structures called polyhedrons. Low-molecular-weight boranes are spontaneously flammable in air, although reactivity generally decreases with increasing molecular weight. Boranes are sources of high-energy fuels for rockets and jet aircraft.



Chemical Structure of Borane

BORATE: Any of the boron-oxygen ionic radicals with boron in oxidation state +3. The simplest borate ion is the trigonal planar, BO_3^{3-}, although many others are known. BO_3^{3-} forms salts with metallic elements.

BORAX (SODIUM BORATE; SODIUM TETRABORATE DECAHYDRATE; DISODIUM TETRABORATE): An important compound of boron, usually a white powder consisting of soft colourless crystals that dissolve easily in water, with the chemical formula $Na_2B_4O_7.10H_2O$, density 1.73 g/cm³ (solid), melting point 743 °C, 1016 K (anhydrous) and boiling point 1575 °C (1848 K). It is a component of many detergents, cosmetics and enamel glazes and also used to make buffer solutions in biochemistry, as a fire retardant, as an anti-fungal compound for fibreglass, as an insecticide, as a flux in metallurgy and as a precursor for other boron compounds.

BORAX CARMINE: A red dye used in optical microscopy that stains nuclei and cytoplasm pink. It is frequently used to stain large pieces of animal tissues.

BORAX-BEAD TEST: Part of qualitative inorganic analysis to test for the presence of certain metal ions in salts. A small amount of the salt is mixed with borax and a molten bead formed on the end of a piece of platinum wire. Certain metals can be identified by the colour of the bead produced in the oxidising and reducing parts of a Bunsen flame. For example, iron gives a bead that is red when hot and yellow when cold in the oxidising flame and a green bead in the reducing flame.

BORDEAUX MIXTURE: A combination of cupric sulfate (copper(II) sulfate) and calcium hydroxide in water used as a fungicide. A common mixture is 40 g copper sulfate and 40 g calcium hydroxide in 5 litre of water.

BORDER GATEWAY PROTOCOL (BGP): It is the *de facto* Standard for inter-Autonomous Systems (AS) routing protocol in today's Internet. It is a standardised exterior gateway protocol designed to exchange routing and reachability information between autonomous systems (AS) on the Internet.

BORDER PLANT: A planting technique that is used to separate a part of the landscape from another. It may be used as a fence or a windbreak.

BORDER-STRIP IRRIGATION (BORDER IRRIGATION): A type of surface irrigation in which water is applied at the upper end of a strip having dykes to confine the water to the strip. This type of irrigation is best suited to large mechanised farms for ease of machine operations. Crops best suited to this kind of irrigation are wheat, barley, fodder crops and legumes. Border slopes should be uniform, with a minimum slope of 0.05% to provide adequate drainage and a maximum slope of 2% to limit problems of soil erosion.

BORDERING: The enlargement of a matrix or determinant annexing a column and a row, especially when all the entries of the annexed row and column are zero except for a 1 in the common entry, so that the value of the determinant does not change.

BORDERLINE MECHANISM: A mechanism intermediate between two extremes, for example, a nucleophilic substitution intermediate between S_N1 and S_N2 and or intermediate between electron transfer and S_N2 (S for chemical substitution, N for nucleophilic, and the number represents the kinetic order of the reaction).

BORE: 1. A high tidal wave that moves along a narrow estuary from the sea linked with the narrowing and shallowing of the estuary. The largest bore in the world occurs on the Amazon, where it moves upstream at a rate of 10 m/s reaching a height of 5 metres. 2. To make a hole, well, tunnel, etc. with a revolving tool or by digging. 3. The hollow part inside a tube or gun barrel.

BOREAL FOREST (TAIGA): A terrestrial subpolar biome occurring across North America and Eurasia. It consists mainly of evergreen coniferous forests growing on swampy ground that is covered with lichens. The predominant trees are pine, fir and spruce. Soil organisms are protozoans, nematodes and rotifers; larger invertebrates such as insects that decompose plant litter are lacking, so humus accumulates very slowly. The taiga is rich in fur-bearing animals such as sable, fox and ermine and is home to elks, bears and wolves.

BOREHOLE: 1. A deep hole made in the ground, either vertically or horizontally for purposes such as the extraction of water, petroleum, gases, for a geotechnical investigation, environmental site assessment, mineral exploration, temperature measurement or as a pilot hole for installing piers or underground utilities. 2. The hollow part inside a gun barrel.

BOREL MEASURE: Any measure defined on the sigma-algebra generated by all the open (or equivalently compact) subsets of a compact topological space (the Borel fold), especially on the unit interval. All continuous functions in a Borel measure are measurable.

BOREL SET (BOREL MEASURABLE SET): Any set derived from the intervals on the real line by repeated application of countable union and intersection; the Borel sets constitute a sigma-algebra. For Borel measure all continuous functions are measurable.

BOREL SUMMATION: An integral that is defined so as to represent the sum of a divergent series, introduced by Emile Borel (1871 - 1956) in 1899.

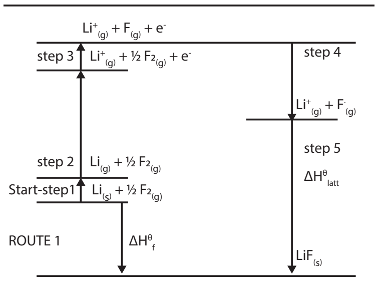
BOREL-CANTELLI LEMMA: The result that, given an infinite sequence of events in a probability space for which the sum of the individual probabilities is finite, then the probability that infinitely many events occur is zero. If the events are independent and the sum of the individual probabilities infinite, then the probability that infinitely many events occur is unity. More generally, if $\{A_n\}$ is a sequence of measurable sets in a measure space whose measures $\mu(A_n)$ have a finite sum, then $\mu(\lim\sup \mu) = 0$; that is, the set of points that are in an infinite number of the given sets has measure zero.

BOREL, FELIX EMILE (1871-1956): French mathematician and politician, who together with Lebesgue and Baire founded the theory of real-valued functions; he also contributed to the development of game theory.

BORIC ACID: Any of the three chemical compounds orthoboric or boracic acid, metaboric acid and tetraboric or pyroboric acid containing boron and oxygen. The term also applies to the compound H_3BO_3, also called orthoboric acid or trioxoboric(III) acid. It is a white or colourless solid that is soluble in water and ethanol. It occurs naturally in the condensate from volcanic steam vents (suffioni). Commercially, it is made by treating borate minerals such as kernite, $Na_2B_4O_7.4H_2O$ with sulfuric acid followed by recrystallisation. It is used as fire retardant, antiseptic, in the manufacture of heat-resistant glass and ceramics.

BORIDE: A compound of boron with a metal, which is a hard, high-melting material with metal-like conductivity. Most metals form at least one boride of the type MB, MB2, MB4, MB6 or MB12. The compounds have a variety of structures; in particular, the hexaborides contain clusters of B6 atoms.

BORN-HABER CYCLE: A sequence of chemical and physical processes used for calculating either the energy required to break down a crystalline solid into its constituent ions lattice energy; or the energy required to break a chemical bond into noncrystalline solids bond energy. It also shows various steps through which reactants are converted to products using energy cycle. It was named after Max Born 1882-1970, German-born British physicist and mathematician Fritz Haber (1868-1934), German chemist. Born-Haber cycle for the formation of LiF can be given as shown in the diagram.



Born-Haber cycle

BORN-OPPENHEIMER APPROXIMATION (ADIABATIC APPROXIMATION): A way to simplify the complicated Schrödinger equation for a molecule. It is used in the Born-Oppenheimer method that the electronic wave functions and energy levels at any instant depend only on the positions of the nuclei at that instant and not on the motions of the nuclei. It is the assumption that the electronic motion and the nuclear motion in molecules can be separated. It leads to a molecular wave function in terms of electron and nuclear positions.

BORN, MAX (1882-1970): German-born British physicist and mathematician, who taught theoretical physics at the University of Göttingen from 1921 to 1933. He fled to Britain and taught at the University of Edinburgh from 1936-53. He was instrumental in the development of quantum mechanics. He also made contributions to solid-state physics and optics and supervised the work of a number of notable physicists in the 1920s and 30s. In 1926, together with his student Werner Heisenberg, they developed the mathematical formulation that would adequately describe Heisenberg's first laws of a new quantum theory. He was awarded the 1954 Nobel Prize for Physics for his work on statistical mechanics with W. Bothe.

BORNITE (PEACOCK ORE; PEACOCK COPPER): An important ore of copper, with the chemical formula Cu_5FeS_4, which occurs widely in porphyry copper deposits along with the more common chalcopyrite and also found as disseminations in mafic igneous rocks, in contact metamorphic skarn deposits, in pegmatites and in sedimentary cupriferous shales. It is important for its copper content of about 63 percent by mass. It has a metallic reddish-brown surface when freshly taken or exposed, which soon turns purplish.

BORON: A hard and brittle, yellow-brown non-metallic element, with the symbol B in Group III of the Periodic Table. It is derived from borax and kernite. Its properties are intermediate between a metal and nonmetal. It is a component of alloys used in the control rods of nuclear reactors due to its high neutron absorption ability, as alloys, glass and ceramics.

BORON CARBIDE (TETRABOR): A black solid compound of boron and carbon, with the chemical formula B_4C, density 2.52 g/cm^3 (solid), melting point 2763 °C (3036.15 K) and boiling point 3500 °C (3773.15 K). It is an extremely hard, black crystalline compound or solid solution with a Mohs hardness of above 9, only cubic boron nitride (Mohs hardness above 10) and diamond (Mohs hardness 10) are harder. It is soluble only in fused alkali. It is produced by the reduction of boric oxide with petroleum coke in an electric furnace. It is used largely as an abrasive, in control rods for nuclear reactors, as a reinforcing filament in composite structural materials; and also for the fabrication of objects using high-temperature powder metallurgy.

BORON COUNTER: A counter tube containing a boron fluoride, used for detecting slow neutrons.

BORON HYDRIDES (BOROETHANE; DIBORANE; DIBORON HEXAHYDRIDE): A colourless, volatile compound that is soluble in ether with the chemical formula B_2H_6, density 1.216 g/dm^3, boiling point -92.5 °C (180.65 K) and melting point -165.5 °C (438.65 K). It mixes well with air and may explode on contact with air. It can be used to produce boranes of higher molecular weight, such as decaborane ($B_{10}H_{14}$) and to form coatings of pure boron for various technological devices.

BORON NITRIDE: A white inert crystalline solid existing both in a graphite-like form and in an extremely hard diamond-like form called borazon with the chemical symbol BN, melting point 2973 °C, (3246.15 K). It is insoluble in cold water and slowly decomposes in hot water. It is used in electrical industries, where its high thermal conductivity and high resistance are of especial value. It is also used as a refractory, high temperature lubricant and insulator and heat shield.

BORON TRICHLORIDE: A colourless liquid, with the chemical formula BCl_3, melting point -107.3 °C (165.85 K) and boiling point 12.6 °C (285.75 K) that reacts vigorously with water to produce hydrogen chloride and boric acid. Boron trichloride is a Lewis acid, forming stable addition compounds with such donors as ammonia and the amines and is used in the laboratory to promote reactions that liberate these donors. The compound is important industrially as a source of pure boron (reduction with hydrogen) for the electronics industry; used for the preparation of boranes by reaction with metal hydrides; and in refining of aluminium, magnesium, zinc and copper.

BOROSILICATE: A salt that is derived from both boric acid and silicic acid, therefore contains both BO_3 and SiO_4 units linked in a network. Borosilicates can be found in a wide range of naturally ocurring compounds with complex structures. Borosilicate is used in glazes and enamels in the manufacture of glass and in the production of glass wools.

BOROSILICATE GLASS: Any of heat and chemical resistant glasses, such as pyrex, prepared by fusing together boron(III) oxide, silicon dioxide and, usually, a metal oxide. It results in a material which does not expand much; hence the glass can be heated or cooled rapidly without cracking. Such glasses have a wider range of applications than soda glasses, which have narrow plastic range and higher thermal expansion or silica with much higher melting point. Borosilicate glass was first developed by German glassmaker Otto Schott in the late 19th century.

BORROW: To redistribute a number between place-values to enable subtraction in each place-value position to be effected within the natural numbers. For example, to subtract 36 from 80, we first try to take 6 from 0 in the units column, but as this is impossible within the natural numbers, we 'borrow' 10 from the tens column, making the 0 now 10, from which 6 can be subtracted permitting us to continue with the process.

BORT (BOART): An impure or discoloured diamond that is as hard as a pure diamond but useless as a gem. It is commonly used as an abrasive in cutting tools or for making drills.

BOSCH PROCESS: An industrial process for manufacturing hydrogen by the catalytic reduction of steam with carbon monoxide. It is used in the production of hydrogen for the Haber process.

BOSE-EINSTEIN CONDENSATION: A phenomenon occurring in a microscopic system consisting of a large number of boson at a sufficiently low temperature, in which a significant fraction of the particles occupy a single quantum state of lowest energy.

BOSE-EINSTEIN DISTRIBUTION: It describes the statistical behaviour of integer spin particles (bosons). At low temperatures, bosons can behave very differently than fermions because an unlimited number of them can collect into the same energy state, a phenomenon called **condensation**.

BOSE-EINSTEIN STATISTICS: The quantum statistics obeyed by integer-spin particles (bosons) in which any number of particles may occupy a given state. Bose-Einstein statistics describe physical systems in which the particle number is nonconservative, particles are indistinguishable and cells of phase space are statistically independent. Such systems allow two polarisation states and exhibit totally symmetric wavefunctions.

BOSON: One of the two basic divisions of elementary particles, the basic units of matter and energy, with integral spin that obeys Bose-Einstein statistics but does not obey the Pauli Exclusion Principle. Examples are alpha particle, photon and atomic nuclei of even mass numbers.

BOSONISATION: The procedure used in quantum field theory and the many-body problem in quantum mechanics in which theories involving fermions have the fermions replaced by an effective field theory with bosons.

BOSS: A projection or protuberance from a surface or organ.

BOSUK-ULAM THEOREM: It states that every continuous function from an n-space into Euclidean n-space maps some pair of antipodal points on the same point. In other words, it is the result that there does not exist any continuous odd mapping of the unit n-sphere into the unit (n -1)-sphere.

BOT FLY: A two-winged fly, belonging to the Oestrus family, whose maggots are parasitic on sheep and horses.

BOT WORM: A self-replicating malware program that resides in current memory (RAM), turns infected computers into zombies (or bots) and transmits itself to other computers. A bot worm may be created with the ultimate intention of creating a botnet that functions as a vehicle for the spread of viruses, Trojans and spam.

BOTANICAL: Relating to plants, especially the scientific study of plants.

BOTANICAL GARDEN: A tract of land for the cultivation of a diversity of plant species, mainly for scientific and educational purposes and also pleasure. The plants in botanical gardens are labelled, usually with both the common and the scientific names and they are often arranged in cultural or habitat groups, such as rock gardens, aquatic gardens, desert gardens and tropical gardens. An **arboretum** is a botanical garden devoted chiefly to the growing of woody plants.

BOTANY: 1. The scientific study of all forms of plant life along with their growth, use, structure and other processes of plant life including evolution, ageing, decaying and demise and also bacteriology and virology. The branches include plant anatomy, plant chemistry, plant cytology, plant ecology including autecology and synecology, plant embryology, plant genetics, plant morphology, plant physiology, plant taxonomy, ethnobotany and paleobotany. The group of plants studied include agostology, the study of grasses; algology or phycology, the study of algae; bryology, the study of mosses; mycology, the study of fungi; and pteridology, the study of ferns. Botany, microbiology and zoology together constitute the science of biology. A scientist who studies botany is called a **botanist** and may specialise in any of the several branches. 2. The biological description of a plant or group of plants. It may also refer to the plant life of a given region.

BOTNET (ZOMBIE ARMY): A number of Internet computers that have been set up to forward transmissions including spam or viruses to other computers on the Internet without the knowledge of the owners. Most computers compromised in this way are home-based.

BOTRYTIS (Botrytis cinerea; BOTRYTIS BUNCH ROT; GREY MOULD): A fungal parasite affecting many plant species. Its most notable hosts may be wine grapes.

BOTS: The larval stage of the bot fly which lays its eggs on horses' forelegs in summer. It gives the horses restlessness.

BOTTLE GOURD (CALABASH GOURDS; *Lagenaria siceraria*): A hardy tropical and subtropical hardy, annual climbing plant in the family Cucurbitaceae, producing stems about 9 metres long. The plant does well in loamy soils and requires a lot of water in the growing season. The stems scramble over the ground and into the surrounding vegetation, attaching themselves to the plants by means of tendrils that grow out of the leaf axils. The gourds come in several variety of shapes: some are huge and rounded, slim and serpentine or small and bottled-shaped. The vine produces fruits which are either harvested young or mature and used as vegetable and for the hard wooden shell that can be used for making bottle, utensil, musical instruments, etc. The seeds contain oil, which is extracted and used for cooking. Medicinally, various parts of the plant have been used to treat many ailments such as fungal infections, boils, toothache, etc.

BOTTLED GAS: A term used for hydrocarbon gases such as propane or butane which are gaseous at standard temperature and pressure (STP) and have been compressed and stored in carbon steel, stainless steel, aluminium or composite bottles known as gas cylinders and used in camping stoves, blowtorches, etc. The atmospheric pressure held by the bottle is equivalent to the number of volumes of standard temperature and pressure of the gas held by the bottle for an ideal gas.

BOTTLENECK: 1. A phenomenon in which the performance or capacity of an entire system is limited by a single or limited number of components or resources. 2. Drastic short-term reductions in population size caused by natural disasters, disease or predators, which can lead to random changes in the population's gene pool. 3. A certain point in the central nervous system through which the passage of information is restricted.

BOTTOM-UP MODELS: A modelling approach that includes technological and engineering details in the analysis.

BOTULIFORM: A structure which is sausage-shaped.

BOTULINUM TOXIN: A nerve toxin produced by the bacterium *Clostridium botulinum*, which can cause fatal food poisoning.

BOTULISM: A type of severe food poisoning in badly preserved foods caused by botulinum toxin, a natural poison produced by the bacteria *Clostridium botulinum*. Botulinum toxin blocks motor nerves' ability to release acetylcholine, the neurotransmitter that relays nerve signals to muscles, resulting in paralysis.

BOUGAINVILLEA: Common name to thorny ornamental climbers or shrubs with colourful petals of flowers, which are modified leaves or bracts; belonging to the genus *Bougainvillea* of the four-o'clock family (Nyctaginaceae) with 18 species that grow between 1 - 12 metres. Bougainvillea is native to tropical South America and grown in subtropical and tropical climates as ornamental plants. The broad leaves are simple, ovate-acuminate and are alternately arranged. The flowers are white and tubular. Each flower is surrounded by three paper-like purple, red, yellow orange or pink bracts, with some cultivars having double bract structures and mixed colours. The flowers and bracts are long-lasting and grow on the end of new branches. There are three species in cultivation that produce many different varieties. The largest species is *Bougainvillea spectabilis*, a shrub or thorny, woody vine reaching up to about 12 metres tall and 7 metres wide. Its bracts vary in colour from magenta and purple to orange, white and yellow. *Bougainvillea glabra*, commonly known as paper flower or lesser bougainvillea is an evergreen, climbing woody vine, which is smaller than *B. spectabilis* and produces flower all along its stems rather than just at the ends. It also has fewer spines. *Bougainvillea peruviana* is a thorny, climbing vine with small, crinkly bracts that are smaller than those of *B. spectabilis* and *B. glabra*, has green stems and oval-shaped leaves. The flowers are yellow and tubular.

BOUGUER GRAVITY ANOMALY: A value that corrects the observed gravity for latitude and elevation variations, as in the free-air gravity anomaly, plus the mass of material above some datum (usually sea level) within the earth and topography.

B

BOUGUER'S LAW OF EXTINCTION (BEER-BOUGUER-LAMBERT LAW; LAMBERT'S LAW; BEER'S LAW): It defines the attenuation of an optical signal travelling through an optically homogeneous medium, based on propagation distance and an extinction coefficient specific to medium and signal. In other words, provided that temperature, pressure and composition are held constant, extinction is linear and independent of both the radiation intensity and amount of matter.

BOULDER: A rock fragment of over 25 cm in diameter. **Boulder clay** is a mixture of rock and powdered rock formed beneath a moving glacier as it drags rocks beneath it.

BOUNCE E-MAIL (BOUNCE MAIL): An electronic mail that is returned to the sender because it cannot be delivered for some reason and usually appears as a new note in your inbox. There are two kinds of bounce e-mail: (i) **Hard bounce e-mail** is permanently bounced back to the sender because the address is invalid. (ii) **Soft bounce e-mail** is recognised by the recipient's mail server but is returned to the sender because the recipient's mailbox is full, the mail server is temporarily unavailable or the recipient no longer has an e-mail account at that address.

BOUNCE RATE: The percentage of visitors to a given website that leave the site after viewing only a single page. A high bounce rate can be interpreted to mean either that the content is not relevant to its target market or that the site is not attracting its target market effectively.

BOUND: 1. In mathematics, it is a number that is greater than all the numbers in a given set of numbers, called **upper bound**, if the number is less than all the numbers in the given set, it is called **lower bound**. If the bound holds uniformly, usually for every member of a sequence, it is a **uniform bound**. 2. An element of an ordering that has the same ordering relation to all the members of a given subset; for example, since it is a subset of every set the empty set is a bound on any family of sets ordered by weak inclusion. 3. In logic, of a variable, occurring within the scope of a quantifier that indicates the degree of generality of the open sentence in which the variable occurs; for example, in $(x)(Fx \rightarrow Gxy)$, x is bound, but y is not.

BOUND CHARGE: The portion of the electrical charge on a conductor that does not escape to the earth when the conductor is grounded because of the inductive action of a neighbouring charge.

BOUND CURRENT DENSITY: The bound current density in a dielectric caused by a magnetisation, M.

BOUND ELECTRON: An electron whose wave function is negligible except in the vicinity of an atom.

BOUND FRACTION: The fraction of the incubation mixture, which after separation, contains the analyte bound to the binding reagent.

BOUND STATE: A system in which two or more parts are bound together that behaves as a single object, which requires energy to split them. An example of a bound state is a molecule formed from two or more atoms.

BOUND-BOUND TRANSITION: A change to the energy of an electron within an atom or more rarely within a molecule, in which the electron remains attached (bound) to the atom or molecule both before and after the change. When the energy is increased, a photon is absorbed. When the energy is reduced, a photon is emitted. Bound–bound transitions produce the emission and absorption lines found in stellar spectra.

BOUND-FREE TRANSITIONS: A change to the energy of an electron within an atom or a molecule in which the electron gains sufficient energy to escape. The electron goes from being bound to being free and leaves behind an ion; hence another name for ionisation. The energy for the change may come from a photon, resulting in the absorption bands known as **ionisation edges** in stellar spectra or from collisions with other atoms or particles, known as **collisional ionisation**. If the energy comes from another excited electron within the atom, the process is known as **autoionisation**.

BOUNDARY (FRONTIER): 1. The official line that divides one area of land from another in ecology. 2. The set of points (frontier-points) that are members both of the closure of a given set and of the closure of its complement. 3.The set of points in the closure but not in the interior of the given set, usually written $\mathrm{Fr}A$, for example, Fr⍰(0, 1)⍰ = {0, 1}. The frontier of the rationals is the set of all real numbers.

BOUNDARY CONDITION: A condition in which a quantity that varies throughout a given space or enclosure must fulfill at every point on the boundary of that space, especially when the velocity of a fluid at any point on the wall of a rigid conduit is necessarily parallel to the wall. Given the boundary conditions of an unknown function helps in its identification since other unknowns, such as variables in integrations, can then be eliminated.

BOUNDARY LAYER: 1. The thin layer of fluid formed around a solid body or surface relative to which the fluid is flowing. Adhesion between the molecules of the fluid and that of the body or surface causes the molecules of the fluid closest to the solid to be stationary relative to it. 2. A thin layer of air around the stomatal pore to prevent excessive loss of water vapour through transpiration.

BOUNDARY SLIP: A boundary condition used in fluid mechanics, which predicts a finite velocity of a fluid-solid interface adequate for molecular dimensions but not for macroscopic flow. The no-slip boundary condition for viscous fluids states that at a solid boundary, the fluid will have zero velocity relative to the boundary. The no-slip boundary condition is when a fluid flows over the surface of a solid and layers of liquid close to the solid are assumed to be stationary relative to the solid.

BOUNDARY SURFACE DIAGRAM: The diagram of the region containing a substantial amount of the electron density, about 90 percent, in an orbital.

BOUNDARY VALUE PROBLEM: A problem, such as the Dirichlet or Neumann problem, which involves finding the solution of a differential equation or system of differential equations which meets certain specified requirements, usually connected with physical conditions, for certain values of the independent variable. An ordinary differential equation or a partial differential equation given together with boundary conditions to ensure a unique solution.

BOUNDED: 1. of a set, having a bound, especially where there is a measure in terms of which all the elements of the set or the pairs of members, are less than some value or else all its members lie within some other well-defined set. In other words, a set of numbers is bounded if there is a number K so that the size of every element in the set is not greater than K. For example, the interval [3,7) is bounded. 2. Of an operator, function, etc., having a bounded set of values. A bounded real function must be bounded from both above and below.

BOUNDED ABOVE: Having an upper bound. For example, a function f is said to be bounded above, when there exists a real number K such that $f(x) \leq$ K; for all x in the domain of f.

BOUNDED AWAY FROM ZERO: Having a bound strictly greater than zero or correspondingly an upper bound strictly less than zero.

BOUNDED BELOW: Having a lower bound. For example, a function f is said to be bounded below, when there exists a real number K such that $f(x) \geq$ K; for all x in the domain of f.

BOUNDED VARIATION: The property of a real-valued function that its variation is bounded; it is then expressible as the difference of two monotone non-decreasing functions.

BOURDON GAUGE: An instrument used to measure fluid pressure. It consists of a pressure gauge containing a curved metal tube which tends to straighten when subject to internal pressure; this movement is translated into readings on a graduated dial.

BOVIDAE: The largest class of even-toed ungulates, including cattle, antelopes, sheep and goats with hollow unbranched permanent horns in males as well as females of most species. They generally live in open habitats such as grasslands, tundra and alpine regions. A member of this family is called a bovid. They are herbivores and usually have a four-chambered stomach that allows digestion with the help of special bacteria, protozoa and fungi.

BOVINE LEUCOSIS (ENZOOTIC BOVINE LEUKOSIS; EBL): A cancerous disease in cattle caused by the retrovirus, bovine leukaemia virus (BLV).

BOVINE SPONGIFORM ENCEPHALOPATHY, BSE (MAD COW DISEASE, MCD): A neuro-degenerative disease that affects cattle's nervous system making it behave abnormally. It is caused by an abnormal form of a cellular protein called prion that is a microorganism, many times smaller than virus. It causes a spongy degeneration in the brain and spinal cord as well as redness of the eyes. There is a human type of the disease, called Creutzfeldt-Jakob disease, believed to be caused by infection of prion.

BOW COMPASS (BOW SPRING COMPASS): A compass whose legs are joined by a flexible metal bow-shaped spring rather than by a hinge, the angle being adjusted by a screw.

BOWDEN CABLE: A device for the flexible transmission of small mechanical movements, generally to the controls of a machine. It consists of a strong steel cable (usually multi-stranded and lubricated) in a flexible outer tube, often made in the form of a long coiled spring. The outer tube is anchored at the ends so that the inner cable can move independently. Bowden cables are commonly used for throttle (accelerator), clutch, choke and sometimes brake controls on cars and motorcycles.

BOWEL: The two segments of intestine in mammals, the small intestine and the large intestine, which form the intestine, which is a segment of the alimentary canal extending from the stomach to the anus. In humans, the small intestine is subdivided into the duodenum, jejunum and ileum; and the large intestine is subdivided into the caecum and colon.

BOWEL MOVEMENT (DEFECATION): The final act of digestion by which organisms eliminate solid, semisolid or liquid waste material or faeces from the digestive tract via the anus. Waves of muscular contraction known as **peristalsis** in the walls of the colon move faecal matter through the digestive tract towards the rectum. Undigested food may also be expelled in this way. This process is called **egestion** or **diarrhoea**.

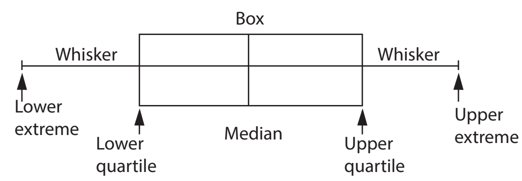
BOW RIDING: The movement of cetaceans, especially dolphins in the ocean, being propelled by the crests of waves, often created by boats and ships. The movements include activities such as swimming, leaping in the air, turning or drifting, etc., alongside the vessels.

BOWEL OEDEMA: A bacterial infection of pigs associated with sudden changes of diet and management. Pigs may become paralysed and in acute cases, piglets die.

BOWMAN'S CAPSULE (RENAL CAPSULE; CAPSULA GLOMERULI; GLOMERULAR CAPSULE): The cup-shaped sac of a kidney nephron that is involved in the initial stages of waste removal by filtration of waste substances and water from the blood and the formation of urine.

BOX: 1. A small evergreen shrub or tree of the genus Buxus, especially the Eurasian species *B. sempervirens*, having opposite, leathery, simple leaves and clusters of unisexual flowers with thick dark leaves used, especially for garden hedges. 2. A set in $\mathbb{R}^n$ consisting of the n-fold Cartesian product of intervals of the form $[a, b)$, $[a, \infty)$, $(-\infty, b)$ or $\mathbb{R}$.

BOX-AND-WHISKER-PLOT: A box diagram showing the distribution of a set of data along a number line, dividing the data into four parts using the median and s. The box-and-whisker plot does not show frequency and it does not display each individual statistic, but it clearly shows where the middle of the data lies. The various divisions are: **lower quartile (Q_1)**, which is the median of the lower half of the data set; **median (Q_2)**, which is the median of the entire data set; **upper quartile** (Q_3), the median of the upper half of the data set; and the **extreme values**, the smallest and largest values in a data set. Quartiles are defined as being the 25th, 50th and 75th percentiles that contain 25%, 50% and 75% of the data of a data set respectively.

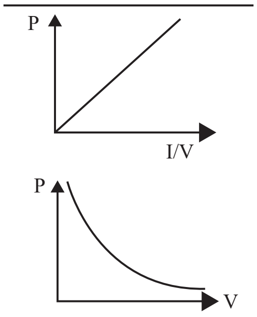


Box-And-Whisker-Plot

BOYLE-MARIOTTE LAW: An empirical law which states that the product of pressure *P* and volume *V* is constant for an isothermal process: PV = [constant].

BOYLE, ROBERT (1627-91): English chemist and physicist, born in Ireland, whose emphasis on experimentation and quantification helped lay the foundation for modern chemistry. He is noted for formulating Boyle's Law in 1662. After moving to Oxford in 1654, together with his assistant Robert Hooke, they devised an air pump to experiment with vacuums and the properties of gases, metals, combustion and sound, enabling him to prove that sound does not travel in a vacuum.

BOYLE'S LAW: One of the three ideal gas laws which states that the pressure of a fixed mass of gas at a constant temperature is inversely proportional to its volume. Mathematically, it is given by: $P \alpha \frac{1}{V}$ or $PV = k$, where k is a constant, it follows that $P_1V_1 = P_2V_2$. It was formulated in 1662 by Robert Boyle (1627–1691), an English chemist and physicist.



Graphical illustration of Boyle's law

BRACES: The symbols { } used for grouping or to represent a set. In other words it is used to indicate that the expressions between them must be evaluated and treated as a single unit. They are usually only used in an expression that already contains both parentheses ( ) and square brackets [ ] and the contents of the brackets or parentheses are evaluated before that of the braces. Brackets are also often employed for set definitions; thus one writes $[a, b, c)$ for the set whose members are a, b and c and $\{x: Fx\}$ for the class of elements that have the property F.

BRACHIAL PLEXUS: The collection of large nerve trunks that pass from the lower part of the cervical spine (in the neck) and the upper part of the thoracic spine (in the chest) down the arm. The nerve trunks of the brachial plexus divide into the musculocutaneous and axillary, median, ulnar and radial nerves which both control muscles in and receive sensations from the arm and hand.

BRACHIOPODA (BRACHIOPODS): A phylum of marine invertebrates, the lamp shells, characterised by a peculiar feeding organ, the iophophore. They live in shallow waters attached to a firm substratum by means of a flexible stalk and are protected by a bivalve shell consisting of dorsal and ventral valve.

BRACHISTOCHRONE (BRACHYSTOCHRONE PROBLEM): The classical and motivating problem of the calculus of variations that sought the path taken by a constrained weighted particle falling between two points not in the same vertical under gravity. The solution was discovered by Johann Bernoulli. It was a problem posed by Johann Bernoulli in Acta Eruditorum (the first scientific journal of the German lands, published from 1682 to 1782) in June, 1696 that given two points A and B in a vertical plane, what is the curve traced out by the point acted on only by gravity, which starts at A and reaches B in the shortest time?

BRACHYSCLEREID (STONE CELL BRACHYSCLEREID): A relatively short, more or less isodiametric, sclerenchyma cell (sclereid). Brachysclereids are usually present singly or in small groups, although they occur in large numbers in some tissues, such as the fleshy part of the pear fruit.

BRACKET: Any of the pair of symbols used to enclose a value or set of values or other expressions to indicate that their contents must be evaluated and treated as a single unit and to be evaluated before proceeding to other parts of the evaluation. For example, in the equation 8 x (3 + 5), 3 + 5 is evaluated first and the result multiplied by 8. Usually, the parenthesis with the symbol ( ) is treated before the square brackets [ ], the contents of the square brackets are evaluated before the braces, if an expression contains all the three. If the three provide insufficient levels of bracketing, a vinculum is used and its contents are treated first before the others. The angle brackets in the expression $\langle a_1, a_2, a_3 \rangle$ and the braces in the expression $\{a_1, a_2, a_3\}$ indicate that the terms between them are to be treated together as a sequence and a set respectively.

BRACHYDACTYLY: A human genetic disorder in which a child inherits shorter digits of both hands and sometimes the toes, relative to the length of other long bones and other parts of the body from one parent.

BRACKEN: A large genus of ferns distributed from the Arctic to the tropics common on hillsides and in woodland.

BRACKETT SERIES: A series of absorption or emission lines due to electron jumps between the fourth and higher energy levels of the hydrogen atom. These lines lie in the infrared with wavelengths from 4.05 microns (Brackett-alpha) to 1.46 microns (the series limit) and are named after the American physicist Frederick Brackett (1896–1980).

BRACKISH WATER: Slightly salty water of 3-22% salinity, which may occur at estuaries as a result of mixing of seawater with fresh water.

BRACT: Modified leaf that is often brightly coloured and showy, which may be mistaken for the petals of a flower due to their brightness, often positioned beneath a flower or inflorescence. Examples are found in bougainvillea.

BRACTEOLE: A reduced leaf that arises from the stalk of an individual flower.

BRACTEOSE: A structure or organ with many bract or conspicuous bracts.

BRADY METABOLISM: Metabolism, characteristic of cold-blooded animals that is sustained at a relatively low resting rate with a high active rate. Brady metabolic animals often undergo dramatic changes in metabolic speed, according to food availability and temperature.

BRAGG ANGLE: The angle between an incident beam of X-rays and a set of crystal planes for which the reflected or transmitted radiation displays a maximum intensity as a result of constructive interference. It is named after the physicist William Lawrence Bragg (1890–1971).

BRAGG CELL (ACOUSTO-OPTIC MODULATOR; ACOUSTOOPTICAL TUNABLE FILTERS): A device that varies the amplitude and phase of a light beam, for example, from a laser or by sound waves. Bragg Cells can replace radio/infrared filter banks by adding a piezoelectric transducer at one end of the cell to convert a downmixed radio signal into an acoustic vibration. They are completely tunable.

BRAGG PEAK (BRAGG SCATTERING): A peak in the scattering pattern in x-ray diffraction of a crystal. The intensity of Bragg peaks is proportional to the square of the number of scatterers.

BRAGG RAFT: A two-dimensional raft of bubbles that is used as an analogue of a three-dimensional atomic crystal lattice.

BRAGG'S LAW (BRAGG RELATION): A law first proposed by William Lawrence Bragg and his father, William Henry Bragg in 1913, which states that when a beam of x-rays with wavelength λ, strike a crystal surface in which the layer of atoms or ions are separated by a distance d, the maximum intensity of the reflected ray occurs when $\sin\theta = n\lambda/2d$, where θ, known as the Bragg angle is the angle between an incident x-ray beam and a set of crystal planes for which the secondary radiation displays maximum intensity as a result of constructive interference. The Bragg law is useful for measuring wavelengths and for determining the lattice spacings of crystals.

BRAILLE DISPLAY: A device, attachable to a computer keyboard that allows a blind person to read the contents of a display one text line at a time in the form of a line of Braille characters.

BRAIN: The enlarged anterior part of the vertebrate central nervous system, which is encased within the cranium of the skull that regulates behaviour as well as sensory and motor functions and the centre for intelligence and memory. It has three main parts: the cerebellum, the cerebrum and the medulla. It functions as the main coordinating centre for nervous activity. It receives information from sensory organs, makes decisions and controls muscles and other effectors.

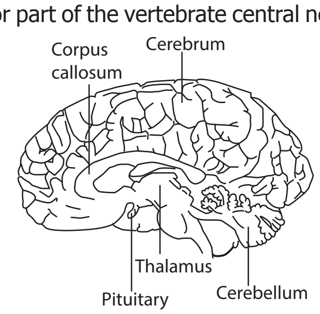


Fig. 1: Half section through a human brain

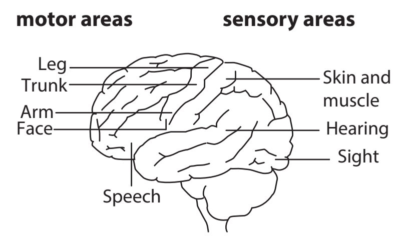


Fig. 2: Lateral view of external features of a human brain

BRAIN DEATH: The irreversible absence of all brain activity including involuntary activity necessary to sustain life due to total necrosis of the cerebral neurons following loss of blood flow and oxygenation. It is marked by cessation of breathing and other reflexes controlled by the brain stem and by a zero reading on an electroencephalogram.

BRAIN-COMPUTER INTERFACE (BCI): A collaboration between a brain and a device that enables signals from the brain to direct some external activity, such as control of a cursor or a prosthetic limb. The interface enables a direct communications pathway between the brain and the object to be controlled. In the case of cursor control, for example, the signal is transmitted directly from the brain to the mechanism directing the cursor, rather than taking the normal route through the body's neuromuscular system from the brain to the finger on a mouse.

BRAINSTEM: The lower part of the brain comprising the medulla oblongata, the midbrain and the pons.

BRAINSTEM DEATH: The irreversible loss of all functions of the brain, including the brainstem characterised by coma, absence of brainstem reflexes and inability to breathe. The brainstem is important in the control of many processes which are essential to life, such as breathing, heartbeat and blood pressure, so that the individual is unable to live without the aid of a ventilator. It may be caused by cardiac arrest, when the heart stops beating and the brain is starved of oxygen; heart attack, when the blood supply to the heart is suddenly blocked; stroke; blood clot in the vessel, impeding flow of blood around the body; by severe head injury; and by brain tumour.

BRAIRD: Fresh shoots of crops such as corn.

BRAKE: A device that slows down the speed or stops a moving body such as a rotating wheel in a vehicle or machine. The most common kind is friction brake, which operates on the principle that friction can be used to convert the mechanical energy of a moving object into heat energy, which is absorbed by the brake.

BRAKE HORSEPOWER: The actual or useful horsepower of an engine, usually determined from the force exerted on a friction brake or dynamometer connected to the drive shaft.

BRAKING DISTANCE: The distance travelled by a vehicle from the point where its brakes are fully applied to when it comes to a complete stop. It is affected by the original speed of the vehicle, its weight, the type of brake system in use and the coefficient of friction between its wheels and the road surface.

BRAN: Husk or covering of cereal grain when milled, which could be used to make dietary fibre.

BRANCH: 1. A stem growing out from the trunk or from a bough. Large branches are known as **boughs** and small branches are known as **twigs**. 2. A continuous section of a curve with an endpoint at which it is continuous but not differential meets another branch so that it is continuous but not differentiable at that point. 3. A continuous selection from an analytical set-valued function such as the logarithm. 4. A path in a tree that either is infinite or has an end point as its final member.

BRANCH POINT: 1. An intermediate in a biochemical pathway that can follow more than one route in subsequent steps. 2. A point at which one can switch from one branch of an analytic function to another.

BRANCH-AND-BOUND METHOD: A tree-based heuristic search method that avoids exhaustive search by using one branch of the tree to set a bound to the desired quantity and eliminating the other branches as soon as they contravene that bound.

BRANCHED CHAIN: Side group(s) attached to the main chain in the molecule of an organic compound.

BRANCHIAL: 1. Of, relating to or resembling the gills of a fish. Most fishes possess five pairs of branchial arches and skeletal support for the gills contained in the branchial chamber. 2. Pertaining to body structures of the face, neck and throat area, particularly the muscles.

BRANCHING: Growth extension of vascular plants. A branch develops from the stem and consists of the growth from the previous year (branchlet) and new growth (twig). New twigs are produced during the next season of growth, both from the terminal bud at the end of the twig and from lateral buds in leaf axils along it.

BRANCHING DECAY: Nuclear decay which can proceed in two or more different ways.

BRANCHING EVOLUTION (ADAPTIVE RADIATION; CLADOGENESIS; DIVERGENT EVOLUTION): The formation of several species with adaptations to different ways of life or evolution of different species in different directions from one from a parent species. This is likely to occur whenever members of a species migrate to exploit new habitats and food sources and new populations emerge, each adapted to its particular environment. Eventually, these become sufficiently distinct to be recognised as new species. Examples are the 13 species of Darwin's finches on the Galapagos Islands, each filling a different niche on different islands but evolved from one ancestral species. Also, early placental mammals, for example, gave rise to modern burrowing, climbing, flying, running and swimming organisms. The vertebrate limb is another example of divergent evolution. The limb in many different species has a common origin, but has diverged somewhat in overall structure and function.

BRANCHING FRACTION: In branching decay, the fraction of nuclei which decay in a specified way.

BRANCHING RATIO: The ratio of alternative products formed in a reaction. An example is the HF/DF ratio in the reaction between F and HD. When a particle decays, it often can decay in several ways. The likelihood of it decaying to a particular mode is known as its branching ratio for that decay mode.

BRIANCHON'S THEOREM: The theorem which states that if a hexagon is circumscribed about a conic then its main diagonals meet at a single point; this is the dual of Pascal's mystic hexagram theorem.

BRANDLING: A small reddish-brown earthworm, *Eisenia foetida* that is often used as bait by anglers.

BRANE: An object that is the analogue of a two-dimensional membrane embedded in three-dimensional space. In other words, an entity with a certain number of dimensions (one-brane, two-brane, three-brane, etc) embedded in the higher-dimensional space of string theory. A 0-brane is just a point particle, one-brane or 1-brane has one spatial dimension, a two-brane has two, etc.

BRANE WORLD: A theory in which the four space–time dimensions of the universe that are apparent make up a surface, called the brane, in a higher-dimensional space–time, called the bulk. One of the appealing features of the brane world is that it can explain why gravity is much weaker than the other forces, with the non-gravitational forces being localised to the brane and the gravitational force not limited to the brane. The theory predicts the existence of tiny black holes about the size of an atomic nucleus but have masses similar to that of a tiny asteroid, seeded throughout the universe, remnants of the Big Bang. Thousands of them should exist in our solar system. However, General Relativity, predicts that such primordial black holes evaporated long ago.

BRANS-DICKE THEORY OF GRAVITATION (JORDAN-BRANS-DICKE THEORY): A theoretical framework to explain gravitation, which modifies Einstein's general theory of relativity. It was developed in 1961 by Robert H. Dicke and Carl H. Brans that combined tensor and scalar qualities and allowed the gravitational constant G to vary with time in a way that is in accord with Mach's principle.

BRASH: Soil containing many stone particles or fragments of rock.

BRASS: An alloy of copper and zinc, which usually contains about 30% zinc. It is an attractive yellow golden coloured material but its colour generally varies from coppery to whitish yellow. It is used for making ornaments, electrical components, plumbing fittings, for applications where low friction is required such as locks, gears, bearings, ammunition and valves; screws and extensively in musical instruments such as horns and bells for its acoustic properties.

BRASSICA: A genus of plants such as cabbage, broccoli, cauliflowers, turnips and sprouts in the mustard family (*Brassicaceae*). This genus is remarkable as it contains more important agricultural and horticultural crops than any other genus.

BRASSIN (BRASSINOSTEROID): A group of steroidal plant hormones (polyoxygenated steroids) that produces plant growth-regulatory effects. They are released by mature cells and act as auxins to encourage leaf elongation and inhibit root growth. Brassinosteroids also protect plants from some insects because they work against some of the hormones that regulate insect moulting.

BRATTAI, WALTER (1902-1987): U.S. scientist born in Amoy, China and educated at Whitman College, at the University of Oregon and at Minnesota, where he obtained his PhD in 1929. He became a researcher at Bell Laboratories in 1929. With John Bardeen and William B. Shockley, he shared the 1956 Nobel Prize for Physics for the development of the transistor and for investigation of the properties of semiconductors.

BRATTING: The process of putting a jacket on sheep to protect them from cold in winter.

BRAUN, KARL FERDINAND (1850-1918): German physicist, inventor and Nobel Prize winner, born in Fulda, Germany, studied at Marburg and, in 1872, received a doctorate from the University of Berlin. In 1885 he became professor of experimental physics at Tübingen and in 1895, professor of physics at Strasbourg. Braun is best known for his invention of the first oscilloscope, providing the basic component of a television receiver. He also contributed much to the study of electricity and telegraphy or wireless communication. He shared the 1909 Nobel

Prize in physics with Italian electrical engineer and inventor Guglielmo Marconi for their work on wireless communication.

BRAVAIS LATTICE: An array of infinite lattice points in crystal systems generated by a set of discrete translation operations, named after Auguste Bravais. A Bravais lattice looks exactly the same no matter from which point one views it. In all, there are 14 possible Bravais lattices that fill three-dimensional space and which can be grouped into 7 crystal systems.

BRAZING: A process in metallurgy whereby two pieces of metal are joined using a nonferrous alloy (usually of copper) of lower melting point. The bonding alloy is either preplaced or fed into the joint as the parts are heated. The process is similar to soldering but is performed at higher temperatures (about 1000 K). Brazed joints are very reliable and are used extensively in the aerospace industry.

BREADTH (WIDTH): Distance or measurement from side to side indicating how wide or broad a given surface or region is.

BREAK: 1. The break of foam is the process involving the coalescence of gas bubbles. The flocculation of an emulsion, which is the formation of aggregates, may be followed by coalescence. If coalescence is extensive it leads to the formation of a macrophase and the emulsion is said to break. 2. To train a horse to wear a saddle.

BREAKDOWN: Disintegrate, decompose, cease to function, fail or collapse.

BREAKING STRESS: The mechanical stress required to fracture a material either by compression, tension or shear.

BREAST: 1. The upper ventral region of an animal's body, particularly that of mammals, including human beings. The breasts of female mammals contain the mammary glands, which contain milk producing cells and a network of tubes or ducts that lead to an opening in the nipple. Breast milk contains all the nourishment a baby needs including antibodies to help fight infection. Breasts are more visible on adult women, but male humans also have breasts which, although usually less prominent, are structurally identical to the female, as both develop embryologically from the same tissues. 2. A part of an animal, corresponding to the human breast, eaten as food.

BREASTBONE (STERNUM): The broad vertical bone located in the centre of the thorax to which the ribs are attached. It connects to the rib bones via cartilage, forming the anterior section of the rib cage and helps to protect the lungs, heart and major blood vessels.

BREATH TEST: A type of test performed on exhaled air. Most commonly, the test is used to measure the amount of ethanol drunk in alcoholic drinks by a person with the help a breathalyser. Above a certain limit, drivers, especially, are accused of committing an offence. Breath tests include the following: (i) **Hydrogen breath test**, a medical test for clinical diagnosis of dietary disabilities such as fructose intolerance, fructose malabsorption and lactulose intolerance. (ii) **Exhaled nitric oxide** is a breath test that might signal airway inflammation such as in asthma. (iii) **Urea breath test**, is a rapid diagnostic test used to identify infections by *Helicobacter pylori*, a bacterium implicated in peptic ulcer diseases. The test is based upon the ability of the bacterium to convert urea to ammonia.

BREATHING (EXTERNAL RESPIRATION): The muscular movements of an animal's body, which draws air into the lungs for the exchange of gases between the alveoli and the blood capillaries to take place.

BREATHING-RATE: The number of times per minute the lungs inhale and exhale.

BRECCIA: A rock composed of large coarse angular fragments with the spaces between them filled with a matrix of smaller particles and a mineral cement that binds the rock together. Breccias are formed in many different geologic processes including tectonic, volcanic, or sedimentary and from a variety of materials.

BREECHING: 1. A leather strap round the hindquarters of a shaft horse, which allows the horse to push backwards and so make the cart go backwards. 2. The passage or conduit through which the exhaust products of combustion are carried to the stack or chimney.

BREED: 1. To keep and take care of animals or plants in order to produce more animals or plants of a particular kind. 2. To produce young animals, birds; or to produce offspring by sexual reproduction. 3. A group of domesticated animals or plants having similar physical characteristics distinct from others of the same species. These characteristics have been produced by artificial selection. When bred together, animals of the same breed pass on these uniform traits to their offspring. For example, **merino** is a breed of sheep. The offspring produced as a result of breeding animals of one breed with other animals of another breed are known as **crossbreeds** or **mixed breeds**. Plant breeds are more commonly known as cultivars. Crosses between animal or plant variants above the level of breed/cultivar are referred to as hybrids. **Breeding** is the crossing and selection of animals and plants to change the characteristics of an existing breed (variety) or to produce a new animal, usually by artificial selection.

BREED IN: To introduce a characteristic into an animal breed or plant variety, by breeding until it is a permanent characteristic of the breed.

BREEDER REACTOR: A nuclear reactor that produces more fissionable materials than it uses. The reactor is built with a core of fissionable plutonium, Pu-239 and surrounded by a layer of U-238. As the Pu-239 undergoes spontaneous fission, it releases neutrons which convert U-238 to Pu-239. After all the U-238 has been changed to Pu-239, the reactor is refuelled. Some disadvantages of this breeder reactor are: (i) Pu-239 is very toxic; (ii) the half-life of U-239 is extremely long, about 24,000 years; this could create an almost impossible disposal problem if large amounts of this material are generated; (iii) because of the nature of the reactor core, water cannot be used as a coolant, so liquid sodium must be used. In the event of an accident a catastrophe could develop because sodium reacts violently with water and air.

BREEDING CRATE: A box designed to take the weight of a bull, when mating with a heifer or cow.

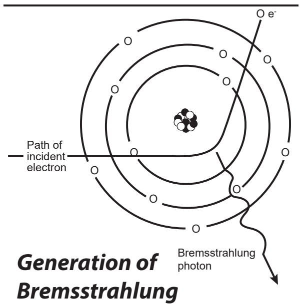
BREEDING GROUND: 1. A place or set of circumstances suitable for or favourable to growth and development, for example for bacteria or mosquitoes. 2. An area where birds or animals migrate to each year to breed. Breeding grounds offer migratory birds and animals the advantage to breed in an area of available food and favourable weather conditions in certain areas during very limited time periods. Many animals and bird populations migrate long distances each year to reach a favourable breeding ground. A **rookery** is a breeding ground or communal living area of some species of gregarious birds or mammals, such as penguins or seals.

BREEDING SEASON (MATING SEASON): A specific season of the year, in which many animals including mammals and birds mate. This ensures that offspring are produced only at a certain time of the year with favourable conditions such as abundance of food for the growth of the young.

BREEDING SYSTEM: The various interbreeding mechanisms within a population. In animals, these include systems such as polygyny (males in the population mate with more than one female); polyandry (some females mate with more than one male during the breeding season); monogamy in which both males and females have only one mate at a time; outcrossing (mating of individuals that are not genetically related); etc. In plants the various means of breeding include outcrossing (cross-fertilisation), autogamy (self-fertilisation) and apomixis (asexual reproduction without fertilisation).

BREEZE: 1. A term sometimes used to describe very fine particles of coke. 2. A wind ranging in strength from light to moderate, with a speed of between 6 - 50 kilometres per hour.

BREMSSTRAHLUNG: Electromagnetic radiation produced when fast moving charged particles are deflected or blocked. Bremsstrahlung is a continuous spectrum, which increases in intensity with increase in the energy



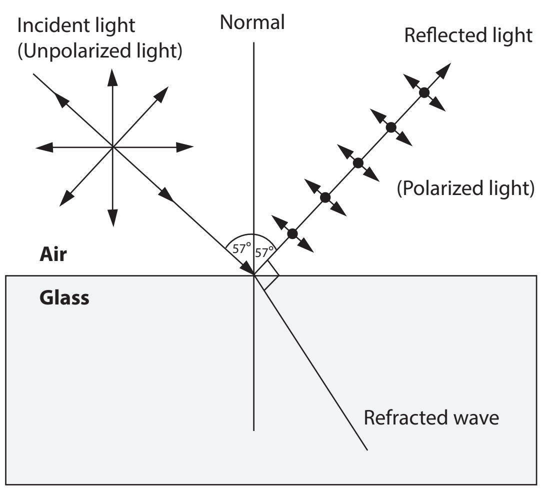
Generation of Bremsstrahlung

of the deflected particles. The phenomenon was discovered by Nikola Tesla during high frequency research he conducted between 1888 and 1897.

BREWING: A process, which involves the fermentation of sugars to alcohol or ethanol or the making of beer and other alcoholic drinks from dried, germinating barley (malt) by steeping (mashing), boiling and fermenting.

BREWSTER'S ANGLE (POLARISATION ANGLE; POLARISATION ANGLE): An angle of incidence (57°) at which maximum polarisation occurs. In other words, it is an angle of incidence of light reflected from a dielectric surface such as glass, at which maximum polarisation occurs. When unpolarised light is incident at this angle, the light that is reflected from the surface is therefore, perfectly polarised. At other angles the reflected light is partially polarised. It is named after the Scottish physicist, Sir David Brewster (1781–1868).

BREWSTER'S LAW: A law named after a Scottish physicist, Sir David Brewster, who first proposed it in 1811. It states that the extent of polarisation of light reflected from a transparent surface is at maximum when the reflected ray is at right angle to the refracted ray. It relates the refractive index n of a substance to the Brewster angle (ϕ) by: $n = \tan\phi$. It defines how an unpolarised light can be plane polarised, using a rectangular block.



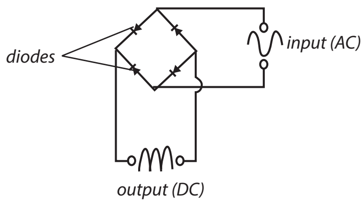
Brewster's Law

BRIDGE: 1. In chemistry, it is a valence bond or an atom or an unbranched chain of atoms connecting two different parts of a molecule. The two tertiary carbon atoms connected through the bridge are termed bridgeheads. 2. In Botany, it is a band of tissue connecting the corolla scales, as occurs in *Cuscuta* (dodder), a rapidly growing, twining, parasitic vine, without chlorophyll. 3. A structure built over a physical barrier such as a body of water, a valley or a road to give passage for pedestrians, vehicles of all kinds, etc. In building bridges, the following forces and how they are distributed through the structure must be taken into account: tension, compression, bending, torsion and shear. Some types of bridges include the following: (i) **Bascule bridge** with a movable span called the leaf that rotates on a horizontal hinged axis called the trunnion to raise one end vertically; (ii) **Box girder bridge** in which the main beams comprise girders in the shape of a hollow box; (iii) **Beam bridge**, built of horizontal beams supported with ends resting on piers or abutments and either classified as a short-span or long-span beam bridge; (iv). **Suspension bridges**, which are suspended from cables; etc. Another type is the **viaduct**, which is made up of multiple bridges, typically with a series of arches, connected into one longer structure, to carry transport or vehicles over a wide valley, a gorge or such obstacles. 4. The upper bony ridge of the human nose. 5. Also known as **bridgework**, it is a removable replacement for one or several but not all of the natural teeth and secured to the surrounding natural teeth. 6. In nautical terms, it is a crosswise platform or enclosed area above the main deck of a ship from which the ship is controlled.

BRIDGE CIRCUIT: An electrical circuit composed of four impedances or resistances connected in series to form a rectangle, with inputs and outputs at pairs of opposite corners. It is used in measuring resistors, capacitors, inductors and in converting signals from transducers to voltage or current signals. An example of a bridge circuit is the Wheatstone bridge, constructed from four resistors, two of known values R_1 and R_3, Rx whose resistance is unknown and to be determined and R_2, whose resistance is variable and calibrated.

BRIDGE INDEX: In a coordination entity, the number of central atoms linked by a particular bridging ligand.

BRIDGE RECTIFIER: A series-parallel arrangement of four or more diodes connected as a bridge circuit. Its output polarity (DC) remains unchanged. It is used to convert an alternating current (AC) input into direct current (DC) output. A bridge rectifier provides full-wave rectification from a two-wire AC input.



A bridge rectifier circuit diagram

BRIDGE SOLUTION: Solution of high concentration of inert salt, preferably comprising cations and anions of equal mobility, optionally interposed between the reference electrode filling and both the test and standard solution, when the test solution and filling solution are chemically incompatible. This procedure introduces into the operational cell a second liquid junction formed usually in a similar way to the first. This definition refers to a bridge solution of a double junction reference electrode.

BRIDGING LIGAND: A ligand attached to two or more, usually metallic, central atoms.

BRIDGMAN EFFECT (INTERNAL PELTIER EFFECT): The phenomenon that when an electric current passes through an anisotropic crystal, there is an absorption or liberation of heat due to the nonuniformity in current distribution.

BRIDLE: The part of a horse's harness, which goes over the head, including the bit and rein.

BRIDLE PATH (BRIDLE WAY): A track along which a horse can be ridden.

BRIGHTENERS (OPTICAL BRIGHTENERS; OPTICAL BRIGHTENING AGENTS, OBAS; FLUORESCENT BRIGHTENING AGENTS, FBAs; FLUORESCENT WHITENING AGENTS, FWAs): Dyes that absorb light in the ultraviolet and violet region (usually 340-370 nm) of the electromagnetic spectrum and re-emit it in the blue region (typically 420-470 nm) and added to detergents or used to treat textiles or paper in order to brighten the colours or, particularly, to enhance whiteness. Blueing agents are used in laundries to give a slight blue cast to white material in order to counteract yellowing.

BRIGHTNESS: The amount of light or other radiation coming from a body. It is the luminance of a body, apart from its hue or saturation, that an observer uses to determine the comparative luminance of another body. Pure white has the maximum brightness and pure black the minimum brightness.

BRIGHTNESS TEMPERATURE: The apparent temperature of the surface assuming a surface emissivity of 1.0. This concept is extensively used in radio astronomy and planetary science. The brightness temperature is obtained by applying the inverse of the Planck function to the measured radiation. Depending on the nature of the source of radiation and any subsequent absorption, the brightness temperature may be independent of or highly dependent on, the wavelength of the radiation.

BRIGHTNESS WAVE: The phenomenom, which results when a modulated laser beam is sent through a cloudy liquid, producing a chain of dark and light patches which acts like an ordinary wave with a characteristic wavelength.

BRILLES: A transparent, immovable, disc-shaped layer of scale or skin that covers the eyes of some creatures, such as snakes and lizards that provides protection. As such they cannot close their eyes.

BRIN PROCESS: A process formerly used for making oxygen by heating barium oxide in air to form barium peroxide and then heating the peroxide at higher temperature to produce oxygen:
$2BaO + O_2 \rightarrow 2BaO_2$; $2BaO_2 \rightarrow 2BaO + O_2$.

BRINE: 1. A saturated solution of sodium chloride in water. It is used in the food processing industry to preserve vegetables, fish and meat. 2. A liquid used in a refrigeration system, usually an aqueous solution of calcium chloride or sodium chloride, which is cooled by contact with the evaporator surface and then goes to the space to be refrigerated.

BRINELL HARDNESS: A scale for measuring the hardness of metals, introduced around 1900 by the Swedish metallurgist, Johann Brinell (1849–1925). A small chromium steel ball is pressed into the surface of the metal by a load of known weight. The ratio of the mass of the load in kilograms to the area of the depression formed in square millimetres is the Brinell number.

BRISKET: The lower front part of the body of a cow, between the front legs.

BRISTLE: 1. A stiff hair or feather. In birds, they are situated near the mouth or eyes. 2. A short stiff hair, wire, plastic, etc., attached to a handle that makes a brush or broom.

BRITANNIA METAL: A silvery alloy consisting of 80-90% tin, 5-15% antimony and sometimes small percentages of copper, lead and zinc. It is used widely in the manufacture of tableware.

BRITISH IMPERIAL SYSTEM (IMPERIAL SYSTEM): A system of weights and measures originally developed in England that is largely based on the foot, pound and second (f.p.s.) system. It is similar to but not always the same as US standard units. In volume measurements, for example, one fluid once in the imperial system is equal to 28.413 ml and in the U.S. system it is equal to 29.57 ml.

BRITISH THERMAL UNIT (BTU): The Imperial unit of heat, being originally the heat required to raise the temperature of 1 lb of water by 1°F. One BTU is now defined as 1055.06 joules, 107.5 kilogram-metres or 0.0002928 kilowatt-hours.

BRITTLENESS: The tendency of a material such as ceramic and glass to fracture or break when subjected to stress or on the application of a relatively small amount of force, impact or shock. Brittle materials absorb relatively little energy prior to fracture, even those of high strength. Rubbers are able to withstand large amounts of deformation without breaking as well as being able to return to their original shape after deformation.

BROAD BEAN (FAVA BEAN; HORSE BEAN; BELL BEAN; ORTIC BEAN): An edible bean, *Vicia faba* with large flat cotyledons of the Papilionaceae family, which grows in most soils and climates. It is a rigid, erect plant 0.3-2 metres tall, with stout stems of a square cross-section. The leaves are pinnately compound with usually 6 large broad leaflets and do not have tendrils for climbing over other vegetation. The flowers are large, white with dark purple markings, borne on short pedicels in clusters of 1-5 in axils of leaves. The pod is 8-20 cm long, 10-30 mm broad, flattened, oblong, obliquely accuminate at both ends and leathery, with green colour maturing to blackish-brown. Each pod contains 3-8 seeds. It is cultivated as a vegetable, for human and livestock nutrition and crop rotation. It is a good source of carbohydrate and protein while containing a low amount of fats.

BROAD NETWORK ACCESS: Resources hosted in a private cloud network that are available for access from a wide range of devices, such as tablets, PCs, Macs and smartphones. These resources are also accessible from a wide range of locations that offer online access.

BROADBAND: Telecommunications across a wide range of frequencies or multiple channels. In particular, it can refer to evolving digital telephone technologies that offer integrated access to voice, high-speed data services and interactive information delivery services.

BROADBAND DIALUP (HIGH-SPEED DIALUP): An Internet service provider (ISP) feature that speeds up data transfer by using a special server, called an acceleration server, to act as a bridge between the user's dialup connection and a Web page.

BROADBAND GLOBAL AREA NETWORK (BGAN): A mobile communications system created to transmit broadband wireless voice and data communications almost anywhere on the earth's surface. It consists of a constellation of geostationary satellites working in conjunction with portable, lightweight, surface-based terminals about the size of a laptop.

BROADCAST: 1. In telecommunications, it is a type of communication in which information is sent from a single source to numerous points that are connected to the source. It is used in radio and Tv transmissions and is supported on most LANs such as Ethernet and may be used to send the same message to all computers on the LAN. 2. In agriculture, to scatter seeds over a relatively large surface of the ground rather than sowing them in rows. It is also to spread fertiliser on a relatively large surface of the soil, or to apply herbicides over a cropped or planted area.

BROADCAST ADDRESS: A special Internet Protocol (IP) address used to transmit messages and data packets to network systems. Network administrators (NA) verify successful data packet transmission via broadcast addresses.

BROADCAST DOMAIN: A logical part or division of a computer network, which are located within a network or multi-network segment. In a broadcast domain, all the nodes can be reached via broadcast at the datalink layer. Multi-network segments require a bridge, such as the networking device.

BROCHUREWARE: Web sites or pages that are produced by taking an organisation's printed brochure and translating it directly to the Web without regard for the possibilities of the new medium. In extreme cases, all the copy in the brochure will be used as-is and visual images will be copied as well.

BROILER: Chicken grown for about 8 weeks old, weighing about 1.4 to 1.6 kg and slaughtered for meat. Broilers typically have white feathers and yellowish skin.

BROKEN SYMMETRY: A situation in which the ground state of a many-body system or vacuum state of a relativistic quantum field theory has a lower symmetry than the Hamiltonian or Langrangian defining the system.

BROMIC ACID (HYDROGEN BROMATE; BROMIC(V) ACID): A colourless liquid with the chemical formula $HBrO_3$ and molar mass 128.91 g/mol that turns yellow at room temperature as it decomposes to bromine. It is made by adding sulfuric acid to barium bromate. It is an oxoacid and exists ony in aqueous solution. It is a powerful oxidising agent.

BROMIDE: A compound of bromine and another more electropositive element. Metal bromides are usually ionic solids, for example, sodium bromide; while non-metal bromides are usually covalent compounds such as hydrogen bromide.

BROMIDE PAPER: A photographic paper coated with an emulsion containing silver bromide. It is used for making black-and-white prints and enlargements.

BROMINATION: A chemical reaction in which a bromine atom is introduced into a molecule.

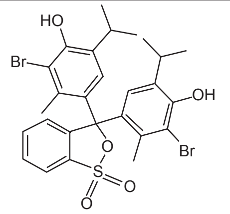
BROMINE: A volatile red or dark reddish-brown liquid (at room temperature), non-metallic element that is a member of Group 17 (VIIA) of the Periodic Table, the halogens group. Its symbol is Br, atomic number 35, atomic weight 79.909, melting point -7.3 °C (265.8 K) and boiling point 59 °C (332 K). It is the only nonmetallic element which is liquid at room temperature. It is very soluble in water or carbon disulfide, forming a red solution. It bonds easily with many elements and has a strong bleaching action. Bromine occurs naturally in the earth's crust and in seawater in various chemical forms. The main commercial source of bromine is ocean water, from which the element is extracted by means of chemical displacement (oxidation) by chlorine in the presence of sulfuric acid. Its compounds are used in photography, in the chemical and pharmaceutical industries, in agriculture, in sanitation and as fire retardants.

BROMINE WATER: A reddish-brown solution of bromine in water. It is mainly used as a reagent to test for unsaturation in organic compounds. The unsaturated organic compounds decolourise the solution as they react with the bromine.

BROMOETHANE (ETHYL BROMIDE): A colourless flammable liquid with the chemical formula C_2H_5Br, density 1.47 g/ml and boiling point 39 °C (312.15 K). It is prepared from ethane and hydrogen bromide and used as a refrigerant.

BROMOMETHANE (METHYL BROMIDE): An organobromine compound, with the chemical formula CH_3Br, density 3.974 mg mL^{-1} at 20 °C (293 K), melting point -93.66 °C (179.49 K), and boiling point 3 °C (276.6 K). It is a colourless, volatile, non-flammable liquid. It is produced both industrially and biologically. It was used extensively as a pesticide until it was phased out in the early 2000s, when it was discovered that it could deplete the ozone layer.

BROMOTHYMOL BLUE (BTB; DIBROMOTHYMOL; SULFONEPHTHALEIN; BROMTHYMOL BLUE): A chemical compound used as an acid-base indicator. It is yellow in acidic solutions and blue in alkaline solutions but green in neutral solution. Colour change from yellow to blue occurs between pH 6 and 8. BTB is also used as a biological slide stain to define cell walls or nuclei under the microscope.



Chemical Structure of Bromothymol Blue

BRONCHIAL-ASSOCIATED LYMPHOID TISSUE (BALT): An organised collection of lymphocytes and lymphoid tissue found in the walls of the airways, where it protects the respiratory epithelium. Each BALT aggregation comprises B and T cells, which are crucial in inducing an immune response to invading micro-organisms and other antigens.

BRONCHIOLE (BRONCHIOLI): A fine, narrow air or respiratory tube in the lungs of reptiles, birds and mammals formed by the sub division of a bronchus. The bronchioles terminate by entering the circular sacs called alveoli.

BRONCHITIS: An inflammation of the mucous membrane inside the bronchial tubes.

BRONCHUS: One of a pair of major air tubes splitting off from the trachea (windpipe) and passing into the lung. It conducts air from the windpipe in to and out of the lung. Its walls are made stiff by rings of cartilage.

BRÖNSTED-LOWRY ACID (BRÖNSTED ACID): A substance capable of donating a proton, for example, hydrogen fluoride (HF), nitric acid (HNO_3) and sulfuric acid (H_2SO_4).

BRÖNSTED-LOWRY BASE (BRÖNSTED BASE): A substance capable of accepting a proton, for example, ammonia (NH_3), fluoride (F^-) and hydroxide (OH^-).

BRÖNSTED-LOWRY THEORY: A concept of acids and bases in which an acid is defined as a substance with a tendency to donate a proton (H^+) and a base as a substance with a tendency to gain a proton. It was named after J. N. Brønsted (1879-1947), a Danish chemist and Thomas Lowry (1874-1936), a British chemist who first proposed the idea.

BRONZE: An alloy of copper, tin, zinc, phosphorus and sometimes small amounts of other elements. It is much stronger than copper and used mainly for gear wheels, engine bearings and other machine parts as well as for bells and medals.

BROOD: 1. All the young birds or other animals produced at one hatching or birth. 2. To sit on eggs to hatch them.

BROOD PARASITISM: The method adopted by one species of organism, especially birds to lay their eggs in the nests of another (the host) and make them responsible for raising the parasite birds' chicks. There are two different types of brood parasitism: facultative parasitism and obligate parasitism. In **facultative parasitism**, the species of birds do not entirely rely on parasitism as they can build their own nests and lay their eggs in other nests only sometimes. Examples of such birds are the bank swallow, and the African weaver. In **obligate parasitism**, the birds cannot build their own nests and rely completely on brood parasitism to raise their young. Examples include African honeyguides, and some species of cuckoos.

BROOD PATCH (INCUBATION PATCH): An area of bare, featherless skin located on the lower abdomen of, especially female birds, consisting of a network of blood vessels in the skin that develops by the shedding of feathers and the consequent thickening of the skin, which serves to transfer warmth from the adult's body to incubate the eggs and keep the young brood of chicks warm.

BROOD REDUCTION: Elimination of the smaller and weaker chicks in a brood by larger ones as a result of competition for inadequate food, or psychological and physical intimidation. Brood reduction can occur through starvation of the weaker chicks or when the larger chick kills and devours the smaller sibling.

BROODER: A container with a source of heat, used for housing newly hatched chicks.

BROODING: 1. In natural brooding, it the lying on eggs that have already been laid by a female bird to provide them with warmth at the right tempeperature in order to incubate them. 2. The practice of protecting and providing warmth to a brood of chicks when they are unable to do so themselves. This can be done naturally or by artificially. In natural brooding, the hen takes care of the chicks. In artificial brooding, such as in poultry keeping, the keeper is responsible for making provisions for the brooding of the chicks by sheltering them, providing and regulating heating to maintain the right temperature, etc.

BROOM: In botany, any of the various brushes of evergreen, semi-evergreen and deciduous shrubs in the subfamily Faboideae of the legume family Fabaceae, which are natives of Europe, north Africa and southwest Asia. They form a tribe, Genisteae, mainly in the three genera Chamaecytisus, Cytisus and Genista, but also in many other small genera. These genera are all closely related and share similar characteristics of dense, slender green stems and very small leaves, which are adaptations to dry growing conditions. Most of the species have yellow flowers, but a few have white orange, red, pink or purple flowers.

BROUTER (BRIDGE ROUTER): A network device that works as a bridge and as a router. The brouter routes packets for known protocols and simply forwards all other packets just as a bridge. A bridge connects one local area network (LAN) to another local area network that uses the same protocol such as Ethernet or token ring. A router connects a network to one or more other networks that are usually part of a wide area network (WAN) and may offer a number of paths out to destinations on those networks.

BROUWER'S THEOREM: A fixed-point theorem stating that a continuous mapping of a compact convex set into itself has a fixed point, for example, in the complex numbers any mapping of the unit disk to itself has a fixed point. This theorem was shown by Schauder and Tychonoff to remain valid for a normed space or a locally convex space. It was named after Dutch mathematician and founder of modern topology, Luitzen Brouwer.

BROWN SOILS: A zonal group of soils in the Chernozemic order having a brown surface horizon found in temperate to cool semiarid climates between 35° and 55° north of the Equator in western and central Europe. It has temperatures of 4°C in the winter to 18°C in the summer. The most common vegetation is deciduous woodland and grassland. Because of its natural fertility, brown earth lands are being converted into growing crops. The brown colour is caused by thin coatings of iron oxides weathered from the parent material.

BROWN ALGAE: Organisms belonging to the phylum Phaeophyta of the kingdom Protista, which are mostly marine. They include the largest of the seaweeds and the kelps. The brown algae have chlorophyll *a* and *c*, as well as carotenes and xanthophylls, including the brown pigment fucoxanthin which masks the green chlorophyll colouration.

BROWN DWARF: An astronomical object with a mass intermediate between the mass of a planet and that of a small star, which is not enough to produce the internal heat that in stars ignites hydrogen and establishes nuclear fusion.

BROWN EARTH: A type of soil that is characteristic of mid-latitude parts of the world (30°- 55° north of the equator) that were originally covered with deciduous woodland. The soil is well-drained and fertile, with a pH of between 5.0 and 6.5.

BROWN FAT (BROWN ADIPOSE TISSUE, BAT): A darker coloured region of adipose tissue found in new born and hibernating animals. Its primary function is to generate body heat in animals or newborns so that they do not shiver. Brown fat cells contain a higher number of mitochondria which contain iron hence the brown colour and more capillaries than white fat cells for more oxygen.

BROWN RING TEST: A test used to determine the presence of nitrates. It is performed by adding iron(II) sulfate to a solution of a nitrate, then slowly adding concentrated sulfuric acid such that the sulfuric acid forms a layer below the aqueous solution. A brown ring will form at the junction of the two layers, indicating the presence of the nitrate ion.

BROWNIAN MOTION: The continuous random movement of microscopic solid particles suspended in a fluid medium, due to the bombardment of the particles by the continually moving molecules of the liquid. It provides evidence of the kinetic theory of matter (Brownian motion) and it is named after Robert Brown, a Scottish botanist.

BROWNOUT RESET: A circuit that causes a computer processor to reset or reboot in the event of a brownout, which is a significant drop in the power supply output voltage. It prevents the processor from malfunctioning or locking up.

BROWSER (WEB BROWSER): An application program that provides a way to look at and interact with all the information on the World Wide Web. It uses HTTP (Hypertext Transfer Protocol) to make requests of Web servers throughout the Internet on behalf of the browser user. Most browsers support e-mail and the File Transfer Protocol (FTP) but a Web browser is not required for those Internet protocols and more specialised client programs are more popular. Some common browsers are Microsoft Internet Explorer, Netscape Communicator and Apple Safari.

BROWSER HIJACKER (HIJACKWARE): A spy or malware, commonly available as a Web browser add-on, that changes a Web user's browser settings in order to change the default home, error or search page. It may redirect the user to unwanted websites while capturing sensitive private data for personal or business gain. When a browser hijacker is not removed, the infected Web browser may default to the malware browser settings after every reboot, even if the settings are manually readjusted.

BRUCELLOSIS (UNDULANT FEVER; MALTA FEVER): Chronic infectious disease of some domestic animals that can be transmitted to human beings through contaminated milk or eating of raw minced meat. The common cause of the infection is the bacterium *Brucella* species.

BRUCITE: A mineral form of magnesium hydroxide, with the chemical formula $Mg(OH)_2$, which crystallises in the trigonal system. Brucite occurs as tabular crystals and as elongated fibres with a hardness of 2.5 on the Moh's scale. Structurally, brucite is a member of the $Cd(OH)_2$ structure type consisting of hexagonal close-packed O atoms with alternate octahedral layers occupied by Mg. The "brucite layer" is an important structural component in the clay and mica mineral groups. Brucite is often used as flame retardant because it thermally decomposes to release water in a similar way to aluminium hydroxide and mixtures of huntite and hydromagnesite. It is a minor ore of magnesium metal and a source of magnesia. It is also used as an additive in certain refractories, as a filler in plastic and paper-making industries, as additive in chemical fertiliser, as furnace additive in steel-making industry.

BRUCK-RYSER-CHOWLA THEOREM: The theorem, proved in 1950, that if D is a symmetric block design on v points, with k points belonging to each block and with each pair of points contained in λ blocks, then if v is even, v - λ is square and if v is odd then the equation $x^2 = (k-\lambda)y^2 + (-1)^{\left[\frac{v-1}{2}\right]}\lambda z^2$. has no non-trivial solution. The converse has not been proved.

BRUNISOLIC: An order of soils with brownish B horizon without much clay whose horizons are developed sufficiently but does not meet the requirements of any of the other soil orders. These soils occur under a wide variety of climatic and vegetative conditions.

BRUNNER'S GLANDS (DUODENAL GLANDS): Any of the small, branched, coiled tubular glands situated deeply in the submucosa of the duodenum that secrete mucus and an alkaline fluid that neutralises the acidic chyme leaving the stomach.

BRUSH: 1. A device that conducts current between stationary wires and moving parts, most commonly in a rotating shaft. It is made of a specially prepared form of carbon and is kept in contact with the moving part by means of a spring. Typical areas of applications include electric motors, alternators and electric generators. 2. A land, which is covered with a dense undergrowth of small trees and bushes. 3. A wild and sparsely populated woodland. 4. An implement with brittles of hair used for scrubbing, sweeping cleaning, painting, tiding the hair etc.

BRUSH BORDER (STRIATED BORDER): A region of surface epithelium that possesses densely packed microvilli, rather like the bristle of a brush. Brush border cells are found in the small intestine tract and in the kidney.

BRUSH DISCHARGE: A luminous discharge from a conductor that forms a luminous branching that penetrates into the surrounding gas. It is a form of corona discharge and it occurs when the electric field near the surface of the conductor exceeds a certain value but is not sufficiently high for a spark to appear. Brush discharges can occur from charged insulating plastics, for example, polythene to a conductor.

BRUSSELATOR: A theoretical model for a type of autocatalytic chemical reaction mechanism that leads to an oscillating reaction involving the conversion of reactants A and B into products C and B by a series of 4 steps: A $\rightarrow x$; $2x + Y \rightarrow 3Y$; B + $X \rightarrow Y + D$; and $X \rightarrow E$.

BRYOPHYLLUM (KALANCHOE): A genus in the Crassulaceae (orpine or stonecrop) family containing about 40 species of mainly drought tolerant, succulent, green vegetative herbs or shrubs, native to Madagascar, South Africa and grown in many parts of the world as an ornamental plant. Plants in the genus have green and fleshy leaves and stems that are covered with shiny layer of wax, that prevents water loss by evaporation. The plants have crassulacean acid metabolism, a system in which carbon dioxide (CO_2) is taken up only at the cooler and more humid night and stored as a four-carbon acid, malate that is released during the day-time to provide the enzyme, rubisco with high concentration of CO_2 to carry out photosynthesis. During the hot daytime, the stomata close to reduce water loss through evapotranspiration. The plants develop tiny plantlets in the notches in the leaves, which are miniatures of the mother plant due to mitotic cell division of the meristematic tissues in the notches. These plantlets drop off easily to the soil, where they quickly develop roots and grow into independent plants. The plants contain isocitric acid, malic acid, free tartaric acid, bufadienolides, alkaloids, calcium oxalate, flavonoids, anthocyans, tannins and considered poisonous to humans and animals. The plants are considered invasive, noxious weed in some countries with negative effect on biodiversity. An example is *Bryophyllum delagoense*, commonly known as mother-of-millions, chandelier plant or finger plant, a long-lived plant with upright stem that grows up to about 30-180 centimetres tall. It is grown in many parts of the world as an ornamental plant.

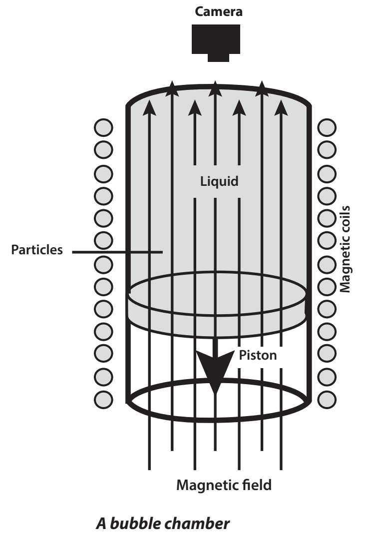
BRYOPHYTA: A division of plants such as mosses and liverworts possessing no vascular tissue and rudimentary root-like organs or rhizoids. They grow in a variety of damp habitats, from rock surfaces to fresh water. Such plants commonly referred to as Bryophytes have separate gamete-bearing (gametophyte) and spore-bearing (sporophyte) forms with the gamete-bearing phase. In liverworts and mosses, the gametophyte stage is the larger and familiar form of the plant, with the sporophyte stage being smaller and found growing on the gametophyte stage. In angiosperms, the sporophyte phase is the larger and independent phase, with the gametophyte phase being small and reduced to pollen grain and an eight-celled female gametophyte situated inside the ovule.

BRYOPHYTE: Any member of the phylum Bryophya that consists of some 23,000 species of relatively small, green, rootless, non-vascular plants, including mosses (Musci), liverworts (Hepacticae) and hornworts (Anthoceratae). Bryophytes grow on damp surfaces exposed to light, including rocks and tree bark, almost everywhere from the Arctic to the Antarctic. The bryophytes display a distinct alternation of sexual and asexual generations; the sexual gametophyte, with a haploid chromosome number, is the more diversified. The spore bearing, diploid sporophyte is reduced in size and structure, attached to the gametophyte and partially or almost completely dependent on it.

BRYOZOAN (POLYZOA): Any member of the phylum Bryozoa (now classified into the phyla Entoprocta and Ectoprocta) forming colonies in seawater and occasionally in fresh or brackish water. They are tiny animals, growing to about 1mm in length, with ciliated tentacles and a hard, box-like, calcium carbonate skeleton. Up to 2 million individuals, called zooids, may be linked to form lacy patterns that occur on the surface of seaweeds.

BUBBLE: 1. A floating spherical or dome-shaped pocket of air or gas formed in a liquid. **Bubbling** is forming or producing spherical or dome-shaped pockets of air or gas in a liquid. 2. Solidified liquid such as glass.

BUBBLE CHAMBER: A device for detecting ionising radiation or for observing the nature and movement of atomic particles and their interaction with radiation. It is a vessel filled with a transparent fluid that is on the verge of boiling (under a pressure and a temperature for which it is on the liquid-gas boundary). For hydrogen this is only a few degrees above absolute zero, -273 Celsius. When an ionising particle passes through a bubble chamber, it initiates a string of bubbles, due to boiling, along its path, which can then be photographed and analysed.



A bubble chamber

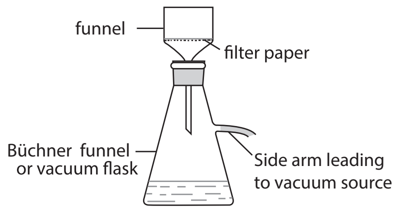
BUBBLE COLUMN: A bioreactor, in the shape of a column, in which the reaction medium is kept mixed and aerated by introduction of air into the bottom.

BUBBLE SORT (SINKING SORT; EXCHANGE SORT): A sorting algorithm that compares adjacent pairs and swaps them to their proper position, if they are in the wrong order, until no swaps are necessary.

BUBBLER: An apparatus used to absorb certain water soluble components in a gas stream for later analysis. Usually it involves the use of a glass fritted tube which forces the air into small bubbles of high surface area during operation.

BUCCAL CAVITY (MOUTH): The beginning of the alimentary canal which leads to the pharynx and in vertebrates to the oesophagus. In mammals, it contains the tongue and teeth, which assist in the mechanical breakdown of food and openings of the salivary duct.

BÜCHNER FUNNEL: A type of funnel used for filtering by suction, consisting of a cylindrical porcelain, glass or plastic filtering funnel that has a perforated plate on which the flat filter paper is placed. It is used together with a Büchner flask, which is a vacuum flask with a side arm underneath the filter to which a vacuum can be applied to allow atmospheric pressure on the sample to force the liquid through the filter paper. It is named after its inventor, Ernst Büchner (1850 – 1924), a German industrial chemist. The Hirsch funnel has a similar design, but used for smaller quantities of material. The main difference is that the plate is much smaller, while the walls of the funnel angle outward instead of being vertical.



Büchner funnel and flask for filtration under reduced pressure

BUCKLING: A state of unstable equilibrium of a thin-walled body when compressive loads are applied on its walls. The resultant deformation may be elastic or permanent. In some cases, it may even lead to collapse of the structure.

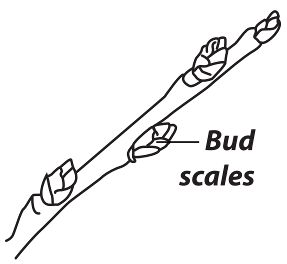
BUCKMINSTERFULLERENE (BUCKYBALL): A spherical hollow cage carbon molecule, with the chemical formula C_{60}, composed of clusters of 60 carbon atoms bonded together in a polyhedral structure composed of pentagons and hexagons. It was first prepared in 1985 by Harold Kroto, James Heath, Sean O'Brien, Robert Curl and Richard Smalley at Rice University for which Kroto, Curl and Smalley were awarded the 1996 Nobel Prize in Chemistry. Buckminsterfullerene was the first fullerene molecule discovered and it is also the most common in terms of natural occurrence, as it can be found in small quantities in soot. It is prepared in helium by passing about 150 amps through a carbon rod and extracting the soot with benzene; the resulting magenta solution contains C_{60} and C_{70}. It was named after the US architect Richard Buckminster Fuller (1895–1983) because of the resemblance of the structure to the geodesic dome, which Fuller invented.

BUCKWHEAT: A herbaceous, erect annual plant and the dry seed or grain, which is used as a source of food and feed. It is not a true cereal and is one of the very few plants, other than those of the Gramineae family, used for their starchy seed, which is processed as a meal or flour.

BUCKYTUBE (NANOTUBE): A fullerene or an allotrope of carbon having a cylindrical or toroidal configuration. It is named buckytube because they were discovered while making Buckyball. They are cylindrical carbon molecules having properties such as excellent strength and unique electrical properties and are efficient thermal conductors making them potentially useful in many applications in nanotechnology, electronics, optics and other fields of materials science, as well as potential uses in architectural fields.

BUD: 1. An undeveloped shoot in plants with a very short stem and numerous undeveloped leaves or flower parts covered with protective scales. 2. The offspring growing on the side of some simple animals or an asexually produced outgrowth from a parent organism that breaks away and develops into a new individual in the process of budding.

BUD SCALES: Modified leaves covering a bud, which act as protection for newly formed leaves and branch growth. In cooler, temperate climates that experience cold and snowy winters, these buds take the form of small, modified leaves that look like miniature scales. The largest of these scaled buds are found at the ends of branches and are called **terminal buds**. The smaller ones, are called **auxiliary buds** and form where the leaf and branch come together.



BUD SPORTS: Buds that produce fruit that is different from the rest of the fruit on the tree; vegetatively propagated by grafting cuttings onto another plant.

BUDDING: 1. A method of asexual reproduction in which a new individual is derived from an outgrowth called a bud that becomes detached from the body of the parent. In other words, it is a term used in horticulture to describe a method of grafting in which a bud of the scion is inserted on the stock. 2. In animals, the process is also called **germination**, common in coelenterates such as Hydra and among fungi. Budding is also characteristic of the yeast.

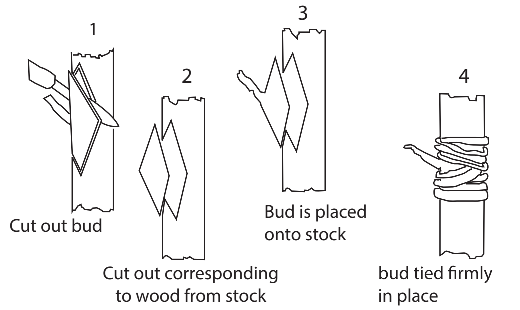


Fig I: ***Budding processes in flowering plants***

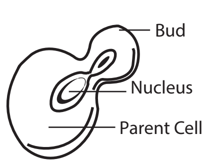


Fig II: ***Budding processes in Yeast***

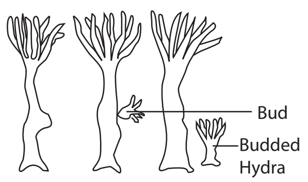


Fig III: ***Budding processes in Hydra***

BUFFER: A data area shared by hardware devices or program processes that operate at different speeds or with different sets of priorities. The buffer allows each device or process to operate without being held up by the other. In order for a buffer to be effective, the size of the buffer and the algorithms for moving data into and out of the buffer need to be considered by the buffer designer.

BUFFER COMPOUNDS/SOLUTIONS: A solution that can resist pH change upon the addition of small amount of acidic or basic components. They are used as a means of maintaining pH at a nearly constant value in a wide variety of chemical applications. Clay, organic matter, and materials such as carbonates and phosphates also enable the soil to resist appreciable change in pH.

BUFFER FLUSH: The transfer of computer data from a temporary storage area to the computer's permanent memory.

BUFFER OVERFLOW: An anomaly that occurs when a program or process tries to store more data in a buffer (temporary data storage area) than it was intended to hold. Since buffers are created to contain a finite amount of data, the extra information, can overflow into adjacent buffers. This may result in erratic program behaviour, including memory access errors, incorrect results, a crash or a breach of system security.

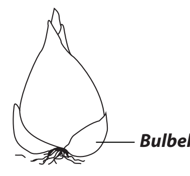
BUFFER SOLUTION: An aqueous solution consisting of a mixture of a weak acid and its conjugate base or a weak base and its conjugate acid that resist changes in the pH, when a small amount of a strong acid or strong base is added to it. Examples are ammonia/ammonium chloride, ethanoic acid/sodium ethanoate, carbonic acid/sodium hydrogen carbonate, etc. Buffer solutions are useful in many chemical applications, where constant pH is required. Many life processes also require constant pH, for example is blood.

BUFFER-ADDITION TECHNIQUE: A technique in which an additive called a spectrochemical buffer is added to both the sample and reference solutions for the purpose of making the measure of the analyte less sensitive to variations in interferent concentration.

BUG: 1. An insect with thickened forewings and mouthparts adapted for piercing and sucking or an insect of similar description, especially one considered to be a pest, for example, an aphid, bedbug, cotton strainer and cockroach. 2. A mild illness caused by an unspecified germ or microorganism. 3. A defect or flaw in a design, machine or system. 4. A concealed electronic device, usually a small microphone that is used for listening to or recording private conversations. 5. An error in computer program or fault in hardware.

BULB: 1. In botany, it is a short underground stem or a modified shoot with a short flattened stem supporting many overlapping, fleshy leaves or containing reserve food supply and with roots growing from its base that enables a plant to survive from one growing season to the next. An example is onion. 2. Also known as **light bulb,** it is a glass bulb containing a gas, such as argon or nitrogen, at low pressure and enclosing a thin metal filament that emits light when an electric current is passed through it.

BULBEL: A small bulb developing from the base of a larger bulb.



BULBIFEROUS: Producing or bearing bulbs or bubils; or of a plant having tiny bulbs, especially if they replace the buds on the stem, as occurs in certain onions.

BULBIL: 1. An aerial, deciduous, fleshy leaf bud, capable of developing into a new individual; or a small bulblike organ of vegetative reproduction growing in leaf axils or on flower stalks of plants such as the onion and tiger lily. 2. Any small bulblike structure in an animal.

BULBLET: A small bulb or a bulblike structure borne above ground, usually in a leaf axil irrespective of origin; or a small bud-like vegetative propagule produced on the leaves of some ferns.

BULBOSE: Any structure or organ, which is bulblike or resembling a bulb.

BULBOURETHRAL GLANDS: Exocrine glands present in the reproductive system of human male. Their function is to secrete a mucus-like substance that is added to sperm and provides lubrication during intercourse. They are homologous to Bartholin's glands in females.

BUILDING BLOCK: A basic unit that combines with others to build up something. For example, the four main building blocks of life are carbohydrates, lipids, proteins, and nucleic acids.

BULGE OF THE EARTH: The extension of the equator caused by the centrifugal force of the Earth's rotation, which slightly flattens the planet's spherical shape. This bulge causes the planes of satellite orbits inclined to the equator (but not polar) to rotate slowly around Earth's axis.

BULK DATA TRANSFER: A software application feature that uses data compression, data blocking and buffering to optimize transfer rates when moving large data files. File Transfer Protocol (FTP) is the most common way to transfer bulk data over the Internet.

BULK DENSITY, SOIL (SOIL BULK DENSITY): The ratio of dry mass of soil to a known bulk soil volume, including both the solids and the pore space. It is expressed in g/cm^3 using undisturbed, oven-dried soil obtained from a core sample with a known volume. The bulk volume is determined before the soil is dried to constant weight at 105 ^{o}C. Soils with a bulk density higher than 1.6 g/cm^3 tend to restrict root growth. Bulk density typically increases with depth since subsurface layers have reduced organic matter, aggregation, and root penetration compared to surface layers and therefore, contain less pore space. Bulk density influences water infiltration and plant root health. Sandy soils are more prone to high bulk density.

BULK FLOW: 1.The movement of organic products or photosynthates in plants from sources to sinks due to the difference in pressure between two locations in the plant as opposed to ascent of water and mineral ions that require forces to be transported up the plant. 2. A process by which proteins with a sorting signal travel to and from different cellular compartments. Proteins often have sorting signals, either transport signals or retention signals, specifying if a protein will translocate to another compartment within the cell or is retained in the current, membrane-bound, compartment in which it is already located respectively. For instance, proteins with the KDEL (Lys-Asp-Glu-Leu) sorting signal are specified to return to the endoplasmic reticulum from the Golgi apparatus. However, proteins lacking a sorting signal will increase in concentration in a specific compartment until it reaches bulk concentration in the donor compartment. At this point, it enters budding vesicles and is transported to an acceptor compartment.

BULK MESOPHASE: A continuous anisotropic phase formed by coalescence of mesophase spheres. Bulk mesophase retains fluidity and is deformable in the temperature range up to about 770 K and transforms into green coke by further loss of hydrogen or low-molecular-weight species. This bulk mesophase can sometimes be formed directly from the isotropic pitch without observation of intermediate spheres.

BULK MODULUS: A numerical constant that describes the elastic properties of a solid or fluid under pressure from all sides; or a numerical constant that measures a substance's resistance to uniform compression, symbol K. It is defined as the pressure increase needed to cause a given relative decrease in volume or the ratio of the compressive force per unit surface area to the change in volume per unit volume of the solid or fluid. It can be expressed as $K = -V\left(\frac{\partial P}{\partial V}\right)$, where P is pressure, V is volume and $\partial P/\partial V$ denotes the partial derivative of pressure with respect to volume. The inverse of the bulk modulus gives a substance's compressibility. Matter that is difficult to compress has a large bulk modulus. Its base unit is the pascal.

BULK POWER SYSTEM (BPS): A large interconnected electrical system made up of generation and transmission facilities and their control systems. A BPS does not include facilities used in the local distribution of electric energy. If a bulk power system is disrupted, the effects are felt in more than one location.

BULK SAMPLE: The sample resulting from the planned aggregation or combination of sample units.

BULK SPECIFIC GRAVITY: The ratio of the bulk density of a soil to the mass of a unit volume of water. It can also be defined as the ratio of the oven-dry weight of the aggregate to the bulk volume of the aggregate particles.

BULK VOLUME: The volume per unit mass of a solid material, including the solids and the pores.

BULL: A male of the cattle family.

BULL LIGHTING: A white or reddish luminous sphere, about 50cm in diameter that sometimes appears at ground level in a thunderstorm. It is slow in movement.

BULLA: The rounded hollow projector of a bone, from the skull, that encloses the middle ear in mammals.

BULLATE: A structure with rounded, blistery projections covering the surface.

BULLDOZER: A kind of tractor with a powerful engine and a curved shovel-like blade at the front which can be lifted and forced down by hydraulic arms. It is an earth moving machine widely used in construction work, for clearing rocks and tree stumps and levelling a site.

BULLET: 1. A small missile usually round or cylindrical with a pointed end, propelled by a firearm, sling or air gun. Usually bullets do not contain explosives, but damage the intended target by impact and penetration. 2. Also known as **bullet point**, it is a small graphical element used to highlight or itemise a list.

BULLET AND NUMBERING: In word, it is a numbering command under the format menu that gives users a number of choices for bulleting, numbering and outlined numbering.

BULLOCK: 1. A castrated bull. 2. A young male ox.

BUMBLE FOOT (ULCERATIVE PODODERMATITIS): A bacterial infection and inflammation of the feet of birds, especially poultry and rodents that end in lameness.

BUMPING: Violent boiling of a liquid caused by superheating so that bubbles are formed at pressures above atmospheric pressure.

BUMPS AND PITS: High and low areas of a compact disc (CD) used to encode data. When data is being written on the smooth side of CDs with a laser beam, it etches bumps called pits into its surface, which represents the number zero, so every time the laser burns a bump into the disc, a zero is stored there. When there is no bump on the surface, it is called a land which represents the number one.

BUNA RUBBER: A type of synthetic rubber based on polymerisation of butadiene (buta-1.3-diene). The name comes from Bu for butadiene and Na for sodium, which was used as a catalyst in the original polymerisation reaction.

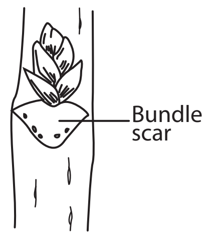
BUNCH-TYPE GRASS: Any grass which grows in clumps or tufts and spread exclusively by the production of tillers. They will form a uniform lawn with sufficient seeding rates. Common cool season lawn grasses with bunch-type growth are perennial ryegrass, tall fescue, hard fescue and chewings fescue.

BUNCHING: 1. In quantum optics, the arrival of photons at a detector grouped more closely together than they would be if grouped randomly. 2. The flow of electrons from cathode to anode of a velocity-modulated tube as a succession of electron groups rather than as a continuous stream.

BUND: A soil wall built across a slope to retain water or to hold waste in a sloping landfill site.

BUNDLE OF HIS (AV BUNDLE; ATRIOVENTRICULAR BUNDLE): The specialised cardiac muscle fibres in the mammalian heart that receive electrical stimuli from the atrioventricular node and transmit them throughout the network of purkyne fibre. This allows the excitation to reach all parts of the ventricles rapidly and initiates a wave of contraction to expel blood into the aorta and pulmonary artery.

BUNDLE SCAR: The remnants on a twig left by vascular bundles when a leaf falls as found on stems of cassava.



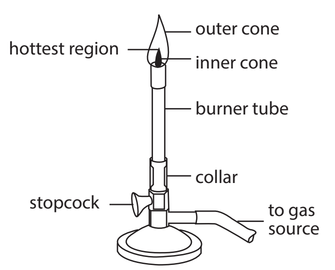
BUNDLE SHEATH CELLS: A layer of cells in plant leaves and stem that forms a sheath surrounding the vascular bundles. It is a part of the transport system in vascular plants and houses the xylem and phloem.

BUNG: A stopper for closing the hole in a barrel or jug or flask, especially one made of cork or rubber.

BUNIAKOVSKI'S INEQUALITY (CAUCHY-SCHWARZ INEQUALITY): The inequality, valid for any inner product, that $\langle x,y\rangle \le \|x\|\|y\|$. In a Euclidean space this can be written as Cauchys inequality.

BUNKER: A container for storing fuel especially on a ship or outside a house.

BUNSEN BURNER: A laboratory gas burner having a vertical metal tube on a base, about 13 cm long through which gas is fed from the cylinder to the burner. A German chemist, Robert Wilhelm Bunsen (1811-99) was credited with its invention and named after him. It has one or more hole (s) in the side of the base which can be regulated with the collar to admit air, which mixes with gas to produce a very hot flame that can reach a temperature of about 1500 °C. The flame has a pale blue primary flame, which is the inner cone and a larger, almost colourless outer cone that results when the remaining gas is completely oxidised by surrounding air. Various types of nozzles can be fitted to the top of the burner to control the flame's shape. The gas used may be natural gas such as methane or a liquefied petroleum gas, such as propane, butane or a mixture of both. It is used in the laboratory for heating, sterilisation, preparing microscopic slides, bending glass tubing and for many other purposes.



Bunsen Burner

BUNSEN CELL: A zinc-carbon primary cell developed by Robert Bunsen (1811-99) in 1843 consisting of a carbon anode in nitric or chromic acid separated by a porous pot from a zinc cathode in dilute sulfuric acid. The cell produces an emf of about 1.9 volts. It was used for powering laboratory instruments, for galvanising different items and powering small electric motors.

BUNSEN, ROBERT WILHELM (1811-99): German chemist, who was a professor at Kassel, Marburg and Heidelberg Universities in Germany. His early researches on arsenic-containing compounds cost him an eye in an explosion. He invented many laboratory tools, including a grease-spot photometer, Bunsen burner, a galvanic battery and an ice calorimeter. Together with Gustav Robert Kirchoff, they invented a spectrometer (1859) that led to his discovery of the elements caesium and rubidium. In 1855, he got the university mechanic, Peter Desaga to build a gas burner that could produce a steady and almost colourless flame for laboratory experiments, which was named the Bunsen burner, which has remained a common laboratory tool, used for heating, sterilisation, preparing microscopic slides, bending glass tubing and for many other purposes. Similar principles had been used in an earlier burner design by Michael Faraday as well as in a device patented in 1856 by the gas engineer, R. W. Elsner.

BUNT: A disease of wheat caused by the smut fungus (*Tilletia controversa*) by converting the interior of the grain into a fetid black powder. At threshing the spores are released and contaminate healthy seed as well as the combine harvester and grain-handling machinery. Contaminated seed in turn gives rise to more infected plants.

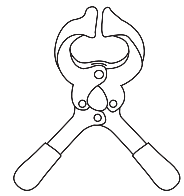
BUOY: A floating object used to mark channels for shipping or warn of hazards to navigation.

BUOYANCY (BUOYANT FORCE): The upthrust or the upward force on an object wholly or partially immersed in a fluid. An object is said to be buoyant if it is hard to sink. The buoyant force F_B acts vertically upward, in opposition to the gravitational force that causes it. Its magnitude is equal to the weight of fluid displaced.

BUOYANT FORCE: The upward force that fluids exert an object. It opposes gravitational force and present whether the object floats or sinks.

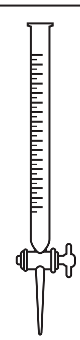
BURALI-FORTI PARADOX: In logic, the paradox that the ordinal of the set of all ordinals must necessarily be larger than any of the members of the set and so would be an ordinal which was not contained in the set of all ordinals, whence the set of all ordinals cannot itself be assigned an ordinal and in fact, the set of all ordinals is not an allowable set. It was named after the Italian mathematician, Cesare Burali-Forti (1861-1931).

BURDIZZO: A plier-like device that is used to crush the spermatic cords of animals being castrated. Once the blood supply to the testicles is lost, testicular necrosis occurs and the testicles shrink, soften and eventually deteriorate completely. The burdizzo is used primarily on farm animals.



An example of a burdizzo castrator

BURETTE (BURET): A piece of apparatus used in titration for the controlled delivery of a measured variable quantity of a liquid. It consists of a long, narrow, transparent, calibrated glass tube fitted with a tap at the bottom leading to a narrow bore exit. It is used to measure out small volumes of liquid accurately. The tube has a scale (usually from 0 – 50 cm^3) that can be accurately read to 0.05 cm^3.



Burette

BURGUNDY MIXTURE: A copper fungicide, similar to but stronger than **Bordeaux mixture** and used especially against rusts. It contains 60 g sodium carbonate and 50 g copper sulfate in 5 litre of water.

BURIED SOIL: Soil profiles that have been covered by a layer of later sediment including alluvial, loessial, or other deposits, usually to a depth of 50 cm or more. Buried soils are important sources of archaeological and environmental information as well as land management practices such as tillage and manuring.

BURN: 1. Destroy, damage, injure or mark an object or something by fire, heat or acid. 2. In computer science, it refers to writing data on a compact disc (CD), digital video disc (DVD) or other recordable disc with a laser. CD-Recordable (CD-R) and CD-Rewritable (CD-RW) are the two most common types of drives that can write CDs, either once (in the case of CD-R) or repeatedly (in the case of CD-RW).

BURN-UP: Induced nuclear transformation of atoms during reactor operation.

BURN-UP FRACTION: The fraction of an initial quantity of a given nuclide that has undergone burn-up.

BURNER: A mechanical device such as Bunsen burner that burns a gas or liquid fuel into a flame in a controlled manner.

BURNING SPLINT TEST: A test for a gas sample such as hydrogen to determine whether it is a combustible gas. The gas is put in a vessel with a stopper. When a burning splint is brought near the vessel and the stopper is partially removed to expose the flame to the gas, it ignites if it is flammable. If it is not flammable, the splint will be extinguished. However, this test cannot be used to identify the gas in a vessel as many other common gases such as carbon dioxide, nitrogen, and argon are not flammable.

BURR (BUR): In botany, it is a type of pseudocarp in which the fruit is surrounded by barbed involucres. Burrs tend to repel some herbivores, in the same way that other spines and prickles do. They also disperse seeds by their hook-like structures that attach the seed or fruiting body to hair and in particular, wool or occasionally feathers. Such plants rely largely on living agents to disperse their seeds. Examples of burrs include burdocks (Arctium) and cocklebur (Xanthium). Often the term refers to any barbed fruit, such as cleavers (*Galium aparine*) and enchanter's nightshade (*Circaea lutetiana*) both of which have hooks borne directly on the pericarp.

BURROW: 1. A hole or tunnel dug by a small animal such as a rabbit and rat, which serves as a living space, shelter or refuge against predators, etc. Burrows are also commonly preserved in the fossil record as a type of trace fossil. 2. To make a hole or passage in, into or under something.

BURSA: A sac or cavity of fibrous connective tissue lined with synovial membrane and filled with synovial fluid. It is found between bones and other tissues, such as skin, ligaments, tendons and muscles. It acts as a cushion between bones and tendons and/or muscles around a joint to reduce friction between the bones and allows free movement. The plural of bursa is **bursae**.

BURST: 1. In data communication, a sequence of data counted as one unit in accordance with some specific criterion or measure. 2. To break open suddenly, often as a result of internal pressure, for example, a seed pod bursting to forcefully disperse its seeds or an artery bursting in the brain to cause stroke. Examples of such plants are okro (*Abelmoschus esculentus*) and the kapok or silk cotton tree (*Ceiba pentandra*) in the mallow family (Malvaceae); Sandbox tree (*Hura crepitans*) and the rubber tree (Hevea spp.) in the Euphorbia family; touch-me-not (Impatiens) in the family Balsaminaceae; Acanthus in the family Acanthaceae; and some plants in the pea family (Leguminosae or Fabaceae).

BURST AND TRANSIENT SOURCE EXPERIMENT (BATSE): One of the four main detectors located on board the Compton Gamma Ray Observatory that detected and located high energy gamma-ray bursts in the sky including observing many other objects between 10 keV and 5 MeV.

BURST PRESSURE: The maximum point at which devices such as a valve or hose will fail as a result of pressure. This is important in selecting the appropriate material in designing systems like a radiator or a pressure vessel that uses pressured systems such as water, gas, and other fluids.

BURST SIZE: The number of phages ejected per infected bacterium by a host cell over the course of its lytic life (phase of the virus life cycle during which the virus replicates within the host cell).

BUS: In computing, it is a subsystem that transfers data between one part of a computer to other computer components. In other words, a bus consists of optical fibres connecting several components of a computer system and allowing the components to send signals to each other.

BUS MOUSE: A mouse that connects to a computer through an expansion board.

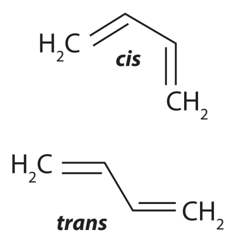
BUSBAR: An electrical conductor held at a constant voltage, used to carry high current between different circuits in a system, especially one that is part of a power distribution system. It usually consists of a thick copper or aluminium strip. The size of the busbar is important in determining the maximum amount of current that can be safely carried. Busbars can have a cross-sectional area of as little as 10 mm² but electrical substations may use busbars of 50 mm in diameter with a cross-sectional area of 1,000 mm² or more. It is used mainly in switchboards, distribution boards, substations or other electrical apparatuses.

BUSH: 1. A low thickly growing plant with crowded branches. 2. A wild uncultivated land.

BUSH BABY: A small nocturnal primate, *Galago senegalensis* about the size of a squirrel that is native to continental Africa, living in woodlands and bushlands. Its generic name is *Galago* or *Galagoides*. It feeds mostly on insects and fruits and has large, round eyes for good night vision and bat-like, delicate ears that enables it to look for prey such as insects in the dark. It produces loud, shrill cries like a human baby, hence its name. Bush babies move about in leaps and bounds, hop like a kangaroo or move on four legs. Their main predators are eagles, owls, hawks and large snakes.

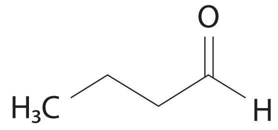
BUSHEL: A unit for measuring dry goods in volumes in the U.S. Customary System. It is equivalent to 35.24 litres. In the British Imperial System, it is equivalent to 36.37 litres and used in dry and liquid measurements.

BUTA-1,3-DIENE (BUTADIENE): A simple conjugated diene with the chemical formula C_4H_6. It is an important industrial chemical used as a monomer in the production of synthetic rubber. It is made by catalytic dehydrogenation of butane from petroleum or natural gas. The name butadiene can also refer to the isomer, 1,2-butadiene, which is a cumulated diene. However, this allene is difficult to prepare and has no industrial significance. It exists in two forms, the cis and trans.



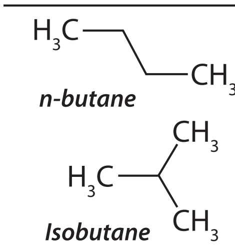
Chemical structure of Buta-1,3-diene

BUTANAL (BUTYRALDEHYDE): A colourless flammable liquid with an acrid smell, miscible with most organic solvents, with the chemical formula C_3H_7CHO, density 0.8 g/ml, melting point -99 °C (174.15 K) and boiling point 75.7 °C (348.85 K). It is the aldehyde analogue of butane as well as an isomer of butanone. It can be produced by catalytic dehydrogenation of 1-butanol, catalytic hydrogenation of crotonaldehyde or via the hydroformylation of propylene. Upon exposure to air, it is oxidised to form butyric acid.



Chemical structure of Butanal

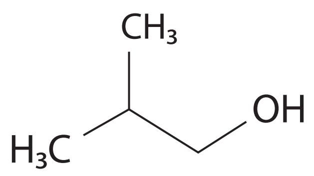
BUTANE: A colourless highly flammable gas obtained by the distillation of crude oil, with the chemical formula C_4H_{10}, density 2.48 mg mL^{-1} (15 °C, 288 K), melting point, -140 to -134 °C (133-139 K) and boiling point -1 to 1 °C (272-274 K). It has two isomers: *n*-butane and isobutane. It is used in refineries both as a fuel and as a starting material for the production of hydrogen and petroleum products. Also, liquefied under pressure, it is used as a fuel for



Chemical structure of Butane

industrial and domestic purposes, refrigerant and aerosol propellant and in the manufacture of synthetic rubber.

BUTANEDIOIC ACID (ETHANE-1,2-DICARBOXYLIC ACID; SUCCNIC ACID): A colourless, crystalline organic compound, with the chemical formula $C_4H_6O_4$, density 1.56 g/cm^3, melting point 184 °C (457.15 K) and boiling point 235 °C (508.15 K). A weak carboxylic acid, it is produced by fermentation of sugar or ammonium tartrate and used as a sequestrant and in making dyes.



Chemical structure of Isobutanol (2-methyl-l-propanol)

BUTANOL (BUTYL ALCOHOL): A colourless liquid alcohol, with the chemical formula C_4H_9OH that exists in four isomeric forms: *n*-Butanol(butan-1-ol), Isobutanol (I-butanol or 2-methylpropan-1-ol), *sec*-Butanol (butan-2-ol) and *tert*-Butanol (2-methylpropan-2-ol).

BUTENE: An unsaturated hydrocarbon containing one double bond and four carbon atoms, with the chemical formula C_4H_8. Butenes are formed during the cracking or breaking down of large molecules of petroleum to produce petrol or gasoline; they can also be prepared commercially by the catalytic dehydrogenation or elimination of hydrogen atoms from the molecule of butanes. There are four isomer forms: 1-butene, *cis*-2-butene, *trans*-2-butene and isobutylene. All four butenes are gases at room temperature and pressure. The most important use of butenes is in the production of octanes, which are important constituents of gasoline.

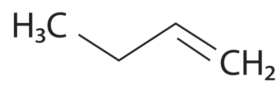


Fig I:
Chemical Structure of 1-butene

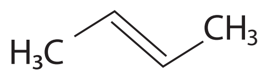


Fig II:
Chemical structure of trans-2-butene

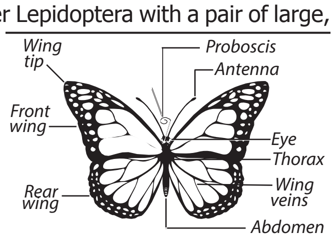
BUTANEDIOL FERMENTATION: A kind of anaerobic fermentation found in Enterobacteriaceae family, in which glucose is fermented and 2,3-butanediol is produced as the main substances.

BUTTE: An isolated hill rising abruptly above the surrounding area and having steep sides and a flat top. Most of the top has been removed by erosion, and it has a smaller summit area than a mesa.

BUTTER: The fatty constituent of milk separated by churning into a soft whitish-yellow food substance for table use and cooking. Butter is a water-in-fat emulsion.

BUTTERFAT: The natural fat of milk from which butter is made, consisting largely of the glycerides of oleic, stearic and palmitic acids.

BUTTERFLY: An insect of the order Lepidoptera with a pair of large, broad and brightly coloured wings at each side of the body (two at the front and another pair at the rear which are attached to the thorax), which are erect when at rest. Its head is spherical which contains a brain, two compound eyes, its proboscis, two antennae, etc. The proboscis is used for sucking liquid foods such as nectar. The antennae are used as senses of smell and balance. It has a segmented abdomen and a thorax that is divided into three segments. Each of the segment has a pair of jointed legs. Most butterflies reproduce through laying eggs and pass through the stages of larvae or caterpillars, pupae and maturity.



External features of a butterfly

BUTTERFLY EFFECTS: In general physics, it is a property of chaotic systems such as the atmosphere, in which small changes in initial conditions can result in far-reaching, large-scale effects on the development of the system. Although this may appear to be an esoteric and unusual behaviour, it is exhibited by very simple systems, for example, a ball placed at the crest of a hill might roll into any of several valleys depending on slight differences in initial position. The butterfly effect is a common trope in fiction when presenting scenarios involving time travel and with "what if" cases where one storyline diverges at the moment of a seemingly minor event resulting in two significantly different outcomes.

BUTTERFLY PEA (*Centrosema pubescens*): A leguminous perennial herb in the family Fabaceae, subfamily Faboideae, native to Central and South America. It is drought resistant and grows to a height of about 40 to 45 centimetres, providing a compact dense cover and suppressing weeds. The root system can reach about 30 centimetres in depth, forming a symbiotic association with a nitrogen-fixing bacteria called Rhizobium. The leaves are trifoliate, pubescent on both surfaces and with an obtuse tip. The leaflets are dark green elliptic or ovate-elliptic, obtuse or shortly obtusely acuminate and slightly hairy, especially on the lower surface. The flowers are pale violet with darker violet veins positioned in axillary racemes. The pods are linear with prominent margins, up to 9 centimetres long and containing up to 20 seeds. It is used as forage for cattle and other animals such as broiler chickens.

BUTTON: 1. A small device on a piece of electrical or electronic equipment which is pressed to operate a door bell, a switch on a machine or any other electrical or electronic device. 2. A knob or disc made of materials such as wood and metal and sewn onto a garment as a fastener or as an ornament.

BUTTRESS: An external structure built against or projecting from a wall which serves to support or reinforce it against the sideways force, called thrust.

BUTTRESS ROOT: Thick woody walls running obliquely between the trunk and the ground around the base of the trunk of many tropical trees such as silk cotton (*Ceibapentandria*) and certain figs (*Ficus*). They are formed by asymmetrical secondary thickening, chiefly on the upper side of the lateral roots. They act as additional support for the trunk.

BUTYL RUBBER (POLYISOBUTYLENE; PIB): A synthetic rubber produced by polymerisation of about 98% of isobutylene with about 2% of isoprene. Its chemical formula is $-(-CH_2-C(CH_3)_2-)_n-$. It is very flexible and highly impermeable. It is used as inner tubes, hose, insulation and as seals for food jars.

BUYS BALLOT'S LAW: A law in meteorology concerning the winds, which states that observers with their back to the wind will experience a lower pressure on the left than on the right in the Northern Hemisphere and the opposite in the Southern Hemisphere. This is because wind travels counterclockwise around low pressure zones in the Northern Hemisphere and is reversed in the Southern Hemisphere.

BY-PRODUCT: A secondary product left from the production of a primary commodity or a substance formed incidentally during the manufacturing of some other substance or something, which is produced in a chemical reaction in addition to the product, which is required. For example, carbon dioxide is a by-product of fermentation process.

BYPASS CAPACITOR (DECOUPLING CAPACITOR): A capacitor providing a low impedance path in an electronic circuit, over a predetermined frequency range. It is used to decouple one part of an electrical network (circuit) from another.

BYPASS INJECTOR: A sample injector by means of which the eluent (carrier gas) may be temporarily diverted through a sample chamber so that the sample is carried to the column.

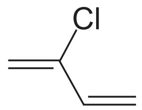
BYRSOCARPUS: A genus of evergreen or deciduous plants in the family *Connaraceae*. Some of the species include *Byrsocarpus coccineus*, commonly known as Shortpod; and *Byrsocarpus puniceus*. *Byrsocarpus coccineus* is a scrambling or climbing shrub or small tree of up to 6 m tall with numerous prominent lenticels. It ranges from Guinea to West Cameroons and some other parts of tropical Africa. It grows in open woodland and bushland complexes, forest margins, thickets fringing water courses and persisting in fire-climax grassland. The flowers are white or pink in racemes, mostly appearing earlier than the foliage and are sweet-lemon-scented. Young leaves are reddish-orange and old leaves are light green. The fruit is an ellipsoid capsule, reddish-brown when ripe, splitting along one side, revealing the seed in a bright orange-red pseudo-aril. The root, bark and leaves are used traditionally to treat several ailments such as muscle pain, rheumatism, piles, inflammation of the pharynx, jaundice, etc.

BYTE: A unit for measuring the size of a computer's memory or the capacity of a disc. It also refers to enough computer memory to store a single character of data such as the number of letters of the alphabet. A single byte can specify 256 values such as the decimal numbers from 0 to 255 and three bytes (24 bits) can specify 16,777,216 values. A group of bits, usually 8, is used to express numbers, letters or instructions in a digital computer.

BYTECODE: An object-oriented programming (OOP) code compiled to run on a virtual machine (VM) instead of a central processing unit (CPU). The VM transforms program code into readable machine language for the CPU because platforms utilise different code interpretation techniques. A VM converts bytecode for platform interoperability, but bytecode is not platform-specific.

C

2-CHLOROBUTA-1,3-DIENE (CHLOROPRENE): A colourless liquid, with the chemical formula C_4H_5Cl, density 0.9598 g/cm^3 (liquid), boiling point 59 °C (332.15 K) and melting point -130 °C (143 K). It is polymerised to make synthetic rubbers such as neoprene. It is obtained from 1,3-butadiene in three steps: (i) chlorination, (ii) isomerisation of part of the product stream and (iii) dehydrochlorination of 3,4-dichloro-1-butene to 2-chloro-1,3-butadiene. Chloroprene is polymerised to form polychloroprene (neoprene), a synthetic rubber used for wire and cable covers, gaskets, automotive parts, adhesives, caulks, flame-resistant cushioning and other applications requiring chemical, oil and weather resistance or high gum strength.



Chemical structure of chloroprene

c: Abbreviation for centi-, which represents one-hundredth. It is normally used in the metric system.

C: 1. A symbol used in the Roman numerals to represent 100. 2. The symbol for degree Celsius, usually written as °C; a scale of temperature in which water freezes at 0° and boils at 100° under standard conditions. 3. A symbol used to represents complex numbers. 4. The symbol for coulomb, the SI unit of electrical charge. 5. A high level general purpose computer programming language that requires less memory than other languages and popular on minicomputers and for personal computer programmers. It is useful for writing fast and efficient systems programs such as operating systems, which control the operations of the computer. 6. The symbol for the element carbon on the periodic table with atomic number 6, mass number 12 with a melting point of 3 500 °C (3 773 K).

C-CELL (PARAFOLLICULAR CELL): Any one of a group of cells in vertebrates that is derived from the terminal of gill pouches. They are incorporated into the thyroid and parathyroid glands, producing and secreting calcitonin.

C-FACTOR (COVER-MANAGEMENT FACTOR): The effect of cropping and management practices on erosion rates that policy makers and farmers can readily apply in order to help reduce soil loss rates. The C-factor is the ratio of the soil loss from land with a specific type of land use/land cover to the soil loss from continuously fallow and tilled land under the same environmental conditions. The C-factor is unitless, with a value between 0 and 1. Some of the land use and management factors that affect soil loss are vegetation, type of crop, and tillage practice.

C-HORIZON: Soil below the B-horizon of a soil profile. It is important for storing and releasing water to the upper layers of the soil.

C-REACTIVE PROTEIN (CRP): An acute-phase protein secreted into blood plasma by the liver in response to inflammation and infection. This response is achieved by binding to the phosphocholine component of lipopolysaccharides in the cell walls of certain bacteria and fungi and thus increasing susceptibility of the target cell to ingestion by phagocytes. This ultimately results in the destruction of pathogens.

CGS SYSTEM OF UNITS (CENTIMETRE-GRAM-SECOND SYSTEM OF UNITS): An outdated system of units using centimetre as the unit of length, the gram as the unit of mass, and the second as the unit of time that also included electrical and mechanical units. For example, the cgs units of energy and force are the dyne and the erg, abampere (abA) for electric current, abcoulomb (abC) for electric charge, abfarad (abF) for electric capacitance, etc. It was introduced by the British Association for the Advancement of Science in 1874 and was immediately adopted by many working scientists. The cgs system was largely replaced by the MKS (metre, kilogram, second) system with base units of the metre, kilogram, second, ampere, kelvin, candela and mole, which has also been replaced by the current International System of Units (SI). The International System of Units (SI) is an extension of MKS (metre, kilogram, second) system with base units of the metre, kilogram, second, ampere, kelvin, candela and mole.

C# (C-SHARP): An object-oriented programming language from Microsoft that aims to combine the computing power of C++ with the programming ease of Visual Basic. C# is based on C++ and contains features similar to those of Java.

C++: An object-oriented programming (OOP) language that is viewed by many as the best language for creating large-scale applications. C++ is a superset of the C language.

C3 PATHWAY (C_3 CARBON FIXATION): The metabolic pathway for carbon fixation followed in the light-independent phase of photosynthesis by most plants. The process converts carbon dioxide and ribulose bisphosphate (RuBP), a 5-carbon sugar into glycerate-3-phosphophate (G3P).

C3 PLANT: Any plant, mostly of the temperate regions that produces, as the first step in photosynthesis, phosphoglyceric acid, which contains three carbon atoms. Examples include wheat, rice and soybeans.They exhibit photorespiration and are relatively inefficient photosynthetically compared to C4 plants. They generally have lower carbon dioxide fixation rates and higher compensation points than C4 plants. C3 plants must however be grown in areas where carbon dioxide (CO_2) concentration is high, temperature and light intensity are moderate and ground water is abundant. In hot dry conditions, C3 photosynthetic efficiency suffers from a process called photorespiration. The stomata are closed to prevent water loss, which results in the rise of molecular oxygen (O_2) level. When this occurs, rubisco reacts with molecular oxygen (O_2) instead of carbon dioxide (CO_2) and leads to photorespiration, which in turn, causes wasteful loss of carbon dioxide (CO_2) in C3 plants.

C3D: A combined hardware/software process that captures a pair of two-dimensional images, objects or scenes and automatically reconstructs them into a digital three-dimensional (3-D) model. In turn, this model can be used to create a virtual representation of the image, object or scene. The C3D process uses a standard PC and camera to produce a realistic photographic model for viewing on a PC at a reconstruction accuracy of 50 microns. C3D can be used with other tools, such as the Virtual Reality Modelling Language (VRML).

C4 PATHWAY (HATCH-SLACK PATHWAY): The metabolic pathway followed in the light-independent phase of photosynthesis in plants, such as sugar cane and maize and in plants that live in arid environments. These plants are known as C4 plants.

C4 PLANT: Any plant that produces, as the first step in photosynthesis, oxaloacetic acid, which contains four carbon atoms. Examples include maize, sugar cane, millet, sorghum, Bermuda grass and many desert plants. C4 plants require 30 molecules of adenosine triphosphate (ATP) and 24 molecules of water to synthesise a molecule of glucose (C3 plants need only 18 molecules of ATP and 12 molecules of water). However, C4 plants produce more glucose for a given leaf area than C3 plants and consequently grow more quickly. They can also continue to photosynthesise at high light intensities and low carbon dioxide concentrations and, most significantly, do not exhibit photorespiration.